



2021 K-Innovation ODA Program with Ghana

Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Centre (1st Year) : Analysis of Technological Innovation Support Programs (TISPs) in Ghana and Sharing Korean Experiences

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Title	Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Centre (1st Year): Analysis of Technological Innovation Support Programs(TISPs) in Ghana and Sharing Korean Experiences
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Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Centre (1st Year) : Analysis of Technological Innovation Support Programs (TISPs) in Ghana and Sharing Korean Experiences

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Executive Summary

The STEPI and MESTI plan to implement the K-Innovation Program for Ghana from 2021 to 2023 under the title “Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Centre.” During the first phase of the project in 2021, the project team has analyzed the Technological Innovation Support Programs (TISPs) as the boundary conditions affecting the establishment and operation of the GIRC Centre, and identified insightful Korean experience-based implications for the government of Ghana.

For the purpose of achieving the aforementioned objectives of the project, the project team dealt with TISPs in nine theme areas including i) tax & finance, ii) human resources, iii) technology development & information, iv) technology transfer & commercialization, v) startup & incubation, vi) intellectual property management, vii) purchase, viii) accreditation & certification), and ix) I-U-R collaboration. The Ghanaian experts analyzed the current state of Ghana’s TISPs based on which the Korean experts presented important Korean experience-based implications for the government of Ghana.

Considering the numerous implications identified in nine theme areas, the successful implementation of the “Ghana Beyond Aid” Initiative, currently being rigorously implemented by the government of Ghana, requires the government of Ghana to focus on the following three factors; i) a vision and strong political will for the realization of ‘STI-led Economic Growth’, ii) all-out efforts to expand government R&D investment and attract R&D investment from the private sector, and iii) instating and disseminating a culture or attitude among government officials promoting “immediate actions once plans are mapped out.”

The entire process of the project was carried out virtually due to travel restrictions caused by the COVID-19 pandemic. Following the preparatory meeting between STEPI experts and their

Ghanian counterparts in March 2021, the project's kick-off meeting was held and the field interview were conducted in April. Then, an interim report was written by the Korean and Ghanian experts from May to August. The Capacity Development Workshop was held in September to share their analyses on the current state of the Ghanian TISPs and Korean experiences. The final report was prepared in October and November and the outcomes of the project was shared and disseminated through the Dissemination Workshop in December.

The project team would like to offer a special thanks to Dr. Kwaku Afriyie, Minister of MESTI, who has provided his trust and support for the team throughout the entire process of the project. Our appreciation also goes to Mr. Oliver Boachie, special advisor to the Minister of MESTI, who has worked in the capacity of the PM, and experts from Heritors People who have supported the project as local consultants. We sincerely hope that this project serves as momentum for promoting STI diplomacy between Korea and Ghana, as well as the bilateral cooperation between STEPI and MESTI.

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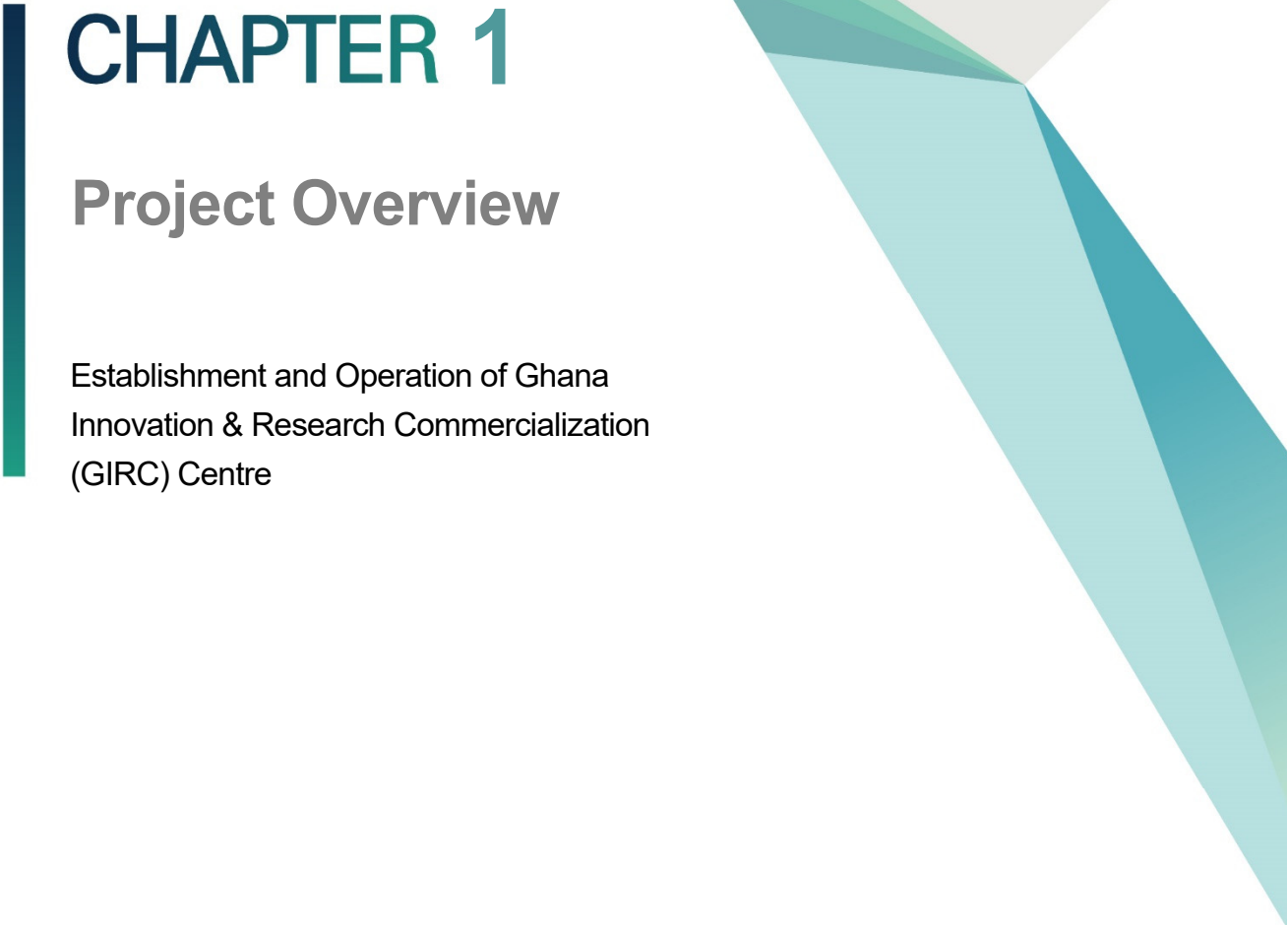
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CHAPTER 1

Project Overview

Establishment and Operation of Ghana
Innovation & Research Commercialization
(GIRC) Centre

Chapter 1. Project Overview

1. Introduction

As of 2020, Ghana's total population was estimated at around 32 million. The country is placed in the rank of the lower-middle income countries (LMICS) with per capita GDP of USD 2,337. Since the inauguration of the new government in 2017, Ghana's economy has grown rapidly at a CAGR of 6.5% in terms of real economic growth rate. The Nana Akufo-Addo government has announced "Ghana, Beyond Aid" as its key national agenda, seeking economic growth without resorting to foreign aid. It is also emphasizing economic growth and job creation by boosting its manufacturing sector. The Ghanaian government has been concentrating its efforts on building physical infrastructures and strengthening human capacity to achieve these goals.

However, Ghana's capacity in some of the key components of its national science, technology, and innovation (STI) system has been extremely limited, such as in such areas as university-industry-research institute cooperation, R&D, technology transfers and commercialization, startup and incubation, and intellectual property (IP) management. To overcome such weaknesses, the Ministry of Environment, Science, Technology & Innovation (hereinafter referred to as 'MESTI') officially announced its plan to establish the Ghana Innovation & Research Commercialization Center (hereinafter referred to as 'GIRC Centre') in July 2018. In 2020, the government of Ghana asked for help from the K-Innovation Partnership Program hosted by the Science and Technology Policy Institute (hereinafter referred to as 'STEPI') out of its aspiration to learn from Korea's development experience in the commercialization of innovative research.

Against this backdrop, STEPI and MESTI finally launched a three-year project (2021-2023) called the 'Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Centre' based on the approval from the two governments. In 2021, the first year of the project, all the project-related activities were performed online due to cross-border travel restrictions amid the COVID-19 pandemic.

2. Project Objectives

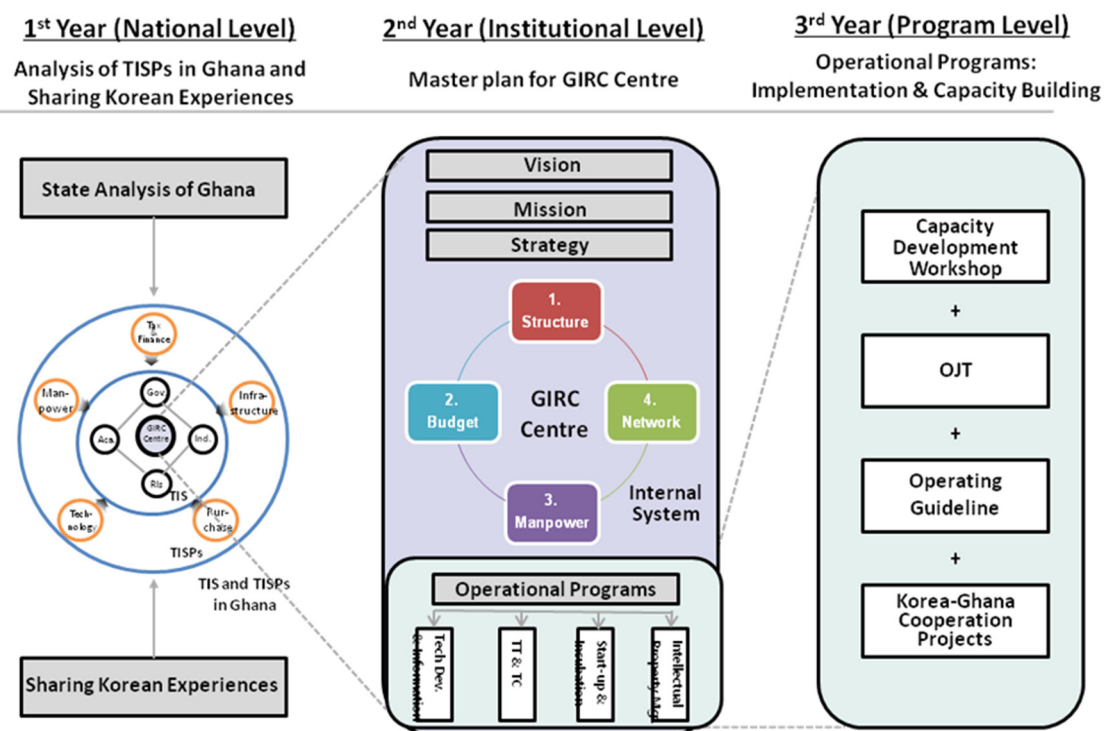
In the process of planning the three-year K-Innovation Partnership Program from 2021 to 2023, STEPI and MESTI have agreed to the following. During the first year, the program will focus on analyzing the current state of Ghana's Technological Innovation Support Programs (TISPs) and sharing Korean experiences. In the second year, the project team will then develop a master plan for establishing and operating GIRC Centre based on the analyses of the first year. For the third year, both parties agreed to concentrate on capacity-building for successful operation of the GIRC Centre.

To be more specific, the 1st year will focus on analyzing Ghana's TISPs at a national level and sharing Korean experiences. TISPs will broadly cover a total of 9 themes including i) tax & finance, ii) human resources, iii) technology development & information, iv) technology transfer & commercialization, v) startup & incubation, vi) intellectual property management, vii) purchase & public procurement, viii) accreditation & certification, and ix) I-U-R (industry-university-research institute) collaboration. Ghanaian experts take responsibility for analyzing the current state of nine TISPs, while Korean experts provide Korean experiences in corresponding areas and their implications for Ghana.

In 2022, the second year of the project, the project team will focus its efforts on developing a master plan for successfully establishing and operating the GIRC Centre at the institutional level. Along these lines, the current state of the GIRC Centre will be analyzed and its future direction will be presented as part of an effort to build its internal system covering its mission, vision, strategy, structure, budget, human resources, and networks. In addition, a roadmap for the operational program by growth stage will also be developed.

In 2023, the third year of the project, a focus will be placed on capacity-building at the program level according to the operational program to be presented as part of the master plan of the second year. For this purpose, the Capacity-Building Workshop will be held, where participants learn fundamental knowledge for their capacity-building. Learning by participation will also be promoted through OJT means (one-the-job training) and drafting

exercises of operating guidelines. Potential Korea-Ghana joint projects will also be explored to facilitate the implementation of the operational programs of the GIRC Centre.



[Figure 1-1] Project Objectives by Year

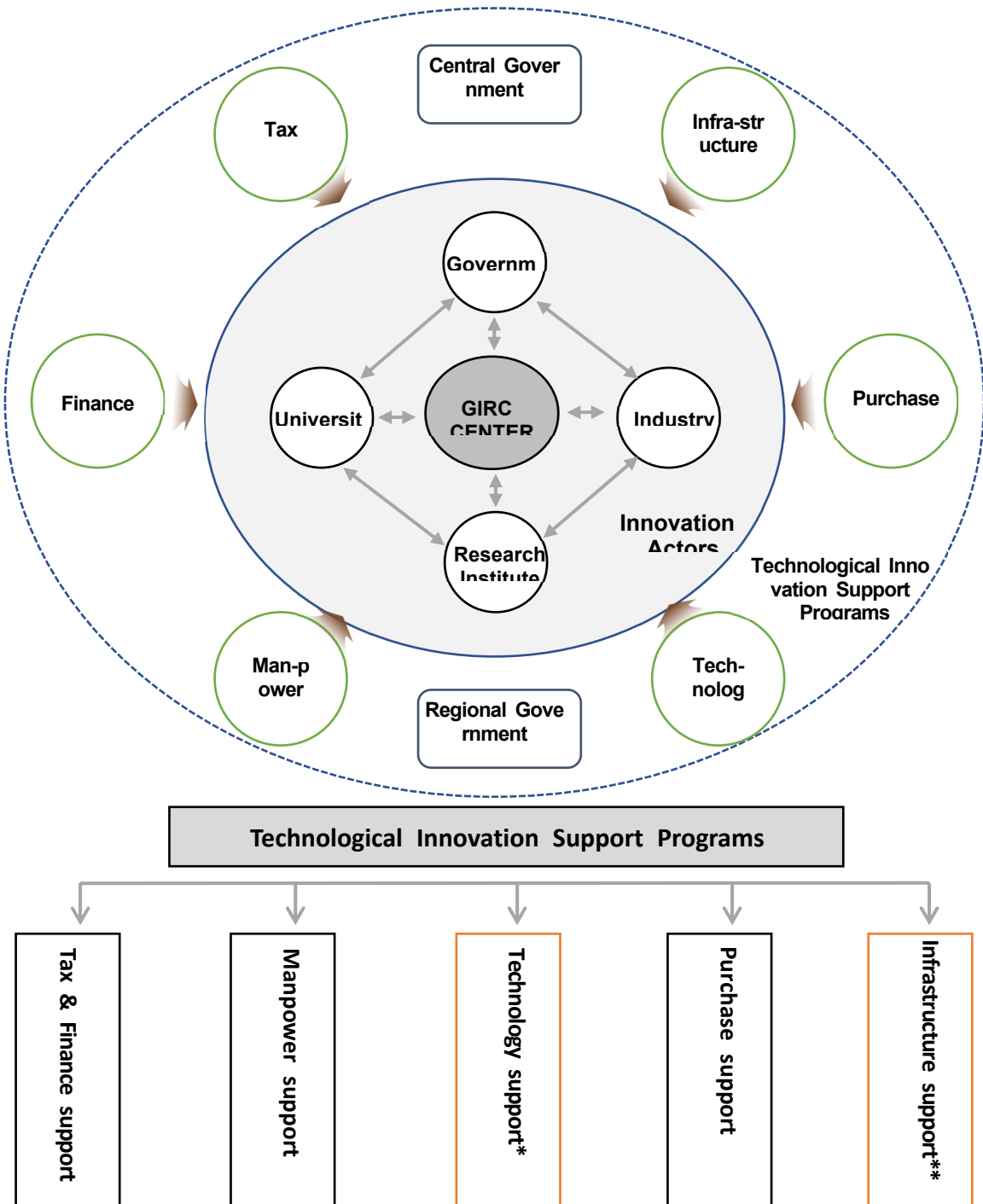
3. Project Framework

A country's competitiveness is often determined by its own technology innovation capacity, which heavily relies on the capacities of its innovation actors, the key constituents of the technology innovation system (hereinafter referred to as 'TIS'). Again, innovation actors' capacities or performances are often determined by the effectiveness of TISPs who support these innovation actors.

The GIRC Centre, a key subject of the present study, forms the core of the Ghanaian TIS, playing a pivotal role as a collaboration hub between universities, industries, research institutes, and the government. In this process, the GIRC Centre's activities are bound to be influenced by various TISPs of the government of Ghana.

During the year 2021, the first year of the project, the project team focuses on carefully reviewing the technology innovation support environment surrounding the GIRC Centre. TISPs are generally categorized into i) tax & finance support, ii) human resources support, iii) technology support, iv) purchase & public procurement support, and v) infrastructure support. The technology support is again classified into i) technology development and information support, ii) technology transfer and commercialization support, iii) startup & incubation support, and iv) intellectual property management support. In the meantime, the infrastructure support is broken down into i) accreditation & certification support and ii) I-U-R collaboration support.

The first year of the project focuses on analyzing nine TISPs affecting the establishment and operation of the GIRC Centre and drawing insightful implications from the analyses. For this purpose, Ghanaian experts will be in charge of analyzing the current state of Ghana's nine TISPs and Korean counterparts will share Korean experience for corresponding themes and present their policy implications for Ghana. The outcomes of the present study are expected to help the Ghanaian government not only design its future TISPs but also explore the operational direction of the GIRC Centre.



* Technology Development & Information, Technology Transfer & Commercialization, Start-up & Incubation, Intellectual Property Management

** Accreditation & Certification, I-U-R linkage

[Figure 1-2] Analytical Framework

4. Project Team

For the first year, the project has been conducted through collaboration between Korean and Ghanaian research teams. Led by Dr. Wangdong Kim, a senior research fellow from STEPI who serves as the project manager (PM) of the project, the Korean research team consists of experts from STEPI and Gyeongbuk Technopark (hereinafter referred to as 'GBTP'). Dr. Myung-Do Oh, Vice-President of the University of Seoul (UOS) kindly offered his support for the research team as a senior advisor. STEPI is in charge of five themes including i) tax & finance, ii) human resources, iii) purchase & public procurement, iv) accreditation & certification, and v) I-U-R collaboration, while GBTP provides research support in four themes such as i) technology development & information, ii) technology transfer & commercialization, iii) startup & incubation, and iv) intellectual property management.

On the Ghanaian side, MESTI has led the Ghanaian research team as the Ghanaian counterpart of the Korean research team. Mr. Oliver Boachie, special advisor to the Minister of MESTI delivered his presentation at the workshop and also provided counseling assistance for the research team. In addition, Heritors People also formed a team of local consultants and provided their support for collecting data and developing the final project report.

5. Main Activities

5.1 Activity 1: Preparation of the Project (January – March 2021)

In January 2021, STEPI appointed the PM of the project and formed the project team. Based on the PCP proposed by the Ghanaian side, STEPI developed a three-year project plan including project activities over three years along with the implementation direction of the first year.

Around mid-March, STEPI also held a video conference with Mr. Oliver Boachie, a counterpart from MESTI, to reach a consensus on the overall direction and key activities of the project plan. In this online meeting, both parties actively reviewed and discussed key contents of the three-year project plan and key activities and implementation procedures of the first-year project plan.

Incorporating the inputs from this meeting, STEPI developed a project proposal for the first year and sent it to MESTI for feedback in late March. Then, the two parties, STEPI and MESTI, agreed on the following schedule of events: i) a kick-off meeting in April or May, ii) the Capacity-Building Workshop in July or August, and iii) the Dissemination Workshop in December.

However, the cross-border travel restrictions amid the COVID-19 pandemic made it impossible for STEPI experts to visit Ghana. Instead, STEPI asked MESTI to recommend a group of local consultants capable of conducting an effective analysis of Ghana's current state. Then, MESTI recommended Heritos People to STEPI, with whom the project team signed a consulting service contract. Heritos People provided support for collecting local data and drafting the final report. It also assisted the project team by circulating the questionnaires for the field interview, prepared by the Korean experts, in advance among Ghanaian interviewees, so that they could be better prepared for the interview.

5.2 Activity 2: Kick-off Workshop and Field Interview (Late April 2021)

On April 29, STEPI and MESTI jointly held the Kick-off Workshop to officially inaugurate the first year of the project. Dr. Myung-Do Oh, Vice-President of the University of Seoul and a senior advisor to the Korean research team, delivered the opening remarks on behalf of the Korean side. On the Ghanaian side, Dr. Kwaku Afriyie, Minister of MESTI, delivered the congratulatory remarks. Then, Dr. Wangdong Kim from STEPI, the PM of the project, presented a project overview. At the Kick-off Workshop, participants finalized the key contents and activities for the first year of the project.

On the same day, a field interview session was also held to elaborate on the current state of Ghana. In the session, Ghanaian experts briefed on the current state of Ghana's TIS and TISPs. Then, Korean experts provided consultations as well as questions on the questionnaire they had sent earlier.

5.3 Activity 3: Writing the Interim Report (May – August 2021)

After the Kick-off Workshop and the field interview, the Korean research team developed an interim report in July by incorporating Korean experiences. The interim report was written based on the report titled 'Questionnaires on Preliminary Findings for Understanding TIS/TISPs in Ghana' prepared by the Ghanaian side. From June to July, the Ghanaian team also wrote up Chapter 2 of the final report based on their analysis of the country's TIS according to the table of contents mapped out by STEPI. The project teams on both sides also prepared materials to be presented at the Capacity Development Workshop based on the contents of the interim report.

5.4 Activity 3: Capacity Development Workshop (September 2021)

To strengthen technology innovation support capacity of the government of Ghana, the Korean research team and MESTI jointly held a three-day workshop titled the Capacity Development Workshop from September 14 to 17. Usually workshops of this kind used to be held offline in Korea, but this workshop was held online due to the COVID-19 pandemic.

The three-day workshop was composed of three sessions. On Day 1, an opening ceremony was held and presentations on nine TISPs were made by Ghanaian experts. On Day 2 and 3, Korean experts shared their experience on nine TISPs and proposed some insightful implications for Ghana.

5.5 Activity 4: Writing the Final Report (October – November 2021)

Once the Capacity Development Workshop was over, the Korean research team developed their version of the final report by incorporating inputs from the workshop. The Ghanaian counterpart also came up with their version of the final report by incorporating Korean experts' comments on Chapter 2 of the interim report and sent it to STEPI. Then, STEPI developed a final report draft by merging the two versions of the Korean and Ghanaian research teams and sent the draft to the Ghanaian side for their feedback. Finally, STEPI came up with the final report incorporating the feedback from the Ghanaian team.

5.6 Activity 5: Dissemination Workshop (December 2021)

In a bid to spread the outcomes of the policy consultation, STEPI held a dissemination workshop in December 16. The workshop was attended not only by the project team members but also by Ghanaian experts from STI related ministries including MESTI, universities, industries, and other relevant organizations in Ghana. At the workshop, the Korean experts presented the outcomes of their year-long research on nine themes. Then, the presentations were followed by questions and comments from the Ghanaian participants who paid careful attention to the presentations given.

5.7 Activity 6: Final Reporting

STEPI wrapped up the final report by incorporating comments to be made in the dissemination workshop and deliver an electronic file of the final report to MESTI by the end of December. Afterwards, STEPI will also send the hard copy of the final report to MESTI.

6. Project Schedule

The following table shows the project schedule over the past year. During this period, project activities have been performed online due to travel restrictions amid the COVID-19 pandemic.

	Activities	Apr. '21	May '21	Jun. '21	Jul. '21	Aug. '21	Sep. '21	Oct. '21	Nov. '21	Dec. '21
Kick-off & field interview (virtual)	Kick-off Seminar									
	Field interview									
Reports on the current status	Ghana, Korea									
Capacity Development Workshop (virtual)	Capacity-Building Workshop									
Draft write-up	Ghana, Korea									
Final report	Dissemination Workshop (virtual)									
	Final reporting									



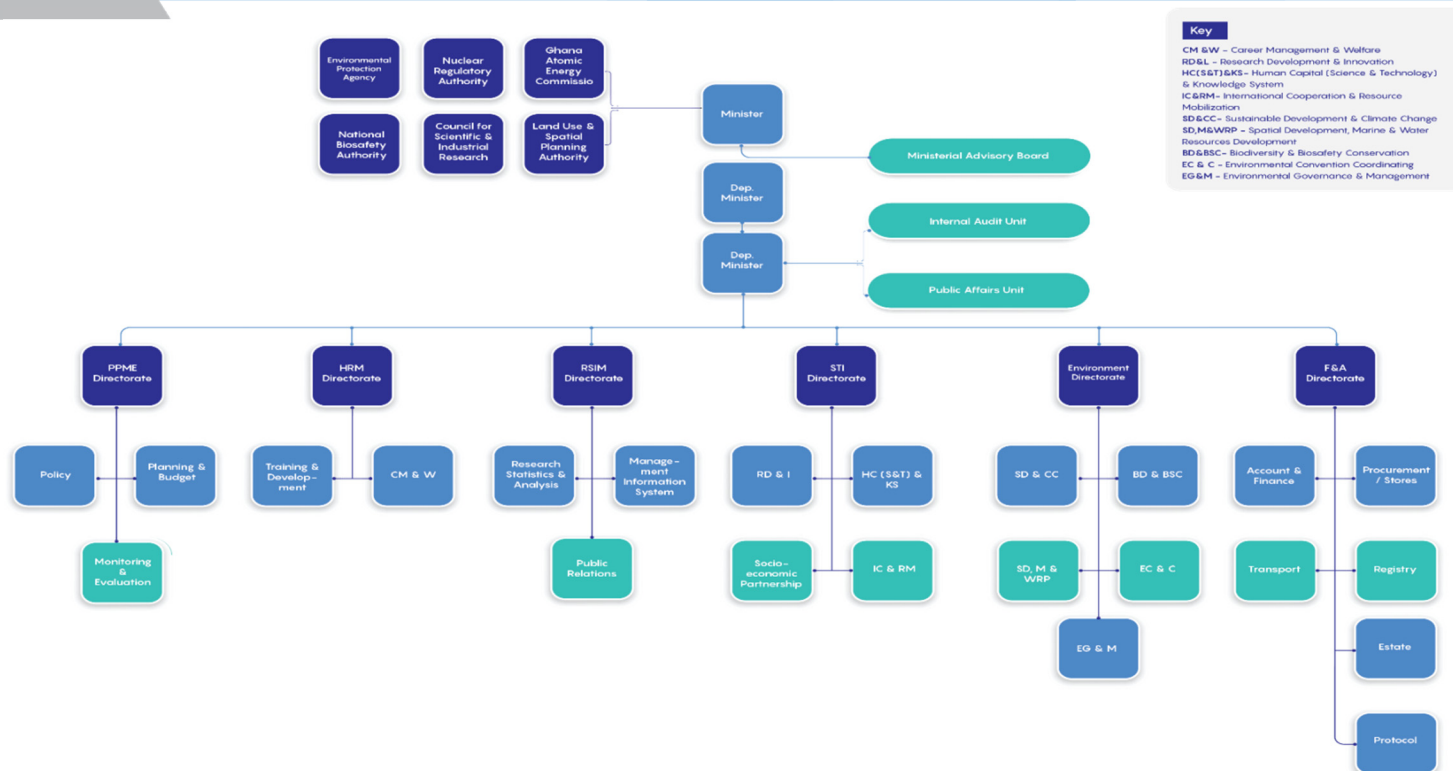
CHAPTER 2

Current Status of Technological Innovation Support Programs in Ghana

Establishment and Operation of Ghana Innovation &
Research Commercialization (GIRC) Centre

Chapter 2. Current Status of Technological Innovation Support Programs in Ghana

1. Legal Framework & Governance in Ghana



[Figure 2-1] STI Governance & Organizational Chart.

1.1 The Processes of Policy Planning, Policy Implementation, Policy Evaluation, Feedback of Government Officials in Relevant Ministries

Policymaking, an essential function of government, is a means by which the government can address critical issues that affect people in society. Over the years, there are ministries, departments, and agencies (MDAs) supporting government in its policymaking mandate, whose policy formulation responsibility is central to achieving overall national development goals and addressing identified issues of national concern. A review of existing public policies developed by the MDAs showed variations in policy formats and structures and thus, it was noted there was the following;

- Use of different processes and procedures;
- Limited stakeholder engagement in some policy formulation processes;
- Weak implementation arrangement;
- Weak policy-ownership;
- Inadequate funding arrangement;
- Policy conflicts, inconsistencies, duplications, and contradictions.

The situation has continued to hampered the effectiveness of some sector policies formulated to address socio-economic challenges and fulfil national development goals and objectives.

So to address these issues in the public policy arena, the National Development Planning Commission (NDPC) developed these public policy guidelines to streamline the process for public policy formulation and approval. This set of guidelines, the first of its kind for public sector institutions at a national level, has sought to standardize public policy formulation and approval processes and procedures, strengthen coordination, collaboration, harmonization, and implementation among key actors, and prevent overlapping duplications, conflicts, and contradictions in public policies.

The steps involved in the preparation of a public policy include the following;

Chapter 2. Current Status of Technological Innovation Support Programs in Ghana



1.2 STI-Related Policy Think-Tanks

A prominent think-tank focused on STI policy is the CSIR-Science and Technology Policy Research Institute (STEPRI).

They conduct knowledge-based research that contributes to the formulation and implementation of policies and programs for socio-economic development. CSIR-STEPRI has been strategically positioned in the policy space to deliver on this mandate. Over the years, they have expertise in agricultural and food policies, climate change and innovation studies, advocacy, and policy engagements. CSIR-STEPRI also deals with the facilitation and organization of international and local workshops.

As stated prior, the Council for Scientific and Industrial Research (CSIR) has thirteen research institutes. STEPRI, which stands for the Science and Technology Policy Research Institute, is one of these thirteen. The Institute was established in 1988 through a UNDP project to address policy gaps relating to the transfer of technology mainly from developed to developing countries. It was named the Technology Transfer Centre (TTC) to reflect the core mandate of the Institute, and then the Overtime Policy Research and Strategic Planning Institute (PORSPI) replaced the name TTC in 1992 to reflect an expected order covering other relevant research themes. Finally, in 1994, the current name was adopted to underscore the diversifying nature of its program activities.

2. Tax and Finance Support Programs in

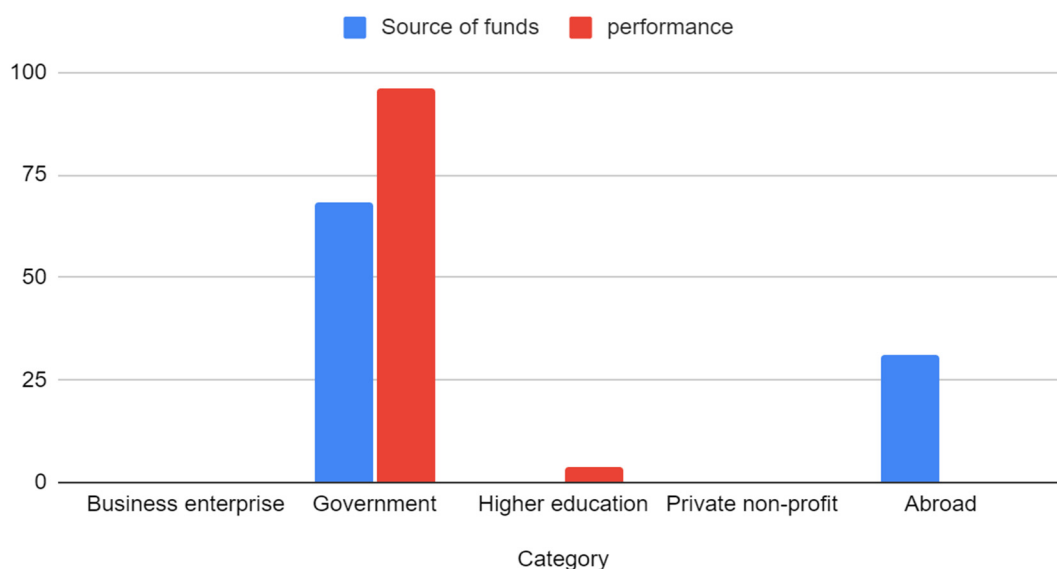
2.1 R&D Investment/ Performances

According to the UNESCO Statistical Institute, the Gross Domestic Expenditure on Research and Development (GERD) in Ghana was 277 million (in PPP\$) as of 2010, representing a GERD per Capita of 11.20 (in PPP\$). GERD was 0.38 percent in 2010 as a percentage of GDP, increasing from 0.23 percent recorded in 2007. The source of funds for GERD was primarily attributed to the government, which accounted for 68.3 percent, whilst funds from outside Ghana contributed 31.2 percent. The government accounted for most of the expenditures at 96.05 percent in performance, with higher education recording a distant 3.8 percent.

[Table 2-1] GERD in Ghana, 2010 (UNESCO Institute of Statistics, April 2015)

	GERD (% of GDP)	GERD per capita (current PPP\$)	GERD per researcher (HC) in current PPP\$ thousands	GERD by source of funds (%), 2010				
				Business	Government	Higher education	Private nonprofit	Abroad
Ghana	0.38	11.3	108.0	0.1	68.3	0.3	0.1	31.2

GERD in Ghana - Source of funds and Performance, 2010



[Figure 2-2] GERD in Ghana (UNESCO Institute of Statistics)

Ghana does not seem to have a dedicated national policy on research and development. However, through MESTI, the government developed the Science, Technology, and Innovation (STI) Policy (2017 to 2020). The policy aimed to build a strong STI capacity to drive social and economic development for the sustainable transformation of the economy. In addition, there are also government-supported institutional policies within specific organizations to promote research and development and create an empowering environment for innovation, with the CSIR being one such institution.

The Council for Scientific and Industrial Research (CSIR) is Ghana's pioneer national science and technology institution. It is mandated to carry out scientific and technological research for national development. The CSIR boasts 13 institutes, which are namely:

- Animal Research Institute Building and Road Research Institute
- Crops Research Institute
- Food Research Institute

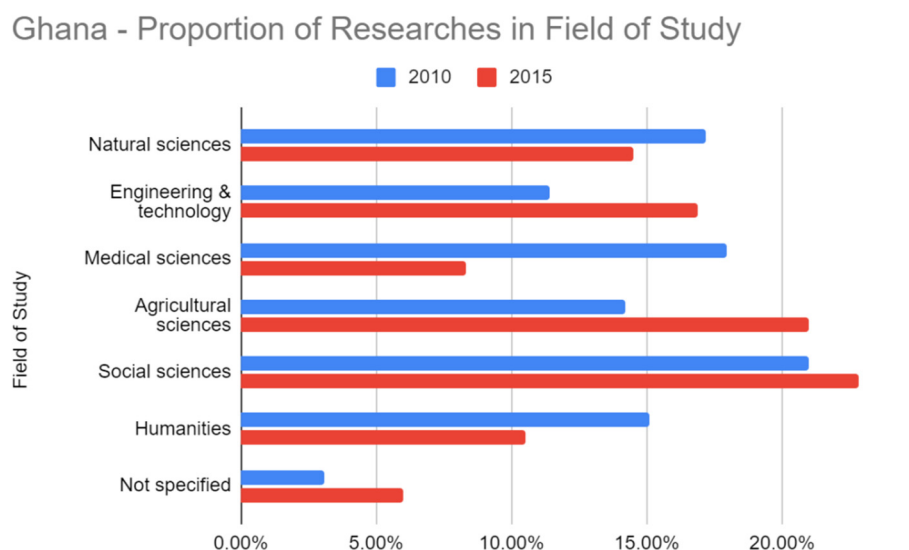
- Forestry Research Institute of Ghana
- Institute of Industrial Research
- Institute for Scientific and Technological Information
- Oil Palm Research Institute
- Plant Genetic Resources Research Institute
- Savanna Agricultural Research Institute
- Science and Technology
- Policy Research Institute
- Soil Research Institute
- Water Research Institute

As a whole, the CSIR operates research and development programs in 7 overall themes as follows:

- Food Security and Poverty Reduction
- Climate Change
- Environmental Conservation and Green Technology
- Material Science and Manufacturing
- Energy and Petroleum
- Biomedical and Public Health; Electronic and ICT;
- Science and People.

2.2 Overview Of Major R&D Performances

Ghana recorded a total number of 5,578 researchers in 2015 compared to 2,542 in 2010, a 119 percent increase. For both years, social sciences accounted for the majority of researchers (21.01% in 2010; 22.84% in 2015).



[Figure 2-3] Proportion of Research in Field of Study (UNESCO Institute of Statistics)

2.2.1 R&D Publications

Concerning publications in international journals, Ghana recorded an average citation of 0.96 between 2008 to 2012 compared to 0.57 recorded for Nigeria in the same period. The output volume from Ghana has steadily increased from 210 publications in 2008 to 579 publications in 2014. Ghana ranks second in West Africa to Nigeria, which recorded 1,961 publications in 2014. Cumulatively between 2008 to 2014, publications in Medical Sciences constituted the majority (782), followed by publications in Biological Sciences (648), with the least being in Psychology (4). There were 160 publications recorded in Engineering. The Research and development (R&D) indicators generally show an increasing trend. The number of researchers in R&D was 38.68 per million people in 2010, increasing from 17.41 per million people in 2007.

Total direct trademark applications recorded in Ghana tallied 3,596 in 2016, rising from the previous year to 3,464. The average number of applications from 1963 to 2016 tallied 562, with the highest number of applications occurring in 2014 at 4,213. However, high technology exports experienced a decline in 2016, recording USD13.89 million compared to USD 59.68 million in 2013. The share of high technology exports to exports was 1.1 percent in 2019

[Table 2-2] Research and Development (R&D) Indicators (<https://www.ceicdata.com> | Worldbank)

PERFORMANCE INDICATORS	LAST	FREQUENCY	RANGE
Research and development expenditure: % of GDP (%)	▲ 0.38	yearly	2007 - 2010
Scientific and technical journal articles (unit)	▲ 982.10	yearly	2003 - 2016
Trademark applications: Direct nonresident (unit)	▲ 2696.00	yearly	1964 - 2016
Trademark applications: Direct resident (unit)	▲ 900.00	yearly	1964 - 2016
Trademark applications: Total (unit)	▲ 3596.00	yearly	1963 - 2016
High-technology exports (USD mn)	▼ 13.89	yearly	1992 - 2016
High-technology exports: % of manufactured exports (%)	▼ 1.51	yearly	1992 - 2016
(DC)Patent applications: Nonresidents (unit)	▼ 17	yearly	1963 - 2016
Researchers in R&D: per million people (Ratio)	▲ 38.68	yearly	2007 - 2010
Technicians in R&D: per million people (Ratio)	▲ 30.08	yearly	2007 - 2010
Industrial design applications: Nonresident (unit)	▲ 116.00	yearly	2008 - 2016
Industrial design applications: Resident (unit)	▲ 453.00	yearly	2010 - 2016
Trademark applications: Nonresident (unit)	▲ 4619.00	yearly	2008 - 2016
Trademark applications: Resident (unit)	▲ 900.00	yearly	2015 - 2016

2.3 Tax Incentives Support Programs

2.3.1 Tax Deductions for Research & HR Development Costs

The 2015 Income Tax Act (Act 896) provides the treatment of Research and Development (R&D) expenses for the determination of assessable income from a business. Section 13 of the Act states that R&D expenses that meet the residual deduction rule ('wholly, exclusively, and necessarily incurred expense' rule) are fully deductible whether or not the R&D expense is of a capital nature. Under the Act, 'research and development expenses' are defined as:

- Including an expense incurred by a person in the process of developing the business of that person and improving business products or processes; and
- Excluding an expense incurred otherwise included in the cost of an asset used in the process referred to in paragraph.

2.3.2 Special Treatment in Taxation on Technology Transfer Income

The regulatory framework that governs a technology transfer in Ghana are as follows:

- The 2013 Ghana Investment Promotion Centre Act (Act 865)
- The 1992 Technology Transfer Regulations (LI 1547)

Under the 2013 Ghana Investment Promotion Centre Act (Act 865), all technology transfer agreements (TTA) must be registered with the Ghana Investment Promotion Centre (GIPC), the government agency responsible for attracting and promoting foreign investments in Ghana. Thus, the GIPC, as part of its mandate, is responsible for registering, recording, and monitoring all technology transfer agreements in Ghana. Act 865 categorizes transfer agreements into four (4) primary forms:

- Agreements covering industrial property rights
- Agreements for the provision of technical services and assistance

- Agreements covering the transfer know-how
- Agreements for the provision of management services

Per Act 865, the TTAs must have a tenure of not more than ten (10) years, not less than eighteen (18) months. There cannot be a renewal for a period exceeding five (5) years.

LI 1547 provides the range of fees that can be charged under a TTA as follows:

[Table 2-3] Allowable Fees for TTA

Technology transfer category	Allowable fees
Royalties	0-6% of transferee's net sales
Technical services and know-how	0-5% of transferee's net sales
Technical services only	0-3% of transferee's net sales
Know-how only	0-2% of transferee's net sales
Management services	0-2% of Transferee's Profit before tax
Technical services, know-how & management services	Not more than 8% of transferee's net sales

The expenses relating to technology transfer agreements generally attract a withholding tax of 20 percent. These expenses are also fully deductible when arriving at tax profit for the year. The deductibility of costs is, however, only granted if the TTA is registered with the Ghana Investment Promotion Council.

2.4 Financial Support Policies

The Ghana Export-Import Bank (GEXIM) was established by the 2016 Ghana Export-Import Bank Act (Act 911). With the mission 'to facilitate the transformation of Ghana's economy into an export one by supporting and developing trade between Ghana and other countries, overseas investments by Ghanaian companies and eliminate critical market failures in the

Ghanaian economy, thereby making Ghana competitive in the global marketplace'. GEXIM was formed from the merger of 3 governmental agencies, namely the Export, Trade, Agricultural and Industrial Development Fund, Export Finance Company, and Eximguaranty Company Limited.

2.4.1 Policy Financing Supporting System(s)

Research and development falls under the overall STI policy funded through the Consolidated Fund overseen by the Ministry of Finance. In addition, there are also the development finance and international public-interest institutions, such as the World Bank, DANIDA, and USAID, which fund STI activity in Ghana through various research programs and projects that they implement in different sectors of the economy. Over the past decade, STI funding per annum has been between 0.25 to 0.3 percent of GDP, below the African Union target of 1 percent¹. Hence, to further support their activities, some STI institutions receive grants and donations representing about 20 percent of their annual budgets. However, some STI institutions generate revenue internally to fund their programs and projects, representing about 10 to 30 percent of their yearly budget.

The government of Ghana has proposed establishing a national STI Fund to raise STI funding to 1 percent of GDP.

2.4.2 Direct Financing (Investments to VC, angel, PEF, etc.) or Indirect Financing (Loan, credit guarantees, etc.)

The private sector, led by commercial banks, savings and loans companies, finance houses, and microfinance institutions, drives Ghana's finance support. Banks lead in the provision of formal capital. Of the USD 32 billion of capital on the books of formal financial sector institutions in 2019, Universal Banks accounted for 70 percent. Non-bank financial

¹ Mapping of Science, Technology, & Innovation Policy Development in Ghana Using the Transformative Change Lens, Quaye et al, 2019

institutions, microfinance institutions and investment funds combined for ~USD 4 billion in total capital assets focused on Ghana², while these patterns are similar to lending.

One of the indigenous players, Oasis Capital, closed a USD 50 million SME fund in 2018 with investors including DGGF, IFC, EIB, Norfund, and PROPARCO. Ghana saw less than 55 private equity deals and 20 recorded venture capital deals between 2014 and 2019.

Formal angel investing is relatively new in Ghana, and there is limited information about its impact on financing SMEs. The investment sizes typically financed by angel investors are low and often agnostic with sector focus. The Ghana Angel Investor Network (GAIN) is a network of Angel investors passionate about investing in early-stage businesses.

2.4.3 The Presence Of a Technology Guarantee System

Ghana has legislation protecting intellectual property (IP), while being a member of the World Intellectual Property Organization (WIPO) and signatory to several WIPO-administered treaties. Accordingly, Ghana's investment laws guarantee that investors can transfer the following transactions in convertible currency out of Ghana including:

- Dividends or net profits attributable to an investment;
- Loan service payments obtained by a foreign loan;
- Charges concerning technology transfer agreements registered under the 2013 Ghana Investment Promotion Act (Act 865)

Ghana is a signatory to the World Bank's Multilateral Investment Guarantee Agency (MIGA) Convention, ensuring insurance against non-commercial risks

² ANDE Financial Constraints to Entrepreneurs in Ghana, 2021

3. Human Resources and Purchase Support Programs in Ghana

3.1 Human Resources Support Programs

Soon after independence (1957), the Ghanaian government built several scientific institutions to help increase the national capacity for STIs. The Ghana Academy of Arts and Sciences (GAAS), Council for Scientific and Industrial Research (CSIR), the Ghana Atomic Energy Commission (GAEC), and the Kwame Nkrumah University of Science and Technology (KNUST) are important institutions that the government set up to address the fundamental challenge of building a scientific and technological base for the socio-economic development programs of the country.

Concerning STI in Ghana, the Ministry of Environment, Science, Technology, and Innovation (MESTI) has, over the years, held the primary responsibility for the Science, Technology, and Innovation (STI) policy. MESTI seeks to ensure accelerated socio-economic development of the nation by formulating sound policies and a regulatory framework to promote appropriate eco-friendly, scientific, and technological practices.

Some institutions that have played significant roles in growing the STI capacity of Ghana over the years include as follows;

The Ghana Atomic Energy Commission (GAEC) has three institutes: the Radiation Protection Institute (RPI), the Biotechnology and Nuclear Agricultural Research Institute (BNARI), and the National Nuclear Research Institute (NNRI). Their operations over the years are dedicated to nuclear research for agriculture, health, and energy purposes. The NNRI employs about 60 scientists, engineers, and 15 technicians. Its primary funding source has been the International Atomic Energy Agency (IAEA), which provides about USD 500,000 per year. In addition, the government of Ghana contributes funds for salaries and some of the running costs.

In Ghana, the Noguchi Memorial Institute has been one of the leading medical research institutes for medical research. A semi-autonomous institute established in 1979 as an integral member of the College of Health Sciences of the University of Ghana. The government of Japan built and donated the Institute to Ghana, which has still been the leading biomedical research centre in Ghana, specializing in communicable diseases and nutrition.

Ghana's human resources support policy expenditure is guided by the Ministry of Employment and Labor Relations version of the medium term expenditure framework for 2019 to 2022. MTEF spells out how much the government is spending and will spend by 2022 on human resources development.

The NMTDPF contains eight (8) policy objectives linked with the sustainable development goals that are vital to the Ministry for Employment and Labour Relations

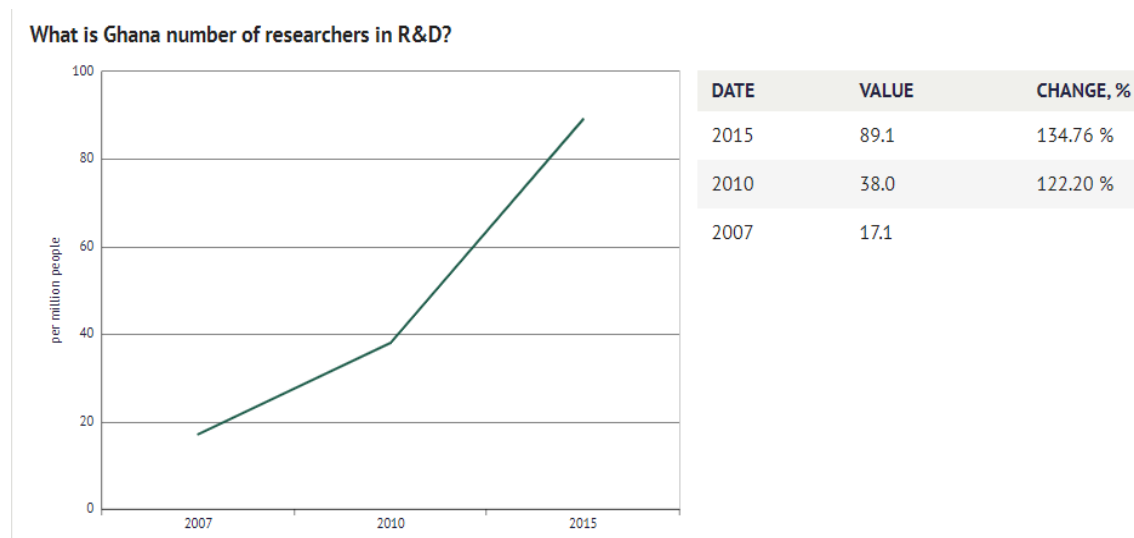
- Ensure improved skills development for the industry
- Achieve gender equality and equity in political, social, and economic development systems and outcomes
- Exploit the benefits of migration for socio-economic development
- Ensure competitive pensions for beneficiaries
- Promote the involvement of PWDs in social and economic development
- Improve human capital development and management
- Promote the creation of competitive jobs
- Promote efficient participation of the youth in socio-economic development

3.1.1 Researchers Per Million Inhabitants

According to the World Bank, Ghana boasts 38.4 researchers in R&D per million inhabitants, lower than the average for Sub-Saharan Africa (95.1). Furthermore, most of Ghana's

research personnel are employed in the higher education sector (59.9%), 38.3% of researchers work in government, and just 1.8% of researchers are employed in business enterprises or the private non-profit sector.

In 2015 reports indicated that the number of researchers in R&D for Ghana was 89.1 per million people.



Source: KNOEMA webpage³

[Figure 2-4] No. of Researchers in R&D

■ WOMEN AS A SHARE OF TOTAL RESEARCHERS

According to the World Bank, just 17 percent of Ghana's researchers are female compared to the African average of 31.6 percent. Through the Western and Central Africa Higher Education Centres of Excellence Project (ACE), the World Bank has sought to address this issue by improving access to funding for female researchers.

³ <https://knoema.com/atlas/Ghana/Number-of-researchers-in-RandD> (Accessed 1st July, 2021.)

■ NUMBER OF PHD, M/S, B.A IN SCIENCE & ENGINEERING SECTORS

According to the World Bank, 34.4 percent of Ghana's researchers have an education up to a PhD or equivalent. As it was previously relatively lower, much of the drive behind the rapid increase of PhD graduates is the ministerial directive that by 2020 lecturers will be required to hold a PhDs to teach at university.

3.1.2 Policies For Nurturing Technicians And Skilled Engineers/High Caliber Scientists And Engineers

There are several initiatives to provide skills development for technicians. These include as follows:

- The Ghana Skills and Technology Development Project (GSTDP): To achieve improved in-demand skills development and increased adoption of new technologies in selected economic sectors.
- Support to Private Sector Development Project (SPSD-II): To support the coordination of different efforts related to national skills development.
- The Development of Skills and Industry Project (DSIP): Support the development of high-quality middle-level technical and vocational skills needed in the Ghanaian economy. It has the following components:
 - Expanding equitable access to TVET
 - Building human and institutional capacity for TVET
 - Improving quality and relevance of (TVET) delivery
 - Technical & vocational education & training voucher program (KfW=Euros 10 million):
 - A project under Ghanaian – German Finance Cooperation Co-financed by the German government through KfW and implemented by the Council (now Commission) for Technical and Vocational Education and Training (COTVET) provides in-demand training

vouchers COTVET-registered master craft persons, apprentices, and workers. So far, there have been about 14,045 beneficiaries. Out of them, 3,625 are males, and 10,420 females. Additionally, stakeholders contracted across the nation include one hundred and sixty (160) centres, sixty-eight (68) trade associations, and one awarding body, Ghana Education, which conducts the technical examination.

- (GET FUND: GHC 3 million from 2011 until the date to provide the youth with the opportunity to gain skillsets in key trade areas leading to improvement in their livelihoods; Provide a pathway leading to self-employment and industrial linkages.
- The Ghana Skills Development Initiative (GSDI) - to build the capacity of the informal sector by improving the traditional apprenticeship system. Helps CBT roll out in the informal sector with in-demand courses for both MCPs and apprentices.

3.1.3 STI-Related Vocational Schools/Colleges/Universities

Some public universities and polytechnics train scientists and engineers with the most notable institutions being the University of Ghana (UG) and Kwame Nkrumah University of Science and Technology (KNUST). In addition, the government established the Commission for Technical Vocational Education and Training that regulates, promotes, and administers technical and vocational training in Ghana. The Commission for Technical and Vocational Education and Training (CTVET) was established by the 2020 Act of Parliament (Act 1023) to regulate, promote, and administer technical and vocational education and training.

There are the following;

- 2 major public research bodies- CSIR -13 institutes and GAEC -6 atomic and nuclear institutes
- 1 Ghana Space Science and Technology Institute under GAEC
- 1 Cocoa Research Institute of Ghana (CRIG)
- 2 medical research institutes- the Noguchi Medical Research Institute and Center for

Scientific Research in Plants Medicine (CSRPM), Mampong

- The GRATIS Foundation
- Ghana Standards Authority
- The Food and Drugs Board
- 10 public universities
- 8 technical universities
- 2 polytechnics
- 38 colleges of education and 8 technical institutes
- Approximately 6 out of the 60+ accredited private universities are into one form of Science, Technology, Engineering, and Mathematics (STEM) related fields

3.2 Purchase Support Programs

The primary legislation that regulates and governs public procurement in Ghana is the 2003 Public Procurement Act (Act No. 663) (the Act) as amended by the 2016 Public Procurement (Amendment) Act (Act914) (the Amendment Act).

All government agencies, institutions, and establishments in which the government has a majority interest are mandated to comply with the Act. The Public Procurement Authority (PPA) is required to ensure that public procurement is conducted fairly and transparently without discrimination.

Ghana has recently launched the Ghana Electronic Procurement System (GHANEPS), a web-based, collaborative system developed following public procurement laws to facilitate general procurement processes. GHANEPS offers a secure, interactive, dynamic environment for procuring all categories, complexity, or value.

It supports processes of procuring goods, works, consultancy, non-consultancy and disposal of assets. It also supports public procurement procedures, including user registration, tender notification, tender preparation and submission, online tender evaluation, contract awarding, creation and management of catalog, creation, and management of framework agreements, auctions and payments.

PPA aims to roll out GHANEPS to all public entities in Ghana by 2023. At present, there are about 197 entities live on GHANEPS. Over 106 training sessions have been organized for about 1,576 procurement officers, 1,307 management and other essential entity staff, over 19,000 suppliers, and about 43 entities have already uploaded their annual procurement plan for the 2021 fiscal year. Five institutions, including; the Ghana Health Service, Tema Metropolitan Assembly, Ghana Cocoa Board, and Department of Feeder Roads, were selected to pilot the usage of GHANEPS. In addition, several other government systems such as the Registrar General's Department, Ghana Revenue Authority, and SSNIT are now running their procurement activities through GHANEPS.

The answer on the likelihood of using products made through R&D in public procurement is very likely. If Ghana as a country can supply what it needs locally, it will increase internal revenue. In Ghana, public procurement represents about 24 percent of total imports, and apart from personal emoluments, public procurement represents 50 to 70 percent of the national budget and 14 percent of gross domestic product (GDP).

The One District-One Factory Initiative has supported the private sector to establish medium and large-scale projects at the district level throughout the country. As of the end of 2020, 232 projects were at various stages of implementation, out of which 76 are currently in operation. In addition, 107 are under construction, whilst 49 are ready to commence construction in 2021. The impact in terms of jobs has been significant. A total of 139,331 direct and indirect jobs were created in 2020, and an additional 285,915 direct and indirect jobs are projected to arise from the projects under construction.

SRN	Ministries / Departments & Agencies	Goods and Services	Capital Expenditure
1	Office of Government Machinery	-	275,882,164
	<i>o/w Special Development Initiatives</i>		275,882,164
2	Ministry of Finance	2,350,375	20,000
	<i>o/w Public Interest and Accountability Committee</i>	2,350,375	20,000
3	Ministry of Local Government, Decentralisation and Rural Development	-	20,000,000
	<i>o/w Regional Reorganisation and Development</i>		20,000,000
4	Ministry of Food and Agriculture	-	8,000,000
	<i>o/w Planting for Food and Jobs</i>		8,000,000
5	Ministry of Fisheries and Aquaculture Development	-	2,000,000
	<i>o/w MoFAD Hqtrs and Agencies</i>		2,000,000
6	Ministry of Lands and Natural Resources	10,000,000	12,000,000
	<i>o/w Ghana Integrated Iron and Steel Development Corp.</i>	10,000,000	12,000,000
7	Ministry of Trade and Industry	-	17,214,681
	<i>o/w One District One Factory Programme</i>		17,214,681
8	Ministry of Energy		8,000,000
9	Ministry of Water Resources and Sanitation	-	14,962,394
	<i>o/w Water and Sanitation Initiative</i>		14,962,394
10	Ministry of Roads and Highways	-	794,897,439
	<i>o/w Roads Infrastructure</i>		794,897,439
11	Ministry of Railways Development	-	120,000,000
	<i>o/w Railways Development</i>		120,000,000
12	Ministry of Education	763,180,024	13,000,000
	<i>o/w Free Senior High School Programme</i>	763,180,024	
	<i>o/w Construction of Regional STEAM Centres</i>		13,000,000
13	Ministry of Health	-	32,425,000
	<i>o/w Health Infrastructure</i>		25,425,000
	<i>o/w e-Health Project</i>		7,000,000
14	Ghana Infrastructure Investment Fund		361,914,186
15	Dist. Ass. Common Fund	129,255,066	
	<i>o/w ABFA</i>	129,255,066	
GRAND TOTAL		904,785,465	1,680,315,864

Source: Ministry of Finance in Ghana⁴

⁴ https://mofep.gov.gh/sites/default/files/budget-statements/2021-Budget-Statement_v1.pdf (Accessed 28th Dec, 2021.)

[Figure 2-5] Breakdown of Projected ABFA Spending (GH¢) – 2021

SRN	ITEM DESCRIPTION	AMOUNT
1	Micro Finance and Small Loans Centre (MASLOC)	48,000,000
2	Nation Builders Corps	672,000,000
3	National Identification Authority	200,000,000
4	Free Senior High School Programme	1,974,021,968
5	Government Communication	4,000,000
6	Planting for Food and Jobs	432,000,000
7	One District One Factory Programme	164,296,000
8	Water and Sanitation	90,000,000
9	Roads Infrastructure	952,587,681
10	Railways Development	136,340,132
11	Fish Landing Sites	29,200,000
12	Special Development Initiatives	852,052,756
13	Zongo Development Fund	83,086,439
14	Livelihood Empowerment Against Poverty	197,534,538
15	School Feeding Programme	488,800,000
16	Regional Reorganisation and Development	252,320,000
17	Teacher Trainee Allowances	169,912,081
	Nursing Trainee Allowances	231,200,000
TOTAL		6,977,351,593

Source: Ministry of Finance in Ghana⁵

[Figure 2-6] Cost of Government Flagship Programmes (GH¢) – 2021

⁵ https://mofep.gov.gh/sites/default/files/budget-statements/2021-Budget-Statement_v1.pdf (Accessed 28th Dec, 2021.)

3.3 Purchase Support Programs

3.3.1 Amount of Participation in Public Procurement by Domestic Companies

The authority's clientele is in both the public and private sectors of Ghana. Those in the public sector are mainly MMDAs, public hospitals, universities, polytechnics, schools, and colleges. Building manufacturing facilities in Ghana to meet the product needs of the public and private sector is vital for domestic companies to compete in the public procurement arena. In addition, foreign companies can help build the capacity of local companies through partnership programs, while the private sector includes suppliers, contractors, consultants, and other service providers.

3.3.2 Public Procurement-Related Laws, Regulations, and Policies

The primary legislation that regulates and governs public procurement in Ghana is the 2003 Public Procurement Act (Act No. 663) (the Act) as amended by the 2016 Public Procurement (Amendment) Act (Act 914) (the Amendment Act).

All government agencies, institutions, and establishments in which the government has a majority interest are mandated to comply with the Act. The Ministry of Finance (MoF) in 2010 established the Public Investment Division (PID) as the Ghana PPP advisory team responsible for taking a lead role over the PPP Programme in Ghana.

3.3.3 Innovation Procurement Program and Supporting Organization

Access to procurement information is readily available in national newspapers. In the past decade, platforms have evolved to alert and provide access to information on public or other procurement needs. The PPA operates a portal to provide information on ongoing procurement requests from all public institutions. Additionally, local and international platforms send alerts via email and SMS to interested parties on ongoing procurements, with one such platform being the website www.tenders.com.gh. As its contribution to transparency and promotion of business opportunities, the World Bank-mandated

information for work in Ghana is made available on their website, including recent contracts awarded, procurement plans for all projects, economic and sector work assessing the implementation strategies of country assistance, and a link to facilitate reporting of suspected fraud and or corruption associated with project implementation.

Domestic industries, however, would need to be prioritized to enable them to thrive. Procuring domestic goods and services has always been required to increase domestic purchasing power and for local companies to be influential. Domestic companies can increase their capacity to replace significant imports, while technological innovation in public procurement methods will help deter corruption since it renders a more transparent approach.

4. Technology Support Programs in Ghana

4.1 Technology Development and Information Support Programs

Over the years, there has been massive restructuring and reorganization of the government ministry responsible for STI to address specific developmental challenges. However, an issue of concern is an inadequate commitment over the years to train more significant numbers of scientists, engineers, and technologists to sustain the national scientific capability. Making matters worse, Ghana lost many of its trained persons due to brain drain, and unlike other countries, the use of experts in the diaspora has not been factored into a national assets utilization plan. In addition, there needs to be a national register of Ghanaian experts and other skilled persons working overseas. Though their remittances have become a significant income base for the economy, their intellectual contributions are even more desired and could be a source of technology transfer. The third area of concern for building Ghana's STI capacity is the inadequacy and failure to develop the capacity of universities to educate and train significant numbers of scientists and technologists at the level of PhDs.

4.1.1 Technology Development Support Programs

Government efforts in financing STI to ensure the availability of needed funds is described as follows.

Government;

- i. Will take stock of existing funding routes established to aid development in science and technology and industry to streamline them to achieve economies in their operations.
- ii. Will establish the National Science, Technology, and Innovation Fund to support innovation in its operations.

- iii. Encourage the private sector to help in the funding for R&D activities, primarily to meet the demands required by the micro, small and medium enterprises (MSMEs), which can be groomed to become top prospects for the commercialization of novel products or processes (that is, products of innovation).
- iv. Boost the allocation of a minimum of 1 percent GDP to support the science and technology sector.
- v. Establish an attractive tax reward mechanism for the instituted fund contributors or directly to R&D activities, but in a manner as not to erode the national tax base.
- vi. Encourage the formation of venture capital (high risk) funds administering authority to commercialize new technologies from scientific and technological institutions.
- vii. Encourage public procurement of products and services from S&T institutions as a means of facilitating their promotion.

The private sector is encouraged to contribute to financing STI application and development in Ghana through various schemes. A key strategy used is the initiation of Public-Private Partnerships (PPPs) in the application and development.

There are other stakeholder institutions, which can contribute to the financing of STI in the country. For example, the Ghana Education Trust Fund (GET-Fund) can contribute a percentage of the total inflow to STI programs, particularly in high-level scientific human resource development.

4.1.2 Information Support Programs

To support the development of SMEs, the government of Ghana established various programs aimed at promoting SME development. GEDC, the Ghana Enterprise Development Commission, the Office of Business Promotion, was the only coordinating body providing financial and technical support to Ghanaian businesses in the early 1970s.

Its primary role was to help in fortifying small-scale industries in general in Ghana. Later on, the National Board for Small Scale Industries (NBSSI), the central government body attending to the affairs of small-scale enterprises, was set up under the umbrella of the Ministry of Industries, Science and Technology under Parliament Act 434 in 1981. Funded by the government of Ghana, the NBSSI was established to address the concern of small-scale businesses by organizing training programs for entrepreneurs, advising them on business operations and providing financial assistance in the form of loan schemes. It also cooperated with NGOs and international establishments to receive support for small businesses. Ironically, this institution wasn't well-funded. The political intrusion of management affected its operations, and there the remuneration of workers was poor.

The government of Ghana established the Ghana Regional Appropriate Technology Industrial Service (GRATIS) as a project under the umbrella of the Ministry of Environment, Science and Technology in 1987. It provided support to small-scale businesses in the various regions of Ghana by establishing intermediate technology transfer units (ITTUS), and aimed to improve employment opportunities and income levels and enhance the expansion of small-scale industries at the grassroots through technology transfers and upgrades.

The Rural Enterprise Project (REP) was another institution that has sought to generate wealth and improve the living conditions of inhabitants of rural areas. Set up to improve the income level of women and vulnerable groups through self-employment. REP provided a list of services to SMEs in technology transfer, business development services, apprenticeship training, and rural financial services. Funded by the International Aid for Agricultural Development (IFAD), the Rural Enterprise Project collaborates with NBSSI, Rural Banks, GRATIS, District Assemblies, and other organizations to support the development of SMEs in rural areas.

The government had also set up organizations to provide financial support to SMEs. For example, under the Package of Action to Mitigate the Social Cost of Adjustment (PAMSCAD), an amount of USD 2million was put aside to be given as credit to develop and train people for self-employment as part of its Entrepreneurship Program (ERP).

Some of the government's information support programs and policies are described below;

- The Ghana Integrated Financial Management Information System (GIFMIS): GIFMIS is part of the e-Ghana project aimed at reforming the public financial management in Ghana
- Ghana Open Government Partnership (OGP): In 2011, Ghana signed the OGP Initiative to improve public services, increase public integrity, manage public resources effectively, create safer communities, and increase corporate accountability.
- e-Immigration: The Ghana integrated e-Immigration system is set up to manage the increased visitor numbers and help regulate and monitor employment and residence activities related to foreigners
- The Ghana Open Data Initiative (GODI): The objective of this initiative includes promoting transparency in government transactions and creating business opportunities for the re-use of open government data.
- The Ghana-Oracle Digital Enterprise Program (GODEP): a partnership between the government of Ghana and the Oracle Government of Ghana Initiative supported by the Accra Digital Centre. A special acceleration program for startups puts modern business resources and technologies behind emerging businesses to help them succeed.

4.2 Technology Transfer & Commercialization Support Programs

4.2.1 Data on the Organization of a Technology Transfer Organization or Consortium

Technology transfer agreements in Ghana is carried out between two enterprises (referred to as the 'transferee') and enterprises outside Ghana (referred to as the 'transferor').

The provisions of every technology transfer agreement in Ghana must conform to and be governed by the following laws:

- The 1992 Technology Transfer Regulations (LI 1547)
- The 2013 Ghana Investment Promotion Centre Act (Act 865)

Under Act 865, all technology transfer agreements must be reviewed and registered by the Ghana Investment Promotion Centre, the government agency in Ghana responsible for registering and monitoring all technology transfer agreements in Ghana.

■ Categories of Technology Transfer Agreements

As stated earlier, Act 865 categorizes technology transfer agreements into four (4) primary forms:

- Agreements covering industrial property rights
- Agreements for the provision of technical services and assistance
- Agreements covering the transfer know-how
- Agreements for the provision of management services

■ Registration of a Technology Transfer Agreement

Prospective applicants for the registration of a technology transfer agreement need to take note of the following:

- The Agreement must have a tenure of not less than eighteen (18) months and not more than ten (10) years. An agreement that is less than eighteen months does not qualify as a technology transfer agreement and will not be subject to Act 865 and LI 1547.
- The lifespan of a technology transfer agreement must not exceed five (5) years for renewal applications.

■ Merits of Registering a Technology Transfer Agreement in Ghana

Subject to the 2006 Foreign Exchange Act (Act 723), an enterprise is assured unconditional transferability in the freely convertible currency of charges regarding a technology transfer agreement registered under the Ghana Investment Promotion Centre Act.

4.2.2 Data Related to the Establishment of the Technology Supply Chain for Universities and National Research Institutes

■ National Supply Chain Assessment (NSCA)

Under MOH's leadership, the TWG, GHS, USAID, and GHSC-PSM supported the requisite fieldwork for the National Supply Chain Assessment (NSCA) in Ghana from August 19 to September 6, 2019. The assessment had four primary objectives as follows:

- Analyze and measure the performance, operational capacity, maturity, and capability of the national public sector–financed health commodity supply chain.
- Identify progress to date on critical technical areas outlined in the 2016–2020 Supply Chain Master Plan, and gaps in implementation progress.
- Develop a comprehensive baseline of stock availability and logistics data quality for the LMIS implementation activity.
- Serve as a 'source of truth' and solid quantitative basis to inform the development of the next Supply Chain Master Plan for 2021–2025.

The NSCA 2.0 includes three distinct elements:

- Results in a visual representation of the country's supply chain
- Measures of the overall capability, resources, processes, and functionality of the country's supply chain
- Collection of site-level data on key performance indicators to measure supply chain performance.

Based on these findings, the government of Ghana (GOG), in collaboration with key supply chain stakeholders, will revisit and refine strategic priorities and operational plans, leverage a shared understanding of the current context to build stakeholder support for collective action, and follow up on flagged areas of poor relative performance with targeted root-cause analyses.

■ Centre for Applied Research and Innovation in Supply Chain-Africa (CARISCA)

The state of supply chain management in Ghana can be better and more effective than the present. Efficient supply chain management will reduce the required resources to provide customer service to a specific segment and improve customer service through increased product availability and reduced order cycle time.

By harnessing a combination of big data AI, machine learning, and predictive analytics, institutions can automate warehouse operations, improve delivery times, proactively manage inventory, optimize strategic sourcing relationships, and create new customer experiences that increase satisfaction and boost sales.

The United States Agency for International Development in 2020 awarded USD 15 million to create a research and training center for supply-chain management in Ghana. In some quarters in a move seen as a signal of a broader effort by the US to respond to the growing economic influence of China, and to a lesser extent Russia, throughout the African continent.

The money from USAID's Global Development Lab helped fund a supply-chain management center for training, research, and innovation in Ghana. Named the Center for Applied Research and Innovation in Supply Chain-Africa (CARISCA), the center is managed by a partnership between Arizona State University and the Kwame Nkrumah University of Science and Technology.

CARISCA leverages local and international governmental, civil society, and industry partners to connect African researchers, practitioners, and businesses to supply chain assets worldwide. The overarching idea is to strengthen local capacity to improve supply chains by increasing efficiency, effectiveness, and inclusiveness, driving national self-reliance.

■ The Financial Trust Corridor

MAS, the Monetary Authority of Singapore, and the Bank of Ghana (BOG) on December 8, 2020 commenced discussions to adopt the Business Sans Borders (BSB) open hub of platforms between Singapore and Ghana under the Financial Trust Corridor. These discussions on close cooperation will benefit small and medium-sized enterprises (SMEs)

and micro-SMEs (MSMEs) in both countries by expanding their connectivity with the counterpart economy while introducing innovative fintech and digital tools to help each SME seamlessly complete its everyday business needs. In addition, MAS and BOG will also work on a complementary Financial Trust Corridor (FTC) initiative to engender greater trust, trade, and recognition between both countries' businesses and financial institutions.

4.2.3 Commercialization Support Programs in Ghana

Between 2013 and 2018, with funding of USD 24million, the Feed the Future USAID Ghana Agriculture Technology Transfer (ATT) Project was introduced. It was a five-year activity funded by the U.S. Agency for International Development's Ghana Mission (USAID/ Ghana).

ATT aims to improve Ghana's agricultural research and extension systems by creating private-sector-led agricultural technology transfer mechanisms, linking research-extension systems and producers in a market-driven approach to seed value chain development and access to critical production inputs, including affordable finance, labor-saving technologies, and dissemination of good agriculture practices (GAPs).

Also, ATT focused on the Ghana Feed the Future (FTF) Intervention Zone, which covered these three regions of Ghana, northern, upper west, & upper east, and the northern parts of Brong Ahafo and Volta Regions.

4.3 Startup & incubation Support Programs

The Ghana Tech and Business Hubs Network (TBHNG) has been established with the central goal of creating thriving innovation clusters across the country. Local tech hubs such as iSpace, Impact Hub Accra, Kumasi Hive, the HOPin Academy, iCode in Takoradi, and Ho Node in Tamale and Ho have been operational in this space.

According to GIZ's Make-IT-Africa, the innovation ecosystem in Ghana has developed enormously over the last three years. Ghana has more than 34 tech hubs becoming increasingly specialized and located in Accra and other parts. Several of the innovation and

incubation hubs are united in the Ghana Hubs Network. The recent surge in startup activity can be attributed to solid ecosystem players such as the Meltwater Entrepreneurial School of Technology (MEST), high-quality talent, and reasonable access to finance. It has led to the creation of several highly valued technology-driven startups, such as Snoocode and Agrocenta. According to Disrupt Africa 2018, investments in Ghanaian startups accounted for 11.4 percent of total African startup investments.

4.3.1 Ghana's SMEs and Startup Support Policies and Programs

There is currently ongoing engagement by the startup ecosystem with solid support from the development community to enact a startup bill, which would become an act passed by the Parliament of Ghana to serve as the legal framework defining what a startup is and creating rewards to motivate the creation and further development of startups in Ghana, thereby facilitating economic growth.

The Ghana Startup Act sets up an incentive framework for the creation and development of startups in Ghana to promote creativity, innovation, and new technologies in achieving a substantial added-value at the regional, national, and international levels. In addition, the Act seeks to provide the legal basis for the introduction and promotion of startups for jobs and wealth creation.

4.3.2 Startup & incubation Support Programs in Ghana

The startup ecosystem is touted as one of the developed and robust startup ecosystems in Africa. It places fifth after Egypt, Kenya, Nigeria, and South Africa, according to a 2018 Venture4Africa Report.

There are several support programs for startups in Ghana. One such support program is the Ghana Startup Capital Fund. This fund is a private sector (non-profit) and a self-help initiative set up in 2011 to promote entrepreneurship, provide the necessary support and resources to entrepreneurs, create jobs, reduce unemployment, and provide skills development and knowledge transfer.

The vision of the fund is to raise 10 million Ghana Cedis in ten years (GH¢10,000,000) through donations and contributions by entrepreneurs, individuals, organizations, local and international donors, investments and fund-raising activities to aid in the development of entrepreneurship and invest in innovative and economically viable ideas and startups.

Boosting Green Employment and Enterprise Opportunities in Ghana (GrEEn) Innovation Challenge is another support program. This program is an annual competition to support SMEs and businesses that have outstanding, innovative green ideas. The challenge forms part of GrEEn's objective of supporting entrepreneurs and green SMEs to increase their operations. For its 2021 edition, a total of EUR 25,000 per SME will be awarded in matching grants to eight selected SMEs.

Its objectives are to promote the green and circular economy in Ghana, promote green product/ service innovation, and contribute to creating green jobs whilst strengthening the business skills of green entrepreneurs.

This program will improve access to the market for green products by highlighting and showcasing green businesses and SMEs, and supporting the operations and sustainability of green SMEs

In Ghana, a startup can be registered as/for;

- Companies limited by shares.
- Companies limited by guarantee.
- Companies with unlimited liability.

Additionally, in regard to the establishment of startups in Ghana,

- Foreigners who want to establish joint ventures with local firms or established wholly-owned companies must register with the Ghana Investments Promotion Council (GIPC).
- Joint ventures with at least 10 percent Ghanaian participation, the foreign partner is

required to contribute not less than USD 200,000 (in either cash or capital goods) relevant to the investment.

- Similarly, for an entity to be wholly owned by a non-Ghanaian, GIPC requires a minimum amount of foreign equity capital of USD 500,000 in either cash or capital goods relevant to the investment.
- A company can also register with the Ghana Free Zones Board (GFZB) to operate in areas other than mining, petroleum, or timber. However, to qualify, the entity needs to export 70 percent of its goods or services.
 - Registering under the GFZB enables the company to enjoy tax holidays for ten years and pay a corporate tax of 25 percent on local sales and 15 percent on exports.

Various startups and incubation support programs have been rolled out. Below are some of the business incubation support programs:

- Meltwater Entrepreneurial School of Technology (MEST). MEST Africa is a private entrepreneurial training program, seed fund, and incubator for African tech startups headquartered in Accra, Ghana.
- The Ghana Startup Capital Fund. The GSC Fund is a government incubator program that provides shared premises, business advisory and support services, access to investors, markets, international networks, mentoring, and full-time hands-on management coaching.
- The National Entrepreneurship and Innovations Fund. This government program provides financial support to 200 scalable businesses per annum and business advisory support.
- Entrepreneurship for Commercial Seed Incubation Business: This is an international business incubator program funded by AGRA- USAID, focused on the business

development capacities of SMEs with the seed value to enhance productivity and profitability.

- The Enterprise Development Center. This center is a Ghanaian government incubator program commissioned to support Ghanaian small and medium enterprises (GhSMEs) to strategically position themselves and take advantage of the many business opportunities in the oil and gas sector.
- The Rural Enterprise Program is an incubator program by GoG to improve the livelihoods and incomes of poor rural small and micro-entrepreneurs. The aim is to increase the number of rural MSEs that generate profit, growth, and employment opportunities.
- The Africa Network for Entrepreneurs (TANOE). This private business incubator focused on making entrepreneurship a sustainable solution to unemployment by training and empowering youth and developing their talents, entrepreneurial abilities, and skills, as well as supporting women and startup entrepreneurs with resources and needed information to sustain and grow their businesses.
- Youth in Agriculture Program (YIAP) is an initiative of the Ministry of Food and Agriculture to motivate the youth in accepting and appreciating farming/ food production as a commercial venture through the provision of tractor services and agro-inputs in crops / block farm, livestock & poultry, fisheries/aquaculture, and agribusiness, as well as supply inputs such as seeds, agrochemicals, & sacks and facilitate marketing of your produce.
- The National Entrepreneurship and Innovation Plan Program is an initiative of the government. It primarily focuses on providing business development services, startup incubators, and funding for young businesses to grow and become successful through the Incubation Programme, NEIP Industrialization Plan, Youth Enterprise Fund (YEF), and business advisory.
- National Youth Authority: The Youth Leadership & Skills Training Institutes (YLSTI)

provide two-year vocational skills training and opportunities to build their enterprises for the youth. They offer welding, dressmaking, masonry, electrical installation, ICT, carpentry & joinery, and general agriculture courses.

- The Kumasi Institute of Tropical Agriculture. KITA provides training in pig, goats and sheep rearing, poultry / turkey/ guinea fowl, rabbit farming, fish farming, grass cutter farming, vegetable farming, crop production and marketing, pineapple farming and processing, nurseries management as business, citrus farming, agro-processing, cocoa agronomy, and many others.

The contributions of startups, particularly technology innovations, underscores the government of Ghana's STI efforts at evolving robust policies, activities of implementing agencies, ecosystem drivers, such as tech hubs, and research institutions that support innovations, through the following:

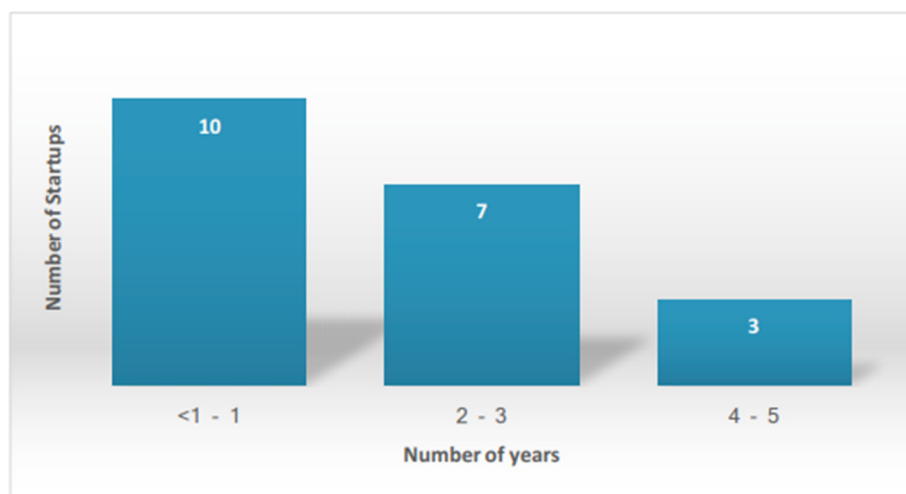
- The 2017 National Science, Technology and Innovation Policy.
- Council for Scientific and Industrial Research (CSIR)
- National quality infrastructure
- Ghana Hub Network
- Fintech and Innovation Office at the Bank of Ghana

4.3.3 Statistical Research Related To Startups in Ghana

An inductive study involving 20 startups in Ghana in the publication with a multiple case study of the technology entrepreneurship ecosystem in Ghana was done by Nickie Besiwa Johnson, whose findings are described as follows:

Graph 1: Years of Operation

Half of the startups had been in operation for less than a year or a year. Seven had been in operation for two to three years, and only three companies were in their fourth or fifth year of operation.



4.3.4 Startup-Related Support Projects

The key cities in Ghana with overt startups culture are Accra, Kumasi, Takoradi, and Tamale. The Ghana Hubs Network synergizes and stimulates collaboration among Ghana's business, technology, and innovation hubs by serving as an umbrella body. Through the GHN, the startup ecosystem presents a strong voice to push and support key stakeholders in formulating policies.

Make-IT (GIZ) is partnering with the Ministry of Environment, Science, Technology, & Innovation (MESTI) to support its vision to build a solid technological and innovation-driven economy.

4.4 Intellectual Property Management Support Programs

On Intellectual property management support programs in Ghana, the two leading public institutions responsible for the administration of IP are the:

- The Copyright Office and
- Industrial Property Office (the Registrar General's Department) for patents, industrial designs, trademarks, geographical indications, and layout -designs.

The two institutions operate under the Ministry of Justice and the Attorney General's Office.

In enforcing IP rights in Ghana, the following institutions are key:

- The courts
- The Copyright Office
- The Ghana Revenue Authority
- The Food and Drugs Authority
- The Ghana police
- Registrar General's Department
- The Ghana Investment Promotion Centre (registration of technology transfer agreements)
- The Ghana Standards Authority

The following laws may be warranted to manage IP in Ghana;

- 2003 Geographical Indications Act (Act 659)
- 2000 Protection against Unfair Competition Act (Act 589)
- 2003 Industrial Designs Act (Act 660)
- 2003 Patents Act (Act 657)
- 2004 Trade Marks Act (Act 664)
- 2005 Copyrights Act (Act 690)
- Layout-Designs (Topographies) of 2004 Integrated Circuits Act (Act 667)

- The passage of the 2020 Plant Breeders' Bill by the Parliament of Ghana
- In addition to local legal frameworks, Ghana is a signatory to international conventions and treaties
- The Paris Convention for the Protection of Industrial Property
- The Berne Convention for the Protection of Literary and Artistic Works
- The Patent Cooperation Treaty (PCT)
- The Convention establishing the World Intellectual Property Organization (WIPO)
- The WIPO Copyright Treaty (WCT)
- Hague Agreement on International Deposit of Industrial Designs

4.4.1 Ghana IP Dynamics by the numbers

According to the World International Patents Indicators 2019, Ghana had 52 applications with 13 residents and 39 nonresidents. Many of the universities in Ghana have IP policies that provide clear guidelines. The policies are consistent with the National Intellectual Property Policy Strategy Ghana and related IP acts. The IP policies in universities help create an empowering environment for creating new knowledge, promoting innovations, and creating opportunities to raise funds.

For example, the University of Ghana's IP objective is to ensure that research, inventive or original technologies created by the university are transferred to industry to develop beneficial and ethically acceptable processes, products, and services.

4.4.2 Government Programs to Support Prior Technology Research For Public/Private IP

IP is protected by law. However, patents, copyright, and trademarks enable people to earn recognition or financial benefit from what they invent or create. This phenomenon helps strike the right balance between the interests of innovators and the general public. It also aims to foster an environment in which creativity and innovation can flourish (WIPO).

Ghana instituted its National Intellectual Property Policy and strategy (NIPPS) in 2016 for introducing internationally acceptable best practices and incorporating universal standards in copyright. The Act also protects works such as computer programs and folklore that were before not expressly protected, and also extends the general term of protection of life plus 50 years after the author's death to life plus 70 years after death. In anonymous works, economic rights are protected by law for 70 years from being published. In cases where a corporate body is vested with the copyright, protection is for 70 years. On the other hand, the state is awarded the copyright in folklore and perpetual protection. Thus, the terms of protection for works in Ghana generally exceed the standard duration of protection under the TRIPs Agreement (Adusei et al., 2010).

The primary intellectual property laws in Ghana are as follows:

- The 2005 Copyright Act (Act 690); the 2003 Patents Act (Act 657).
- The 2004 Trademarks Act (Act 664); 2003 Industrial Designs Act (Act 660); and 2000 Protection Against Unfair Competition Act (Act 589).
- The 2020 Plant Breeders Bill.

There also are international IP associations such as;

- The National Association of Manufacturers (NAM)
- The International Intellectual Property Alliance (IIPA)
- The International Trademark Association (INTA)
- The Coalition against Counterfeiting and Piracy
- The International Anti-Counterfeiting Coalition (IACC)
- Pharmaceutical Research and Manufacturers of America (PhRMA)
- Biotechnology Industry Organization (BIO) Selling to the Public Sector

4.4.3 Ghana's National Intellectual Property Policy and strategy (NIPPS)

On January 21, 2016, Ghana launched its National Intellectual Property Policy and Strategy (NIPPS). The vision was for Ghana to be among the leading countries in utilizing intellectual property as a tool for rapid national development. The mission was to enhance Ghana's competitiveness by improving and increasing the use of intellectual property systems.

NIPPS's long-term goal is to exploit intellectual property rights for accelerated growth in technological and industrial development in Ghana. Accordingly, the objectives of this policy are as follows:

- To bolster the legal framework for the protection of intellectual property (IP) rights.
- Promote creativity and innovation to improve IP generating activities in Ghana
- To promote and facilitate commercial exploitation of IP rights and technology transfer
- Strengthen the legal and institutional framework for the enforcement of IP rights
- To develop adequate human resource capacity in the administration, protection, commercialization, and enforcement of IP rights.
- Fortify the institutional framework for the administration and management of IP rights.
- To create public awareness on IP issues for the general public and identifiable groups.
- To develop and promote the IP services industry in Ghana.
- To promote research related to IP.

5. Infrastructure Support Programs in Ghana

5.1 Certification Support Programs

5.1.1 GIRC Centre's Focus

The Technical Innovation fields that the GIRC-Centre has already planned to focus on include the following:

- I. Agriculture and Food Processing,
- II. Environment, Waste Management, and Circular Economy,
- III. Manufacturing
- IV. Information and Communications Technology (ICT)
- V. Renewable Energy,
- VI. Health and Pharmaceuticals
- VII. Mining & Minerals Processing
- VIII. Oil and Gas

5.1.2 Status of the Conformity Assessment System in Related Areas Based on Ghana Laws.

Ghana has many conformity assessment bodies (CABs) that test, calibrate, or certify products and services. Some of the products are certified to national standards. However, national standards are often not enough to avoid the quality issues that hinder exports; examples are as follows;

- Rejection of consignments due to product failures caused by questionable test results,
- Re-testing of products due to inaccurate measurement linked to poorly calibrated or non-calibrated equipment,
- Rejection of orders by not supporting product claims with an internationally recognized

accreditation certificate or test reports.

Very few CABs in Ghana are accredited to international standards. Until recently, Ghana did not have a national accreditation body to assess the competence of these CABs to international standards. Ghanaian CABs seeking internationally recognized accreditation had to refer to foreign accreditation bodies. This phenomenon makes the accreditation process cumbersome and affordable to only a few.

5.1.3 Status of Accreditation Bodies in Ghana

The Ghana National Accreditation Service (GhaNAS) was established to provide accreditation services to conformity assessment bodies. It is an associate member of the African Accreditation Cooperation (AFRAC) AFRAC and recognized by the ECOWAS Accreditation System (ECORAS).

GhaNAS is in the process of becoming a signatory of the mutual recognition agreements of the International Accreditation Forum (IAF)/ International Laboratory Accreditation Cooperation (ILAC). As a result, it is working towards having its accreditations recognized in many other countries. This recognition will allow Ghanaian companies to sell their goods and services more easily in the international market.

5.1.4 Status of Standardization Organizations in the Relevant Field (Ghana Standardization Organization)

The Ghana Standards Authority (GSA) is responsible for metrology, standards, testing, inspection, and certification. The GSA operates under the 1973 Standards Authority Act with the following mandate:

The mandate of the Authority:

- Establish and promulgate standards to ensure the high quality of goods produced in Ghana, whether for local consumption or export.

- Provide quality assurance through inspection, testing, and metrology.
- Assist operators in both the manufacturing and service sectors to improve their competitiveness by establishing effective quality management systems along with ISO/IEC 9001: 2008 and 22000: 2005.
- Promote standardization in industry and commerce
- Promote standards in public and industrial welfare, health, and safety

The certification directorate of GSA is made up of three departments, namely:

- Product certification
- Management system certification
- Industrial support

5.1.5 GIRC's Testand Certification Promotion Budget and Investment Plan

The GIRC-Centre will have budget allocations from the government through the National Research & Innovation Fund that Parliament is currently considering. The GIRC-Centre will also look at other sources from funding, including revenues that may accumulate from its operations.

5.1.6 Status of Innovative Institutions (i.e. Universities & Research Institutes) That Can Cooperate When Promoting GIRC Test Certification

Most universities and research will be closely collaborating with the GIRC-Centre. There is already close collaboration with KNUST and UG, and they are represented on the Ghana team working with STEPI under the K-Innovation Partnership Program.

5.1.7 Company and Product Development Status in Need of Test Certification

There is much demand for testing and certification, especially from the industry. A testing unit under the Ghana Atomic Energy Commission (GAEC) is dedicated to non-intrusive testing, and the GIRC-Centre will be working with GAEC once the appropriate organs are established.

5.2 Industry-University-Research Institute(I-U-R) Linkage Support Programs

5.2.1 HR Development Status for Industry-University Collaboration in Ghana

The private sector plays a crucial role in bridging the industry-academia divide through internships, digital skills development, and other infrastructure programs. DreamOval is an example of a technology firm in Ghana that offers internships to university students while helping them build industry knowledge through the DreamOval Foundation.

In the same vein, Huawei also works with academia and the government to support digital skills development and through its methods such as:

- Seeds for the Future Programme aims to develop skilled, local ICT talent and bridge communication between countries and cultures.
- ICT Academy, where they have trained over 3,500 students in digital skills
- Engagement with universities where they set up ultramodern ICT Labs

Another support program is the EPA program. Education Partnership in Africa (EPA), which is a three-year program (2008-2011) funded by the U.K. government to help African universities improve entrepreneurship & social enterprise and enhance graduate employment. This program aims to build institutional capacity and quality, and ensure the employment of students upon graduation. In a partnership between Ho Polytechnic in

Ghana and the City College of Brighton and Hove, the program managed to improve the matching up of engineering graduates with the modern workplace.

5.2.2 Supporting System of Tech Development, Transfer & Startup for I-U Collaboration in Ghana

In the University of Ghana (Legon), the Intellectual Property and Technology Transfer Services (TTIPS) team manages and leads the university's intellectual property and technology transfer activities, emphasizing the development of innovative mechanisms for its engagement with industry.

Their activities are geared towards internal stakeholders (researchers, staff, students) and external stakeholders (entrepreneurs, industry, government agencies, investors).

5.2.3 Capacity Building and Infra for Collaboration in Ghana

The recognition of the need to support development through capacity-building and infrastructure collaborations has been a theme emphasized, if not overemphasized, since the early 1990s. Aid agencies, private industries, and other stakeholders have played significant roles in fortifying this idea with various programs and action-based solutions to steer Ghana's development onto the yellow brick road.

One such collaboration is the GNPC Professorial Chair Program. The GNPC Professorial Chair is a novelty initiative championed by the leadership and management of GNPC. The University of Ghana College of Basic & Applied Sciences (Department of Earth Science and School of Physical and Mathematical Sciences) is one such beneficiary of the initiative.

The drive behind this initiative is to further deepen educational research in these institutions and build broad-based capacity to ensure sustainable development. The initiative is forecasted to bridge the gap in both local content and capacity-building in Ghana. The outcomes of these chairs are tailor-made to produce graduates who will participate in and spur developing the oil and gas sector in a relevant manner. In doing so, structures are being

put in place for the idea of GNPC to be a stand-alone corporation, in line with exploratory development, production, disposal, pricing, and capacity-building to be a reality in no time.

5.2.4 Industry Campus

Regarding bridging the gap between industry and academia, several relevant industries have partnered with various universities and campuses to make that possible through capacity-building and internships. Shared infrastructure, research partnerships, and services, to name a few, have been the desirable way to go over the years but have become even more popular since the turn of the millennia. Collaborations between public and private universities and the Council for Scientific and Industrial Research – Ghana (CSIR) tell the hands-on, ground-level story.

5.2.5 Governance and Structure of Industry-University Collaboration

Over the years, the government of Ghana has held several stakeholder meetings with regards to improving I-U collaborations and structure, showing in no small way its importance. Universities the world over have been the high mountain and gateway to research and education. The closest Ghana has to a national policy for research is a comprehensive policy framework for science, technology, and innovation (STI). This policy in 2017 identified key barriers to innovation and commitment to supporting the adoption of foreign innovation as a mammoth economic driver. The government of Ghana has made investments in research training and creating of new universities. Also, the government research funding mechanism (the Research and Book Allowance) provides a monthly allowance to academics and college teachers.

The most typical form of partnerships between Ghanaian universities and industry has been the collaboration to provide internships and attachment opportunities where academic institutions make special provisions to have some of their students momentarily employed by employers to provide them with practical on-field experience through the vacation period. In addition, most tertiary institutions make it mandatory for students to take on an

internship, usually ranging from a few weeks to a year, which creates employment opportunities for students after university.

Also, a one-year national service program, which is mandatory, instituted by the government of Ghana, has been in place to foster labor collaborations at government and industry levels. Furthermore, the national service scheme instituted by law (Act 426 of 1980) and enacted by the National Service Secretariat gives new graduates the chance to gain hands-on experience both in the public and private sectors as part of their service to the country.

5.2.6 Industry-University Collaboration Foundation

The Institute of Applied Science and Technology (IAST) of the University of Ghana, on August 10, 2020, signed an MOU with the Global Entrepreneurship Network, Ghana (GENG), which was intended to identify opportunities in entrepreneurship, mentorship, product development, & innovation and facilitate access to markets. GENG runs in over 160 countries to activate young people to initiate and scale-up sustainable businesses.

The Research and Grant Institute of Ghana (REGIG) has also supported and funded university research over the years. This Institute is an independent, non-profit organization devoted to sponsoring the generation, dissemination, and utilization of research to enable Ghana's socio-economic development and stimulate the global scientific discourse.

In partnership with some Ghanaian universities, the World Bank, United Nations, and USAID also undertake joint developmental and research projects to expose real-world scenarios and direct access to knowledge. They also offer practical experience and knowledge transfers between staff, students, and the communities involved. The University of Ghana has also collaborated with the Ghana Stock Exchange (GSE) to research and develop the Ghanaian financial markets.



CHAPTER 3

South Korean Experiences

Establishment and Operation of Ghana Innovation &
Research Commercialization (GIRC) Centre

Chapter 3. South Korean Experiences

Chapter 2 reviewed the current state of Ghana's nine TISPs, so now, Chapter 3 specifically presents Korean experiences and cases related to each of these nine TISPs. For each of the nine themes, an overview of Ghana's current state is presented and Korean policies and programs applicable to Ghana are selectively reviewed.

1. Tax & Finance Support Programs in Korea

1.1 Current State of Ghana

The general corporate income tax rate in Ghana stands at 25 percent, though effective corporate tax rates differ across industries. Exports including horticultural products and artifacts are taxed at a rate of 8 percent. In the meantime, manufacturing businesses pay corporate tax at a rate of 12.5 or 18.75 percent depending on their locations. Hotel businesses also pay a reduced corporate tax rate of 22 percent. The deduction period of loss carryforward generally spans three years, which has been extended to five years in some priority sectors such as ICT, manufacturing, tourism, agricultural processing, farming, petroleum, and mining.

In compliance with the 2009 Ghana Revenue Authority Act or Act 791, the Ghana Revenue Authority (GRA) supervises the country's tax system, covering taxes from corporate, withholding, value-added, national health insurance contribution, education trust fund contribution, employment, dividend, customs taxes to consumer and telecommunication service taxes.

Ghana's financial sector consists of commercial banks, as well as savings and loan, lease-purchase financing, and microfinance companies. The government of Ghana is pushing forward an initiative to establish the Development Bank of Ghana (DBG) which will be playing the role of a large-scale loan provider for small businesses. This DBG initiative has garnered financial support worth EUR 170 million from the European Investment Bank (EIB), and the Board of the World Bank has approved a financial support package of USD 250 million for this initiative.

Ghana needs to build and solidify its tax and finance support system to facilitate corporate R&D investment, which will accelerate the country's industrial development in the capacity of a developing nation.

1.2 Evolutionary Path of the Korean Tax & Finance Support Programs

1.2.1 Phase 1 (1961-1979)

In the 1960s, the Korean economy was heavily dependent on labor-intensive industries, so the policy demand for technology adaptation and improvement was far greater than that for direct R&D. To meet such demand, a provision on 'tax exemption for technology import expenses' was inserted into the 1966 Foreign Capital Inducement Act, which marked the beginning of Korea's tax support program for R&D.

With the enactment of the 1972 'Technology Development Promotion Act', the Korean government actively started to implement technology development support measures. In 1975, a program called the 'deduction of R&D reserve fund' was introduced to allow companies to deduct planned R&D expenses from taxation in advance, so that they could stably secure R&D funds.

For financing support, the Industrial Bank of Korea (IBK), formerly known as the Small and Medium Business Bank, was established in 1961 and the "Lending Quota Program for SMEs" was introduced to ensure uninterrupted funding for SMEs with poor credit records and limited collateral capacity.

1.2.2 Phase 2 (1980-1999)

From the 1980s, private companies started to lead R&D investment in Korea. In 1981, the ‘Corporate R&D Center Reporting System’ was introduced to recognize 53 corporate R&D centers nationwide. In 1982, a series of tax support measures were actively introduced to boost R&D investment including the ‘tax credit for R&D and HR development expenses (current expenditure)’, ‘tax credit for R&D facility investment (capital expenditure)’, ‘special taxation on income from technology transfer’, and ‘tax credit for technology acquisition expenses’. These measures made significant contributions to supporting R&D activities of Korean corporate R&D centers in their early stages.

In the meantime, Korea’s finance support programs were mostly implemented in the 1980s. With the focus of the government’s policy financing being placed on technology development and productivity enhancement, the government-led financial support programs were also restructured shifting their focus from large companies, heavy and chemical industries, and exporting companies to SMEs. In 1989, the Korea Technology Finance Corporation was established to meet Korean companies’ R&D funding needs. In 1994, the ‘bank-intermediated loan facility’ was introduced to support financial institutions’ intermediary loan services for SMEs.

1.2.3 Phase 3 (2000-Present)

In the 2000s, Korea started to exit the era of rapid growth over the past four decades. In this process, the country had to face a transitional dilemma of shifting its focus from input to innovation-driven growth. In the meantime, Korea’s tax support program for R&D became stabilized after a trial-and-error period of over three decades since its introduction in 1966 and the Korean government further expanded its tax support for next-generation growth engine industries. In 2009, the sunset rule defining the expiry date of the ‘general tax credit for R&D and HR development expenses’, which accounted for the biggest share in the total R&D-related tax credit, was eliminated, making this category of tax credits for R&D expenses a permanent one. In 2010, the Korean government newly instituted the ‘tax credit for R&D

and HR development expenses specifically for new growth engine industries and innovative technologies’.

For financing support, the Korean government upgraded its financial support for R&D by introducing such programs as the ‘Technology Commercialization Fund’ in 2000 and ‘Investment and Loan Hybrid Fund’ in 2012. In 2020, the ‘support for COVID-19 pandemic-affected businesses’ was added as part of the ‘bank-intermediated loan facility’.

[Table 3-1] Evolutionary Changes by Phase

	Phase 1 (1961-1979)	Phase 2 (1980-1999)	Phase 3 (2000-Present)
Tax support	<Introduction of tax support for R&D> • Tax exemption for technology import expenses (1966) • Deduction of R&D reserve fund (1975)	<Growth of tax support for R&D> * Full-fledged enforcement from 1982 • Tax credit for R&D & HR development expenses (general) • Tax credit for R&D facility Investment • Special taxation on income from technology transfer • Tax credit for technology acquisition expenses	<Upgrade of tax support for R&D> • Tax credit for R&D & HR development expenses (general) became permanent (2009) • Tax credit for R&D and HR development expenses (new growth engine/ innovative technologies) (2010) • Special taxation on income from technology licensing (2015)
Financial support	<Introduction of financial support for R&D> • IBK established (1961) • New Technology Development Fund Support (1978) • Lending quota program for SMEs (1965)	<Growth of financial support for R&D> • Korea Technology Finance Corporation established (1989) • Bank-intermediated loan facility (1994) • Startup Fund (1998)	<Upgrade of financial support for R&D> • Technology Commercialization Fund (2000) • Investment and Loan Hybrid Fund (2012) • Support for COVID-19 pandemic-affected businesses (2020)

1.3 R&D Tax Incentive Programs

1.3.1 Tax Exemption for Technology Introduction Payments

The ‘tax exemption for technology introduction payments’ is applied to parties who introduce advanced foreign technologies critical to the international competitiveness of Korean industries. This is a program designed to exempt foreign parties from corporate or income taxes for five years from the date of the first payment for technologies they provide according to the technology introduction contract stipulated in the ‘Foreign Investment Promotion Act’. After it was first instituted in 1966, the program was abolished in 2009. The program is based on Article 121-6 of the ‘Restriction of Special Taxation Act’.

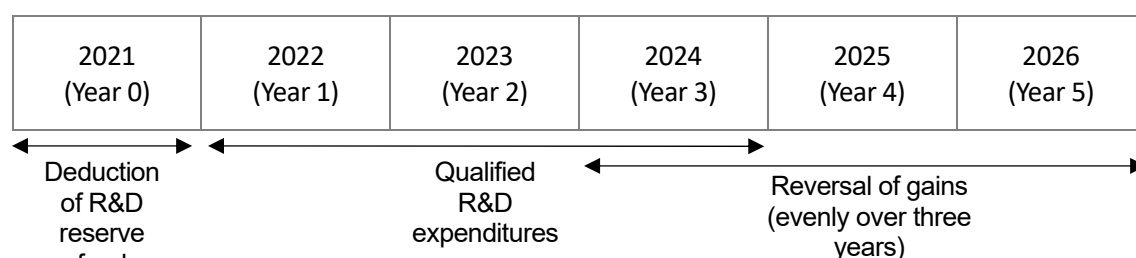
Technology introduction payments are classified into three types as follows: i) Fixed royalty, ii) running royalty, and iii) initial payment combining the two previous types. The fixed royalty is again split into i) a one-time royalty (a lump-sum single payment) paid at the time of contract and ii) an installment royalty (fixed-amount multiple payments of royalty) paid annually in fixed amounts regardless of sales of the contracted products. A running royalty is paid at a certain rate of the net sales or purchased volume or quantity. An initial payment of a certain amount can be made in either a one-time payment or installment payments separately from a running royalty.

1.3.2 Deduction of R&D Reserve Fund

The ‘deduction of R&D reserve fund’ is a program designed to support corporate R&D by deducting R&D expenses from taxation in advance and reversing them as gains later. Accordingly, the program offers a tax deferral effect, and was first instituted in 1975, then abolished in 2006. It was reinstituted in 2009 but abolished again in 2013. The program is based on Article 9 of the ‘Restriction of Special Taxation Act’.

For Year 0, this program allows a deduction of R&D reserve fund within the limit of maximum 3 percent of sales on the condition of a secured R&D reserve fund. For the following three years (i.e. years 1, 2, & 3) after the termination of the taxable year of the deduction, an R&D reserve fund is spent and treated as R&D expenditure, which is then

reversed as gains evenly for three consecutive years (i.e. years 3, 4, and 5). The leftover amount from R&D expenditures during years 1, 2, and 3 is reversed as gains for the taxable years that fall on years 3, 4, and 5. For the unused amount, the reversal of gains should include additional interest equivalent to daily rate of 0.03 percent or annual rate of 10.95 percent.



[Figure 3-1] Deduction of R&D Reserve Fund

1.3.3 Tax Credit for R&D and HR Development Expenses (general)

The ‘tax credit for R&D and HR development expenses (general)’ is a program deducting a certain portion of R&D and HR development expenses from corporate income tax. After its introduction in 1982, the program became permanent in 2009. The program is based on Article 10 of the ‘Restriction of Special Taxation Act’.

The amount to be deducted is calculated based on either the total tax amount or increased tax amount. When based on the total tax amount, the deductible tax amount is calculated by multiplying the total tax amount by the tax deduction rate. When using an increase in tax amount as the basis, the tax amount to be deducted is calculated by multiplying the tax amount gap between the current and previous years by the tax deduction rate.

- (1) Based on total tax amount: Tax amount of the current year x tax deduction rate
- (2) Based on an increase in tax amount: (Tax amount of the current year – tax amount of the previous year) x tax deduction rate

The tax deduction rates differ across companies of different sizes. The tax deduction rate for large companies remains in the range of 0 to 2 percent based on the total tax amount and 25 percent based on the increased tax amount. The figures for mid-ranked companies stand at 8 to 15 percent and 40 percent, respectively and they further rise to 25 and 50 percent, respectively, for SMEs.

[Table 3-2] Tax Deduction Rate for R&D and HR Development Expenses

Classification	Based on total tax amount	Based on increased tax amount
Large companies	0~2%	25%
Mid-ranked companies	8~15%*	40%
SMEs	25%	50%

* For the first three years after a company's graduation from SME status, a 15% tax deduction rate is applied. Then, 10% is applied for the following two years and 8% afterwards.

The deductible expenses include R&D labor costs, R&D current expenditures (including the purchase of raw materials, parts, sample products, and test reagents), and expenses for joint and contract R&D. The unused deduction amount can be carried over for 10 years. SMEs' R&D and HR development expenses are exempted even from the minimum tax requirement scheme.

1.3.4 Tax Credit for R&D and HR Development Expenses (new growth engine/ innovative technologies)

The 'tax credit for R&D and HR development expenses (new growth engine/ innovative technologies)' is a program to deduct a certain portion of R&D and HR development expenses in the areas of new growth engine and innovative technologies from corporate or income tax. The deduction rate for new growth engine and innovative technologies is higher than that for a general tax credit. This program was instituted in 2010 based on Article 10 of the 'Restriction of Special Taxation Act'.

The amount to be deducted is calculated by multiplying the current year's R&D and HR development expenses by the deduction rate, which differs across companies of different sizes. For example, the maximum tax deduction rate of medium and large companies stands at 30 percent while that of KOSDAQ-listed mid-ranked companies and SMEs reaches 40 percent.

[Table 3-3] Tax Amount Deducted for R&D & HR Development Expenses (new growth engine/ innovative technologies)

Classification	Deduction amount
SMEs	R&D & HR development expenses of the current year $\times$ [30% + maximum 10%*]
KOSDAQ-listed mid-ranked companies	R&D & HR development expenses of the current year $\times$ [25% + maximum 15%*]
Medium & large companies	R&D & HR development expenses of the current year $\times$ [20% + maximum 10%*]

* (R&D expenses in new growth engine & innovative technologies/ sales) $\times$ 3

R&D expenses for new growth engine and innovative technologies should be tracked in separate accounting from that of general R&D and HR development expenses. The deduction amount can be carried over for 10 years. SMEs' R&D and HR development expenses are exempted even from the minimum tax requirement scheme.

1.3.5 Integrated Tax Credit for Investments

The integrated tax credit for investment is a program to deduct investments in assets used for research & experiment, vocational training, energy saving, environmental preservation, or employees' wellbeing from corporate or income taxes. This program was first launched in 1982 under the title 'tax credit for R&D facility investment'. In 2021, it was redesigned to integrate other existing investment-related tax credit programs. This program is based on Act 24 of the 'Restriction of Special Taxation Act'.

The total amount to be deducted is calculated by combining the basic and extra deduction amounts. The basic deduction amount is calculated by multiplying the investments in the

relevant assets by the basic deduction rate. When the investment amount of the current taxable year exceeds the average amount of investments or asset acquisitions for the past three years, an extra deduction of 3 percent of the gap can be applied.

(1) Basic deduction amount = amount of investments in the relevant assets of the current year $\times$ basic deduction rate

(2) Extra deduction amount = (investment of the current taxable year – average amount of investments or asset acquisitions over the past three years) $\times$ 3%

The tax deduction rates differ across companies of different sizes. In case of large companies, a basic deduction rate of 1 percent is applied, whereas 3 percent is applied for investments in new growth engine and innovative technologies. For mid-ranked companies, the basic deduction rate stands at 3 percent, while the deduction rate for investments in new growth engine and innovative technologies increases to 5 percent. For SMEs, the figures even reach 10 and 12 percent, respectively. Among companies of different sizes, SMEs are subject to the highest deduction rates.

[Table 3-4] Tax Deduction Rate for Investments

Classification	Basic deduction rates	Deduction rate for investment in new growth engine/ innovative technologies
Large companies	1%	3%
Mid-ranked companies	3%	5%
SMEs	10%	12%

The deductible investments do not include maintenance fees for existing facilities or capital expenditures. The deduction amount can be carried over for 10 years, but it is subject to the minimum tax requirement scheme.

1.3.6 Special Taxation on Technology Transfer & Licensing

The special taxation on technology transfers and licensing is a program to exempt corporate or income taxes for income generated from the transfer or licensing of patent rights to local parties. The special taxation on income from technology transfer was first instituted in 1982. It was abolished in 2005 and then reinstituted in 2014. The special taxation on income from technology licensing was instituted in 2015, and this program is based on Article 12 of the 'Restriction of Special Taxation Act'.

Technologies subject to special taxation include patents, model utility rights, technological know-how developed by Korean nationals in the science and technology field, and other technologies that meet the requirements specified by relevant laws. For income generated from technology transfer, mid-ranked companies and SMEs are subject to a tax deduction rate of 50 percent, while for income from technology licensing, a tax deduction rate of 25 percent is applied to SMEs.

1.3.7 Tax Credit for Technology Acquisitions

The tax credit for technology acquisitions is a program to deduct a certain portion of expenses spent for acquiring patents created and possessed by Korean nationals from their corporate or income taxes. This program was first instituted in 1982 and then abolished in 2018. The program is based on Article 12 of the 'Restriction of Special Taxation Act'.

The tax deduction rates applied before the abolishment of the program varied across companies depending on their size. For example, SMEs were subject to a 10 percent deduction rate for the amount they paid for technology acquisitions, whereas non-SMEs were subject to a 5 percent deduction rate. The deduction amount was limited to a maximum 10 percent of corporate or income taxes of the current taxable year. However, for technology acquisitions by non-SMEs, a tax credit was provided only for those who acquired technologies from SMEs. Technology acquisitions from specially related parties were excluded from the tax credit program.

The Tax Incentive Structure in Korea

Korea's corporate income tax rates vary across different ranges of taxable income as follows: 10 percent for the range of KRW 200 million or less, 20 percent for the range of over KRW 200 million to 20 billion, 22 percent for the range of over KRW 20 billion to 300 billion, and 25 percent for the range of over KRW 300 billion. Additionally, local income tax is levied at a rate of 10 percent of corporate income tax. This means local income tax rates also vary across different ranges of taxable income: 1 percent for the range of KRW 200 million or less, 2 percent for the range of over KRW 200 million to 20 billion, 2.2 percent for the range of over KRW 20 billion to 300 billion, and 2.5 percent for the range of over KRW 300 billion.

Despite numerous tax exemption measures for policy purposes, the Korean government is operating the minimum tax program, making corporate taxpayers liable for the minimum tax. This program is intended to ensure fair taxation while securing the government revenues. A minimum tax rate of 7 percent is applied for SMEs. For companies which have graduated from the status of SMEs, a minimum tax rate of 8 percent is applied for the first three years after graduation and 9 percent for the following two years. The minimum tax rates also vary depending on the tax base ranges. For the tax base of KRW 10 billion or less, a minimum tax rate of 10 percent is applied. A minimum rate of 12 percent is applied for the range of over KRW 10 billion to 100 billion and the rate reaches 17 percent for the range of over KRW 100 billion.

The unused R&D tax credits can be carried forward for 10 years, but a tax refund is not allowed.

1.4 Financial Support Programs

1.4.1 Industrial Bank of Korea (IBK)

The Industrial Bank of Korea (IBK), formerly known as the Small and Medium Business Bank, was established in 1961 based on the ‘Small and Medium Business Bank Act’ for the purpose of promoting SMEs’ self-supporting economic activities and their economic status through establishing an efficient credit system for SMEs (Article 1 of the Small and Medium Business Bank Act). The scope of IBK’s support covers i) SMEs as defined in Article 2 of the ‘Basic Act on SMEs’ and ii) SME cooperatives and associations who serve the interests of SMEs according to Article 3 of the ‘Small and Medium Business Cooperatives Act’ (Article 2 of the Small and Medium Business Bank Act).

IBK supports SMEs’ technology acquisitions, import and export, and facility investment by providing them with capital and liquidity through loans. IBK also performs an array of supportive activities for SMEs including loan underwriting, bill discounting, issuing certificates of savings, deposits, securities and other types of liabilities, stock subscriptions and purchases, corporate bond subscriptions, purchases and guarantees, and research and counseling on SMEs (Article 33 of the Small and Medium Business Bank Act). As of the end of the 2nd quarter of 2021, IBK’s loan balance was estimated at KRW 197.2 trillion.

1.4.2 Lending Quota Program for SMEs

The ‘lending quota program for SMEs’ is a program to allocate a certain portion of a banks’ loan increase to SMEs. This program intends to ensure uninterrupted funding for SMEs with poor credit records and limited collateral capacity. It was introduced in 1965 based on Article 2 of the ‘Regulations on Financial Institutions’ Loan Operation’ of the Bank of Korea. The current lending quotas have been maintained since their revision in 1997.

This program is applied to commercial and regional banks’ loans to SMEs. When SME lending quotas are fulfilled by banks (i.e. 45 percent for commercial banks, 60 percent for regional banks, 35 percent for Housing & Commercial Bank, Peace Bank of Korea⁶, and

⁶ Housing & Commercial Bank and Peace Bank of Korea are now integrated into the Bank of Korea.

Korean branches of foreign banks, and 25 percent for Korean branches of foreign banks who are not servicing aggregate credit ceiling loans), these banks are qualified for receiving the bank-intermediated loan facility. For banks that do not fulfill SME lending quotas, a certain portion of the gap amount is deducted from the amount of the bank-intermediated loan facility. As of the end of March 2021, the credit ceiling of the bank-intermediated loan facility was set at KRW 43 trillion.

1.4.3 Korea Technology Finance Corporation (KOTEC)

The Korea Technology Finance Corporation (hereinafter referred to as 'KOTEC') was established in 1989 based on the Korea Technology Credit Guarantee Fund Act to help SMEs whose access to credit was limited due to a lack of collateral capacity despite their substantial technological competency. KOTEC extends credit guarantees for SMEs based on the evaluation of their future values through technology and business viability assessments.

SMEs in new technology areas are eligible to receive KOTEC's credit guarantee support. A business viability assessment is conducted for a comprehensive evaluation of SMEs' technological competency, business prospects, and management competency. Based on such an evaluation, appropriate amounts of credit guarantee support are provided for companies with stable business operations which are deemed technologically competent and viable.

KOTEC's guarantee facilities include the following: i) A guarantee for loans, ii) guarantee for commercial bills, iii) guarantee for contract performance, iv) guarantee for trade financing, and v) guarantee for e-commerce transactions. For loan guarantees, KOTEC provides guarantees for loans extended to companies by financial institutions. Through a guarantee for commercial bills, KOTEC guarantees the payment of the amount on the promissory notes issued by companies in connection with their commercial transactions. Through a guarantee for contract performance, KOTEC extends guarantees for a contractors' liability to pay various deposits as a security to ensure faithful performance of the contract. Included in this category are a bid deposit, contract deposit, payment deposit, and maintenance deposit to be paid in relation to construction contracts and supply of goods and services. KOTEC's guarantee for trade financing provides exporting companies with a guarantee for

raw material purchases. KOTEC's guarantee for e-commerce transactions is provided as a security for the buyer's liability to pay the loans or account payables arising in e-commerce transactions. As of 2021, KOTEC's guarantee balance was estimated at around KRW 26 trillion.

1.4.4 Bank-Intermediated Loan Facility (Bank of Korea)

The bank-intermediated loan facility is a lending facility that provides funds for financial institutions' intermediated lending to SMEs according to Article 2 of the BOK Regulations on Financial Institutions' Lending. When first introduced in 1994 based on the Bank of Korea Act, the program was called an 'aggregate ceiling loan', which was then renamed as the 'bank-intermediated loan facility' in 2013.

The recipients of the program include banks extending loans to SMEs who are eligible to receive the bank-intermediated loans. These banks are supported by the Bank of Korea with a low interest of 0.25 percent for a certain portion (up to 100 percent) of their loan performance. This special lending program offers support for i) trade financing, ii) startups, job-creating companies, and materials/ components/ devices companies (new growth engine/ job-creating businesses), iii) regional SMEs contributing to balanced regional development (stabilization of SME loans), and iv) COVID-19 affected SMEs' (including small and self-employed businesses) working capital loans with a maturity of less than one year. As of the end of March 2021, the aggregate ceiling of this program stood at KRW 43 trillion.

Program	Time of introduction	Ceiling (KRW trillion)	Interest rates
Trade finance support	1994. 3.	2.5	0.25%
Regional SME support	1994. 3.	5.9	
New growth & employment support	2013. 6.	13.0	
SME loan stabilization	2017. 9.	5.5	
Support for COVID-19 pandemic-affected businesses	2020. 3.	13.0	
Support for small businesses	2020.10.	3.0	
Total	-	43.0*	-

* Including reserved ceiling of KRW 0.1 trillion

1.4.5 Innovative Startup Loans

Composed of support schemes for both startup and technology commercialization, the innovative startup loan aims to promote the startup of technologically competent and viable SMEs which are lacking funds, so that they can contribute to job creation. The startup support scheme was introduced in 1998, which was then followed by the technology commercialization support scheme in 2000. The recipients of the startup support scheme include SMEs whose business history is less than 7 years from the start date of their operation and pre-startup entrepreneurs. In the meantime, the recipients of the technology commercialization support scheme include the following SMEs as defined in the Basic Act of SMEs: i) Inno-biz, main-biz, and venture companies planning to commercialize their indigenous technologies, intellectual property management companies certified by the Korean Intellectual Property Office, and companies which have attracted investments of a minimum KRW 100 million through crowdfunding, and ii) companies planning to commercialize technologies they have successfully developed (or completed their development) through participation in R&D projects funded by the central or local governments.

This program provides three types of loans, such as facility loans, investment loans, and working capital loans. Facility loans cover funds required for acquiring manufacturing facilities and test and inspection equipment, installing manufacturing processes, operating safety tests, establishing distribution and logistics facilities, and acquiring business sites (via purchase, auction, or public auction). Investment loans are provided to meet companies' capital needs for acquiring intellectual property rights (i.e. patents and trade rights), paying for outsourcing services (i.e. design and application development) including research, and acquiring IT related services (i.e. software purchase). Working capital loans cover funding requirements for startup, production, corporate management, and technology commercialization including raw materials purchases and market development expenses.

The interest rate of this special loan is lower by 0.3% points than the key interest rate (floating) of government policy loans. The period of the loan varies across the types of loans, such as within 10 years for facility loans, within 7 years for investment loans, and within 5

years for working capital loans. The loan ceiling is a maximum KRW 6 billion per borrower, which extends up to KRW 7 billion for companies located in local provinces. The total size of the loan is KRW 2.25 trillion.

1.4.6 Investment and Loan Hybrid Fund

The Investment and Loan Hybrid Fund was introduced in 2012 to support technologically competent SMEs with high-growth potential by accelerating startup activities and expediting the growth of startups.

This program targets technologically competent companies with growth potential and IPO possibility. The program specifically provides scale-up finance for SMEs with high-growth potential which want to have a direct access to the financial markets through the issuance of corporate bonds.

The program provides two types of loans, facility and working capital loans. Facility loans cover funding needs for introducing production facility and test & inspection equipment, upgrading IT systems, and paying for other relevant services (excluding funds for constructing or purchasing the business sites). Working capital loans meet the funding requirements for operation costs including raw materials, market development, and production costs.

The loans are extended in the forms of convertible bonds (CB), bonds with warrants (BW), or redeemable convertible preferred stocks (RCPS). The coupon rate of this loan is 0.5 percent with its maturity guaranteed interest rate set at 3 percent. The loan period is within 5 years in general. For startups with a business history of less than 7 years, the loan period is extended to within 7 years. The loan ceiling is a maximum KRW 6 billion per borrower, which extends up to KRW 7 billion for companies located in local provinces. The total size of the outstanding loans is KRW 140 billion.

2. Human Resources Support Programs in Korea

2.1 Current State of Ghana

According to the World Bank (2010), Ghana boasts 38.4 researchers in R&D per million population, which stands at only 40.1 percent of the average for Sub-Saharan Africa. Just 17 percent of Ghana's researchers are female compared to the African average of 31.6 percent. The majority of Ghana's research personnel are employed in the higher education sector (59.9 percent) and 38.3 percent of researchers work in government. Merely 1.8 percent of researchers are employed in business enterprise or private non-profit sectors.

In 2015, the number of researchers in R&D for Ghana was 89.1 per million people. Between 2007 and 2015, the number of researchers in R&D of Ghana grew substantially from 17.1 to 89.1 per million people rising at an increasing annual rate that reached a maximum of 134.76 percent. 34.4 percent of Ghana's researchers are educated to PhD level or equivalent, which is higher than the average of seven major African countries assessed. A ministerial directive mandating university lecturers to hold doctoral degrees by 2020 is considered to be a key driving force behind such an increase.

The government of Ghana's policy for developing engineers and scientists is focused on building TVET (Technical & vocational education & training) capacity aimed at developing intermediate technologies through i) technology development projects in selected economic sectors, ii) R&D support for the private sector, and ii) industrial projects for technology development. The government of Ghana also introduced the vocational training voucher program to support national skills assessment and national apprenticeship initiatives. The Ministry of Environment, Science, Technology and Innovation (MESTI), in particular, has been in charge of managing and implementing the country's STI policies over the years. MESTI is also offering jobs for the country's workforce by establishing the National Construction Corporation as well as national research institutes including the National Nuclear Research Institute. The ministry is also supporting vocational training and employment of the youth for sustainable development by enacting the Youth Employment

Act and launching the Youth Employment Agency. Ghana's educational institutions are training scientists and engineers through public universities and polytechnical institutes, while the government has established the Council for Technical and Vocational Education and Training (COTVET) to regulate, promote, and manage technical and vocational training of the workforce.

In summary, while Ghana possesses a pool of R&D experts based on its higher education system, the country has a weak linkage between the private sector and educational institutions and is lacking in R&D funds. It is also faced with challenges in hiring and supplying young researchers in R&D, let alone its underdeveloped R&D infrastructure such as insufficient R&D equipment, ineffective maintenance structure, and poor Internet coverage.

Taking Ghana's current state into account, the country's human resources training and support policies need to focus on i) bringing up skilled talents such as intermediate-level technicians and doctoral degree holders, ii) developing measures to train and utilize young researchers, and iii) building a linkage between the private sector and educational institutions. Particularly, these policies need to be geared towards i) strengthening human resources training tailored to the needs of the agriculture-related food industry and jobs, ii) translating R&D outcomes into actual products or services through linkages with local industries, and iii) expanding software development-related education and training programs.

2.2 Evolutionary Path of Korea

Korea carefully planned the development of scientists and engineers based on its national economic development plans and labor market conditions. Accordingly, Korea's plan for developing scientists and engineers was intricately designed, considering the need for an efficient supply of industrial workforce required for the country's industrial transformation and economic advancement. These were the prime objectives of its five-year economic development plans, which started in the 1960s. Around that time, however, the Korean government was faced with severe limitations in establishing institutions for technical

education and training and acquiring equipment necessary for apprenticeship. Korea addressed such problems by mobilizing support from major developed countries such as Germany, Belgium, and the U.S. It also received support from a group of international organizations including the International Labor Organization (ILO), the United Nations Development Program (UNDP), and International Bank for Reconstruction and Development (IBRD). In particular, the UNDP and German Gesellschaft für Technische Zusammenarbeit (CTZ) provided equipment for education and training for free of charge or at cost. Korean instructors, researchers, and technicians were also invited to participate in overseas training programs offered by some developed countries such as Germany and Australia. They were also provided with useful advices on policies and institutions for developing technological human resources.

Table 3-5 presents an overview of the evolutionary path of Korea's economic development, TVET policy and system, and its industrial human resources development targets by period based on the size of its GDP.

[Table 7-5] Evolution of Human Resources Support in Korea

GDP(\$) by year	147 (1960s)	753 (1970s)	2,991 (1980s)	9,707 (1990s)	16,740 (2000s)	25,409 (2010s)
Economy	Labor intensive	Heavy industry		ICT & high tech	Fusion convergence	Industry 4.0
TVET policy	VT Act	NQ Act	Levy system	EI Act	Private infrastructure innovation	New training era
Human resources system	Vocational training system	Vocational training obligations system	Establishment of KAIST & Science high schools	Graduate schools of Science & Engineering specialization	Fostering creative science & technology and convergence education	
Target of human resources	Mono skills	Multi-skills advanced		Technicians	Technicians & technologist	
		Science & technologist				

2.3 Introduction of Technical Human Resources Training System and Cultivation of Technical Human Resources in Korea (1960s-1970s)

In 1961, when Korea's Gross National Product per capita remained low at USD82, the Korean government established the Economic Planning Board, with the first Five-Year Economic Development Plan being instituted in the following year. The foremost objective of the plan was to build the foundations for a self-supporting economy. In order to achieve such an objective, the government focused on acquiring energy resources, improving agricultural productivity, reinforcing its key industries, improving its trade balance with increase in exports, and promoting technological advancement.

The Korean government instituted its second Five-Year Economic Development Plan to continuously drive the nation's economic development. With the aim of modernizing the country's industrial structure and building a self-supporting economy, the plan placed its priority on securing human resources capable of implementing a myriad of initiatives including the achievement of food self-sufficiency, development of chemical, steel and

machinery industries, improvement of the trade balance, improvement of farmers' income, and promotion of science, technology, and management skills. Toward this end, the Korean government enacted and promulgated the Vocational Training Act in 1967. It also set up a dual system of industrial human resources development, composed of i) public vocational training in the public sector and ii) corporate in-house vocational training in the private sector. Corporate in-house vocational training organized and hosted by companies is again divided into i) training for supervisors and ii) basic training for technicians. When the training programs were deemed to meet certain requirements and standards, they were authorized by the government and became eligible to receive government subsidies for training fees.

The 1970s witnessed an increase in both the number and size of businesses, which led to a dramatic increase in the demand for a technical workforce. This also expanded the scale of human resources training. As a result, the Korean government encouraged business owners to provide in-house training programs for their employees by offering government subsidies and introducing the Mandatory Vocational Training Scheme that requires businesses exceeding a certain size to provide vocational training for their employees. It also enacted the Basic Vocational Training Act. In 1974, the Korean government enacted the Act on Special Measures for Vocational Training, which mandated businesses with over 300 employees to train a certain percentage of their employees to become technical personnel. The percentage varied across industries within the limit of 10 percent. The Act also required businesses not complying with the Act to pay their contributions to the government's vocational training programs. As a result, the Mandatory Vocational Training Scheme induced businesses to focus on in-house training of their own personnel rather than paying the contributions. Such a development became the major driving force behind an exponential quantitative growth of corporate in-house vocational training.

In the process of developing a technical human resources training system, the Korean government introduced the National Qualifications System to objectively assess and certify the quality of Korean workforce. For this, the Korean government enacted the National Technical Qualifications Act in 1973 and established the Korea Testing Institute of Technical Qualifications under the purview of the Ministry of Science and Technology as an organization dedicated to the operation of national technical qualifications tests. Based on

this Act, technical assessments to acquire national technical qualifications were administered, in which eligible applicants took written and skills tests along with interviews to acquire qualifications. To become eligible to take the tests, applicants were required to meet certain criteria including both academic credentials and work experience.

2.4 Solidifying the Foundation for Developing Technicians and High-Caliber Scientists and Engineers (1980s-1990s)

Following the quantitative growth of vocational training, the Korean government started to focus on improving the quality of its vocational training programs. In recognition of such shifts, the Korean government established the Korean Vocational Training Management Agency (KVTMC) in 1982. Its responsibility was to link vocational training, qualifications assessment, and vocational support, so that it would contribute to improving the quality of vocation training, thereby ensuring high-quality jobs for qualified human resources. The agency was comprised of 37 departments under 13 divisions. Under the headquarters, the agency operated 24 vocational training centers nationwide. It also operated 13 regional offices, located in various cities and counties throughout the nation, and had a total of 39 affiliated organizations. While the agency directly administered vocational training through its affiliated vocational training centers, it also engaged in developing and circulating vocational training related standards and textbooks.

The following years witnessed an increase in demand for technologically competent ‘technicians’ rather than manually skilled ‘craftsmen’. Responding to such changes, the Korean government launched joint vocational training centers to develop and train skilled talents through cooperation between specialized high schools, vocational training centers, educational training institutes and businesses. The budget required to open such centers was partly provided by the central government through the vocational training promotion fund, with the rest being funded by a number of local governments. In the meantime, the operational budget of these joint centers was funded by local governments and regional businesses.

In the 1980s, the Korean government instituted a system for developing gifted and talented persons in science and technology by establishing science high schools, the Korea Institute of Technology (KIT), and Korea Advanced Institute of Science and Technology (KAIST). During this period, as the consensus emerged that the old way of relying on inexpensive labor would no longer guarantee Korea's future competitiveness, the demand for skilled talents capable of conducting future-oriented innovative research increased. After its establishment in 1973, the Korea Advanced Institute of Science (KAIS) played the important role of supplying high-caliber scientists and engineers by producing over 1,000 master and doctoral degree-holders. By the late 1970s, however, as the country's industrialization led by heavy and chemical industries accelerated and numerous government-funded research institutes were established, it became apparent that KAIS alone could no longer meet the soaring demand for high-caliber scientists and engineers. This explains the rationale behind the establishment of a new system for developing high-caliber scientists and engineers.

In a bid to enhance the linkage between education and research, the Korean government established KAIST in 1981 by merging KAIS and the Korea Institute of Science and Technology (KIST). Since then, KAIST has placed its focus on doctoral programs. In addition, KAIST introduced a part-time doctoral program, where researchers affiliated with industries or government-funded research institutes could obtain degrees while working in their regular jobs, thereby improving the link between education and research.

The Korea Institute of Technology (KIS), which was eventually merged into KAIST, was established in 1986 with the aim of discovering and training scientifically gifted and talented youths. Upon its establishments, KIS provided appealing offers such as free tuition, free dormitory, scholarships, and overseas training for a total of 540 students to be enrolled in 4 departments and 16 majors. Under its mission statement, KIS instituted a flexible academic program with no designated majors or school years, allowing students to graduate early. KIS also accepted those who had completed two years of the three-year course at science or ordinary high schools. Until it was finally merged into KAIST, KIS had maintained over 1982 enrolled students. After the merger, a significant number of early graduates of KIS moved onto KAIST master's program. This clearly shows the intermediary role KIS played in developing high-caliber scientists and engineers.

For the development of high-caliber scientists and engineers, the Korean government also pushed forward the establishment of science high schools as an educational institution dedicated to developing scientifically gifted and talented youths at the high school level. Starting from Gyeonggi Science High School for the Gifted which opened in 1983, the Korean government established three more science high schools including Daejeon Science High School, Jeonnam Science High School, and Gyeongnam Science High School in 1983. By 1993, there were 13 science high schools under operation in 12 different cities and provinces. Following its mission statement of discovering and training scientifically gifted and talented youths at an early age, science high schools operated science-focused curriculums compared to ordinary high schools. In addition, a fast-track program was introduced to allow students to graduate early from science high schools and take courses at KIT. Between 1983 and 1992, a total of 3,151 graduates of science high schools nationwide were admitted to KIT. Of those, 1,861 were early graduates who entered KIS after completing their high school curriculum in two years, and also accounted for 85% of KIT's total enrollment. In fact, this highlights the importance of science high schools as a hub for developing scientifically gifted and talented youths.

The Korean government also expanded its support for university-based R&D programs for basic sciences in order to reinforce science and engineering education at the graduate level. Examples of such support are the Excellent Research Center Support Program by the Korea Science and Engineering Foundation (KOSEF) and the Support Program for University-Affiliated R&D Centers by the National Research Foundation of Korea (NRF). The government also expanded special cases of military service and scholarships for graduate students of science and engineering.

2.5 Developing Creative and Convergent Talents in Science and Technology (2000s-Present)

In the 2000s, the increased emergence of new technologies and widening scope of creative basic sciences highlighted the importance of developing core S&T talents capable of creating original knowledge and lead technology innovation in the nation's S&T human resources development policy. In developing such policies, the Korean government emphasized the

need to enhance autonomy at universities, promote the strategic linkage between education and research, discover scientifically gifted and talented youths, and expand the international exchange of S&T talents. In particular, the government reinforced programs for developing convergent talents and instituted life-cycle-based support policies to help talented students in science and engineering become nationally recognized scientists in their respective fields.

During this period, the education of scientifically gifted and talented youths was further reinforced as the percentage of those receiving GT (gifted and talented) education relative to the entire number of students enrolled in elementary, middle, and high schools increased from 0.77 percent in 2008 to 1.59 percent in 2011. A plan to promote four specialized universities in science and technology, including KAIST, was mapped out to promote research-oriented universities. In addition, the government also introduced the World Class University (WCU) Program to invite foreign talents from abroad, spending USD 699.15 million between 2008 and 2013.

For promoting SMEs and venture companies' human resources development, the Korean government reinforced its support for SMEs which hired high-caliber talents holding master or doctoral degrees. It also implemented a series of policies for promoting university-industry cooperation while encouraging universities and government-funded research institutes to provide support for SMEs and venture companies.

2.6 Key Policies and Programs in Korea

2.6.1 Developing Skilled Technicians

1) Specialized High Schools and Meister High Schools

In Korea, vocational training often starts at the high school level through a three-year program led by specialized and meister high schools. According to Article 91 Section 1 of the Enforcement Decree of the Elementary and Secondary Education Act, specialized high schools were established to provide specialized field training and practice-oriented education for students with similar traits, aptitudes, and abilities, so that they would

become skilled talents in specific fields. Article 91 Section 1 of the same Decree stipulates that meister high schools are 'high schools tailored to the needs of the industries'. And so these schools provide customized curriculum directly linked with the needs of the industries. Meister high schools are designed to meet the needs of the industries as they are aimed to train young 'meisters' according to individual students' talents and aptitudes. Their eventual goal is to bring up top-class experts in technology, technology transfer agents, entrepreneurs, and business managers. While students of specialized high schools can choose between work and study upon graduation, graduates of meister high schools have the sole option of work immediately after graduation.

Both types of high schools provide full scholarships that cover admission fees and three-year tuitions, so students can focus on study without any financial burdens. As of the end of 2016, there were 287,684 students enrolled at 471 specialized high schools nationwide. The figure for meister high schools was 17,781 at 40 schools.

2) Korea Polytechnics

Korea Polytechnics (KOPO) is a public vocational training institution devoted to the holistic development of vocational competencies. Based on the Vocational Training Act, KOPO has been operated with government funding since its establishment in 1968. In March 2006, 24 vocational colleges and 19 vocational training centers were merged together to create the current version of KOPO. Under the purview of the Ministry of Employment and Labor, KOPO is operating two-year and four-year degree programs as well as other programs for both the unemployed and employed at 39 campuses throughout the nation. It has 240 different majors and enrolls about 60,000 students each year. When the regionally specialized programs for the employed are taken into consideration, the annual number of trainees enrolled in KOPO's vocational training programs reaches about 100,000.

KOPO's mission over the past 50 years has been developing and training technicians and craftsmen in the nation's backbone and strategic industries by tailoring its programs to the needs of industries, so that they can lead the country's industrialization and regional economic development. While promoting job creation tailored to the needs of regional

industries, KOPO is operating various vocational training courses to lead the country's vocational training and education. KOPO is also famous for its field-linked system called 'Factory Learning', which promotes synergies with industry.

Included in its degree programs are the industrial associate's degree program and intensive major course program. The industrial associate's degree program is a two-year course for technicians and aims to produce intermediate technicians linking field technician and skilled technician courses. The program runs a total of 145 majors across 26 different campuses. As of 2021, the number of enrolled students is around 13,000. In the meantime, the intensive major course is an advanced technical educational program targeting engineers who are graduates of two-year colleges including vocational colleges. The degree of this program is considered equivalent to a four-year university's bachelor's degree. This two-year program is being offered at nine campuses as an evening course, and as of 2021, the program has 500 enrolled students.

As a non-degree program, KOPO is also operating the master technician course, a one- or two-year program designed to train highly skilled craftsmen who can supervise skilled technicians in the field. Currently, this program has 275 enrolled students. In the meantime, the Vocational Training Program offers vocational training in the form of field training for a period between three to twelve months for young job-seekers as well as the socially vulnerable. This program is being operated at 39 different campuses, and as of 2021, the program has 8,500 enrolled students.

2.6.2 Developing Science and Engineering Professionals: Science and Engineering Specialized Universities

The Korean government has been steadily funding science and engineering specialized universities since their establishment. KAIST, the first Korean science and engineering university founded in 1971, is aimed at cultivating leaders of state-of-the-art scientific and technological innovation. Following KAIST, several other science and engineering universities were established in the 1990s in major cities, such as the Gwanju Institute of Science and Technology (GIST) in 1993, the Daegu Gyeongbuk Institute of Science and

Technology (DGIST), and Ulsan National Institute of Science and Technology in 2004 and 2007 respectively.

Their research infrastructure is being operated based on the government's budget support. Students enrolled in these universities hardly pay any tuition and can qualify for various scholarship opportunities. Students or graduates of these universities are often praised for having unparalleled competencies in their respective fields. Those who earn a doctoral degree from one of these universities are even exempted from mandatory military service.

[Table 6-6] Science and Engineering Universities

Category	Year established	No. of students enrolled* (2016)		
		Bachelors	Masters & doctorates	Total
Korea Advanced Institute of Science and Technology (KAIST)	1971	3,932	6,701	10,633
Gwangju Institute of Science and Technology (GIST)	1993	665	1,257	1,922
Daegu Gyeongbuk Institute of Science & Technology (DGIST)	2004	533	378	911
University of Science and Technology (UST)	2004	-	1,235	1,235
Ulsan National Institute of Science and Technology (UNIST)	2007	3,184	1,291	4,475
Total		8,314	10,862	19,176

2.6.3 Professional Skills Development Program for Science and Engineering Majors

Since 2003, the Korean government has been providing vocational training programs by industry for the unemployed science and engineering majors. It has been also brokering job opportunities to graduates of these programs. The applicants of the programs must be aged 32 or under. The Ministry of Science and ICT provides funds for technical training and field practice. While the program is being managed by the Korea Industrial Technology Association (KOITA), the program courses are being hosted by various government-funded research institutes, national and public research institutes, specialized production

technology research institutes, colleges and universities, non-profit organizations, and other organizations specialized in developing professional technicians.

The host institutions receive the government funding for the overall course operation and trainees receive a monthly stipend of USD 254-423. The program runs for six months, with four months of technical training and two months of on-the-job training.

[Table 6-7] Overview of Professional Skills in the National Sciences & Engineering Training Program

Category		Year established	No. of students enrolled* (2016)
Duration of program		6 months	
		4 months	2 months
State	Education institute	Overall costs related to operation of educational program (wages for staff in charge of programs, instructor fees, etc.)	
	Trainees	Trainee allowances (KRW 300,000 monthly)	Trainee allowances (KRW 500,00 monthly)
Program description		• Theory and practice in specialized fields • Training on basic job functions and theory required at companies	• Practical job training at companies recommended by the host organization

2.6.4 Developing and Supporting Local Scientists and Engineers

The Korean Ministry of SMEs and Startups has promoted a ‘wage subsidy’ support program. Through this program, the Korean government is providing financial support for SMEs and mid-ranked firms who hire R&D personnel. According to this program, a certain percentage of wage of these newly recruited R&D personnel is subsidized by the government. The government subsidy is often granted after hiring, but not every hire is eligible for the government subsidy, so businesses are inevitably competing to hire researchers who are eligible to receive such subsidies. For example, even though 900 researchers are hired by companies after the government’s announcement of its plan to subsidize the wages of 300 researchers, the government subsidy is paid only to cover the wages of 300 researchers.

3) Hiring Support Program for Newly Graduating Master's or Doctoral Degree Holders

Since 2004, this program has been supporting SMEs and mid-ranked companies' hiring of high-caliber R&D personnel, master's or doctoral degree holders by subsidizing their wages. Only those who have attained master's or doctoral degree in science and engineering within three years from the time of hiring are eligible for the subsidy. For SMEs and mid-ranked companies, subsidies are paid for two people per company for up to three years. The average annual salary of master's degree holders is estimated at USD 16,949, while the average labor cost per employee stands at USD 21,186. The salary paid by the companies must be at least on par with the minimum standard pay.

4) Hiring Support Program for R&D Personnel with Prior Experience

Since 2006, the Korean government has been supporting SMEs and mid-ranked companies who hire R&D personnel with prior experience. Those eligible for this program are bachelor's degree holders with 10 years of prior experience, master's degree holders with 7 years of prior experience, or doctoral degree holders with 3 years of prior experience. For SMEs and mid-ranked companies, subsidies are paid for one person per company for up to three years. Regardless of the types of academic degrees, the government subsidizes half of the actual salary paid to eligible R&D personnel, and the annual subsidy cap per person is USD 4,237.

2.6.5 Invitation Programs of Foreign Scientists and Engineers

1) Brain Pool Program

The Brain Pool Program was instituted in 1994 as a part of the government policy to create productive and favorable research environment by overcoming the barriers of technology protectionism (i.e. developed countries' reluctance to transfer their technologies), and thus inviting renowned foreign scientists to Korea to be engaged in local R&D initiatives. Under the program, foreign invitees work at various R&D related organizations including

government-funded research institutes, universities, corporate R&D centers, and other related institutions.

Those eligible for invitation are foreign nationals holding doctoral degrees with more than five years of R&D experience in foreign countries. For foreign scientists to be invited by corporate R&D centers, they should have at least 5 years of R&D experience at overseas companies, though a doctoral degree is not a prerequisite. The period of invitation is between three months and one year in principle, which can be extended to up to three years. Invited foreign scientists receive financial support for all expenses to be incurred during their stay in Korea including research expenses, salary, living expenses, relocation expenses, and insurance fees. Since the start of the program in 1994, the Korean government has supported a total of 1,798 foreign scientists, of whom 1,657 (92.5%) have worked for universities and government-funded research institutes.

2) Hiring Program of Foreign Science and Technology Professionals

The hiring program of foreign S&T professionals was instituted in 2012 for the purpose of attracting and utilizing prominent skilled foreign talents, thereby improving the technological competence and global competitiveness of Korean SMEs and mid-ranked companies in the ICT field, which especially faces personnel shortages. Applicants must have prior job experience at ICT firms, research institutes, or universities (including graduate programs) in their native countries. As long as they meet these requirements, anyone is eligible to apply for the program regardless of their academic degree, age, or nationality.

Short-term programs last for less than 12 months, whereas long-term programs can be extended to over a year to a maximum of five years. Once candidates are selected, the government allocates 70 percent of the contracted annual salary as a part of labor costs. The yearly cap on government subsidies is USD 127,118, and round-trip flights and housing stipends are also provided for invitees regardless of the period of their stay. Applicants for long-term programs are eligible to receive additional government subsidies to pay for their children's international school enrollment fees, tuitions, and medical expenses.

2.6.6 Job and Career Support Program for Science and Engineering Majors

The Career Service Center for Science and Engineering Majors analyzes and provides job-related information for science and engineering majors and introduces them to hiring opportunities offered by leading R&D companies. The center was established in 2008 based on the Special Act on the Support of Scientists and Engineers for Strengthening National Science and Technology Competitiveness.

The Career Service Center for Science and Engineering Majors is operating a job portal (www.rndjob.or.kr) for scientists and engineers. This website provides the aforementioned information on R&D jobs, connects hirers to science and engineering graduates, and provide recruitment information for scientists and engineers. It also provides customized recruitment search and job arrangement services for its users. As of 2014, 25,960 companies and 68,293 job seekers are using this job portal.

The center also holds job fairs for leading scientists and engineers, and provides hiring services. It also hosts job seminars for job-seekers, sponsors lectures and employment consulting by HR personnel and hiring experts, and also conducts surveys on hiring trends and employment status, while providing statistics on corporate R&D centers and unemployed scientists and engineers.

3. Technology Development & Information Support Programs in Korea

3.1 Current State of Ghana

As of 2010, Ghana's gross domestic expenditure on R&D (GERD) is USD 277 million in terms of its purchasing power, and the ratio of GERD to its gross domestic production (GDP) is 0.4 percent. If we look closer at the total R&D investment by source of funds, the government sectors account for 68.3 percent of GERD, which is more than two-thirds of the total R&D investment in the country and is followed by the overseas sectors taking 31.2 percent. In other words, the funds for R&D investment in Ghana are raised in large by the government and overseas sectors. In particular, the dependence on the government sectors is very high, which implies that for Ghana, the government drives the national R&D system, and the country gets actively involved with international organizations to secure R&D funds. In contrast, funding for R&D from higher education sectors in Ghana, private for-profit and non-profit sectors remain very low. Looking into Ghana's GERD by sector of performance, the government sectors account for about 96 percent of the total, whereas non-government entities use merely 4 percent of the total GERD. While the private sectors lead the national R&D in developed countries, making advanced R&D ecology, most developing countries have an ecological structure for R&D where the government sectors play a key role for the national R&D, with Ghana following the latter case. For Ghana's GERD by R&D stage, development research accounts for 49.4 percent, followed by applied research (33.8%) and fundamental research (16.8%). As the higher educational institutions (college/university) engaging in fundamental research in Ghana grow slowly, most of its R&D is carried out with imported technology.

Looking into the current status of digital infrastructure and information support, only 35 percent or so of the population in Ghana has access to the internet and broadband speed is at 9.9kb/s, which is very slow. This low level of digital infrastructure becomes a high entry barrier to smooth access to research data and academic publications. Therefore, it is critical

that Ghana began to establish an information support system at a national level to build global high-tech networks such as a science and technology research network. Moreover, only a small number of the research institutes in Ghana collect and store research data on their own at the level of an organization, while the country does not have an R&D data management system at a national level. In Ghana, it is known that only two research institutes are properly managing research data at the level of an organization: the Innovation Organization for Poverty Elimination and the Research Information Center of the University of Ghana. The lack of a research data management system at a national level makes it virtually impossible for research performers to access and exchange their research-related information. Therefore, it is urgent to establish a system in which domestic and overseas information can be collected, managed, and shared for common use.

3.2 Evolutionary Path of Korea

3.2.1 Stage I (1960s-70s)

Korea's technology development policy has changed through three stages. The first stage began in the 1960s and 1970s, which was the foundation set-up and ramping up period. As the fundamental policy direction in the 1960s was set to 'building the foundation and groundwork for national science & technology advancement', the central government began to develop science and technology policies to free Korea from a dependence on imported technology support. Above all, the central government organized the Korea Scientific Documentation Centre (KORSDOC) within the Korean National Commission for UNESCO (KNCU) to build a science and technology information infrastructure. And it reformed and expanded the organization into the Korea Scientific and Technological Intelligence Center (KORSTIC) to start in earnest the information activity of science and technology. In addition, the central government established the Korea Institute of Science and Technology (KIST), a national research institute, to support industries with research institutes and foster scientific and technological capabilities. The birth of KIST, which has played a role as a general research institute for Korea's science and technology sectors, required the expansion of government expenditure for R&D, naturally leading to the establishment of the Ministry of Science-Technology, a government department

responsible for science and technology. In the 1970s, the keynote of the policy for science and technology development was referred to as 'Establishing the Foundation for Technological Independence and Shaping the National Science and Technology System'. As economic development accelerated in Korea, the central government newly organized the General Science and Technology Council (GSTC), which is a coordinating body exclusive for science and technology, to strengthen science and technology policies. The new organization began mediating science and technology policies and businesses systematically, reinforcing the technical support for companies by enacting laws related to technology development. In addition, the central government embodied the national science and technology system in the human development and research management sector, which aimed to elevate the capability of freeing domestic R&D from a dependence on overseas support. And it newly formed many government-funded research institutes to scale up the research institutes for public sectors. In summary, during the build-up and reinforcing period (stage 1), the government departments and public research institutes accounted for a considerable portion of national R&D investment. And the government has sought both the public benefit and profitability, focusing on the application of foreign technologies and development research with an aim to economic growth. Meanwhile, pursuing a tight connection with public research institutes, the business sectors using government subsidies scaled up their R&D activities on applied and development research, focusing on profitability. Academia (colleges and universities) concentrated on fundamental research on their own, seeking public profit, but didn't make much progress. Therefore, they seized a small portion of national R&D.

3.2.2 Stage II (1980s-90s)

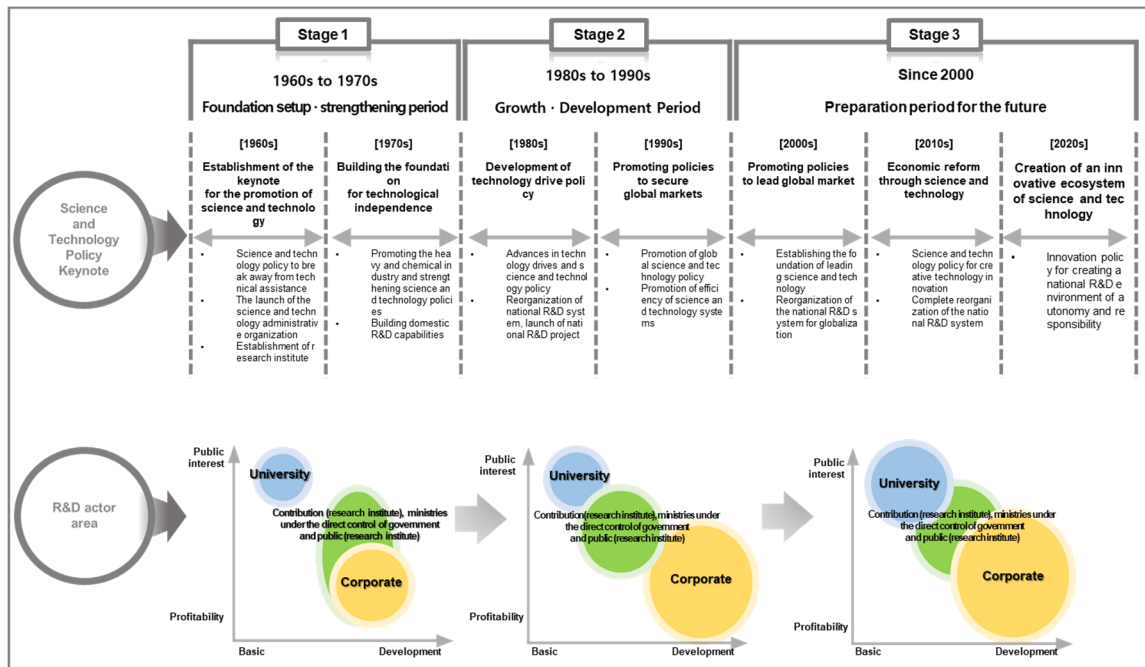
The second stage spanned the 1980s and 1990s, which was the period of growth and development. In the 1980s, the keynote of policy was called the 'Development of Technology-Driven Growth and Expansion of the National R&D System'. As the national economy turned to minus growth due to the triple whammy of (i) industrial imbalance caused by rapid economic growth, (ii) aggravation of the global economy, and (iii) technology protectionism maintained by developed countries, the central government

initiated a technology-driven growth policy to attempt industrial structure advancement through technological innovation. In addition, it expanded the national R&D system to support its technology-driven growth policy more effectively and improved the efficiency of national R&D investment and research organizations. In the 1990s, the keynote of policy was called the 'Promotion of Science and Technology to Develop High Tech and Secure the Global Market'. Aiming to advance domestic science and technology to the level of G-7 by 2001, the central government embarked on the guidance technology development business project (G7 Project in short), activating privately led technology development and leading the boom of establishing company research institutes. In addition, it devised 「The Comprehensive Plan for Fundamental Research Advancement」 and attempted to reorganize the roles and directions of key support directions for university research. Furthermore, it enacted 「The Special Act for Scientific and Technology Innovation」 to promote science & technology innovation at the governmental level. In summary, during the growth and development period of national R&D, the government departments and public research institutes pursued balanced and aligned development throughout the whole areas of research by system integration and adjustment. Making a remarkable growth in scale with the technology-driven growth policy and G7 Project of the government, the business sectors accounted for most of national R&D. The R&D by the academia (colleges/ universities) also expanded in size thanks to the government subsidies for the promotion of fundamental research.

3.2.3 Stage III (since 2000)

Stage III spans from the 2000s until the present. It is the preparation period for the future. The keynote of policy in the 2000s was characterized as 'Advancement of Science and Technology System to Leap into Global Leadership'. Therefore, the central government newly formed the national technological innovation system, so that it could be easy to acquire, create, use, and spread knowledge suitable for a knowledge-based society and advanced science and technology information infrastructure. In addition, as universities became the main body for research and development, the government launched the Ministry of Education, Science and Technology to smoothly connect higher education policy

with science and technology policy. Moreover, it organized the National Research Foundation of Korea to unify the research support systems, which had been dispersed, scaling up national R&D projects. In the 2010s, the keynote of policy was referred to as 'Strengthening the Capability of Leading Science and Technology Based on Creativity and Innovation for an Economic Resurgence. To realize this, the central government newly organized the Ministry of Science, ICT and Future Planning and set (i) science and technology and (ii) information and communication technology as two core axes for the national growth engine, and it started to promote the science and technology policy for creative technology innovation. Moreover, to keep abreast of the enlargement and convergence of fundamental science in the 21st century, the central government opened the Center for Creative Economy and Innovation nationwide, while the centers supported startups and SMEs venture companies and strengthened fundamental science research. Then in the 2020s, the keynote of policy is referred to as 'Creating the Ecology of Science and Technology Based on Self-Regulation and Responsibilities for Coping with the Future.' To respond to the changes in the economic and social environment, which feature the 4th Industrial Revolution and ESG, the central government puts as the top priority: creating the national R&D environment in which the self-regulation and responsibilities of R&D performers are emphasized. Accordingly, it sets up the comprehensive government R&D management standard regarding the spending of a research grant, agreement, and the like and relieves the administrative burden accompanied with an R&D project, while innovating a research support system by getting tougher on unfair practices. In summary, during the preparation period for the future, all the government departments, public research institutes, business sectors, and academia scale up their R&D activities. In particular, the business sectors take the biggest portion of national R&D. In addition, Stage III is distinguished from the other stages in that the policy for open research and research convergence and promotion reinforced the connection and mutual cooperation among the government departments, public research institutes, business sectors, and universities.



[Figure 3-2] Step-by-step transition process of major policies and programs

Material: Ministry of Science and ICT (2017), 50-year history of science and technology, Korea Institute of Science and Technology Evaluation and Planning (2021), a study on the current status investigation and analysis and development plan for R&D innovation for public research institutes for the 2020 science and technology innovation policy support project

3.3 Major Policies and Programs in Korea

3.3.1 Public Sector-Led R&D Projects

Public sector-led R&D projects are the R&D projects supported with the national budget or a fund on the R&D assignment or business project designated by the central administrative agency in compliance with the laws to solve the challenges of science and technology that require R&D at a national level. A public-led R&D project includes a preliminary feasibility test and evaluation in its planning stage to budget and manages a fund for a large-scale new business. With this, it aims to improve the efficiency of financial operation, strengthen

national competitiveness through investment in science and technology R&D, and attain innovation in science and technology to improve the life quality of the public.

Looking into it more closely, the head of the central administrative agency selects the target projects for a preliminary feasibility study among public sector-led projects requiring the study. When determined 'suitable', the selected projects go through a preliminary feasibility study. The candidates for a preliminary feasibility study are limited to a new R&D project whose total cost is more than KRW 50 billion, national subsidies are more than KRW 30 billion, and that is classed as a national R&D project specified in Article 11 of the Framework Act on Science and Technology. For the projects selected for a preliminary feasibility study, a project manager, etc. and an advisory committee are formed by project. And they are also reviewed by the assessment advisory committee of national R&D projects. After that, a preliminary feasibility study is carried out in earnest on the projects and the results are notified to the project-ordering department and related agencies. To be selected as a project subject to a preliminary feasibility study, the candidate projects are evaluated for their technology on 4 categories and 10 items by project.

[Table 3-8] Categories and Items for Technology Evaluation

Evaluation category	Evaluation items
Necessity & urgency of science & technology development	• Urgency of R&D implementation • Maturity of project implementation conditions • Ripple effect by the project
Suitability for national subsidy	• Validity of national subsidy • Efficiency of national subsidy
Differentiation from/ and association with existing project(s)	• Differentiation from existing project(s) • Plan to associate it with existing project(s)
Materiality of project plan	• Systematicity of project idea • Adequacy of implementation system • Materiality of operation plan

Source: Korea Institute of Science & Technology Evaluation and Planning (2021)

According to the performance analysis of 216 public sector-led R&D projects that completed a preliminary feasibility study until December 2020 on 216, the total project cost proposed was KRW 112 trillion and 401.9 billion. And the total projects costs were adjusted through

the studies to KRW 38 trillion and 147.4 billion, which is 33.9 percent of the originally requested amount by the managing departments. That is, those preliminary feasibility studies saved KRW 74 trillion and 254.6 billion, which proves that public sector-led R&D projects prevent national budgets from being wasted and secure the efficiency of financial management while leading science and technology R&D to strengthen national competitiveness.

3.3.2 Private Sector-Led R&D Project (1): Cultivation of Locally Specialized Industry + Project (R&D)

With the objectives of policy to balance development for the sake of equality and strengthen local competitiveness for efficiency's sake, the central government establishes the policy of cultivating locally specialized industries, which holds the nature of both 'local policy by the central government' and 'industrial policy by a local government', and implements the policies with both the same and different means by region. As part of the policy implementation to promote local industries, the central government realizes a local industry support policy to cultivate locally specialized industries, which helps local communities stand on their own, increases local jobs, and activates the local economy. The cultivation of locally specialized industry + project (R&D) is a kind of R&D-oriented local industry support project, and is defined as a business project led by a local community to advance the balanced national development and solidify local competitiveness, and intensively cultivate locally strong businesses. In the late 1990s, the concentration of industries in metropolitan areas got worse. As a result, the central government announced the 1st Plan for Promoting Local Economies and began to build infrastructure for local industries and implement R&D local support projects to expand the basis of technological innovation. Those industries were directed toward the advancement and activation of convergence-oriented locally specialized industry; securing of sustainable local growth base; building of a supportive base for SMEs firms with convergence-intensive technology innovation in the field of locally specialized industries in non-metropolitan areas weak in innovation capability; creation of a future growth engine for local communities through cultivation of locally focused industries based on regional balance new deals, invigoration of

local ecology, and creation of quality jobs for local companies through the intensive nurturing of local strong companies.

Looking at it more closely, the cultivation of locally specialized industry + project (R&D) is planned and supervised by the Ministry of SMEs and

Startups (the Korea Institute for Advancement of Technology is entirely responsible for the project). And local governments (supporting Techno Parks) make an implementation plan and carry it out according to the guidelines of the central government). Then local business project evaluation groups assess and manage the projects of 14 cities and provinces. Those cultivating projects are divided into two types: the cultivation of (i) locally-focused industries and (ii) locally strong companies. Firstly, the cultivation of a locally-focused industry is an R&D project aimed at commercializing the new products developed by SMEs in locally-focused and convergence-oriented (product + ICT, products + services) industries of 14 cities and provinces. This project supports the successful candidates having at least one (head office, factory, and R&D lab) in 14 cities and provinces through the application system for item designation or public competition. By project, about KRW 200 million is subsidized for a year (maximum two years) However, creating one new job per KRW 200 million (on average) is the precondition of the subsidy. The cultivation project of locally strong companies is a commercialization R&D support project for the locally strong companies (designated by a local government) to enter into the global market support. Through the application for public competition, successful local strong companies (selection criteria: sales more than KRW 5 to 40 billion and excellent in social contribution such as job creation. 200 companies are selected every year) are added with about KRW 200 million by project for a year (maximum two years) However, creating one new job per KRW 200 million (on average) is the precondition of the subsidy.

The overall performance review of the cultivation of locally specialized industry + project (R&D) for 2020 shows its positive effect on the development of technology competitiveness and economic vitalization. However, the performance of patent registration is too early to conclude because not many projects, which began in the 2020s, have not been yet

completed and nor have the patents of the projects. Nonetheless, it is positively expected that the number of patent registration will increase as the application increases. However, the performance of patent registration is too early to conclude because not many projects, which began in the 2020s, have not been yet completed and nor have the patents of the projects. Nonetheless, it is positively expected that the number of patent registration will increase as the application increases.

[Table 3-9] Overall Performance (2020) of the Cultivation of Locally Specialized Industry + Project (R&D)

Type	National Subsidy (KRW 100 million)	No. of SCI Theses		No. of Patents Applied		No. of Patents Applied Registered		Project Success Rate	Project Sales (KRW 100 million)		No. of New Jobs	
		No. of theses	Per subsidy of KRW billion	No. of patents	Per subsidy of KRW billion	No. of patents	Per subsidy of KRW billion		Sales	Per subsidy of KRW billion	No. of workers employed	Per subsidy of KRW billion
Cultivation project of locally-focused industry	792.5	2	0.0	107	1.4	0	0.0	-	106.1	1.3	462.5	5.8
Cultivation project of locally strong companies	258.1	0	0.0	38	1.5	0	0.0	-	118.1	4.6	189.5	7.3

Source: Korea Institute for Advancement of Technology (2021)

3.4 Private Sector-Led R&D Project (2): World Class 300 Projects Technology Development (R&D) Project

The central government seeks the balanced development of the national economy through the advancement of industrial structure. As a part of its effort, it promotes the policy of fostering Korean-style hidden champions, so that the mid-sized companies can leap to globally specialized companies through both quantitative expansion and qualitative growth. The World Class 300 Projects Technology Development (R&D) Project is part of this policy, which selects mid-sized companies and potential companies of the size with growth potential and technological capability in the global market and supports their mid-and long-term future strategy and original technology development. To propose the desired role model for SMEs, the policy was conceived and introduced as the innovation to shift the industrial economy, which has been mainly led by large conglomerates, to a SME company-based creative economy has emerged as the mission of the age and their role is stressed as hidden champions that will lead the qualitative growth of the national economy and the 4th Industrial Revolution. This project aims to select 300 SMEs with the volition and potential to grow into global companies and nurture them into world-class companies that will lead the 4th Industrial Revolution and new growth sectors through strengthened R&D capability. In addition, the policy aims to improve the weak middle layers of Korea's industries through the aforementioned support and create quality jobs through the ongoing expansion of growth engines.

More specifically, the Ministry of Trade, Industry and Energy supervises the World Class 300 Projects Technology Development (R&D) Project and makes its R&D support plan. And the Korea Institute for Advancement of Technology performs substantial roles in supporting, managing, and operating it including selection, evaluation (including interim evaluation), and post management of the target companies. It is a general-type project: a single task to fulfill by one managing department alone or a managing department in concert with other participating agencies (joint project). The candidates for the World Class 300 are evaluated with the statement of growth strategy they submit. Of the qualified companies for the project, those mid-sized companies with sales of over KRW 70 billion are given the annual government subsidy of a billion won (ceiling) for four years (max.) to develop core and

applied technology. However, the portion of government contribution cannot exceed 50 percent of the total annual development costs. More specifically, the applicant for the World Class 300 Project should be a manufacturer (including some other industries) having a closed record(book) of KRW 40 billion (KRW 10 billion from SW) ~ KRW one trillion in sales as of the previous year with an average ratio of investment in R&D to the total sales at over 2 percent for the last 3 years, or CAGR at over 15 percent for the last 5 years.

The performance review of the World Class 300 Projects Technology Development (R&D) Project shows that the selected companies enjoyed the increase in sales (35.6%p), exports (12.0%p), and R&D investment (14.1%p) through participation in this project. In particular, this R&D investment support turned out to exercise remarkable net increase effect on the following areas: No. of patents applied for both domestic and overseas market (27.0%p), No. of overseas countries penetrated (7.1%p), No. of local customers (6.7%p), technological capability (4.7%p), and price competitiveness (3.6%p). In addition, the net increase effect was also found in employment (4%p). Therefore, this project contributes to job creation.

[Table 3-10] Business Performance Review of World Class 300 Project (2011~2014)

Classification		Participation (A)	Non-participation (B)	Net increase effect (A-B)
Business performance	Sales	35.3	-0.3	35.6
	Exports	33.4	21.4	12.0
	R&D investment	25.9	11.8	14.1
Globalization intellectual property	No. of overseas countries penetrated	11.9	4.7	7.1
	No. of local customers	10.8	4.1	6.7
	No. of patents applied for both domestic & overseas market	54.2	27.2	27.0
	Price competitiveness	5.5	1.9	3.6
	Quality competitiveness	6.7	2.3	4.4
	Technology capability	7.4	2.8	4.7
Employment	No. of total employees	8.3	4.3	4.0
	No. of full-time employees	9.1	4.6	4.5
	No. of part-time employees	-9.3	-3.1	-6.2
	Workers below 29 (years old)	11.1	0.0	11.1
	Workers btw 30~50	8.4	4.4	3.7
	Workers btw 30~50	8.1	4.3	3.7
	Workers above 50	9.6	3.4	6.2

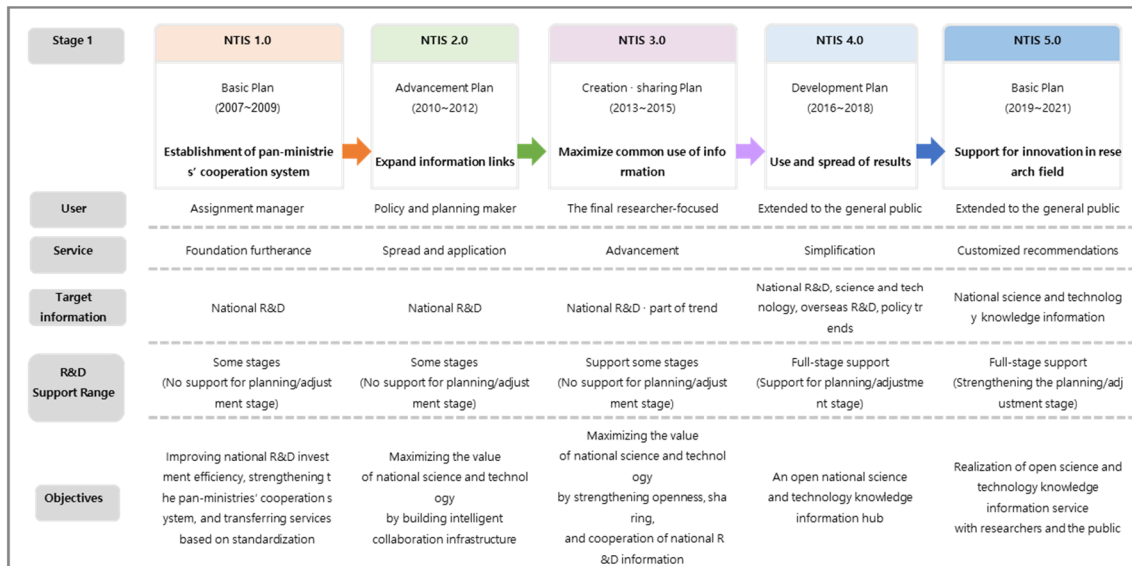
Source: Korea Labor Institute (2015)

3.5 National Science & Technology Information Service (NTIS)

National Science & Technology Information Service (NTIS) is a national R&D knowledge portal. It integrates and provides information (projects, performance, workforce, equipment, etc.) related to the government R&D business projects. Any users can access and analyze the contents of science and technology knowledge information (national R&D projects, participants, papers, patents, research reports, overseas R&D). As science and technology became more important for national development and competitiveness in the 2000s, the central government gradually increased the budget on R&D. The central

government recognized the need to develop a framework that links the performance DB of patents and papers distributed throughout government departments and offices, beyond the existing KORDI, where the entire information of national R&D tasks and projects were gathered and listed. Accordingly, it embarked on building NTIS to prevent overlapped government R&D investment in a project, research facility/ equipment, etc. and support the entire R&D cycle (utilization and dissemination of R&D outputs; establishment of a science and technology policy, allocation and adjustment of budgets, and the planning, management, and evaluation of R&D).

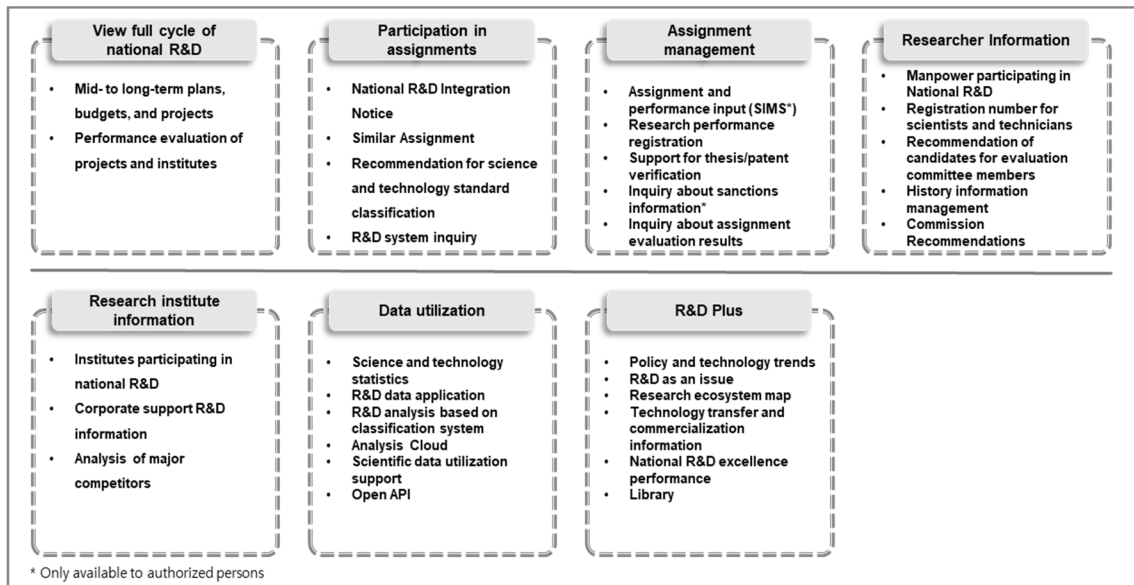
Looking more closely, NTIS conducts a project based on smooth cooperation among a managing research institute, a joint research entity, and the Ministry of Science and ICT (MSIT). And it has an inter-department cooperation system such as a research system council and NTIS experts' council. Beginning with NTIS 1.0 in 2007, NTIS has currently evolved to NTIS 5.0 with the vision of creating an ecology where anyone can easily access science and technology information.



Material: Korea Institute of Science and Technology Information(2020), establishment of national R&D planning · management support service

[Figure 3-3] The flow of NTIS development

At the present, NTIS provides the Korean public with various services, improving the efficiency of national R&D investment and research productivity. NTIS is recognized worldwide for its excellence and exported to Costa Rica, leading scientific and technological innovation in developing countries.



※ Material: National Science and Technology Knowledge Information Service website (2021), NTIS usage guide

[Figure 3-4] NTIS Service

4. Technology Transfer & Commercialization Support Programs in Korea

4.1 Current State of Ghana

The Ghanaian government emphasized the adjustment, absorption, and adeptness of technology in its STI (Science and Technology Innovation) policy. According to the World Economic Forum, the scores of Ghana's FDI & technology transfer and technology absorption were 4.3 of 7 and 4.2 of 7, respectively. As a result, , as specified in the national policy documents, Ghana had the average capability of using the latest technology and modernizing the production and distribution of products & services.

It turned out that the Ghanaian government has been aware of the importance of knowledge exchange and technology transfer. While the STI policy focused on commercializing and disseminating research outputs, the national IP policy placed top priority on research. However, this plan was proved to be far from practical implementation, and it was found that no national institution in Ghana was dedicated to knowledge exchange or technology transfer.

Ghana doesn't yet seem to have enough capability of leading knowledge exchange and technology transfer at a national level. And its most research institutes don't appear to have a department or a policy to handle them either. Many of those concerned said that the government officials and university staff/ faculties had insufficient expertise required for knowledge exchange or technology transfer. At the present, Ghana's industry-academic cooperation activities got 3.5 of 7.

In the cases where research production and commercialization are directly associated, it seems they are limited to the research institutes affiliated with the government departments. The University of Ghana has a technology transfer team, but we suspect that the rest of the universities (including colleges) do not have one. Other plans for technology transfers at a national level neither have obtained a fund, while the group that specializes

in disseminating research to other research institutes, knowledge exchange, or technology transfer have yet to be observed.

4.2 Evolutionary Path of Korea

4.2.1 Laws Related to Technology Transfer & Commercialization

The Korean government enacted the 「The Technology Transfer Act」 and established and operated the Korea Technology Exchange in January 2000 to stimulate domestic technology transfers and commercialization. The Technology Transfer Act was amended to 「The Technology Transfer & Commercialization Act」 in December 2006 to more effectively respond to the change in technology transfers and the industrial environment and instate related government policies. The key laws related to technology commercialization in Korea are as follows:

[Table 3-11] Korea's Key Laws Related to Technology Transfer & Commercialization

Year	Summary
1999	The enactment of 'The Technology Transfer & Commercialization Act' (TTCPA, the Ministry of Trade, Industry and Energy) laid the basis for establishing organizations exclusively for technology transfers such as the Public Technology Licensing Office (TLO).
2001	The enactment of 'The Framework Act on Science and Technology' (FAST, the Ministry of Science and Technology) paved the way to disseminating national R&D performance, technology transfer & commercialization, and revitalization of a technology-based business startup.
2007	The enactment of 'The Industrial Education Enhancement and Industry-Academia-Research Cooperation Act' (IEEIARCP, the Ministry of Education) laid the basis to establish and operate a university technology holdings company.
2012	The enactment of 'The Framework Act on Intellectual Property' (TFAIP, the Ministry of Science and Technology) prepared the basis to create superior intercultural properties (IP) and strengthen national competitiveness based on it.
2015	The enactment of 'The Act on Special Measures for Developing Venture Businesses' (ASMPVB, the Ministry of SMEs and Startups) and the revision of 'The Enforcement Decree of the Act on Special Measures for Developing Venture Businesses' laid the basis for a

Year	Summary
	technology holdings company to exercise and manage micro-VC (Venture Capital) funds and directly invest it in its subsidiaries.
2016	The revision of 'The Enforcement Decree of the Special Act on Designating Special Research and Development Zones' (SAPSRDZ, the Ministry of Science and Technology) paved the basis to activate a technology business startup in which the main actor of a national project under the Ministry of Science and Technology could also be established and run as a research company.

Source: Korea Institute for Advancement of Technology (2020)

The current status of the laws relate to technology commercialization by government department is summarized as follows.

[Table 3-12] The Current Status of the Laws Related to Technology Commercialization by Government Department

The Ministry of Trade, Industry and Energy	The Ministry of Science and Technology	The Ministry of SMEs and Startups	The Ministry of Education
• The Technology Transfer & Commercialization Act • The Industrial Technology Innovation Act	• The Framework Act on Science and Technology • The Framework Act on Intellectual Property • The Framework Act on Intellectual Property • The Special Act on Designating Special Research and Development Zones	• The Act on Special Measures for Developing Venture Businesses • The Support For Small and Medium Enterprise Establishment Act	• The Industrial Education Enhancement and Industry-Academia-Research Cooperation Act

Source: Korea Institute for Advancement of Technology (2020)

4.2.2 Transition of the Technology Transfer & Commercialization Plans

1) The 1st Technology Transfer & Commercialization Plan (2000~2005)

The gist of this plan was to lay the basis for establishing and invigorating a technology transaction market. During this period, establishing venture business startups was actively

driven due to the government policy for nurturing venture enterprises in Korea. To respond to the keynote of the policy, the government put out its 1st technology transfer & commercialization plan to form and activate the technology transaction market. The objective of the plan was to create the virtuous cycle of technology development by ramping up technology transfers and commercialization. To attain its goal, the plan proposed the details as follows: brushing up the laws and systems to establish the technology transaction market, supporting to activate the technology transaction market, and laying the ground to activate technology transaction and commercialization. The outputs of the plan were the Technology Transfer Act, establishment of the Technology Transaction Exchange, the National Technology Bank (NTB), etc.

2) The 2nd Technology Transfer & Commercialization Plan (2006~2008)

The gist of this plan was to actively implement business projects, aiming to expand the mechanism for technology transfers commercialization. The Korean government made its 2nd technology transfer & commercialization plan and proposed the programs to increase the commercialization and success rate of venture technology. The objective of the plan was to build a system for growing technology-innovative companies by ramping up technology transfers and commercialization. To attain its goal, the plan proposed the details as follows: making a credible technology assessment system, broadening the supply of technology funds, ramping up the commercialization of national R&D performance, transferring public technology to the private sector, promoting technology transaction, and accelerating technology transfers and commercialization through international technology cooperation. The outputs of the plan were the formulation of R&BD projects, emergence of leading TLOs, and fundraising for new growth engines.

3) The 3rd Technology Transfer & Commercialization Plan (2009~2011)

The keywords of the 3rd promotion plan were the 'global market' and 'full technology cycle system'. At that time, the government aimed for a 'global market' to meet the keynote characterized as 'creation of 17 new markets for materializing new growth engines', and the 3rd promotion plan also reflected these policy trends by aiming to foster technology-based global companies. The goal of the plan was to build up a full-cycle promotion system for

technology transfers and commercialization by making and fostering technology-based global companies. To realize its goal, the plan featured the following details: expanding the supply of technology funds and building its system to discover and strengthen national technology resources, making a full promotion cycle for technology transfers & commercialization, supporting global market entry, and expanding the mechanism for technology transfers & commercialization. The outputs of the plan include the recommendation and evaluation of performance of R&D project's commercialization, new establishment of a strategic planning group, emergence of invention capital, introduction of green certification, operation of a trust system, etc.

4) The 4th Technology Transfer & Commercialization Plan (2012~2014)

The vision of the 4th plan was to advance the performance of the full promotion cycle of the R&BD system and its keyword was 'business model' (BM). For global market entry, the 4th promotion plan also strived to embody the full promotion cycle for the R&BD system and improve its performance through the reinforced capability of the main actor (mediator) that performed technology commercialization. The objective of the plan was to make substantial market performance through the creation of an ecosystem for consumer-oriented technology commercialization. The detailed implementations included strengthening the activities to link technology and market, enhancing the capacity of the main actors of technology commercialization, promoting convergent and open innovation, and scaling up the infrastructure for the operation of the market mechanism. The plan resulted in ramping up technology transfers and commercialization by public research institutes and expanding technology financial support.

5) The 5th Technology Transfer & Commercialization Plan (2015~2017)

The gist of the plan was to create an ecosystem for technology transfers and commercialization to materialize a creative economy. Accordingly, to invigorate the economy and create jobs, the government announced as the keynote of the related national policy as 'realizing a creative economy where creative ideas develop into technology commercialization'. The objective of the plan was to stimulate the growth of companies by sharing and expanding R&D outputs and increasing the possibility of technology

commercialization. For this, the government decided to support the following details specified in the plan as follows: Activating the operation of the technology transaction market, elevating a public research institute's marketing capability, providing a customized technology whose potential of commercialization was high, and preparing for the conditions in which early commercialization companies could grow. The expected performance of the plan included advancing the mediating system for technology transaction, reforming the support organizations of technology transfer & commercialization in public research institutes, helping technology transfer & commercialization companies occupy the early-stage market, etc.

6) The 6th Technology Transfer & Commercialization Plan (2017~2019)

The gist of the plan was to extend the 'Buy R&D' diffusion strategy that aimed to develop additional technology after the introduction of external technology (technology transfer, M&A, etc.). After gathering the plans from the relevant central administrative agencies, the Minister of Trade, Industry and Energy established and carried out the 「Technology Transfer & Commercialization Plan」. The government had drawn up five plans and carried them out until 2016, while it also made its 3-year 6th plan (2017 ~ 2019) and attempted to activate technology transfers and commercialization across the government departments. The 6th plan consisted of 3-year mid-term promotion items and annual promotion items. This plan's objective was to set up a strategy of building the cooperative system of technology commercialization across the ministries, which aimed to create an ecosystem to promote open innovation. The 6th plan had details to implement and prepare the criteria for the calculation and approval of the cost of technology introduction (the amount paid by a company adopting technology) among the 'in-kind' burdens on the companies participating in a national R&D project. As a key strategy, the plan suggested the convergence between technology and industry and activation of external technology introduction (Buy R&D) to create an ecosystem for promoting open innovation in response to the 4th Industrial Revolution, in which the speed of innovation is crucial. Four core promotion strategies were implemented as follows: Expanding the base for the demand (Buy R&D) from the buyer's viewpoint and for the supply (Buyable R&D) from the producer's perspective, narrowing the gap between buyers and producers in terms of infrastructure,

and building the cooperation system for technology commercialization across the government departments.

The main agenda activities by strategy are as follows:

- **(Open innovation R&D)** The introduction of B&D (Buy and Develop)” system which improves the productivity of the government R&D with external technology.
- **(Expansion and Discovery of Buyer’s Groups)** The introduction of a coaching system for SMEs that introduce technology, activation of technology exportation, and expansion of the basis for discovering the demand for technology through cross-government department cooperation.
- **(Collaborative Model)** The expansion of a private sector-led innovation model for technology commercialization, which is more suitable for commercialization than the existing R&D system, to advance the commercialization performance of R&D outputs.
- **(Technology Financing)** The expansion of policy funds and advancement of credibility in technology evaluation to reduce the failure risk of companies lacking in capital required during the ‘Death Valley’ period.

[Table 3-13] Transition of Commercialization-Linked Technology Development Projects by Transition of Technology Commercialization Policies

Changes in technology commercialization policies		Changes in commercialization-linked technology development projects
1st Promotion Plan ('00~'05)	Founding the basis for establishing and invigorating the technology transaction market.	-
	The 1st plan was mapped out to form and activate the technology transaction market.	
2nd Promotion Plan ('06~'08)	Implementing actively business projects, aiming to expand the basis for technology transfers	The implementation of R&BD projects for new business startups and early-stage companies to nurture venture enterprises
	The 2nd plan was drawn up to expand the basis for commercializing the R&BD projects for early-stage companies by proposing technology commercialization programs.	
3rd Promotion Plan ('09~'11)	Global market and a full-cycle system	Implementing global common projects (focused on a few large projects)
	The 3rd plan was made to implement global common projects to nurture global companies.	

Changes in technology commercialization policies		Changes in commercialization-linked technology development projects
4th promotion plan ('12~'14)	Improving performance by establishing the R&BD system on a full-cycle basis - BM (Business Model)	The implementation of global common (BM planning) projects (focused on importance of BM)
	The 4th plan was prepared to embody the full promotion cycle for the R&BD system and improve its performance through the improved capability of the main actor (mediator) that performed technology commercialization.	
5th Promotion Plan ('15~'17)	Creating an ecosystem for technology transfers & commercialization to materialize a creative economy	Institute-linked projects (support for technology transfers commercialization of public technology), promotion of BI commercialization support business (BI commercialization to realize a creative economy)
	The 5th plan was mapped out to add business incubators' (BIs) commercialization support projects to help commercialize the ideas for promoting the institute-linked projects—in other words, to support the technology transfer & commercialization of the public research institutes.	
6th Promotion Plan ('17~'19)	Forming an ecosystem where open innovation is promoted.	The implementation of cross-government department projects (to increase cooperation among the ministries for open innovation)
	The 7th plan was initiated to implement cross-minister projects to advance the capability of each department and support commercialization.	

Source: Korea Institute for Advancement of Technology (2020)

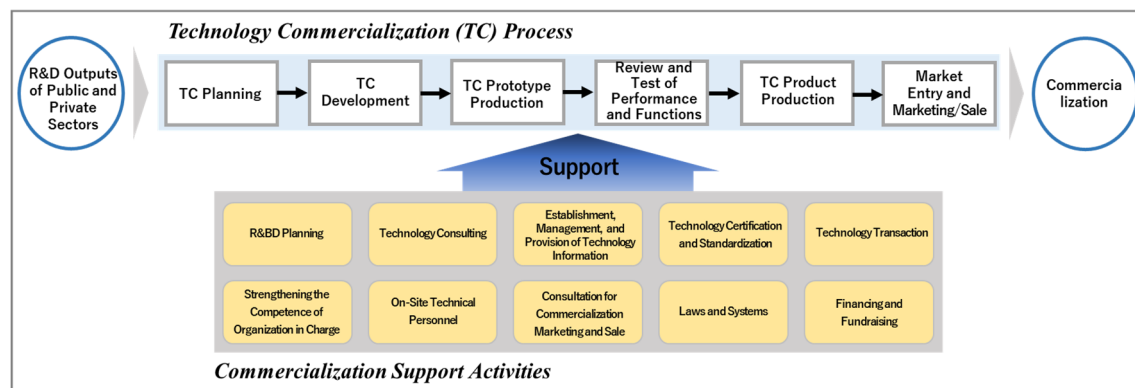
4.3 Major Policies and Programs in Korea

4.3.1 Scope of Technology Commercialization Policy

As for the scope of a budget to support technology commercialization, it covers both R&D and non-R&D budgets. Even at the commercialization stage, improvement R&D, demonstration plant, prototype production, and some pilot production are covered by both government R&D and non-R&D budgets.

The scope of policy projects for technology commercialization may also include R&D planning, technology expert consultation, technology development, and marketing for market entry. The technology commercialization of government-led R&D projects includes the following types: Planning of technology subject to commercialization; the direct activities from technology development to product production and sales; and cooperative

support activities to ramp up technology commercialization besides technology development-oriented direct support.



Source: Korea Institute for Advancement of Technology (2020)

[Figure 3-5] Technology Commercialization Process & Commercialization-Linked Technology Development Project

1) Introduction of Commercialization-Linked Technology Development Project

In September 2004, the Minister of Commerce, Industry and Energy reported to the president the 'Comprehensive Plan for Development Commercialization Promotion' (draft). Also, as part of the technology development commercialization program introduced in the plan, the government implemented a commercialization-linked technology development project (the commercialization planning project was received in 2005, and the project was selected and supported from 2006). It corresponds to 3-1 'The Establishment of Business-Friendly R&D System' of the '5th Technology Transfer & Commercialization Plan'. As a result, the support system referred to as 'BM First and R&D Later' was introduced for a service R&D and product-open R&D projects. Creative Economy Town (online, (www.creativekorea.or.kr)) and BI support agencies (offline) discovered excellent BIs with high potential and helped them plan BMs. The main direction of implementation in 2016 was 1. Early Creation of New Industry Performance (Ministry of Industry, January 2016).

2) The Characteristics of Key Support Methods

① (BA-Led Type) Business Startup Support (*BA: Business Accelerator)

Period	2005 ~ 2010
Support targets	(Planning) A corporation that wants to be a BA (SME enterprises, Techno Parks, universities, public research institutes, etc.) (Development) TBC (Techno-Biz Company: new business startups)
Scope of support	Support for the commercialization of excellent technologies developed by the public and private sector
Characteristics of Project	By-stage (planning and development) support and improvement of success rate in connection with venture capital investment
Limitations	• More technology development SMEs applied for BA than experts, which undermined the by-stage principle. • Difficulty in discovering the target companies for commercialization and helping them start a business within two years. • Insufficient attraction of private investment

② (BA-Led Type) BM Support (*BM: Business Model)

Period	2012 ~ 2016
Support targets	(BM planning) BA → SME enterprises or BAs, (BM development) SME enterprises
Scope of support	Support for BA's business planning activities and technology development companies with additional development funds
Characteristics of project	• Re-introduction of by-stage support projects (1st stage: BM planning → 2nd stage: BM development) • Deleted the provision regarding the foundation of a new corporation that belongs to a new business startup. • Gave a BA the authority to decide a company to which it will transfer a BM during the planning period.
Limitations	• Insufficient project management in public BA-led projects. • Although BA participation is essential, it is virtually impossible for a company to dedicate itself to commercialization because it has to carry out both development and marketing activities during the support period. Thus, a BA's roles are limited.

③ (Investment-Linked) Early-Stage Company Support

Period	2007 ~ 2011
Support targets	SME enterprises with service years of fewer than 7 years and core technology and capability for commercialization of core technology and competency
Scope of support	Support with an additional development fund for core technologies and support for private fund attraction
Characteristics of project	• Support for development stage without verifying project plan. • Individuals and angel investors are included in the pool of private investment in early companies.
Limitations	• This was the most successful type* but there were also many problems. The main cause was a high chance of failure due to omission of a feasibility test at the planning stage and insufficient project management.

④ (Investment-Linked) Consortium Support

Period	2011 ~ 2016
Support targets	SMEs and industry-academia-research joint research institutes
Scope of support	Additional development and mass-production of core technology and private fundraising for marketing
Characteristics of project	• Required: Industry-academia joint researches and attraction of private investment • In-depth evaluation of investment institutions
Limitations	• Much time is required to manage a project evaluation due to handling of the attraction period of private investment. • The (private investment) fund can be divided into and used for commercialization planning & development (government subsidies), promotional marketing, and mass production, but effectiveness is not certain and its performance is hard to track down.

⑤ (Blended Type) BA-Led & Investment-Linked

Period	2017 ~ Now
Support targets	(Run-up platform ToP) SME enterprises (Project Corporate Support SPARC) Project Corporation Bodies (newly established corporation)
Scope of support	• (ToP): Whole process of commercialization including the link to private investment, technology development, commercialization, testing, and certification • (SPARC) Support for the commercialization of newly established or fledgling corporations
Characteristics of project	• Required: Consortium with a technology commercialization BD to ramp up commercializing applicable companies of an applicant. • Required: Attraction of investment from venture capital, commercialization BD, etc.

⑥ (Indirect Support) Institution-Supported

Period	2010 ~ 2014
Support targets	• (Institute-linked) Public research institutes hoping to commercialize transferred technology within 3 years (government-funded research institutes and specialized research institutes) → expansion of support to academia-industry coordination groups from 2012. • (Support for BI commercialization) BI commercialization support agencies for TP and private BA
Scope of support	• (Institute-linked type) Transfer of technology (research outputs of a public research institute) • (BI commercialization support) Laying the basis for commercialization through the designation of agency to support a BI, BI discovery, and laying the basis for commercialization such as BI discovery and entitlement.
Characteristics of project	• (Institute-linked type) A selected managing agency supports the commercialization of a company to which public technology will be transferred. • (BI commercialization support) Expanding commercialization support to TRL level 1 to 8 and equalizing the recipients (BI commercialization support organizations and creative economy towns) to stimulate participation.
Limitations	• Public BAs lacking in the capability of planning and managing commercialization

⑦ (Indirect Support) Institute-Recommended & BI-Linked Type

Period	2014~ 2017
Support targets	Organizations applying for BI commercialization, Youth Startups Academy, creative economy towns, etc.
Scope of support	Support for the discovery and commercialization of excellent BI for small and medium-sized enterprises
Characteristics of project	- BI discovery → One-stop system operation for commercialization support. - Introduction of competition system for the discovery of excellent BIs and commercialization development support.
Limitations	Since it is a project to support a full cycle of commercialization from the business idea creation stage, it is difficult to see the performance (results) of commercialization within the support period.

3) Performance of Commercialization-Linked Technology Development Project

Beginning with a new business startup project in 2005, the government had supported 622 projects until 2017, spending KRW 325 billion. The average subsidy per project was KRW 520 million, and a total of 848 patents were created from the projects selected and supported during 2013 through 2017. Of them, 603 patents (420 domestic, 107 overseas, and 76 PCT patents) were applied and 245 were registered (209 domestic and 36 overseas). The number of average patents per project was 3.4 in total (2.4 applied and 1.0 registered). Of the applied projects, 117 projects were in preparation for commercialization. For the estimated time, 37 projects (31.6%) were less than 24 months, and 23 projects (9.7%) fewer than 12 months. The accumulated sales of R&D were KRW 56.61 billion in 2018 and those of commercialization were KRW 199.13 billion.

The sales target of an applicant for commercialization was KRW 230 million per billion won of government contribution. The target for new job creation was 1.90 persons and 12.76 people were actually hired. During the period between 2011 and 2018, when private investment became mandatory, 117 companies were invested with a total of KRW 203.7 billion (CV investment). Therefore, the average investment per company was KRW 1.74 billion. According to the confirmed sales amount of 211 companies that applied for commercialization-linked technology development projects, they showed that the average annual growth rate until 2016 was 7.6 percent higher than when not supported. The analysis found that the main contributors to performance were the by-stage support that lasted throughout the whole period of a project, BA participation, and private investment-linked support system.

5. Startup & Incubation Support Programs in Korea

5.1 Current State of Ghana

5.1.1 Strengths of Ghana's Business Startup Incubation

The key strengths of Ghana's incubation-innovation hubs firstly lie in its ability to foster entrepreneurs and small companies even from idea generation to market entry. The incubation-innovation hubs provide customized training to support incubators (businesspeople and small businesses) to turn their ideas into products or services and grow their businesses.

Secondly, they provide entrepreneurship training, digital technology, skills training, fundraising approaches, and financial management training, and nurture future business startups with mentoring and coaching. Among the business startup incubation hubs, Kumasi Hive has a facility in which startups can make a prototype to inspect and test before commercializing it, but not every hub has such a space.

Thirdly, EQWIP Hub in Tamale supports incubation candidate companies with a free business startup incubation training in connection with 'The Global Affairs of Canada' and provides 'The Youth Innovation Fund' in which startup companies are given initial equity capital for business promotion. Some of the local incubation-innovation hubs allow a shared space and facility in which small companies and entrepreneurs can start their own businesses.

5.1.2 The Weaknesses of Ghanaian Business Startup Incubation

The biggest weakness of Ghana's incubation-innovation hubs is firstly the difficulty in raising the funds for startups. Most of the hubs but some, which are capable of financing startups with funds from a parent company, are weak in fundraising.

Secondly, the incubation-innovation hubs take care of startups or entrepreneurs until they leave the hubs to scale up their business. But a majority of them close business or do not grow as much as expected due to insufficient funds afterward. Even if funds are provided, they are still far from being sufficient for a capital-intensive business.

Thirdly, the incubation-innovation hubs are not research institutes. Therefore, they are not good at data analysis. An incubation hub does not usually have its own experts inside but rather hires expert consultants when needed. The hubs are supposed to retain researchers and data analysts in the organization to keep expertise, but most of them provide free-of-charge services and don't have enough financial support, which makes it difficult for them to retain professional researchers inside. To provide incubation targets with the training of higher quality, the incubation hubs have to strengthen their own capabilities.

5.2 Evolutionary Path of Korea

The most fundamental concepts of startup incubation go with early business nurturing and startup activation. A business startup incubation project is aimed to support a business founder or early startups with business capability and resources that they usually lack and contribute to the growth of their companies and economic development of the nation. The gist of the business startup incubation includes the provision of startup space, prototype production, investment links, etc.

Korea's business startup incubation business projects can be divided into two main stages as follows: The 1990s for the establishment of related laws & infrastructure and the 2000s for the foundation of startup specialization and detailed contents.

5.2.1 Stage 1 (1990s)

Firstly, the 1990s was the period for the establishment of related laws and infrastructure. The legal framework was paved for business startup incubation as 「The Policy for the Establishment and Operation of a Business Startup Incubation Center」 was announced in 1991. In 1991, in addition, the TBI (technology business incubator) system was implemented,

and the Ministry of Trade and Industry (currently the Ministry of Commerce, Industry and Energy) helped establish supporting agencies such as Techno Parks and supported technology. It is then that the concept of a business startup incubation center (or herein 'business incubator (BI)') was first introduced in Korea.

In 1996, as the Small and Medium Business Administration (SMBA) was established, domestic business startup projects started to actively expand. The SMBA was promoted to the Ministry of SMEs and Startups (MSS) in 2017. As Korea underwent the Asian foreign-exchange crisis, a voice was raised to shift the Korean economic structure from traditional industries to knowledge-based ones. At that time, employment was a very urgent issue facing Korean society due to the serious problem of mass unemployment in 1998. Moreover, the entire government departments started to establish and operate BIs to activate new technology-based business startups and create jobs.

5.2.2 Stage 2 (since 2000)

Beginning from 2000, Stage 2 was the period for the foundation of business startup specialization and content specification. In 2000, 'startup net (www.changupnet.go.kr)' was established, and was changed to the present business startup support web portal (www.k-startup.go.kr). This web portal has an internet information network that guides all the business establishment procedures from startup to fund support. Also, it has a database with the information of the BIs across the nation and plays a role as business startup info. provider for prep entrepreneurs and investors. In 2002, 'the Plan for Reforming Venture Companies' was announced and became mandatory to hire a manager for a BI. Then in 2008, the Korea Institute of Startup & Entrepreneurship Development (KISED)', which is an affiliated organization of the SMBA and managing agency exclusively for business startups, was established. With this agency, the government executes the following tasks: The support of TBI operation, assistance of new technology development research, support of venture investment, etc.

In 2011, the SMBA reorganized the 'Business Startup Incubation Center (BI)' into an advanced 3rd generation platform and Techno Parks into company growth support centers,

providing a variety of supports with the aim to help the SME enterprises of excellent technology with IPOs. In 2017, a makers' space (business place) and a business startup support system based on innovation technology to keep abreast of the 4th Industrial Revolution were prepared. At the present, Korea has the sites specializing in business startups as follows: the Ministry of SMEs and Startups (www.mss.go.kr), Business Startup Support Web Portal (www.k-startup.go.kr), Korea Institute of Startup & Entrepreneurship Development (www.kised.or.kr), Korea Business Incubation Association (www.kobia.or.kr), One-Person Creative Company Business Center (www.k-startup.go.kr), Korea Technopark Association (www.technopark.kr), Korea SMEs and Startups Agency (www.kosmes.or.kr), and TIPS Program (www.jointips.or.kr).

5.3 Major Policies and Programs in Korea

5.3.1 The Functions of a BI

The BI is an organization that provides companies in their early stage of business, when their autonomy is weak, with a variety of resources and services, and helps them grow by effectively linking a founder's capabilities, technology, capital, and know-how. By providing startup companies with various resources, entrepreneurship training, network, etc., the center plays the main function to help a startup company cultivate the capability of autonomously operating a business after it leaves the center. The operations of the center are various depending on its originator as are its expected impacts by the purpose of establishment.

5.3.2 Development Stages of BI

The BI has been changing along 5 development stages so far. The 1st-generation BI provided a business location for a startup, shared service, and passive business support services.

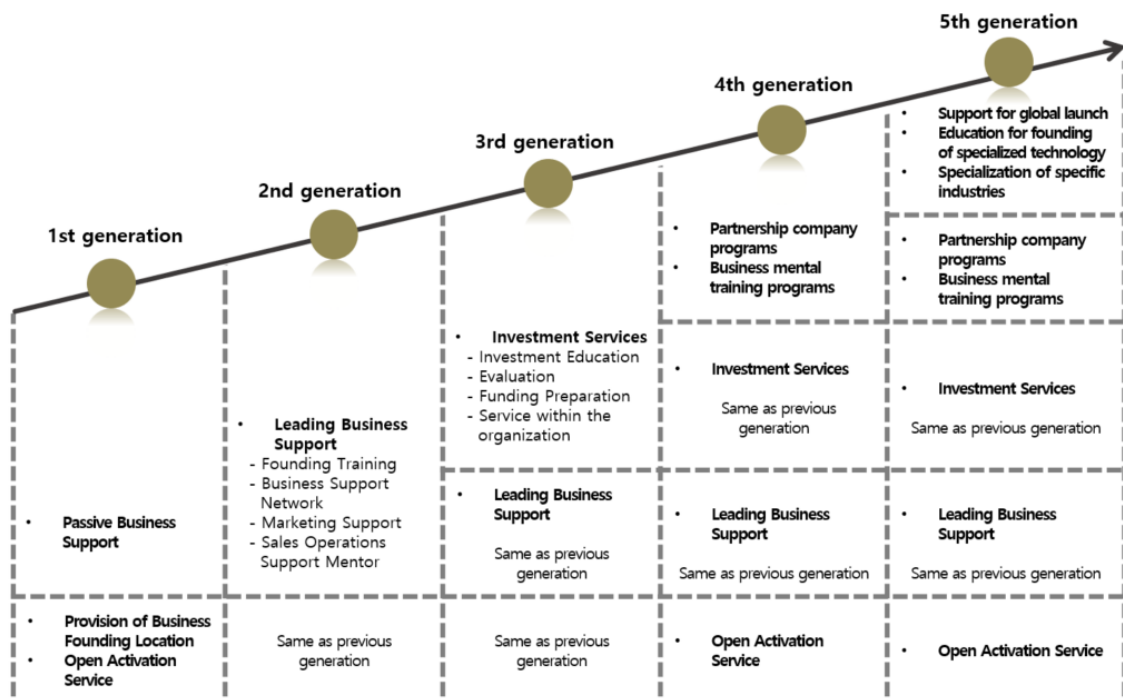
The 2nd-generation BI evolved to provide active (leading) business support services from passive ones and added new services to the list such as startup training, business support network, marketing support, sales business, and mentoring. In addition, startup coaching,

which was added to Stage 2, enabled a startup company to contact target customers and have a cooperative program with a target company.

The 3rd-generation BI added the investment services of the residential companies to the functions of the 2nd-generation BI. For example, it provides investment training, evaluation, funding program, legal defense, and in-house services.

The 4th-generation BI stopped providing spaces for business. Instead, it provided partnership programs and pre-incubation services through which a startup company can participate in the support and program of the center. That is, the 4th center added entrepreneurship programs focusing on company planning and management.

The 5th-generation BI features and provides the key function(s) of each stage as follows: investment services (Stage 3), entrepreneurship training programs (Stage 4), and support for global market entry (Stage 5).



Source: Korea Business Incubation Association (2013)

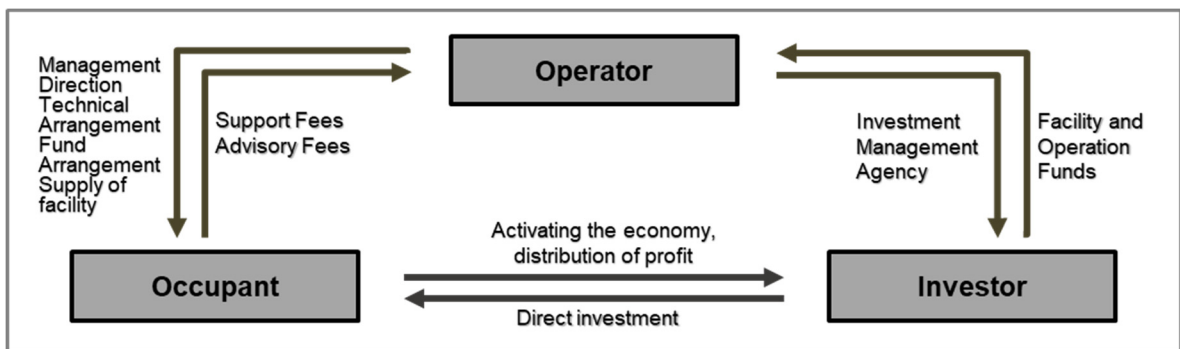
[Figure 3-6] Development System of BI

5.3.3 The Operation System of Startup Incubation Center (Manual for a Startup Incubation Manager, 2016, the Small and Medium Business Administration)

Startup incubation is defined as an activity of providing a business place for a startup company and prep entrepreneurs as well as a variety of services necessary for a business startup and growth such as joint technology R&D, guidance, consultation, fund support, arrangement, management, accounting, taxation, and legal affairs.

1) Basic Model

In general, the BI operating system can be divided into operators, occupants, and investors. Figure shows the components of the BI basic model and their relationship.



Source: Korea Business Incubation Association (2013)

[Figure 3-7] The Basic Model of Business Startup Incubation System

2) The Operation Policy of the BI

The main operating plans of the BI are occupancy, selection, withdrawal, graduation, reviewing resident companies, intellectual property rights/ certification, and linkage between graduated companies with the BI. For its operation, consistent management is required from occupancy to graduation and post management.

3) The Organizational Structure of the BI

The BI consists of professional operating personnel (head of the center and managers), a steering committee/ board of directors, and advisory committee. Among them, the center head and managers take the most important roles.

5.3.4 Cases of Startup Support Program

In Korea, several ministries, including MSS, MSIT, and ME, promote startup support programs. The largest startup support programs are the pre-entrepreneur package, initial startup package, and jump startup package, which are promoted by the Korea Institute of Startup & Entrepreneurship Development (KISED).

1) Prep-Entrepreneur Package

- **Eligibility:** before registering the business license; the prep entrepreneurs of all ages (from 2020)
- **Budget and Scale:** KRW 100 billion and 246 million and 1,530 people (as of 2021)
- **Implementing agency:** MSS, KISED, university, public institution, etc. (about
 - Supports: Startup training, networking, prototype production cost, etc.

2) Initial Startup Package

- **Eligibility:** Companies not older than 3 years
- **Budget and Scale:** KRW 100 billion and 260 million and 960 companies (as of 2021)
- **Implementing agency:** MSS, KISED, university, public institution, etc.
 - Supports:
 - Commercialization fund (max. KRW 100 million): - for the costs of prototype production, acquisition of intellectual property rights, marketing, etc.
 - For the operation of specialized programs (market research, etc.) and growth support programs (marketing, etc.)

3) Jump Startup Package

- **Eligibility:** Startup companies whose service year is 3 to 7 years and in leap(jump) stage as of the date of the announcement
- **Budget and scale:** KRW 102 billion and 1,080 companies (as of 21 years)
- **Implementing agency:** MSS, KISED, university, public institution, etc.
 - Supports: Commercialization fund (max. KRW 300 million) and growth promotion fund (max. KRW 100 million)

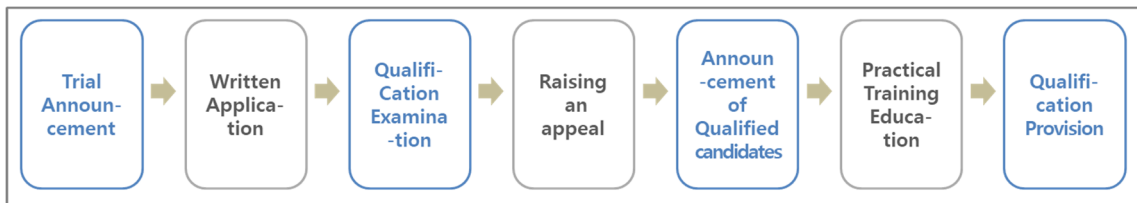
4) Startup Manager

- **Concept of Business Startup Manager**

A business startup manager (hereinafter referred to as ‘the manager’) is a person who holds a license of ‘Professional Business Startup Manager’ issued by the Korea Institute of Startup & Entrepreneurship Development (KISED) or the Korea Business Incubation Association (KOBIA), which are non-profit organizations, according to the Framework Act on Qualifications. In Korea, the KOBIA issues the license.

■ Application Scope of the License (the Manager)

- A person who has acquired the license can work at the following institutions: BI, foundation or association related to a business startup, venture capital firm, government-funded institution and research institute, local government-funded organization, management consulting firm, SME consulting company, etc.
- Enforcement Procedure for Qualification Test



5) Fostering the Managers

■ Qualification Test (Unrestricted eligibility)

The test takers can sit for the exam (test for the manager) twice a year. Its acceptance criteria is as follows: 40/100 for each subject and an average mark of 60/100 for all the subjects.

Applicants take 4 subjects for the exam: The Introduction of Technology Startup, Technology Startup Practice, Technology Startup Incubation Practice, and Technology Startup Development Practice. Afterward, the successful applicants must complete the course ‘The Apprenticeship Training for Hands-On Practice’ to attain the license (the manager).

■ Refresher (Supplementary) Training

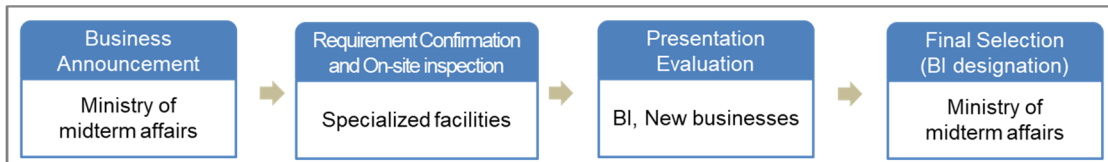
The purpose of the refresher training, which is required to be taken every 5 years after acquiring the license, is to improve the hands-on skills and quality of license holders. This training is expected to help them keep up and improve their expertise levels.

6) The Startup Business Incubation Center

Korea has continued to establish and renovate BIs to provide (fledgling) entrepreneurs with the space for commercialization, and thus help them stably operate and accelerate the growth of their startup businesses.

A candidate for a BI should submit an application form to MSS. When it becomes a successful applicant through screening, it can benefit a support fund up to 80% of the total cost (max. KRW 300 million) for remodeling a BI.

Application for a designated BI has to meet the following requirements: ① A facility larger than 500m² (spacious enough to accommodate more than 10 startup entrepreneurs, ② more than 2 professionals, ③ testing apparatus and production equipment that startup entrepreneurs can use, and ④ a suitable startup incubation business plan.



6. Intellectual Property Management Support Programs in Korea

6.1 Current State of Ghana

Ghana is a member of WIPO and its national IP office is an affiliated organization of the Ministry of Justice. Intellectual property in Ghana is managed by the Ghana Copyright Office and Industrial Property Office (Family Register) affiliated with the Ministry of Justice and Prosecutors' Office.

[Table 3-14] Current Status of Intellectual Property Rights in Ghana

Year	Patent	Trademark	Industrial design	GDP (constant 2017 USD)
2015		1,027		130.07
2016	116	1,255	482	134.55
2017	21	1,120	735	145.51
2018	15	1,008	797	154.62

Source: World Intellectual Property Organization⁷

Looking at the application trend of the intellectual property rights in Ghana, 116 patents (2016), 21 (2017), and 15 (2018) were applied. Also, about 1000 trademarks and 500-800 designs are applied for the IPs annually. As seen, the applications for trademarks and designs are more active than for patents. The number of Ghana's annual patent applications is smaller than that of one university in Korea. Therefore, it seems necessary for Ghana to have a technology licensing office (TLO) in a university or the Regional Intellectual Property Centers (RIPCs) as a supportive organization in the early stage.

As its legal protection system for intellectual property rights, Ghana enacted the Patent Act in 2003 and the Intellectual Property Rights Act in 2005, and also established the Ghana Innovation Research Commercialization Centre (GIRC), which manages public research

⁷ https://www.wipo.int/members/en/details.jsp?country_id=65 (Accessed 28th Dec, 2021)

achievements to generate intellectual properties and placed science, technology, and innovation (STI) at the center of the social and economic agenda, setting STI as the central axis of national growth.

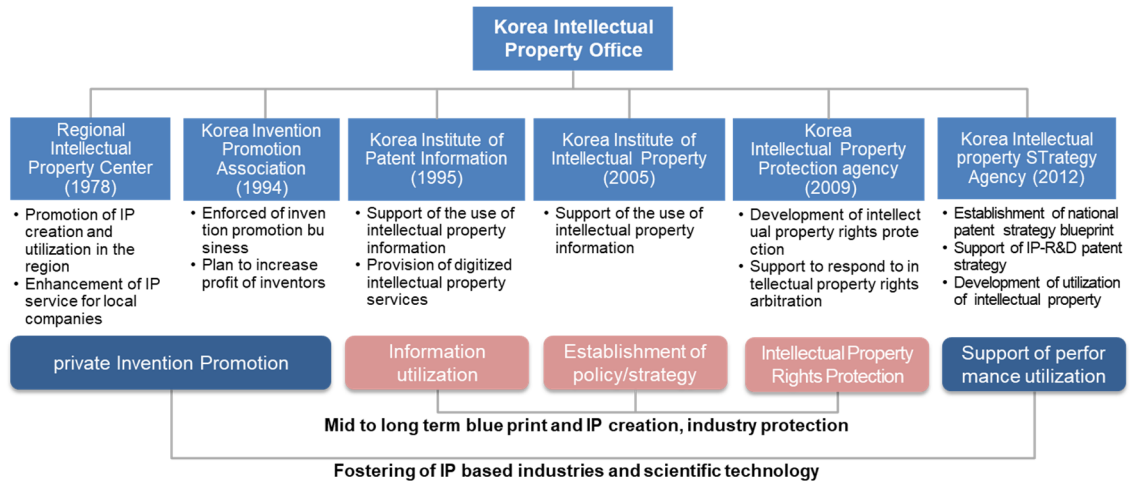
In addition, Ghana organized the Science and Technology Policy Research Institute (STEPRI) in 1909, which handled the policy issues related to technology transfer as part of the UNDP Project. STEPRI is now an affiliated organization of the Council for Scientific and Industrial Research (CSIR), carrying out projects related to Ghana's national science and technology policies.

6.2 Evolutionary Path of Korea

Korea opened the Korean Intellectual Property Office (external office of the Ministry of Industry and Commerce) in 1977. This office handles the administrative tasks (application and registration) of patents, utility models, designs, and trademarks, and affairs involving industrial intellectual property rights such as patent examination, trial, and defending (patent lawyers).

As industries have grown on a large scale, the Korean Intellectual Property Office (KIPO) cooperates with various government departments and agencies for creating and retaining IPs, so that IPs can be used throughout the industries in Korea. In addition, as a government agency, it has several affiliated organizations, some of which are the Korea Invention Promotion Association, Korea Institute of Patent Information, Korea Institute of Intellectual Property, Korea Intellectual Property Protection Agency, and Korea Intellectual Property Strategy Agency. These agencies were established on time and according to the demand of each situation. And so they are operated to create IPs, protect them, set up related policies and strategies, and utilize the output of IPs.

As far as the agency for Ghana's IP management is concerned, it needs to be an intensive organization at first in consideration of country-specific realities and then divided later on according to the expertise required.



[Figure 3-7] KIPO's Affiliated Organizations and Agencies

To protect and activate the attained IPs, Korea has developed IP-related policies as below throughout a full IP cycle from IP generation to commercialization in the sector of relevant organizations, human resources management, support projects, etc.

As for the key points at each step, for the first stage, the sense of need for creating invention was instilled by giving compensation and responsibility for the creation of individual inventions for employee inventions with the enactment of related laws, the Invention Promotion Act, while the establishing a regional intellectual property center in the Chamber of Commerce and Industry and the Invention Promotion Association according to the Invention Promotion Act are thought to have inspired the justification of intellectual property creation at the government level. For the second stage, as the core functions, research on prior art for IP creation through establishing the Institute of Patent Information and a patent map and IP information service are being built through establishing the Institute of Patent Information, and this has laid the groundwork for creating higher-quality IPs by providing various services for researchers' IP creations, and at this time, the enactment of the Technology Transaction Promotion Act played a key role in creating a virtuous cycle structure for the created IPs, and establishment of the TLO organization at the universities was the stepping stone for promoting the use of public technologies. For the 3rd stage, it is thought that the commercialization of IPs was accelerated based on the

institutions established through establishing the Patent Strategy Development Institute and related laws and regulations, and it is considered to have been the impetus for IP-centered, high-value-added and technology-intensive industrial developments.

[Table 3-15] History of Intellectual Property Rights Policies in Korea

Category	Stage 1 (1970~1995)	Stage 2 (1995~2010)	Stage 3 (2010~)
Infrastructure Established	KIPO ¹⁾ Established (Ministry of Industry and Commerce oversees) → Foreign office of Ministry of Industry and Commerce) - Evaluation and judgment of industrial property rights Establishment of the KIPA ²⁾ - Promotes duty invention and inventions	Establishment of the KIPI ³⁾ - Support for intellectual property rights - Operation of an IP information system Establishment of the Koipa ⁴⁾ - Development of domestic and foreign intellectual property rights protection - Support to respond to intellectual property protection arbitration Establishment of RIPC ⁵⁾ - Establishment of local patent Establishment of KIIP ⁶⁾ - Establishment of business strategy and domestic policy related to intellectual property rights	Establishment of the KISTA ⁷⁾ - Materialization of the strategic and systemic spread of intellectual property
Main Policies/ Businesses	Invention Promotion Act Enacted: Support of private enterprises Enactment of special law on science and technology innovation	Enactment of facilitation techniques on technology transfer Establishment of facilitation plans on the commercialization of technology transfers Introduction of Industry-Academy Cooperation and fostering of technology transfer Promotion of intellectual property business transactions centered on the consumer	Establishment of national patent strategy blueprint Fostering of patent business and IP-R&D strategy experts Private · IP-R&D strategic support for public and Expansion of support businesses of technology transfer businesses based on core departments

Establishment of policy to acquire the creation of excellent patents and patent competitiveness
 → Expansion of IP creation

Policy on patent enhancement
 Promotion of business and infrastructure using IP information
 → Acquiring original technology

Strengthening of competitiveness and acceleration of IP usage
 Establishment of R&D re-investment virtuous cycle

- 1) KIPO: The Korean Intellectual Property Office
- 2) KIPA: The Korea Invention Promotion Association
- 3) KIPI: The Korea Institute of Patent Information
- 4) KOIPA: The Korea Intellectual Property Protection Agency
- 5) RIPC: The Regional Intellectual Property Center
- 6) KIIP: The Korea Institute of Intellectual Property
- 7) KISTA: The Korea Intellectual Property Strategy Agency

6.2.1 Stage 1 (1970~1995)

KIPO was established in 1977 and the special accounting system was introduced for patent management. It set the precedence for autonomously promoting the patent administration and informationalization project, invention promotion project, etc. In addition, the foundation was laid for modernizing patent administration such as shortening the process of patent evaluation and judgment, building the patent administration and information system, and paving the framework for invention.

In 1979, KIPO became a member of the WIPO (the World Intellectual Property Organization) and joined the Paris Convention, the Patent Cooperation Treaty (PCT), the Nice Agreement, and the Madrid Protocol to promote international cooperation for an intellectual property rights system and expanded international cooperation with the U.S., Japan, EPO, and China through bilateral agreements for mutual exchange of patent examination and information.

In 1994, KIPO enacted the Invention Promotion Act, which set up a milestone to systematically implement an invention promotion project, and it expanded and reformed the department of industrial IP promotion (newly organized) to have a function of making IP policy.

In particular, KIPO established the KIPA as its affiliated agency and attempted to increase profits for inventors through invention promotion projects. Above all, the activation of the duty invention system contributed to building an infrastructure and system where multiple IPs can be created through sharing profits that are generated from the technology that researchers in a private or public institution develop.

In addition, as the practice of patent lawyers was activated and related industries emerged, it encouraged a small number of patent lawyers and many patent specification writers (patent researchers) to create more patents.

6.2.2 Stage 2 (1995~2010)

The second stage was to secure IPs more systematically by securing original patents and strengthening the connection with industries.

KIPI was established in the public domains to seek systematic IP management. Therefore, KIPI built the base system for creating patents by supporting the use of intellectual property information, operating IP information systems, conducting a prior art search, and drawing a patent map.

As a result, the number of industrial property rights applied rose from about 100,000 in 1989 to about 270,000 in 1996. Consequently, the period of a patent examination extended to

36.9 months, accordingly. In turn, outsourcing projects for prior art search increased, the handling target per patent examiner was adjusted, and more patent examiners were hired. Thus, the period of a patent examination got back to 28.1 months by late 1998.

In 1996, the Korea Industrial Property Business Arrangement Center was established to promote IP commercialization such as the arrangement for transfers and selling/buying of patented technology and facilitation of a license to use the patented invention, etc. In 1997, the Intellectual Property Rights Research Center (presently KIIP) was established to work exclusively on IP issues including patents, trademarks, etc.

In 1999, KIPOnet was established, and it stabilized an electronic patent application system. As a result, a whole process of patent administration was computerized in 2002. In addition, as the Technology Transfer and Commercialization Act was enacted and the Korea Technology Exchange was established in 2000, the mechanism for activating and ramping up technology exchange was founded.

Moreover, as in 2003, specialized organizations related to technology transfers and commercialization were expanded to operate and manage support projects according to the laws under the jurisdiction of each ministry. Starting with the Korea Technology Exchange, technology transfer centers for universities, research institutes, and regions, as well as the Promotion Agency for each department's related projects were established.

In particular, the Regional Intellectual Property Center (RIPC) was established in 2004, which provided patent information by base region. RIPC paved the way forward for expanding the potential of creating patents in local communities and actively supported patent commercialization through the Patented Technology Commercialization Council.

As the public institutions launched academia-industry collaboration groups and RIPC were established, the main actor that creates patents expanded to the public agencies and SMEs companies beyond large companies. Also, various supporting systems were planned and implemented to enhance the usability of the created IP.

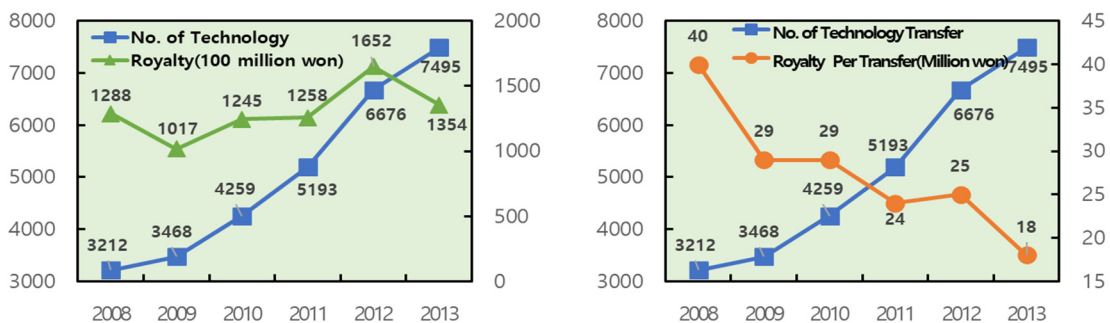
In addition, the KIIP was organized in Stage 3 to improve national competitiveness, which prepared for a launchpad of advancement of IP creation, protection, and use.

6.2.3 Stage 3 (2010~ present)

In 2012, the Korea Intellectual Property Strategy Agency (KISTA) was established as its affiliated organization of the KIPO. KISTA was an agency specializing in IP strategy, and promoted a variety of supporting projects throughout a full IP cycle as follows: Supporting IP-oriented R&D strategies, drawing the national patent strategy blueprint, conducting patent technology trend surveys, supporting an IP-R&D patent strategy for private-public sectors, managing patent performance, facilitating IP transactions, and paving the way for its use.

The key governmental departments expanded technology transfer and startup support projects for the commercialization of patents, which was the output of R&D activities. To create an ecology where technology and market make a virtuous circle, the Ministry of Trade, Industry and Energy built a technology transfer commercialization system through technology transfer commercialization support projects (i.e. technology business follow-up support project, R&BD support project, technology financing, and link to investment), as well as strengthened the specialty of TLO organization, expanded technology financing to small and mid-ranked companies, and solidified cooperation governance for technology commercialization. The Ministry of SMEs and Startups operated the projects to nurture the capability of small and mid-ranked companies' technology commercialization.

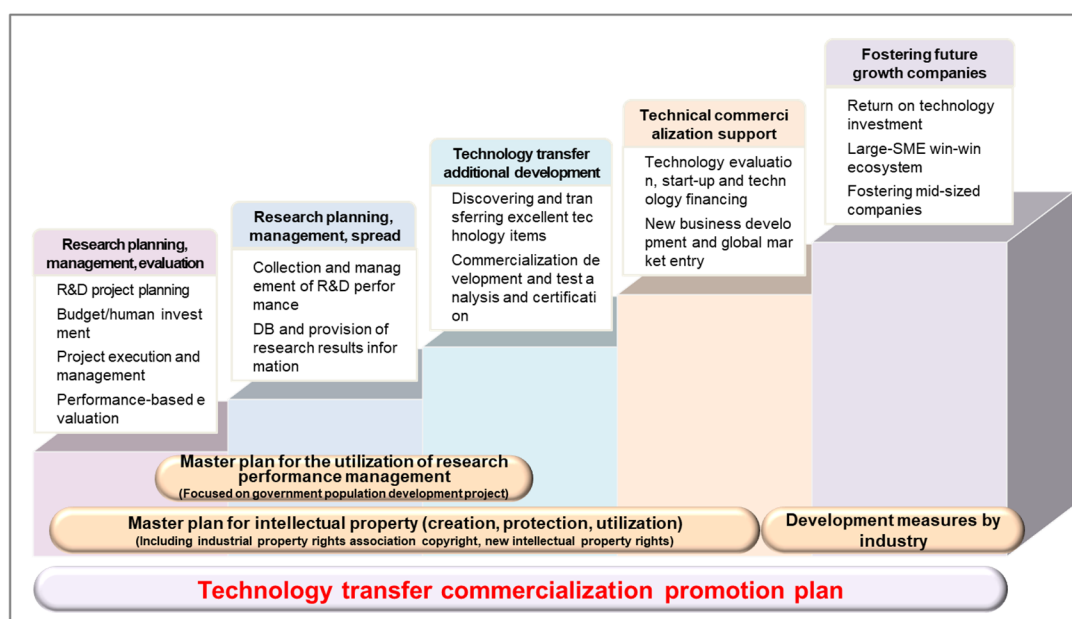
In Stage 3, the number of technology transfers between the domestic universities and public research institutes sharply increased from 2009 through 2013. It was merely 3,212 in 2008 but grew remarkably to 7,495 (133%) in 2013.



Source: Korea Institute of Intellectual Property (2015)

[Figure 3-8] The Performance of Technology Transfers between Universities and Public Research Institutes

In this aspect, Stage 3 was the period during which Korea changed from the country of many IP applications into one boasting homegrown original technology. That is, it was the period when Korea created excellent patents (through IP-R&D support for the creation of original technology and technology useful for industries and IP commercialization support) and was making a virtuous circle (through connecting technology transfers and business startups).



Source: Korea Institute for Advancement of Technology (2020)

[Figure 3-9] Technology Transfer Commercialization Plan

6.3 Key Policies and Programs in Korea

6.3.1 Administrative System for Industrial Property Rights and The Invention Promotion Act (To strengthen IP creation)

1) Domestic Situations

Korea's economic situations in the 1980s are similar to present Ghana's in terms of GDP per capita (USD 1,714), economic growth rate (7 to 10%), and the ratio of GERD to its gross domestic production (less than 1%). By the 1980s, 73 percent of about 20,000 patents

registered belonged to foreigners because Korea had heavily relied on light industries for economic growth, for which patented rights were not important during the 1970s. But it was changing in the 1980s when Korea was shifting its economic focus to heavy chemical industries and exports for which patented rights were essential.

2) Key Policies

For some key policies, KIPO was established in 1977 and the agency overtook the obligations and responsibilities to examine inventions and give a patent. It also reviewed and proposed the laws and systems related to protecting rights such as patent rights. Of them, KIPO enacted the Invention Promotion Act to prepare a plan to link the research performance of a university and public research institute to intellectual property (the 1990s). In addition, it applied the regulation of a duty invention when applying for and registering intellectual property rights in an attempt to give a subsidy to an inventor and manage the number of IP creations by university.

In addition, public institutions such as a college and university introduced a department exclusively for patents within the TLO. The Invention Interview System was designed to discover the technology whose commercialization is promising (based on technological excellence, marketability, and business feasibility) and thereby secure strategic IP early, while accelerating public IP creation. From 1978, the (designated) Chamber of Commerce & Industry in each city and province provided patent information. Later on, RIPC was set up as an affiliated organization of the KIPO and supported IP creation nationwide.

3) Performance and Implications

The duty invention system, which was designed to encourage individuals to apply for a patent, promoted passion for research, improved the quality of invention, and built the capability for applying for multiple PI rights through ongoing R&D investment and reinforced technology competitiveness. In particular, this system served as a launchpad for a virtuous cycle of technology innovation where the profits from duty inventions created from the system are invested again in R&D. In addition, the establishment of RIPC was considered to increase companies' awareness of invention and lead their effort to create IPs, which

eventually advanced technology competitiveness. In the long run, this can be seen as a driving force in securing the foundation for the IP industry such as IT/BT/NT and to improve national competitiveness.

In the case of Ghana, the level of IP creation is significantly lower than that of Korea in the 1980s, which had similar GDP levels to Ghana now, so there is a need to strengthen the current level of IP creation. The Korean Intellectual Property Office was opened in 1977, and efforts to create local intellectual property began with the establishment of the Chamber of Commerce and Industry in 1978, and through this, it became the seed of intellectual property creation in connection with local industries. The establishment of Korea Invention Promotion Association in 1994 led by the private sector and the state coincided with a point in time when the importance of intellectual property was emphasized throughout the industries, and it can be considered as a period of full-scale interlocking of industries and intellectual properties. In addition, a specialized institution has been established that specializes in policy, education, protection, and information services for the creation of core and original patents from multiple applications, and this is part of an effort to overcome administrative limitations of national institutions. Through the establishment and operation of specialized institutions that enhance professionalism with the purpose of improving the quality of intellectual property, it showed the result where the quality of intellectual property creation in Korea has moved up by one level. Finally, the establishment of specialized institutions to increase the utilization of the intellectual properties already created is likely to accelerate the virtuous cycle of the intellectual property ecosystem.

6.3.2 DB System for IP Information (To provide IP information)

1) Domestic Situations

KIPO was organized and the framework for creating patents was prepared with the enactment of the Invention Promotion Act. 6.3.2 DB System for IP Information (to provide IP information).

2) Key Policies

The Korea Institute of Patent Information (KIPI), which is a specialized agency to support the use of IP information and provide IP informationalization service, initiated its KIPRIS (www.kipris.or.kr) service in 1996. Accordingly, the provision of IP information started to strengthen the activities of patent creation such as IP information search, prior art search, and patent map on its web portal <http://www.kipris.or.kr>. In particular, it opened the 'KIPOnet' system, which processes patent administrative tasks such as patent application, examination, and trial online, and realized a computerized patent administration service.

3) Performance and Implications

This DB system became the platform on which researchers who seek to create a patent with patent information can explore prior art and produce a patent. It also helped attempt to quantitatively and qualitatively expand IPs. In addition, it built up both quality domestic and overseas IP data, supported an efficient patent examination and trial, maintained IP data errors existing throughout patent administration, improved data quality, advanced trust in patent administration, provided IP data of high usability at home and abroad, set the direction for IP policy and research, and created various added-values.

6.3.3 Establishment of Agency Specializing in IP Strategy and Operation of Support Projects (To create original patents)

1) Domestic Situations

In 2011, Korea recorded USD 25,000 for GDP per capita, 4 percent for economic growth rate, and 3 percent for the ratio of GERD to its gross domestic production (GDP). As of the end of 2011, Korea held more than 110,000 patents (cumulative): 49,711 by the universities and 66,728 by the public research institutes.

2) Key Policies

In 2009, the R&D Patent Center, which was established as an affiliated organization of KIPO, designated government R&D patent performance management agencies. Later on, it became a separate corporate body named the Korea Intellectual Property Strategy Agency (KISTA),

which supports IP-oriented R&D strategy to efficiently promote both quantitative and qualitative growth of patents and R&D, and operates the projects to adopt an independent and virtuous cycle of technology transfers led by universities and public research institutes.

KISTA covers a broad range of tasks throughout a full IP cycle, which consists of drawing the blueprint for the national patent strategy, conducting patent technology trend surveys, supporting IP-R&D patent strategy for private-public sectors, managing patent performance, and facilitating IP transactions. In particular, its supportive project for R&D patent strategy helps set IP-oriented R&D direction, support the strategy of creating quality patents, and improve the patent performance of the government R&D.

[Table 3-16] R&D Patent Strategy Support Projects

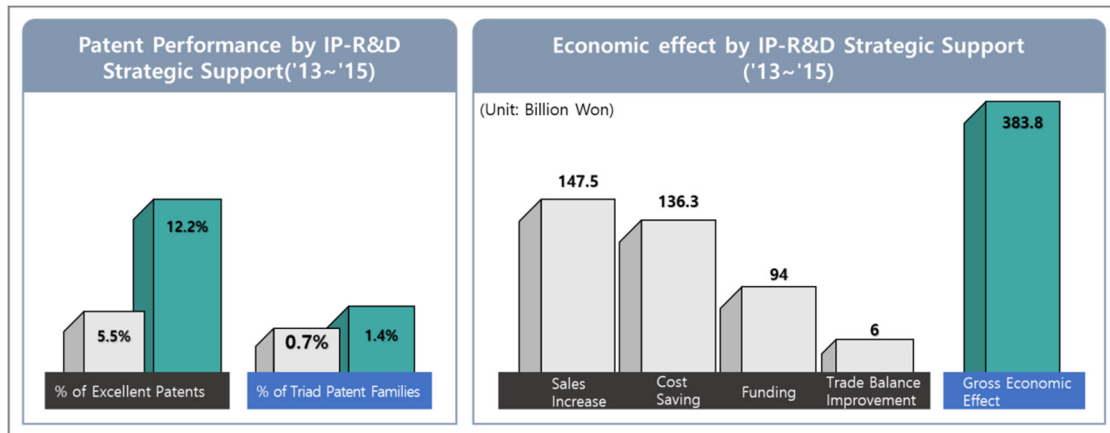
Private-sector R&D strategy support	• Support project for IP-linked R&D strategy • Support project for technology innovation IP convergence strategy • Startups IP voucher project • Support project for exporter's R&D full-cycle strategy
Government R&D strategy support	• Project for patent big-data-based industrial innovation • Project for public R&D patent technology trend survey • Support project for government R&D patent strategy • Support project for government R&D patent plan • Support project for standard patent creation • Support project for NST's convergence research right/commercialization
Business expansion strategy support	• Project for government R&D patent performance management • Support project for public institution-held patent diagnosis • Support project for demand-based invention interview • Support project for product-based IP package buildup • Support project for patent business expert operation • Project for research note utilization
Patent examination support	• Support project for patent examination

Source: Korea Intellectual Property Strategy Agency (KISTA)⁸

⁸ https://kista.re.kr/usr/com/prm/ContentView.do?cld=PRGM_000000000000163&menuNo=11000&upperMenuId=11
(Accessed 28th Dec, 2021)

3) Performance and Implications

The main actor of IP creation expanded from companies to public institutions, while homegrown original technology was discovered early and reprocessed to increase its industrial usability and exclusivity, which reduced the initial burden of R&D on companies. The project for supporting SME companies' IP-R&D strategy, which was part of the efforts to narrow the gap of technology between companies and universities resulted in the creation of 2 to 2.5 times more excellent patents and 11 times as high economic effect (KRW 383.8 billion) over input budget (KRW 355.5 billion) than those by non-recipients of the project.



Source: Korea Intellectual Property Strategy Agency (KISTA)⁹

[Figure 3-10] Effects of R&D Patent Strategy Support Projects

6.3.4 The Technology Transfer Act and the Technology Transfer Commercialization Plan

1) Domestic Situations

As of 2000, Korea recorded USD 12,261 for per-capita GDP and 9.1 percent for economic growth rate. But as ITC technology developed and high-tech products emerged, technology

⁹ https://kista.re.kr/usr/com/prm/ContentView.do?cld=PRGM_000000000000100&menuNo=11006&upperMenuId=11
(Accessed 2nd May, 2021)

got more complicated. Accordingly, the companies increased their open innovation strategies, conducting more R&D projects with several others as part of the strategies.

2) Key Policies

The Technology Transfer Act was enacted and a technology transfer commercialization plan was made. The Korea Technology Exchange was established and TLOs were formed within public research institutes to nurture technology transfer agents and form and activate the technology transaction market through technology transfer commercialization support projects. These support projects paved the way for nurturing the key players of the technology transaction market, activated the market, and laid the mechanism for technology commercialization.

3) Performance and Implications

The effects of the Technology Transfer Act include laying the groundworks for technology commercialization policies, securing related infrastructure and financial resources at a national level, fostering experts, and above all, making a pivot for promoting a technology transaction and cultivation mechanism through ‘the Korea Technology Exchange’. Leading LTOs were organized to activate the private sector-led technology market and the test operation began for outsourcing the public technology transfer to a private sector such as a regional technology transfer center. The Technology Transfer Act was revised to the Technology Transfer and Commercialization Act to strengthen policy support for initial commercialization through reinforced technical evaluation and technology financing.

7. Purchase Support Programs in Korea

7.1 Ghana's Public Procurement System and Implications of Korean Experience for Ghana

7.1.1 Current State of Ghana's Public Procurement System

In Ghana, public procurement is considered not merely a method of purchasing goods but a tool that could lead the country's development through public finances¹⁰. Currently, Ghana places a particular emphasis on the efficiency and transparency of its operating objectives.

Public procurement generates 24 percent of Ghana's total revenue and accounts for 50 percent of the national revenue. With public procurement accounting for 17 percent of the country's GDP, it can induce participation from the private sector, while promoting economic growth. In particular, Ghana anticipates to create more jobs and improve the capacity of its domestic market necessary for its economic development through public procurement.

The current state of Ghana's public procurement system can be found on the website of the Public Procurement Authority (PPA) at (<http://tenders.ppa.gov.gh/tenders>), which contains information on the list of current and planned government procurement orders.

However, PPA does not manage all the orders placed by corresponding government organizations. Since there are cases where each government organization places its own procurement orders, one must check the websites of corresponding government organizations as well as the advertisements printed in the local daily newspapers to find out the details of such orders.

Ghana's public procurement has been operated based on the 2003 Public Procurement Act (Act 663). While the Act was amended in 2016 according to the 2016 Public Procurement Act (Act 914), it still uses the same title number, 663.

¹⁰<https://www.europeanceo.com/world-view/ghanaian-government-seeks-to-restore-public-faith-in-procurement/>

The Act applies to i) the procurement of goods, works, and services, financed in whole, or in part, from public funds, ii) the disposal of public storages, vehicles and equipment, and iii) the procurement with public funds including loans procured by grants, government or foreign aid funds, and internally generated funds. It also applies to i) central management agencies, ii) ministries, departments, and agencies, iii) subvented agencies, iv) governance institutions, v) state-owned enterprises to the extent that they utilize public funds, vi) public universities, public schools, colleges, and hospitals, v) the Bank of Ghana and financial institutions including public trusts, pension funds, insurance companies, and building societies which are wholly owned by the Republic or in which the Republic has a majority interest, and vi) institutions established by the government of Ghana for the general welfare of the public or community.

In May 2017, President Nana Akufo-Addo appointed the Minister of State in Charge of Public Procurement (MOPP) to improve transparency and information-sharing in public procurement. The government of Ghana has also opened a separate website¹¹ for that same effort and established the National Procurement Commission in a bid to establish and solidify the public procurement market structure.

7.1.2 Key Operational Characteristics of Ghana's Public Procurement System

While Ghana has PPA in charge of operating the overall public procurement system, it is not functioning as a centralized public procurement entity like the Korean Public Procurement Service. Ghana's public procurement system is decentralized as the execution function of procurement lies in individual agencies and organizations.

However, an analysis of PPA's key functions reveals that the institution's main function is to develop and manage the country's public procurement policies, while also carrying out functions such as developing and coordinating technology support plans in public procurement.

¹¹ Office of Minister of State in charge of Public Procurement (MOPP)

Unlike Korea which operates a separate entity for public procurement, Ghana runs PPA in the form of a committee. PPA is being managed and supervised by its board of directors composed of 9 members nominated by the national president.

As stipulated in Ghana's relevant laws on public procurement, the country's procurement procedure is not much different from that of a Korean procurement contract system. Ghana also has in place a single-source procurement system with a view to prioritize certain bidders.

[Requirements for single-source procurement contract in Ghana]

- In a case where goods, works, or services are only available from a particular supplier or contractor;
- In a case where a particular supplier or contractor has exclusive rights over the goods, works or services, and a reasonable alternative or substitute does not exist;
- When the procurement entity seeks to enter into a contract with a supplier or contractor for research, experiment, study or development;
- Other requirements similar to those in Korea

According to the MOPP website, the government of Ghana is making efforts to link its support policies with public procurement. Examples of such efforts include the allocation of 30 percent of its public procurement orders to businesses owned by women, youths, and those with disabilities.

7.2 Key Characteristics of the Korean Public Procurement System and Its Evolutionary Path

7.2.1 Korean Public Procurement System

The Korean public procurement system can be approached in two dimensions. The first dimension involves contract-related processes that engage government organizations in the act of purchase or contracting to acquire items (goods, services or works) necessary for conducting their administrative functions. The second dimension involves procurement support programs that facilitate the enforcement of the government policies on public procurement. An example of the latter includes the priority purchase of goods manufactured by SMEs. The rationale for this kind of a dual approach lies in the complexity

of the public procurement-related system, while the public procurement market is run strictly by the relevant laws and regulations. Without this dual approach, new comers in public procurement, both businesses entering the public procurement market for the first time, and personnel newly assigned to procurement-related work at public institutions will have hard time understanding the public procurement system and its operational mechanism.

In some countries, government organizations have their own internal procurement units in charge of handling contract-related processes. Other countries have a centralized entity like Korea's Public Procurement Service, which administers all kinds of contract-related works at a national level. These centralized entities consider high quality, reasonable price, and contract transparency as their core values.

Considering that the public procurement market is directly controlled and operated by the state, some governments occasionally use the public procurement market to meet their policy objectives. Common examples include i) preferential treatment of SMEs by guaranteeing their share in bidding and ii) special benefits for SMEs to ensure their advantage in competitive bidding. In such cases, government agencies in charge of supporting SMEs are often administering and supervising relevant laws and programs.

7.2.2 Public Procurement as Tools to Achieve Policy Objectives in Korea

There are various ways to use public procurement as tools for policy implementation, but their basic mechanism can be summarized as follows.

The most common method involves incorporating certain policy objectives into the contract structure. For example, policy objectives can be achieved through preferential treatment of certain bidders by giving them extra points in the bidding process or requiring the policy objectives as basic procurement requirements. If the public procurement market is structured in a way that competition is the norm, companies will strive to take advantage of factors that will allow them to achieve a high ground in the merits system, such as affordable price, technology and, more recently, factors that reflect social values. Another way the government can use public procurement to meet its policy objectives is through

‘priority purchase’. Now the last method that governments may use is rather broader in scope. It involves regulating industries relevant to goods traded in the public procurement market, regulating the eligibility of bidders to participate in the bidding process, and conducting the evaluation of public organizations’ procurement practice to facilitate the implementation of public procurement policy.

[Table 3-17] Types of Support Using Public Procurement

Type	Description
Incorporating policy objectives into the contract structure.	Reflecting government policy objectives in the bidding process by operating a merit/demerit-based points system, organizing bidding indicators, easing competition, and organizing preferential price
Promoting purchase of certain items	Compulsory purchase of technologically certified products or green products; reflecting certain technology or certain standards requirements
Prioritizing certain companies	Allowing SMEs, women-owned businesses, or social enterprises to have advantage in the market; priority purchase of goods manufactured by those companies.
Incentivizing demand	Reflecting the amount of goods supported by policy into the management evaluation or exempting those responsible for such purchases from disciplinary action
Regulatory or support schemes	Reflecting regulatory factors per industry or business into purchase requirements; organizing a selection process based on the merit/demerit system; improving the efficiency of advertisement and purchasing methods

7.2.3 Case of Policy Support Using Public Procurement

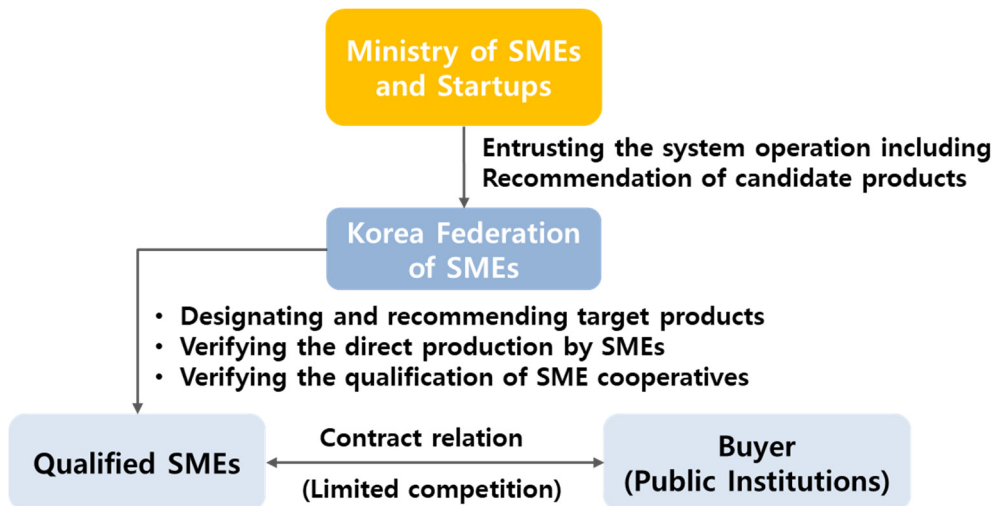
1) Bidding Competition Limited Only to SMEs (SME-Limited Competition)

Korea’s industrialization and economic development has been traditionally led by large corporates. However, economic development centered around large businesses has its limitations. Since support and assistance for SMEs are considered key to achieving a balance between industries and discovering future industries, the Korean government has been running various support schemes for SMEs in the public procurement market that it operates.

A prime example of a system that gives preferential treatment to certain business groups, particularly SMEs, is the SME-limited competitive bidding scheme. This program allows only SMEs to compete with each other in the bidding process.

The operational mode of this program is as follows: The government designates once in every three years goods that can only be supplied by SMEs and allows only qualified SMEs to participate in the bidding process of the designated items.

The Korean government stipulates this program in the Act on Facilitation of Purchase of SME-Manufactured Products and Support for Development of Their Markets. Related policies and programs are regulated by the Ministry of SMEs and Startups. Actual day-to-day operation of these programs is entrusted to a private organization called the Korea Federation of Small and Medium Business, who designates purchase items and is in charge of day-to-day management.



[Figure 3-11] Operational Mechanism of SME-Limited Competition System

2) Preferential Treatment of Technology Development Products (Priority Purchase of Technology Development Products)

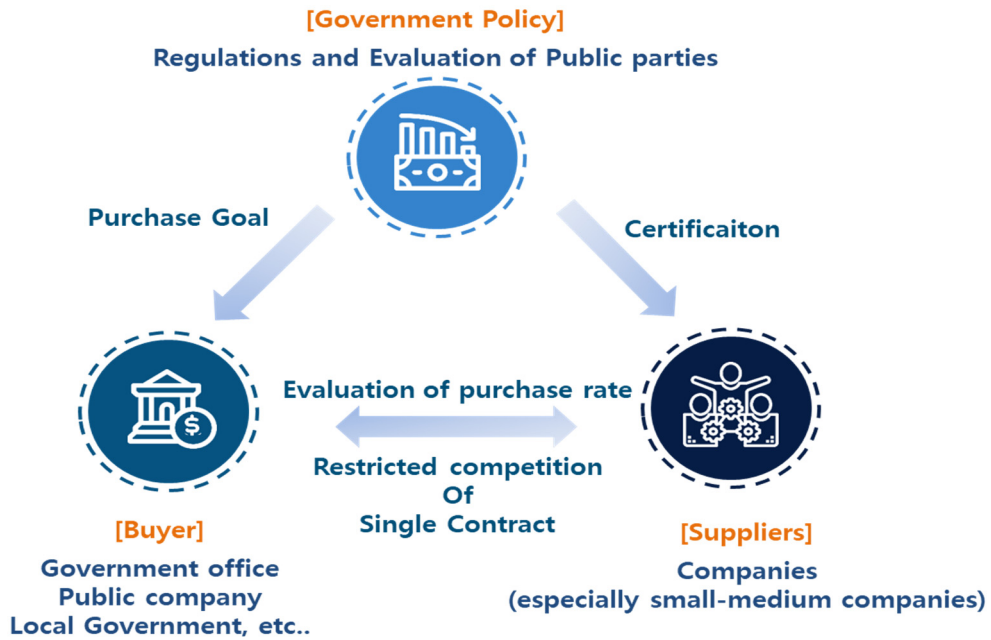
Of various programs supporting SMEs, one that does not limit the type of businesses eligible for bidding, but instead promotes SMEs' technology development efforts and allows for the purchase of technologically outstanding goods from these suppliers in the public sector is called the 'priority purchase' scheme.

The law mandates that the government purchase a certain percentage of those technologically outstanding goods developed and manufactured by SMEs. Accordingly, this scheme is considered a useful tool for expanding R&D support for SMEs and spurring technology innovation.

This preferential purchase scheme is used not only as a tool for invigorating technological development but also as a means to expand the purchase of green products to solve environmental problems. The priority purchase scheme is also being applied to innovative products as new policies on innovative procurement have been recently implemented.

As the following Figure shows, the priority purchase system in Korea is being operated through the linkages of three different perspectives.

Firstly, from a policy perspective, when the government devises a policy to support eco-friendly products or technology development products, it specifies such products as purchase items through certification or designation. In other words, companies strive to have its products certified or designated as eco-friendly, and the government certify such efforts. Occasionally, the government sets the requirements for single-source procurement or offers limited competition in order for public buyers such as government departments and agencies to actively buy such products. It also mandates them to buy a certain percentage of such products by law if the government deems it important to expand purchases of such products.



[Figure 3-12] Operational Mechanism of Priority Purchase System

7.2.4 Korea's Economic Growth and Evolutionary Path of the Korean Public Procurement System

7.2.5 Changes in Public Procurement Policy from the Perspective of the History of the Korean Economy

The Korean public procurement function was first initiated for the purpose of efficiently distributing overseas aid received from international donors in the process of reconstructing the devastated country after the Korean War. During its economic development, Korea prioritized quality-assured goods in the public procurement market in order to ensure competitiveness of its export-oriented industries. This policy of prioritizing high-quality goods has expanded and evolved into the current system of purchasing technology development products. At the present, public procurement is used as an effective tool for spearheading Korea's innovation-led growth.

[Table 3-18] Evolution of Korean Public Procurement Policy

Period	Key national agendas	Objectives of public procurement	Key government policy measures
Post-Independence to the Second Republic, 1945-1961	National reconstruction and stabilization of people's livelihood	• Efficient distribution of foreign aids	• Foreign Capital Service established (February 1955)
Military government (Third-Fifth Republic) 1961-1993	Establishing industrial and economic foundations	• Efficient spending of government revenue to build and expand social overhead capital (SOC) (upgrading construction contracts)	• Public Procurement Service established (October 1961) • 1st Five Year Economic Plan launched (January, 1962) • Framework Act on Small and Medium Enterprises enacted (December 1966) * Support for marketing and sales channels of SME-manufactured products
Pres. Kim Young-sam Administration, 1993-1998	Promoting the development of industrial technologies and their globalization	• Securing quality competitiveness, opening the procurement markets, and supporting the export of SME-	• Industrial Standardization Act enacted (June 1993) * Priority purchase of standardized products

2021 K-Innovation ODA Program with Ghana

Period	Key national agendas	Objectives of public procurement	Key government policy measures
		manufactured goods (upgrading domestic purchasing contracts)	• WTO Government Procurement Agreement (GPA) signed (December 1995) • Small and Medium Business Administration (SMBA) established (February 1996)
Pres. Kim Dae-jung Administration, 1998-2003	Overcoming the financial crisis & promoting digitalization	• Achieving transparency & efficiency in procurement administration	• Electronic bidding launched (January 2001) • 'Nara Market' opened (October 2002)
Pres. Roh Moo-hyun Administration, 2003-2008	Laying the foundations for innovation and procurement of social good	• Supporting the government's economic policy through public procurement	• Policy for expanding public procurement of SME-manufactured technology development introduced (March 2006) * Target purchase rate (5%) introduced • Policies to support companies run by the socially vulnerable expanded
Pres. Lee Myung-bak and Park Geun-hye Admin., 2008-2016	Upgrading the grassroots economy	• Stagnating development in policies and systems	• 5% target purchase rate introduced for goods produced by companies run by female CEOs (2009) • 1% target purchase rate introduced for goods produced by companies run by those with disabilities
Pres. Moon Jae-in Administration, 2016-Present	Promoting innovation-led growth	• Innovating public service through innovation-oriented public procurement	• Designation of innovative goods and setting the target purchase rate (1% of total purchase price) • Integrated public procurement system established

Source: Public Procurement Service (2007)

7.2.6 Technology Development Driven by Public Procurement from the Perspective of Industrial Development (Focus on Quality Assurance)

In their attempt to promote industrial and technology development, ministries have introduced the priority purchase scheme. Nonetheless, a list of certified goods (including designated goods) for priority purchase differs across ministries because each ministry has taken a tailored approach to delivering its policy outcomes.

Since the priority purchase system plays an important role in facilitating the policy implementation through public procurement, a market run by the government, the government has been emphasizing support for SMEs' consistent efforts to upgrade their technologies. At this time, the Korean government enacted the Act on Promotion of Small and Medium Enterprises and Encouragement of Purchase of Their Products, which laid the legal basis for the priority purchase of products from technology development.

Enacted in 1994, the Act on Promotion of Small and Medium Enterprises and Encouragement of Purchase of Their Products inserted a new clause on the prioritized purchase of technology development products. At the outset, the scope of products subject to prioritized purchase included technologically innovative goods produced by SMEs and goods that were quality-assured by group standards.

As for quality-certified goods, standardized products became subject to prioritized purchase after the 1961 Manufacturing Standardization Act was entirely revised into the 1993 Industrial Standardization Act. If the state, local governments, government-invested institutions or public institutions wish to purchase goods, they should purchase goods with the required certification labels per law. If there aren't any such, they should prioritize purchasing goods with quality certification labels. In other words, when the prioritized purchase system was first introduced, the industrial policy back then reflected the Korean government's efforts to improve productivity through standardization and secure competitiveness in an export-driven economy through quality assurance.

The Act on Promotion of Small and Medium Enterprises and Encouragement of Purchase of Their Products was amended into the Act on Promotion of Small and Medium Enterprises

in 2009. The section on the encouragement of purchase and development of their markets was reenacted into the 'Act on Facilitation of Purchase of SME-Manufactured Products and Support for Development of their Markets', with a new clause on the prioritized purchase of goods manufactured by SMEs. According to this Act, a broader range of certification-related regulations are now being applied.

7.2.7 Full-fledged Implementation of Public Purchase System for SME Innovation (Phase 1: Supplier-driven innovation)

The inauguration of a new administration in 2006 was followed by a full-scale revamping of the public procurement system with an emphasis on innovating SMEs. A foremost characteristic of such change is that priority purchase of technology development products, which had so far been only recommended, was set as a concrete objective. Such an objective included mandatory purchase of 5 percent of the total purchase price of goods manufactured by SMEs. After two years, the target purchase rate was raised to 10 percent.

In addition, in order to promote the purchase of R&D products or newly developed products, whose quality or performance have yet to be proved, the then-Small and Medium Business Administration introduced the performance assurance scheme, whereby the government provided a guarantee for possible losses to be incurred from the purchase of new technology products, thereby facilitating the purchase of those goods.

In fact, the purchase price of technology development products, which stood at around KRW 1 trillion before the target purchase rate was presented, increased by five fold reaching KRW 5 trillion. This implies the target purchase rate clearly played the important role of inducing SMEs' technology development efforts.

However, as the system has now been operated for over 10 years, some side-effects such as the buyer institutions' excessive focus on the achievement of target rates and concentrated purchases of certain product items began to emerge.

7.2.8 Introduction of Innovative Procurement for Improved Public Services (Phase 2: Demand-driven innovation)

The government support policies such as priority purchases may have advantages in the early stage as they can create an early-stage market for a small number of goods. However, if such policy continues for an extended period, it may cause some side-effects such as markets disproportionately benefiting certain goods.

In particular, when a purchase target is presented, buyer institutions may make blind purchases simply to achieve the target. This may distort the intended policy objectives, so particular caution must be taken in the operation of the system.

In 2019, the Korean government announced the ‘Innovation-Oriented Public Procurement Policy’ for the purpose of enhancing the livelihood of its people through improved public services, going beyond the immediate tasks of i) overcoming problems that have arisen in the process of operating the target purchase of technology development products and ii) supporting SMEs.

This policy encompasses a set of beliefs as follows. Firstly, it takes the view that public service innovation needs to be achieved through public procurement as the demand for government innovation has increased following the expansion of public finance. This implies that such services as welfare, environment, public safety, security and administrative services that the government offers to the public through public procurement as part of their everyday lives now need to be recognized as public services. Thus, the new public procurement policy aims to introduce solutions necessary for improving these existing services and providing a set of new services.

Secondly, this policy also intends to realize the so-called ‘active procurement administration’, which expands the basic concept of ‘demand-based innovation’ and encourages public participation. This will ensure a new perception of the role of the public procurement market and its efficient use. The Korean government as well as other governments in developed countries such as the U.S. and EU also emphasize the active engagement of a purchasing team or personnel as a prerequisite of innovative procurement.

For a long time, Korea's procurement administration has pursued a stable supply of off-the-shelf products whose performance has been already proven while following the rigidly structured regulations. Since public procurement is funded by taxpayers' money, it is reasonable not to waste funding resources. But recent policy reflects increasing demand for flexibility in public procurement, which will allow stakeholders to make better choices, thereby ushering in a new wave of changes.

Thirdly, the new policy also implies the need for a new system for innovation as the priority purchase of technology development products, which is an offshoot of existing innovative procurement, has led to innovation being concentrated on only certain goods or businesses, thus halving the innovation efforts. In particular, this new policy is designed to solve problems such as the buyers' hesitancy in purchasing SMEs' innovative products out of concerns over the high price and unproven performance. The new policy also helps solve the suppliers' reluctance to make R&D investment due to the high risk associated with R&D and uncertainty of demand. While the nationwide R&D is on the rise, R&D outcomes do not necessarily lead to increased demand or actual purchase. Therefore, this policy aims to address these problems by taking advantage of the purchasing power of the public sector.

Lastly, the new policy was developed in response to the need to develop an integrated procurement system as maintenance and improvement of the public procurement support programs, including the improvement of the contract structure, was deemed urgent. As reviewed in the previous sections on the key concepts and characteristics of public procurement, the public procurement policy of the Korean government has a dual structure of i) managing public procurement contracts and ii) supporting the enforcement of government's SME and industrial policies as a policy implementation tool. Of course, while these two functions of the public procurement policy tend to be merged at the contract stage, its operational mechanism to manage and coordinate the procurement process is decentralized to the extent that it prevents the intended policy outcomes.

Therefore, a new policy initiative has been developed to identify public procurement as an important branch of national policy agendas and set up a government control tower that

can coordinate and preemptively fine-tune the public procurement policies in the wake of changes in surrounding environments.

In summary, the new policy aims to develop a new procurement structure that allows the buyers in the public sector to preemptively present their needs while helping suppliers quickly deliver their innovative products. The new policy has also established an integrated mechanism for inter-ministerial cooperation, so that improvements of any related systems or programs can be made swiftly.

8. Accreditation & Certification Support Programs in Korea

8.1 Current State of Ghana

Among strategic industries Ghana is currently targeting for its technology innovation drive are agricultural and food processing, environmental waste recycling, manufacturing, ICT, healthcare, mining, and crude oil. However, the limited number of technology patent applications filed in Ghana every year (e.g. 21 patent applications in 2017) implies that Ghana's technology innovation capacity remains at a significantly low level.

As a part of the conformity assessment system the government of Ghana has established according to the relevant laws, Ghana is operating various test, calibration, and inspection organizations in certain industries in close linkage with state-run ministries. Nevertheless, these institutions have limited capacity as they are lacking proper facilities, human resources and systems to perform their assigned responsibilities. Despite the efforts of GhaNAS, a national accreditation body of Ghana, to establish mutual recognition agreements (MRAs) with IAF (International Accreditation Forum) and ILAC (International Laboratory Accreditation Cooperation), its current operation has limitations to become an internationally recognized accreditation system that complies with international standards. The government of Ghana also established the Ghana Standards Board (GSB) according to the Standards Authority Act enacted in 1973 as a body in charge of setting standards for conformity assessments. In the early stage, the National Research and Innovation Fund may be used for funding the annual budget of KRW 60 billion (approximately USD 51.86 million) required for the establishment of Ghana's test and certification system. Such budget requirements may later be met with profits from routine testing and certification services. Currently, Ghana has 13 government-funded research institutes and numerous universities available for technology and policy support for building the country's test and certification system, but the country is apparently lacking R&D human resources needed for building and operating such a system. As of 2015, Ghana's R&D workforce per million population remained at 89.1.

8.2 Evolutionary Path of Korea

Until the end of the 1990s, Korea's test and certification industry was recognized mainly as infrastructure assisting the production function of the country's manufacturing sector. In the 2000s, intensified competition in international trade made countries worldwide use test and certification as a non-tariff barrier, which in turn generated increased demand for testing and certification services. Consequently, the test and certification field started to emerge as an independent industry. In fact, the test and certification industry is labor-intensive requiring a large pool of highly skilled workforce, so it has a high potential of creating jobs for high-caliber scientists and engineers. With this in mind, the Korean government came up with a national plan for advancing its test and certification industry as a high value-added service industry in 2014. Since then, the Korean government has vigorously implemented the following key strategies as part of its plan; i) identifying promising areas with high-growth potential and focusing government support on the identified areas, ii) developing and disseminating an integrated brand of testing and certification services, iii) strengthening test and certification related capacity by building up equipment, developing skilled human resources capable of providing specialized services, and promoting smart testing and certification services, iv) expanding new markets, and v) easing regulations and upgrading legal institutions.

Even before the 1990s, the ministries concerned of the Korean government had already established their own test and certification system (i.e. conformity assessment system) covering all industrial sectors based on relevant laws. In the 2000s, the introduction of the legally optional certification scheme for new technologies and products paved the way for transforming Korea's test and certification industry. For this, the Korea certification (KC) mark, an integrated certification mark, was introduced in 2009 to improve the conformity assessment system required by the law. Since 2010, the legally optional certification scheme led by the private sector has been promoted. With the enactment of the 2020 Conformity Assessment Management Act, uniform and consistent test and certification criteria were set forth for the legally mandatory and optional certification schemes across the entire range of industrial sectors, which have been under operation since then. As for accreditation, the Korean government introduced the accreditation system for test organizations in 1992

and also laid the legal ground for the Korea Laboratory Accreditation Scheme (KOLAS) by enacting the 1999 Framework Act on the National Standards. In 1998, the Korean government also signed the ILAC MRA. Since the 2000s, the Korean government has expanded the scope of accreditation for testing, calibration, and inspection organizations to include manufacturers of reference materials, standard substance producers, medical laboratories, and proficiency testing providers. It has also broadened the scope of accreditation performance by conformity assessment organizations. In its bid to upgrade the product certification system, the government also established the Korea Accreditation System (KAS) in 2001 and joined the International Accreditation Forum Multilateral Recognition Arrangement (IAF MLA) in 2007. Before 2010, each ministry used to operate its own designated test and certification bodies, which were then integrated into the KOLAS accreditation body in 2010.

Years		~1990	2000~2009	2010~
Test and Certification Policy	Change of Concept	Manufacturing infrastructure industry	Independent service industry	High value-added service industry
	Industry Characteristics	Supporting industry for production	A highly skilled, labor-intensive industry that can act as a non-tariff barrier	Industry creating quality jobs with rapid growth of the test and certification market
	Key Promotion Policy	1) Establishment of conformity assessment system (legal compulsory certification by department), 2) Promotion of KS certification (introduced in 1961)	1) KC integrated mark (2009) launched 2) Privatization of KS certification and launch of voluntary certification based on law (NEP/NET, etc.)	1) Preparation of a strategy for fostering the testing and certification industry (2014) 2) Support for voluntary certification promotion in the private sector and establishment of a certification standard information center
	Legal System	1) Establishment of legal basis for conformity assessment in each ministry and industry 2) Enactment of Framework Act on National Standards (1999)	1) Establishment of legal basis for voluntary certification (Ex: NEP based on the Industrial Technology Innovation Act)	1) Enactment of Conformity Assessment Act (2020)
Test and Certification Governance	Accreditation (Designation) Body	1) Establishment and operation of designation system for each department 2) Introduction of accreditation system ('92) and preparation of legal basis for KOLAS 3) Signed KOLAS ILAC MRA ('98)	1) Expand the scope of KOLAS accreditation 2) Established KAS ('01) and signed IAF MLA ('07)	Replacing the designation system for each government department with the KOLAS accreditation body
	Certification System Status	Operation of a total of 211 certification systems (as of 2020): legally mandatory 85, legally voluntary 126		

8.3 Major Test and Certification Programs in Korea

Certification is a way of ensuring that specified requirements for products, systems, qualifications, and services are met. Depending on the type of the certification needs, it is classified into product, system, service, and qualification certifications, respectively. Various

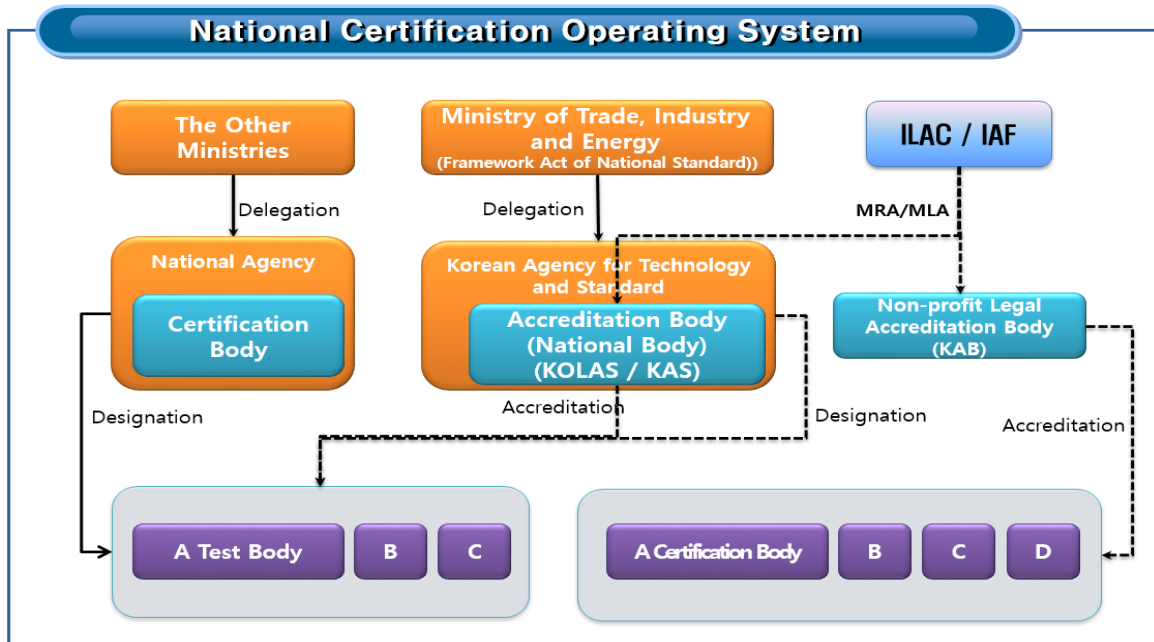
terminologies such as type certification, validation, and authorization are interchangeably used to refer to certification. Certification schemes are then divided into legal certification and private certification depending on the existence of their legal basis. Legal certification schemes are then divided again into mandatory and voluntary schemes. Most of the certification procedures are carried out by test (or evaluation) organizations accredited or designated by the government authorities. Certification is conducted in close linkage with standard, inspection, and test-related activities as well as accreditation or designation activities of test (or evaluation) organizations.

8.3.1 Current State of Certification-related Laws in Korea

The Korean government is operating numerous certification schemes by ministry according to relevant laws in relevant industrial sectors. For example, there is the Framework Act on National Standards, an overarching act governing the certification system. In addition, the Conformity Assessment Management Act regulates matters concerning legally based certification scheme as an enforcement ordinance. The Act was enacted in April 2020 to secure credibility of conformity assessment and enhance the national competitiveness and safety and welfare of the Korean public. The Act includes provisions on i) policy development for conducting conformity assessment, ii) the development of conformity assessment-related regulations, iii) establishment of accreditation organizations, accreditation activities by such authorities, regulations on the expiration date operated by authorization (accreditation) organizations, use of certification marks, & regular inspections, iv) penalties and fines imposed on certification bodies, and v) policy support for strengthening conformity assessment capacity through the development of high-caliber human resources, international cooperation, and technology development.

8.3.2 Operational Mechanism of the National Certification System in Korea

The following Figure shows the operational mechanism of Korea's national certification system.



[Figure 3-13] Operational Mechanism of Korea's National Certification System

Korea's national certification system is run by key players such as KOLAS, KAS and KAB. KOLAS is in charge of the accreditation/certification of standardization bodies, manufacturers of standards, substance producers, and medical laboratories as well as testing, calibration, and inspection bodies. In the meantime, KAS is responsible for the accreditation of product certification bodies and KAB (Korea Accreditation Board) accredits certification bodies in 18 system management fields.

The following Figure shows Korean accreditation organizations providing accreditation services for the certification bodies.

National Accreditation Body

Body	Korea Laboratory Accreditation Scheme KOLAS	Korea Accreditation System KAS	Korea Accreditation Board KAB
Accreditation Mark			
Acts	Articles 23 of the Framework Act on National Standards	Articles 21 of the Framework Act on National Standards	Articles 24 of the Framework Act on National Standards and Article 32 of the Civil Law
Office	Korean Agency for Technology and Standard	Korean Agency for Technology and Standard	(Foundation)Korea Accreditation Board
Accreditation Scope	· test bodies, calibration bodies, · inspection bodies, · standard reference material production bodies, · medical inspection bodies	· product certification bodies	· 18 fields such as quality management/environmental management
MRA Status	ILAC MRA APLAC MRA	IAF MLA PAC MLA	IAF MLA PAC MLA
Accreditation Results	962 bodies	22 bodies	99 bodies

Source: National Technology Standards White Paper (2020)

[Figure 3-14] Korean Accreditation Organizations

8.3.3 Major Optional Certification Schemes in Korea

1) KS Certification Scheme

The KS (Korea Standard) Certification Scheme was introduced in 1961 for the purpose of enhancing industrial competitiveness and advancing the national economy by improving product and service qualities, simplifying and standardizing their transactions, and rationalizing their consumption. The legal basis of this certification scheme is the Industrial Standardization Act. Under this scheme, KS certification bodies evaluate companies' (including factories and business sites) capabilities of producing and supplying KS certified or higher-grade products and services in a stable and steady manner. Based on the evaluation results, KS certification bodies issue a certificate for the certified companies.

Once certified, the KS certified companies are allowed to advertise their products or services by posting a KS certification mark (㉿) on their products, packages, containers, invoices, or

letters of guarantee, indicating their offerings meet the Korean industrial standards. They also receive the benefit of priority purchase in the government's public procurement. Since the transition into the 'competitive system of KS certification bodies' and the introduction of the 'inter-ministerial participatory national standards operation system' in 2015, multiple organizations including the Korea Standards Association have been additionally designated as the KS certification bodies.

As for the scope of the KS Certification Scheme, the president of the Korean Agency for Technology and Standards (KATS) designates certain products and services as items (categories) to be KS certified through a screening process by the Industrial Standards Review Committee. The KATS president then issues a KS certificate, when they are deemed necessary for promoting and disseminating industrial standards. The KS certification procedure starts from the submission of a certification application by a product or service provider whose offerings are subject to the items to be KS certified. Once the application is submitted, the Certification Screening Committee reviews the items to be certified by product or service category according to the specified certification review criteria. When the review results show the reviewed product or service meets the KS certification or higher requirements, the head of the certification body issues a KS certificate. Concurrently, the certification body signs a certification contract with the certified company.

2) NEP and NTP Certification Schemes

The NEP (New Excellent Product) and NTP (New Excellent Technology) Certification Schemes were first initiated under the lead of the Ministry of Trade, Industry and Energy (MOTIE). Following MOTIE, other ministries such as the Ministry of Science and ICT, Ministry of Land, Infrastructure and Transport, Ministry of Environment, and Ministry of Health and Welfare introduced the NTP Certification Scheme in their corresponding areas, which has been implemented ever since. This scheme is intended to promote technology development and commercialization in the corporate sector, which will, in turn, help companies sustain their competitiveness and stabilize their business operations. For technologically competent SMEs which have developed new products but are lacking

marketing capacity, the NEP and NTP Certification Schemes are attractive as NEP and NTP certified items have advantages in making inroads into the market in the early stages with the committed purchase by public institutions through public procurement.

a. NEP Certification Scheme

This scheme is specifically applied to products with excellent quality and performance as well as high economic and technological impacts, selected from commercialized products using locally developed new technologies or other alternative technologies that are equivalent to the new technologies. The certification procedure for the submitted applications involves a four-staged review process, which consist of the first stage of document review, second stage of field due-diligence, third stage of product evaluation, and fourth stage of a comprehensive review. When any of the stakeholders has objections to the review results, the case is sent to the Comprehensive Review Committee for approval after a separate session of adjustment review. This scheme is legally based on Article 16 of the Industrial Technology Innovation Act, Article 18-5 and Article 25 of the Enforcement Decree of the same Act, and the Operational Guide on NEP Certification and Purchase Promotion. Basically, the NEP certification is effective for three years, which can be extended for a further three years. NEP certified companies are subject to the following benefits; i) 20 percent mandatory purchase of NEP-certified products by public institutions, ii) single-source purchasing allowed for NEP-certified products, iii) priority purchase of NEP-certified products developed by SMEs, iv) designation of NEP-certified products as excellent procurement products by the Public Procurement Service, v) preferred treatment of NEP-certified products when applying for the government projects, vi) the beneficiary of the preferential technology guarantee program, and vii) preferential application for technology finance support project.

b. NET Certification Scheme

Basically, technologies whose development has been successfully completed are subject to the NET Certification Scheme. In other words, this scheme is applied to technologies that have secured quantitative evaluation data through manufacturing, testing, and operation of prototypes using the established technological theories. Thus, the main targets of this scheme are technologies that can be commercialized or can significantly improve the

performance of the existing products within two years from the date of certification. Processing technologies with a potential to significantly improve productivity or product quality are also subject to this scheme. As a part of this scheme, a total of 24 specialized review committees are being operated in 6 technology fields. The applicants are required to choose one of these committees depending on the contents of their technologies when applying for this certification scheme. The certification procedure for the submitted applications involves a three-staged review process, which consists of the first stage of a document review, second stage of a field due-diligence, and third stage of a comprehensive review. Once this review process is completed, a formal notification is released to announce successful technologies to be certified as NET and receive any possible objections to the review results. In a case of technologies for which objections are raised, the final decision is made after an additional session of adjustment and arbitration review.

In the meantime, the following four criteria are used for the NET certification review; i) new and original technologies that are locally developed in Korea but are as excellent as or even superior to those of the developed countries and can be commercialized, ii) technologies whose technological and economic impacts are big enough to enhance the country's technological competence and global competitiveness, iii) existence of a quality management system capable of maintaining the target product performance for quality and stability specified as technology development goals, and iv) the need for the NET certification and its impact. The effective term of the NET certification, spanning a maximum of three years, is determined considering the technology development trends, competition dynamics, and expected impact. For the extension of the effective term, an application for the extension should be submitted at least four months prior to the expiration date of the NET certification. Only technologies that have yet to be commercialized at the time of the expiration of the NET certification are subject to the application for the term extension. The effective term can be extended for a maximum of three years. NET-certified companies are provided with the following benefits; i) single-source contracts with the government, ii) priority purchase of NET certified items by public institutions, iii) designation as excellent

government purchase items, iv) funding support through initiatives such as R&D support projects, technology financing support project and Technology Commercialization Fund, v) preferred treatment of NET-certified technologies when applying for government projects (bonus points in evaluation), and vi) tax incentives for investment assets.

3) Corporate Certification Scheme

Among representative corporate certification schemes in Korea are the Inno-Biz Certification Scheme and Venture Business Certification Scheme. In addition, the Korean government also operates a system called the 'Reporting and Accreditation Schemes for Corporate R&D Centers and R&D Dedicated Units'. This section offers a brief introduction of the Inno-Biz Certification Scheme and Venture Business Certification Scheme.

Based on Article 2 of the Act on Special Measures for Developing Venture Businesses, the Venture Business Certification Scheme provides various policy benefits including tax exemption for qualified venture companies selected among SMEs with excellent technology and management innovation capacity.

The Venture Business Certification Scheme issues the five types of certifications to qualified SMEs as follows; i) Venture investment company, ii) R&D company, iii) technology evaluation and finance company, iv) technology evaluation and loan company, and v) preliminary venture company. The scheme offers a wide range of benefits including support for business incubation and startup, financial support, human resources support, technology support, tax incentives, and locational benefits. An online application is available using the Venture Business Open Notification System at the website (www.venturein.or.kr).

In relation to the Inno-Biz Certification Scheme, the term 'Inno-Biz' is coined by merging innovation and business, and refers to technology innovative SMEs which have secured competitiveness using their technology competence and technology innovation capacity. This scheme provides these so-called 'Inno-Biz' SMEs with an array of benefits including technology, funding, market development (globalization) support. The targets of this scheme are SMEs under normal operation, whose business history spans at least three years.

SMEs to be certified as Inno-Biz are selected through a three-staged evaluation process, which consists of i) an online self-diagnosis in the first stage, ii) an onsite field evaluation by the Korea Technology Finance Corporation (KIBO) in the second stage, and iii) the evaluation of technology innovation system and evaluations of corresponding technologies in the third stage. Key benefits of this scheme include financing support from KIBO, funding from the SME Policy Fund, and preferred treatment of Inno-Biz SMEs when applying for SME support programs.

Classification	Inno-Biz Certification Scheme	Venture Business Certification Scheme
Overview	For technology innovative SMEs which have secured competitiveness using their technological competence and technology innovation capacity, this scheme provides an array of benefits including technology, funding, and market development support.	For qualified venture companies selected among SMEs with technology and management innovation capacity, this scheme provides policy benefits including tax incentives.
Legal basis	SME Technology Innovation Act	Act on Special Measures for Developing Venture Businesses
Procedure	1) 1 st Stage: Online self-diagnosis 2) 2 nd Stage: Onsite field evaluation (KIBO) 3) 3 rd Stage: Evaluation of Technology Innovation System and evaluations of corresponding technologies	1) Certification types The qualified companies are certified in the following types: ① Venture investment company, ② R&D company, ③ Technology evaluation and finance company, ④ Technology evaluation and loan company, ⑤ Preliminary venture company 2) Application procedure: Online application using the Venture Business Open Notification System
Key benefits	Financing support from KIBO, funding from the SME Policy Fund, and preferred treatment when applying for government support programs	Numerous benefits including support for business incubation and startup, financial support, human resources support, technology support, tax incentives, and locational benefits

8.4 Key Achievements of the Korean Certification System

8.4.1 Accreditation Performance by Accreditation Organization

As of December 31, 2020, KOLAS accredited a total of 962 organizations in six service areas including test, inspection, and calibration. In the meantime, KAS, a product accreditation body, accredited 22 product certification bodies, while KAB, an accreditation body of system certification, accredited 99 organizations in 18 system management areas.

Classification	2012	2013	2014	2015	2016	2017	2018	2019	2020
KOLAS accreditations	665	722	741	777	793	849	889	919	962
KAS accreditations	16	17	20	20	20	19	19	20	22

Source: National Technology Standards White Paper (2020)

KAB provides accreditation services in the following 18 system management areas.

Quality mgmt.	Environmental mgmt.	Telecom	Occupational health & safety	Food safety	Information security	Green mgmt.	Energy	FSSC
ISO 9001	ISO 14001	TL 9000	ISO 45001	ISO 22000	ISO 27001	GMS	ISO 50001	FSSC 22000
Roads/transportation	Business	Forest mgmt.	Anti-corruption	Aerospace	Educational institutions	Medical devices	Qualifications	Compliance mgmt.
ISO 39001	ISO 22301	KFCC	ISO 37001	AS 9100	ISO 21001	ISO 13485	ISO 17024	ISO 37301

Source: National Technology Standards White Paper (2020)

There are two different ways of designating certification bodies. The first method is the direct designation of certification bodies by the ministry concerned. The second method is the indirect designation by ministries via KOLAS and KAB, who accredit certification bodies. The following is the list of Acts and Statutes related to the use of KOLAS accredited certification bodies by each ministry.

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No.	Ministry	Laws
1	Ministry of Employment and Labor	Laws related to occupational safety and health (2)
2	Fair Trade Commission	Basic Consumer Act (1)
3	Ministry of Land, Infrastructure and Transport	Construction Technology Promotion Act (1)
4	Ministry of Land, Transport and Maritime Affairs	Act on the Construction and Management of Geospatial Information (1)
5	Meteorological Administration	Meteorological Observation Standardization Act, etc. (3)
6	Ministry of Strategy and Finance	Tobacco Business Act, etc. (2)
7	Ministry of Health and Welfare	Public Health Management Act, etc. (2)
8	Ministry of Trade, Industry and Energy	Quantification Act, etc. (17)
9	Fire Service Administration	Laws and ordinances related to quality control of firefighting supplies (3)
10	Ministry of Food and Drug Safety	Act on Test and Inspection in the Food and Drug Field (1)
11	Public Procurement Service	Administration Procurement Business Act (2)
12	Ministry of Oceans and Fisheries	Ship Safety Law, etc. (7)
13	Ministry of Public Administration and Security	Elevator Safety Management Act, etc. (6)
14	Ministry of Environment	Construction Waste Recycling Promotion Act (15)
Total	14 Ministries	Total 62 Laws

Source: National Technology Standards White Paper (2020)

8.4.2 Statistics on Key Certification Schemes

As of December 31, 2020, Korea has a total of 81 legally mandatory certification schemes and 106 legally optional certification schemes.

Classification	2012	2013	2014	2015	2016	2017	2018	2019	2020
Legally mandatory certification	68	69	73	73	69	71	70	80	81
Legally optional certification	115	132	136	130	97	104	104	107	106

Source: Korean Standards and Certifications¹²

¹² <https://standard.go.kr/KSCI/crtfcSystem/searchCrtfcSystemList.do?menuId=60373&topMenuId=536&upperMenuId=537>
(Accessed 6th Sep, 2021)

The following Table shows the current state of certification schemes under operation by individual ministries.

No	Ministry in charge	No. of registered certification	Type		No	Ministries in charge	No. of registered certification	Type	
			Mandatory	Voluntary				Mandatory	Voluntary
1	Ministry of Land, Infrastructure and Transport	32	17	15	13	Ministry of Food and Drug Safety	5	4	1
2	Ministry of Trade, Industry, and Energy	26	15	11	14	Ministry of Employment and Labor	4	2	2
3	Ministry of Oceans and Fisheries	21	10	11	15	Korea Meteorological Administration	3	2	1
4	Ministry of Agriculture, Food and Rural Affairs	18	2	16	16	Defense Acquisition Program Administration	4	2	2
5	Ministry of Environment	15	11	4	17	Ministry of SMEs and Startups	1	0	1
6	Ministry of Science and ICT	9	2	7	18	Fair Trade Commission	2	0	2
7	Korea Forest Service	11	2	9	19	Ministry of Gender Equality and Family	2	0	2
8	Ministry of the Interior and Safety	6	3	3	20	National Fire Agency	5	3	2
9	Ministry of Culture, Sports and Tourism	6	3	3	21	Korea Customs Service	1	0	1
10	Cultural Heritage Administration	1	0	1	22	Korean Intellectual Property Office	1	0	1
11	Ministry of Health and Welfare	9	2	7	23	Korea Coast Guard	2	1	1
12	Ministry of Education	2	0	2	24	Korea Communications Commission	1	0	1
Total							187	81	106

Source: Korean Standards and Certifications¹³

¹³ <https://standard.go.kr/KSCI/crtfcDeptPot/searchSubList.do?menuId=945&topMenuId=536&upperMenuId=549> (Accessed 6th Sep, 2021)

8.4.3 KS Certification Statistics

The KS Certification is used for the industrial standards in Korea. As of December 2020, Korea has a total of 20,282 KS standards. The following Table shows the yearly distribution of KS certification standards for the past three years.

Year	Statistics by year				No. of standards in possession (as of year-end)
	Established	Revised	Confirmed	Abolished	
2018	441	1,392	2,733	216	20,507
2019	364	1,356	2,549	137	20,734
2020	309	1,325	2,068	127	20,860

Source: National Technology Standards White Paper (2020)

The Table below shows the recent trend of KS certifications by year.

Classification year	No. of KS certified items	No. of KS certified plants (Korea/overseas)	No. of KS certifications (Korea/overseas)
2018	692	7,376 (6,937/ 439)	12,441 (11,827/ 614)
2019	689	7,585 (7,110/ 475)	13,055 (12,381/ 674)
2020	682	7,834 (7,331/ 503)	13,395 (12,696/ 699)

Source: National Technology Standards White Paper (2020)

8.4.4 Performance of Legal Optional Certifications in Korea

In 2020, a total of 50 new products were certified as a New Excellent Product (NEP). The following Table shows the NEP certification performance by year.

Classification Year	Carry-over from the previous year	Applied	Sub-total (A)	Certified (B)	Disqualified and rejected	Carry-over to the next year (C)	Certification rate (%)*
2012	31	171	202	68	105	29	39.3
2013	29	171	200	54	119	27	31.2
2014	27	166	193	28	121	44	18.8
2015	44	227	271	61	210	40	26.4
2016	40	252	292	47	202	43	18.8
2017	43	295	338	41	224	73	15.4
2018	73	374	447	39	188	220	17.2
2019	220	395	615	46	569	0	7.5
2020	0	362	362	50	311	1	13.9

Source: National Technology Standards White Paper (2020)

As of December 31, 2020, a total of 86 new technologies were certified as a New Excellent Technology (NET). The NET certification performance by year can be summarized as follows.

Classification Year	Applied (A)	Certified (B)	Disqualified & rejected	Under process (C)	Certification rate (%)
2012	372	157	215	0	42.2
2013	376	103	273	0	27.4
2014	479	111	368	0	23.1
2015	427	72	355	0	16.8
2016	511	77	434	0	15.1
2017	441	83	358	0	18.8
2018	465	54	411	0	11.6
2019	425	57	368	0	13.4
2020	499	86	413	0	17.2

Source: National Technology Standards White Paper (2020)

8.4.5 Korea Certification (KC) Mark

Currently, Korea has a total of 81 different kinds of legally mandatory certification schemes. Under the same category of ‘product safety’, such as different certification marks are being used by the ministries concerned, thereby causing inconveniences of obtaining overlapping

certifications. This has not only wasted time and resources but has undermined Korea's international credibility. Furthermore, a lack of cross-border mutual recognition of certifications has increased the need for re-certification by foreign counterparts, causing the outflow of national wealth. Under these conditions, 13 legally mandatory certification marks were merged into a single national certification mark called the Korea certification (KC) mark in November 2011.

In July 2009, the KC mark was first launched through a merger of 10 certification marks separately run by the then Ministry of Knowledge and Economy and the then

Ministry of Labor. Later 13 certification marks of 5 different ministries were also integrated into the KC mark. As of September 2017, the KC mark has been adopted by 23 legally mandatory certification schemes run by 8 different ministries.

9. I-U-R Collaboration Support Programs in Korea

9.1 Current State of Ghana

According to Mamudu and Hymore (2016), university-industry collaboration (hereinafter referred to as 'I-U collaboration') in Ghana has the following characteristics. Firstly, collaborations between industries and universities in Ghana are conducted mostly in informal modes through conferences, workshops, seminars, and training programs hosted by universities or through publications of journals and research reports for the public. Ghanaian universities are actively taking part in the training and transfer of human resources through internship and placement programs.

Secondly, the I-U collaboration in Ghana tends to emphasize the teaching side of the collaborative endeavor by placing its priorities on the curriculum development and student internships that are often preferred by the least developed countries. In addition to the teaching side, the research dimension of the I-U collaboration is also considered important with a focus on appropriate technologies to respond to local needs. Thus, the Ghanaian industries and universities also operate research consortia, a collaboration mode often preferred by the developed countries, while maintaining strategic partners to expand their research specialties. Though it is less common, the I-U collaboration sometimes focuses on strengthening universities' entrepreneurial capacity by providing business incubation services or entrepreneurship educational programs. These modes of cooperation are also those preferred by the least developed countries.

[Table 3-19] Priorities of U-I Collaborations for 10 Selected Universities

Nature of University	Priorities of UI collaborations	University #										Total
		1	2	3	4	5	6	7	8	9	10	
Teaching University	Focus on curricula development	√	√	√	√	√	√	√	√	√	√	10
	Student internships	√	√	√	√	√	√	√	√	√	√	10
	Private participation in graduate programmes	√		√								2
	Joint supervision of PhD. students											0
Research University	Building absorptive capacity to adopt and diffuse already existing technologies	√	√	√		√						4
	Focus on appropriate technologies to respond to local needs	√	√	√	√	√			√			6
	Research Consortia	√	√			√						3
	Strategic partnerships to conduct research frontiers			√	√							2
Entrepreneurial University	Business incubation services	√		√					√			3
	Entrepreneurship education								√	√		2
	Creation of spin-off companies	√							√			2
	Patent licensing											0

Source: Mamudu and Hymore (2016)

Thirdly, unlike a vast majority of the developed countries where the government or local organizations sponsor the I-U collaboration, the sponsoring role for the I-U collaborations in Ghana is being mostly played by developmental agencies or international partners. This implies a need to promote the government of Ghana's role in the I-U collaboration, such as developing the I-U collaboration related policies or providing funding support for the I-U collaborations.

Fourthly, the I-U collaborations in Ghana often show a low to medium intensity level. For example, the Ghanaian universities are frequently engaged in informal low-intensity interactions such as conferences, seminars, social groups, and scientific publications. Most of them use scientific publications as means to apply their scientific knowledge to industries. They also tend to be involved in medium-intensity interactions with industries through employee internships, industrial training, and invitation of industry experts as adjunct professors. In the meantime, high-intensity interactions such as research services, research partnerships, and shared infrastructure programs are less frequently made than the aforementioned collaborations.

[Table 3-20] Intensity of U-I Collaboration in 10 Selected Universities

Intensity of UI collaborations	University #										Total
	1	2	3	4	5	6	7	8	9	10	
Informal interactions (conferences, seminars, social groups)	√	√	√	√	√	√	√	√	√	√	10
Scientific publications (use of scientific knowledge within industry)	√	√	√	√	√	√	√	√	√	√	10
Commercialisation of IP (transfer of local patents and licenses)											0
Training & transfer of HR (employees, internships, industrial training, adjunct faculty from industry)	√	√	√	√	√	√	√	√	√	√	10
Academic entrepreneurship (dev't and commercialisation of technologies designed by academic inventors)	√							√	√		3
Shared Infrastructure (business incubators, technology parks, use of university labs & equipment by firms)	√		√		√			√			4
Research Services (Research-related activities commissioned to universities by industrial clients, contract research, consulting, quality control, etc.)	√	√	√		√			√			5
Research partnerships (inter-organisational arrangements for collaborative research, research consortia, joint projects)	√		√		√						3

Source: Mamudu and Hymore (2016)

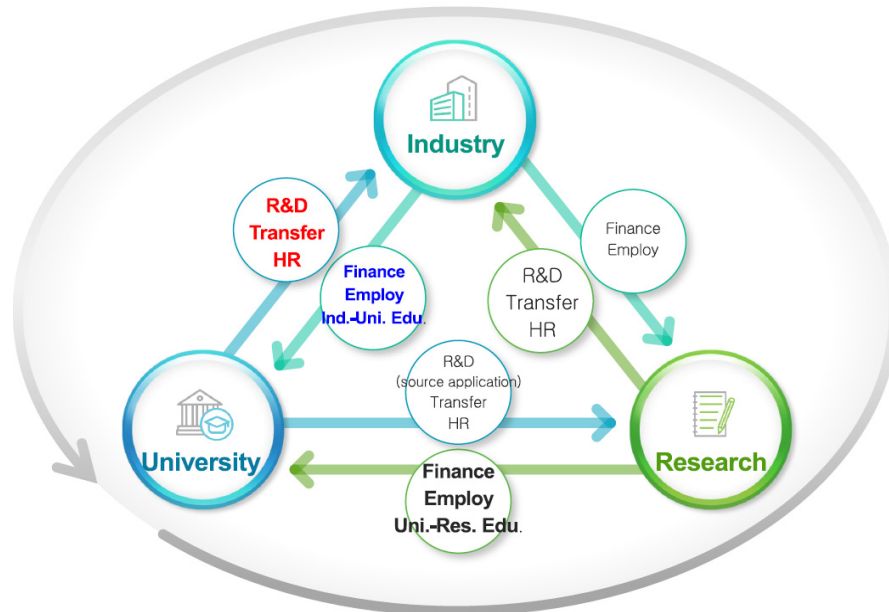
Fifthly, most of the Ghanaian universities have established collaborative networks with industries for the purposes of operating business incubators, developing curriculum, conducting contract research, maintaining research partnership, and providing consulting services. Most of these networks are medium to long term in nature.

Sixthly, most of the frontier research and entrepreneurial innovations at the Ghanaian universities occur in the SME or informal sector of the Ghanaian economy. This creates a unique opportunity for knowledge transfers between the workforce employed in these sectors and Ghanaian universities.

9.2 Evolutionary Path of Korea

The following Figure shows the relationship between the three key stakeholders of the industry-university-research institute collaboration (hereinafter referred to as 'I-U-R

collaboration') system. While universities and research institutes are providing R&D services and human resources for industries, industries are supporting universities and research institutes in terms of employment and funding. In the I-U-R collaboration system, universities and research institutes work together as collaborative research partners for knowledge creation and technology development. For example, universities supply high-caliber human resources to research institutes, which in turn, provide university faculties and students with expensive research infrastructure (i.e. facilities and equipment), funding, and employment opportunities. A balanced development of these three key innovation actors based on mutual trust ensures a nation's economic growth and its enhanced national competitiveness.



[Figure 3-15] I-U-R Collaboration System

9.2.1 Importance of I-U-R Collaboration

The full-fledged arrival of the Fourth Industrial Revolution is expected to accelerate digital transformation and usher in unlimited competition between countries. This new era now requires countries worldwide to build a system of a virtuous cycle. This system creates unique new knowledge and develops new technologies through I-U-R collaborations, which then lead to the development of creative talents. In the 1960s, Korea's GDP remained as

low as that of Ghana, but Korea has now emerged as the world's tenth largest economy. Analysts cite successful I-U-R collaborations as one of the key contributors to Korea's remarkable growth. In fact, I-U-R collaborations offer an array of benefits to corresponding collaboration partners as the following Table shows.

[Table 3-21] Impacts of I-U-R Collaboration on Key Actors

Impacts on universities	Impacts on industries	Socio-economic impacts
• Discover and conduct practical research • Secure quality resources including funding for conducting research • Build research-friendly environment • Provide practical education applying research outcomes • Expand employment opportunities for graduates	• Overcome limited research function • Address the shortage of expensive research equipment • Acquire practical knowledge and technologies • Secure new products and technologies • Hire high-caliber talents and experts	• Improve corporate and national competitiveness • Minimize uncertainty of the future society • Address issues to be caused by reduced workforce • Create jobs and improve youth unemployment rate • Complete the construction of an ecosystem for national development

Source: Kim, W., Namgung, M. and Choi, J. (2018)

The Korean government's I-U collaboration promotion policies and support programs have enabled universities to operate I-U collaborative training courses and conduct R&D that considers technology application, allowing universities to build and stabilize a I-U collaboration-friendly environment and culture. In the meantime, industries are now approaching universities in an open-minded manner to secure a stable supply of creative talents and original technologies that are immediately applicable to the field. This approach has so far produced some significant outcomes, but universities and industries need to work closely together to achieve effective I-U collaborations.

The I-U collaboration can be defined as a phenomenon of interactions and cooperation between stakeholders such as universities, industries, and research institutes in the process of technology development or human resources development. Here, the letter 'I' denotes the private sector in a broader meaning including private companies and non-profit

organizations (NPOs), whereas the letter 'U' refers to universities and research institutes which develop talents and create new knowledge.

The old concept of the I-U collaboration used to focus only on the bilateral cooperation between universities and industries, the two key stakeholders of I-U collaboration, so it was limited to universities' support for industries. However, a more recent trend has been to expand the concept of the I-U collaboration to cover both the formal interactions through technology transfers and joint R&D and informal interactions through university graduates' employment and academic seminars, as well as to emphasize the importance of these informal aspects of collaborations.

As the Figure below shows, the Act on Promotion of Industrial Education and Promotion of Industry-University-Research Cooperation defines the I-U-R collaboration as the following activities jointly carried out between industrial educational institutions, the central & local governments, research institutes, and industries including; i) human resources development according to industrial demands and future industrial development, ii) research, development and commercialization for the creation and diffusion of new knowledge and technology, iii) technology transfers and counseling to industries, and iv) joint use of tangible and intangible resources such as human resources, facilities, equipment and R&D information. This legal definition of I-U-R collaboration suggests that this concept be focused on both formal and informal multilateral activities including human resources development, research and development, and technology transfers.

Article 2 of 「Act on Promotion of Industrial Education and Promotion of Industry-University-Research Cooperation」

1 “Industrial education” means education provided by industrial educational institutions to acquire knowledge and skills necessary for engaging in industry or starting a business and to cultivate entrepreneurial spirit.

6 “industry-university-research cooperation” means the following activities carried out in cooperation between industrial education institutions, the State, local governments, research institutions, and industries

- a. HR according to industrial demand and future industrial development
- b. Research, development, and commercialization for the creation and diffusion of new knowledge and technology
- c. Technology transfer and industry advice to industries
- d. Joint use of tangible and intangible resources such as manpower, facilities and equipment, and R&D information, etc.

[Figure 3-16] Legal Definition of I-U-R Collaboration in Korea

9.2.2 Types of I-U-R Collaboration

The I-U-R collaboration can be classified into various types depending on the mode of exchange, sustainability, directional orientation, policy orientation, and collaborators.

Firstly, it can be classified into formal and informal collaborations according to the mode of exchange. The OECD report (2002) also adopted this classification, which underscores informal collaborations as an important factor that drives formal collaborations (i.e. tangible collaborative actions as a result of contracts).

Classification	Informal collaborations	Formal collaborations
Activities	• Informal advice/ consulting • Participation in industry related academic events • Sharing of research facilities • Students' learning opportunities through corporate internships • Publications and patent applications as outcomes of research collaboration with corporate researchers	• Contract-based consulting • Education/ training services for employees hired by industries • Joint research with industries according to contracts • More intensive participation in industries as their employees and executives • Startup by researchers

Secondly, the I-U-R collaboration can be divided into sustained and one-time collaborations. It can also be classified into regular vs. one-time temporary collaboration or sustainable vs. unsustainable collaboration.

Thirdly, the directional orientation divides I-U-R collaboration into bi-directional and single-directional collaborations. It is often assumed that I-U-R collaborations be conducted in a bi-directional manner, but single-directional collaborations also take place in cases of research outsourcing and industry counseling. These single-directional collaborations are mostly performed based on contracts. Especially, single-directional collaborations tend to be chosen when the capacity of one party of the collaboration is superior to that of the other party.

Fourthly, the I-U-R collaboration can also be classified into four types of i) R&D-focused, ii) education and training-focused, iii) technology transfer-focused, and iv) human resources exchange-focused depending on the I-U-R collaboration policies and objectives.

[Table 3-22] Types of I-U-R Collaboration Depending on I-U-R Collaboration Policies and Objectives

Collaboration types by objectives	Programs
Focus on R&D	Joint research (government R&D), research outsourcing, dispatched research, and invitational research
Focus on education and training	Certificate in engineering education, customized education, field training and internship of the enrolled students, re-education of employees hired by industries, and scholarship support by industries
Focus on technology transfer and production support	Technology transfer by universities and research institutes, solving bottleneck technologies at the production sites, joint use of research facilities, and operation of startup incubators
Focus on human resources exchange	Exchange and mobilization of researchers and holding joint seminars

Fifthly, the I-R-U collaboration can also be classified into different types depending on the key collaborators, key industrial bases, the status of collaborators in the society, and their research capacity. For example, I-R-U collaboration can be i) industry-led, ii) university-led, or iii) government-led, depending on the lead of the collaboration partners. Nevertheless, their activities may overlap to a certain extent in terms of the scope and level of collaboration. I-R-U collaborations are also divided into different types depending on which channels or platforms these key stakeholders use for their collaborative activities, such as i) research university-based, ii) regional economy-based, iii) startup incubator-based (affiliated with universities and public institutions), and iv) university-affiliated company-based.

[Table 3-23] Types of I-U Collaboration based on Collaboration Partners and Key Collaboration Activities

Classification	Types of I-U Collaboration
Collaboration partners	• University-led • Industry-led • Central and local government-led
Collaboration activities	• Research clusters based on regional economy • Basic research focused collaboration based on research universities • Collaboration through business incubators affiliated with universities or public institutions • Collaboration in support of spinoff companies • Collaboration through university-affiliated companies

The following Table shows the expected impacts of I-U-R collaboration on universities and industries by the mode of collaboration.

[Table 3-24] Expected Impacts on Universities and Industries by Mode of I-U-R Collaboration

Mode of collaboration	Key activities	Expected impacts on universities	Expected impacts on industries
Operation of educational courses	• Operate industry-customized academic departments and contract-based academic departments. • Outsource training services for current employees and provide industry-linked educational programs. • Operate short- and mid-term educational courses, i.e. the post-employment support program for college entrance and the match-up program formerly known as the Korean-style Nano Degree Program.	• Improve social conformity and employment rate through tailored demand-oriented education. • Expand resources by operating various educational courses considering trainees' life-cycle.	• Secure high-caliber talents. • Secure competitiveness and enhance business transformation potential through re-training of current employees.

Mode of collaboration	Key activities	Expected impacts on universities	Expected impacts on industries
Human resources exchange	• Operate field training • Dispatch researchers and faculties to industries. • Engage industrial experts in universities' education and research activities.	• Strengthen field capacity through field experience at industrial sites. • Strengthen field capacity through field training of knowledge and skills.	• Address bottleneck technologies and secure new technologies by capitalizing on high caliber talents. • Improve company's public image and secure chances to attract high-caliber talents earlier than rivals.
Joint research & research outsourcing	• Conduct joint research and research outsourcing. • Implement the Capstone Design Program as I-U collaboration.	• Achieve the goal of need-based industrial education through research and education on practical issues in the field.	• Address bottleneck technologies and secure new technologies.
Exchange of knowledge and information	• Exchange knowledge and information • Provide technology consulting • Perform technology transfer and commercialization	• Establish user-oriented educational courses reflecting industries' needs in the field	• Promote technology commercialization and strengthen corporate competitiveness by introducing advanced knowledge and problem-solving technologies
Exchange of physical resources	• Share infrastructure such as facilities, equipment and laboratories.	• Utilize equipment in the field owned by companies or research institutes.	• Utilize costly advanced resources possessed by universities.

Source: Kim, W., Namgung, M. and Choi, J. (2018)

9.2.3 Evolutionary Path of Korea's I-U-R Collaboration Policy

Korea's I-U-R collaboration has evolved through several periods as follows; i) the first period focused on vocational high schools and colleges, ii) second period led by the government-research institutes (GRIs), iii) third period centered on universities, iv) fourth period centered on industries on the demand side, and v) fifth period focused on establishing an ecosystem of the autonomous I-U-R collaboration.

[Table 3-25] Evolutionary Path of Korea's I-U-R Collaboration Policy

Period	Characteristics by period	Key activities
1960s – 1970s	Focus on industrial education by vocational high schools and colleges	• Pursued collaboration as a channel for developing high-caliber talents and absorbing imported technologies. • Established vocational schools and colleges to develop skilled technicians. • Provided education tailored to corporate needs through vocational high schools specialized in engineering and commerce.
1980s	Collaboration led by GRIs	• Established the Daeduck Research Complex to lead industrial technology development. • Promoted the boom of establishing corporate R&D centers. • Implemented R&D programs under the lead of GRIs.
1990s	Full-fledged collaboration by universities	• Strengthened universities' researcher development function in response to increased R&D investment by the private sector. • Strengthened universities' leading role in the venture boom for technology startups. • Universities emerged as a key stakeholder of I-R-U collaboration in response to increased government R&D investment.
2000s	Demand-based collaboration	• Promoted demand-based I-U-R collaboration reflecting the needs of industries. • Introduced 'I-U cooperation center system' to universities (2003). • Implemented 'New University for Regional Innovation (NURI) Project' (2004). • Introduced "I-U-R Technology Holding Company System" (2008).
2010s	Establishing an ecosystem for autonomous collaboration	• Implemented LINC (Leaders in Industry-University Cooperation) Support Program (2012 – Present). • Implemented the BRIDGE (Beyond Research and Innovation Development for Good Enterprises) Project for supporting the commercialization of creative assets owned by universities (2015 – Present). • Launched the National I-U-R Cooperation Council (2018). • Developed the 'Basic Plan for Industrial Education and I-U-R Collaboration' for establishing an ecosystem for creative and autonomous I-U-R collaboration (2018).

Source: Park, M. et al. (2013)

A classical model of the I-U-R collaboration focused on practical education was initiated in the 1960s with the establishment of vocational high schools specialized in engineering and commerce-related education. These vocational high schools educated their students with curriculums tailored to corporate needs, which enabled companies to capitalize on the trained workforce without shouldering much burden.

The establishment of GRIs in the 1970s and 1980s made it possible for this classical model to be transitioned into a research-focused modernized model which specifically promoted cooperation between industries and GRIs. Following the establishment of the Korea Institute of Science and Technology (KIST) in 1966, numerous GRIs were founded in the 1970s in the fields of science and technology. When the internal R&D capacity of Korean companies was still limited, GRIs made significant contributions to Korean companies' technology acquisition and learning from the developed countries.

The 1990s witnessed an explosive growth of demand for highly skilled workforce possessing technological expertise as a result of the advancement of Korean industries. Expanded government R&D investment and an increased number of corporate R&D centers also raised the demand for high-caliber S&T human resources capable of conducting R&D activities. This also highlighted the role of universities as institutions for higher education and promoted their research function.

With regard to the role of universities, they became a key stakeholder of R&D in the 2000s, going beyond the traditional boundary as educational institutions. In the 2000s, universities started to newly emerge as a key R&D collaboration partner as the government increased its investment in university-led R&D activities and implemented diversified R&D programs. Several important issues also surfaced as a result of increased investments in university-led R&D programs, such as the diffusion of R&D outcomes delivered by universities, technology transfers to industries, and I-U collaboration.

In the 2010s, the government provided full-fledged financial support for I-U-R collaboration. In 2012, the LINK Program was created by merging the three programs of the Talent Development for Leading Industries in the Metropolitan Areas, the Hub University for Industrial Collaboration (HUNIC) Program, and the Regional Hub Research Center Program.

In 2015, the BRIDGE Project was initiated for supporting the commercialization of creative assets owned by universities. In 2018, the National I-U-R Collaboration Council was established to function as a control tower in charge of the overall coordination of I-U-R collaboration policies. Due to the silo approach to the I-U-R collaboration policies by the related ministries without a shared vision across ministries, there rose a need to enhance policy effectiveness through inter-ministerial policy coordination and linkage. This finally led to the revision of the Act on Promotion of Industrial Education and Promotion of Industry-University-Research Cooperation on November 28, 2017, which laid the legal foundation for the establishment of the National I-U-R Collaboration Council under the Prime Minister's Office. The revised Act also provided a legal framework for the effective implementation of I-U-R collaboration policies at the pan-ministerial level through the council by facilitating policy planning and coordination for the promotion of industrial education and I-U-R cooperation. Along with these efforts, the Basic Plan for Industrial Education and I-U-R Collaboration (2019-2023) was mapped out to create an ecosystem for autonomous I-U-R collaboration.

As aforementioned, Korea's I-U-R collaboration has evolved through a top-down approach, which differed from the bottom-up approach of other developed countries such as the U.S. and European countries. In the U.S. and Europe, the government only provided legal support for I-U-R collaboration while universities and local stakeholders took the initiative for the collaboration projects, which were then disseminated throughout the country. Unlike their counterparts in the U.S. or Europe, the Korean government set a policy goal of promoting I-U-R collaboration and played a leading role in various projects by establishing related systems and infrastructure.

9.3 Key Policies and Support Programs for I-U-R Collaboration in Korea

9.3.1 Basic Plan for Industrial Education and I-U-R Collaboration (2019-2023)

Through the revision of the Act on Promotion of Industrial Education and Promotion of Industry-University-Research Cooperation on November 28, 2017, the Korean government

laid the legal foundation for the establishment of the National I-U-R Collaboration Council under the Prime Minister's Office, and provided a legal framework for the effective implementation of I-U-R collaboration policies at the pan-ministerial level by facilitating policy planning and coordination for the promotion of industrial education and I-U-R cooperation. At this time, the Korean government developed the Five-Year Basic Plan for Industrial Education and I-U-R Collaboration (2019-2023), presenting a basic direction for the I-U-R cooperation over a five-year period.

Based on the diagnosis that past I-U-R collaborations had been focused on quantitative growth and expansion, the basic plan presents a future direction for diversifying and upgrading the I-U-R collaboration system.

	Past direction		Future direction
Education and human resources development	Industrial changes not fully incorporated Quantitative growth of I-U-R collaboration in education	⇒	Develop talents in newly emerging industries . Incorporate users' (corporates') perspective in education and strengthen linkage between education and employment.
Technology transfer and commercialization	Quantitative growth of technology transfers	⇒	Improve efficiency by supporting the discovery and use of external potential technologies.
Promoting startups	- Entrepreneurship training separately conducted by period. Entrepreneurship training focused on theories . Faculties and researchers' entrepreneurial activities focused on research .	⇒	- Provide a life-cycle support package covering the entire educational cycle from elementary school to graduate school. Strengthen administrative & financial support that can lead to actual startups . Promote faculties and researchers' technology startup and provide them with life-cycle support.
I-U-R infrastructure	Focus on building basic infrastructure including the operation of I-U cooperation centers.	⇒	Upgrade infrastructure including the diversification of the I-U-R collaboration system and strengthening of I-U cooperation centers.

[Figure 3-17] Directional Orientation of Korea's I-U-R Collaboration

A vision of ‘promoting I-U-R collaboration as an engine for innovative growth’, was presented in the basic plan. This vision expressed the Korean government’s determination to drive innovative growth by promoting I-U-R collaboration through linkage and cooperation with ministries concerned that had long been missing due to the competitive relations between these key stakeholders.

The basic plan also set the goal of ‘building an ecosystem of creative and autonomous I-U-R collaboration’. In other words, the basic plan is aimed at establishing a new and innovative ecosystem, where the stakeholders of collaboration can grow together by transforming the current system, perception, culture, and underlying mechanism. The new ecosystem is expected to drive the innovative growth of Korean society through a new paradigm of interactions between the key stakeholders of collaboration.

In a bid to achieve the vision and goal, the basic plan presents the following four key strategies and twelve initiatives in four areas including i) human resources development, ii) technology transfers & commercialization, iii) startups, and iv) infrastructure building for I-U-R collaboration.

In four strategic areas, the following initiatives have been suggested.

■ **Strategy 1: Developing creative talents through diversification and enhancement of industrial education**

- (Developing talents for newly emerging industries) Strengthen talent development reflecting future demand based on the HR needs analysis of future industries conducted by the ministries concerned.
 - Improve policy effectiveness by reviewing and analyzing the relevant ministries’ HR development programs for newly emerging industries targeting undergraduate & graduate students and unemployed graduates, and identifying the improvement needs of these programs.
- (Strengthening vocational training) Strengthen employment-linked vocational education & training in response to the increased demand for continued post-graduation in-service

training with the arrival of the newly emerging industries.

※ Expanded vocational education & training with the introduction of the K-MOOC Program (Ministry of Education), National Voucher Program for Future Learning (Ministry of Labor and Employment), and the Contract Academic Departments for Early Employment (Ministry of Education).

- (Curriculum) Enhance the quality of industrial education by developing and operating curriculums focused on problem-solving through a systematic linkage with industries and providing field training.

■ **Strategy 2: Creating high added value through transfer and commercialization of potential technologies**

- (Promoting the introduction of external technologies) Promote technology transfer from universities and GRIs to industries and pursue follow-up technology development to foster industries' active use of these technologies.
 - Expand support for inter-university technology convergence centered on promising creative assets that align with the recent technological and industrial trends in the Fourth Industrial Revolution.
- (Promoting technology exchange) Identify technology demand, promote the intermediation between supply and demand, and promote data sharing and linkage between public technology databases possessed by the ministries concerned.

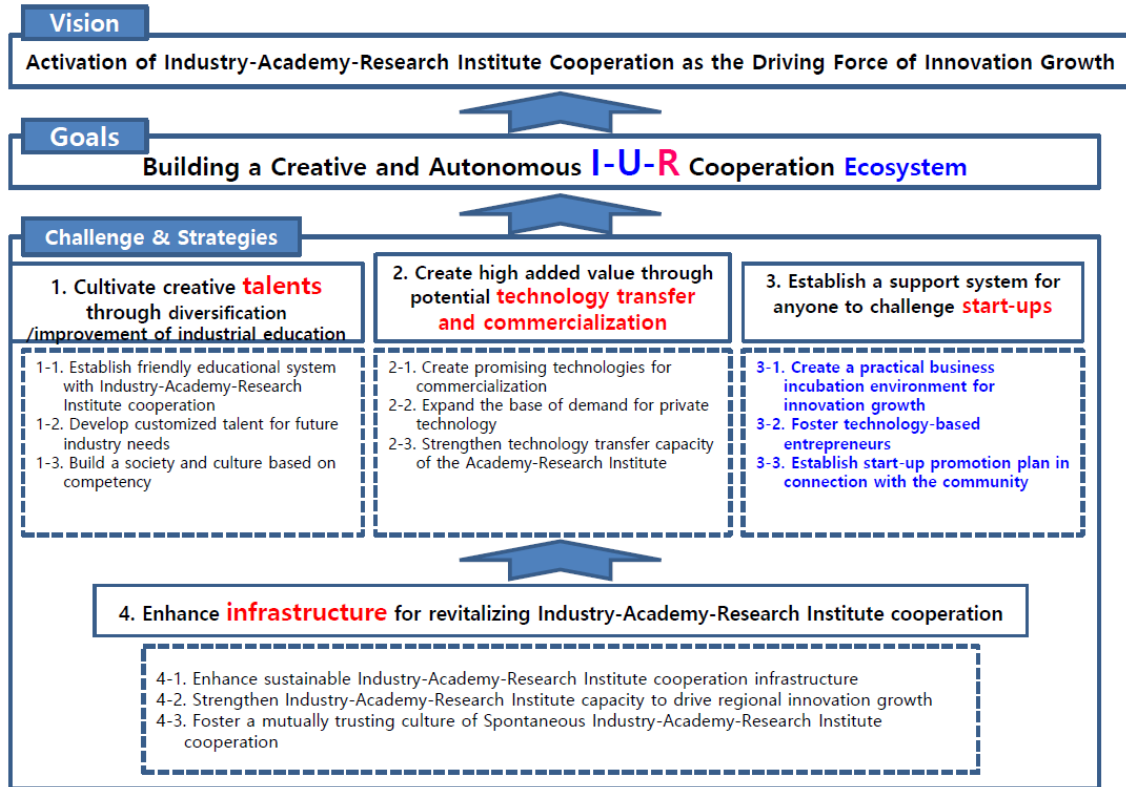
■ **Strategy 3: Establishing a startup support system to help anybody kick-start a business**

- (Creating environment conducive to startups) Strengthen life-cycle based entrepreneurial education, promote the shared use of facilities & equipment owned by universities, strengthen the linkages between startup support programs, and provide expanded follow-up support for early-stage startups.
 - Establish a startup support system considering the entrepreneurs' lifecycle and minimize their career void.

- Link universities' ideas with public technologies owned by tenant companies in the Special R&D Complex and support R&D efforts of startups including technology holding companies located on the university campus.
- (Building a startup friendly culture) Encourage the startup activities of faculties and researchers by disseminating faculty HR and academic systems favorable for startups and introducing the startup promotion program for researchers.

■ **Strategy 4: Upgrading infrastructure to promote the I-U-R collaboration**

- (Regional cooperation) Develop talents in the region based on universities and hire them through joint participation by industries, research institutes and local governments, address regional problems, and promote regional industries.
- (Creating environment for convergence) Use universities as a hub for I-U-R collaboration by setting up special space or facilities on university campuses for I-U convergence activities, sharing research equipment, and providing technology counseling for companies.
 - Establish the I-U-R Complex (Ministry of Education) on university campuses to make universities a hub for I-U-R collaboration that promotes regional innovation and corporate growth, and improve the effectiveness of the complex through cooperation with the I-U Convergence Complex (Ministry of Trade, Industry and Energy) and the R&D Special Zone (Ministry of Science and ICT).
 - Identify the institutional improvement needs of the university-affiliated I-U cooperation centers in charge of I-U-R collaboration and implement such improvements as strengthening the expertise of these centers and expanding the workforce hiring with industry experience.



[Figure 3-18] Korea's Basic Plan for I-U-R-Collaboration, 2019-2023

9.3.2 LINC Program (2012-2017) and LINC+ Program (2017- Present)

In 2012, the Ministry of Science and ICT of the Korean government launched the LINC (Leaders in Industry-University Cooperation) Program for the purpose of restructuring the university system to be I-U collaboration-friendly, swiftly adopting the I-U collaboration culture at universities, and promoting their specialties through I-U collaboration. The LINC Program made a total investment of about KRW 1.83 trillion (approximately USD 0.9 billion) to support 57 universities over five years from 2012 to 2017. The program's comprehensive support for I-U collaboration contributed significantly to adopting the I-U collaboration-friendly culture at Korean universities.

The key contents of this program are as follows. Firstly, universities selected for the program were asked to choose their specialty areas linked to regional industries according to their own capacity and the circumstances of their regional communities, and then to concentrate

their capacity on the development of technologies as well as human resources in their selected specialty areas.

Secondly, the selected universities were also required to establish the so-called 'Field Training Support Center', an organization dedicated to the support of their students' field training, and operate the center with full-time employees, which ultimately enhanced the quality of the job training programs run by the universities. The center plays an important role in linking students with industries and also provides orientation sessions of the field training programs for the I-U collaboration-dedicated professors and other professors from the academic departments in charge of I-U collaboration.

Thirdly, the selected universities had to strengthen their 'entrepreneurship education' for their students. The program asked the selected universities to build the 'creative education system' on their own initiatives to cultivate well-prepared entrepreneurs over the long term, going beyond the simple task of running field training programs for their students' passive learning. For this purpose, the selected universities were required to establish the 'Entrepreneurship Education Center' to promote the interest of faculties, students, and the public in the area of startups. This center plays an important role as a hub organization in promoting the startup boom at universities by offering startup-related courses and expanding its support for startup-related club activities.

Fourthly, for the purposes of making the university system friendly to I-U collaboration and strengthening I-U collaboration, the program required selected universities to introduce a support program called 'university enrollment after employment', allowing graduates of special vocational high schools to continue their college education even after their employment. Universities need to make both 'hardware' and 'software' improvement efforts to support this program of 'university enrollment after employment' as follows; i) establishing related academic departments and improving academic courses in terms of hardware improvement and ii) operating special college application process for current employees, expanding weekend and evening classes, recognizing industrial field experience as academic credits, and operating I-U collaborative bachelor's & master's integrated degree programs (5-year programs) in terms of software improvement.

Fifthly, the selected universities were also required not only to provide support for their students by establishing various student support centers and educational courses but also to strengthen their support for collaborating with industries. Universities dispatch professors on their sabbatical leave to industries and conduct joint research projects engaging their professors and students in collaboration with industries. This enables universities to effectively support industries' technology development in their bottleneck areas, provide management and marketing support customized to industrial needs, and develop talents tailored to industries' human resources needs. These efforts will ultimately promote the mutual understanding between universities and industries, help them strengthen their collaboration system, and even link students' employment opportunities with universities' support for industries.

In short, the LINC Program can be understood as a comprehensive plan for redefining the roles of universities as an educational institution well-positioned to promote I-U collaboration, and for adopting the I-U collaboration culture through strengthened linkage between universities and industries. The following Table presents a list of detailed activities to be considered by universities in their implementation of the program.

[Table 3-26] Detailed Activities of LINC Program (Example)

		Key activities (Examples)	
System	Restructure university system	Restructure university system to be I-U collaboration-friendly.	
		Mandatory	• Reflect the I-U collaboration performance into faculty performance assessment and promotion. • Increase the hires of 'I-U Collaboration Dedicated Professors'. • Pursue the specialization * restructuring of universities considering their capacities and regional circumstances.
		Specialty	• Promote education in convergence related fields. • Introduce the U-R collaboration-dedicated professor program. • Introduce the leave of absence program for professors participating in I-U collaboration. • Dispatch professors to industries during their sabbatical year. • Operate bachelor's & master's integrated degree programs.
	Strengthen I-U cooperation center's roles and position	Establish the I-U cooperation center as a hub of I-U collaboration in terms of collaborative resources and networks.	
		Mandatory	• Overall planning of I-U collaboration activities & employment and startup-related activities on campus. • Hire and operate experts in I-U collaboration (including patent lawyers).
		Specialty	• Technology Transfer Center • Support technology transfer and commercialization.
	Launch incubation centers and field training support centers	Operate a dedicated organization for supporting entrepreneurship education and field training.	
		Mandatory	• Establish the Entrepreneurship Education Center and the Field Training Support Center and secure full-time employees to run them.
	Expand infrastructure for I-U collaboration	Develop human resources for upgrading I-U collaboration and expand infrastructure for technology development.	
		Mandatory	• Link shared equipment on campus and build a cluster of shared equipment.

		Key activities (Examples)	
		Specialty	• Operate the I-U collaborative labs. • Build shared equipment for technology development. • Secure joint labs and experimental equipment. • Secure facilities for training industry employees.
Components	Restructuring and operation of educational courses	Restructure and operate educational courses according to the regional industrial needs.	
		Mandatory	• Operate educational courses by reflecting industrial needs of the specialty fields. • Operate the field training credit program and the Capstone Design Program.
		Specialty	• Operate I-R linked educational courses. • Operate the customized and contract-based academic departments. • Develop teaching methods for problem-solving. • Launch undergraduate research programs. • Support faculty training focused on field experience.
	HR development & support for employment and business startups	Support specialized employee and start-up education depending on the regional industries and the I-U collaboration types.	
		Mandatory	• Operate the I-U collaboration program to strengthen employment capacity. • Strengthen entrepreneurship education and support the operation of startup-related clubs.
		Specialty	• Conduct I-U joint dissertation supervision and joint projects • Provide education on patents/IPs and management/accounting • Support university enrollment after employment
Linkage	Strengthen corporate linkages	Strengthen support for industries to respond to regional and corporate needs	
		Mandatory	• Form and operate an I-U-R collaboration consultation body
		Specialty	• Support the development of frontier technologies. • Provide technological guidance and address bottleneck technologies. • Provide tailored business services including management consulting, design, and marketing.

Source: Park, S. (2012)

The following Table shows the remarkable outcomes of the LINC Program as of 2017 in comparison to 2011, the year before the program launch.

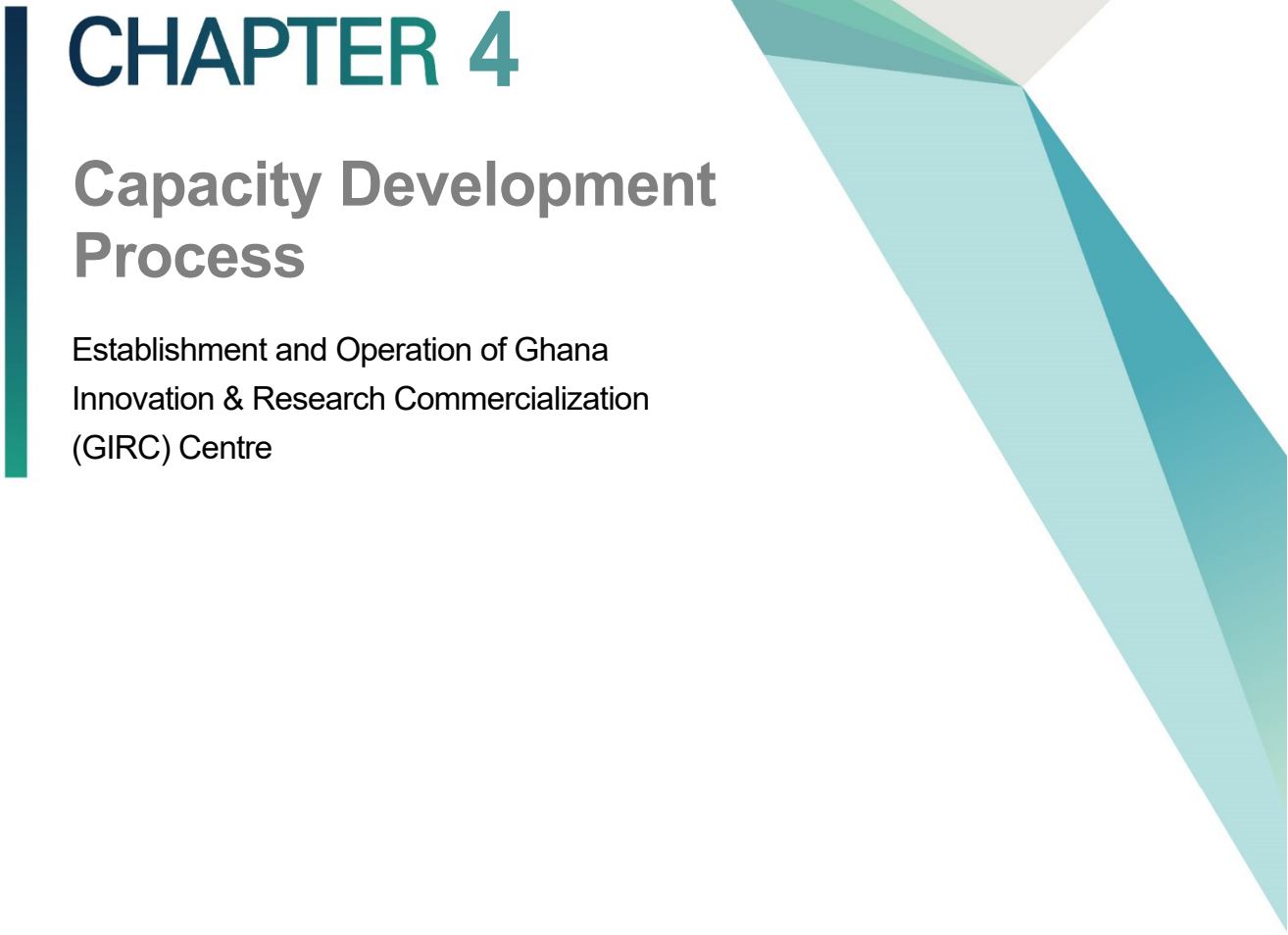
[Table 3-27] Key Outcomes of the LINC Program, 2012-2017

Key indicators	Before the LINC program (2011)	February 2017
Percentage of I-U collaboration in total research performance	70%	92.5%
No. of students who have completed field training	12,000	40,000
No. of students who have completed Capstone Design Program	42,000	10,4000
No. of educational courses tailored to industrial needs	670 courses	2,068 courses
No. of hours on startup education	41 hours	123.2 hours
Income from the operation of shared research equipment	KRW 23.5 billion	KRW 35.2 billion
No. of family-owned businesses	24,000	64,000
No. of technology transfers & income from technology transfer	603 cases/ KRW 10.4 billion	2,551 cases/ KRW 30.8 billion

In 2017, the LINC+ Program was initiated as a follow-up program of the LINC Program, and was especially designed to respond to the social needs. As a part of the LINC+ Program, academic departments tailored to social needs were newly introduced to universities, numerous focused projects were newly launched, and the scope & focus of collaborators were shifted from businesses or industries to local communities. During the 1st Phase (March 2017 – February 2019) of the program, a total budget of KRW 396.8 billion was invested in 55 universities in a bid to upgrade I-U collaboration. The portion of I-U collaboration performance in faculty performance assessment increased from the pre-program level of 119 percent to 175.5 percent in two years, and so all these improvements attest to a quantitative growth of I-U collaboration during the two-year period.

[Table 3-28] Overview of Phase 1 Performance of LINC+ Program by Key Performance Indicator: Focused on the Upgraded I-U Collaboration Type

Key Performance Indicators		Base (Feb. 2017)	2 nd Year (Mar. 2018 – Feb. 2019)
Portion of I-U collaboration performance in faculty performance assessment (%)		119.0	175.5
No. of faculties mainly engaged in I-U collaboration (points)		15.8	20.2
No. of full-time employees involved in I-U collaboration (persons)		31.9	42.3
Percentage of students who have completed field training (%)		27.1	32.9
Percentage of students who have completed the Capstone Design Program (%)		38.8	47.6
No. of companies using shared equipment and operational income	No. of companies using shared equipment	227.4	289.6
	Operational income from shared equipment (KRW million)	559	718
No. of I-U collaborative research and research fund per faculty	No. of I-U collaborative research (through local networks) per faculty	0.4	0.5
	Amount of I-U collaborative research (through local networks) fund per faculty (KRW million)	12.2	15.4
No. of technology transfers & related income per faculty	No. of technology transfers per faculty	0.2	0.3
	No. of technology transfers income per faculty (KRW million)	2.2	3.5
No. of innovations applied to local community		26.0	46.4

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CHAPTER 4

Capacity Development Process

Establishment and Operation of Ghana
Innovation & Research Commercialization
(GIRC) Centre

Chapter 4. Capacity Development Process

1. Kick-off Meeting and Field Interview

1.1 Background

In February 2021, the STEPI team was organized by consisting various experts both inside and outside of STEPI. In March 2021, STEPI and MESTI had an online meeting to discuss the specific directions and procedures of the project for the first year (2021). At the meeting, both parties shared in-depth opinions and suggestions regarding the subject of the first year, 'Analysis of TISPs in Ghana and Sharing Korean Experiences'.

Based on the meeting outcomes and discussion with Korean experts, the STEPI team prepared an interview questionnaire for nine sub-topics of TISPs and sent it to MESTI in early April in order to provide in-depth consulting and develop strategic plans. The selected sub-topics for the analysis and policy suggestion of the TISPs were (1) Tax&Finance Support Program (2) Human Resources Support Program (3) Technology Development & Information Support Program (4) Technology Transfer & Commercialization Support Program (5) Startup & Incubation Support Program (6) Intellectual Property Management Support Program (7) Purchase Support Program (8) Accreditation & Certification Support Program (9) I-R-U Collaboration Support Program.

On April 29th, Kick-off meeting and Field interview was held online due to COVID-19 pandemic. MESTI introduced the TIS/TISPs in Ghana and presented the current status of the GIRC Centre. The field interview was followed to grasp the current status of TISPs in Ghana along with actual and critical demands from the GIRC Centre.

1.2 Overview of the Kick-off Workshop and Field Interview

- **Date** : April 29, 2021 (Fri) 6:00 p.m. – 8:30 p.m. (Korean time)
- **Venue** : Video Conference (due to COVID-19 situation)
- **Purpose**
 - To establish a specific direction for the project through interviews with local experts from Ghana.
 - To review the TIS/ TISPs in Ghana and the current status of the GIRC Centre.
 - To identify the specific demand for the 2021 Ghana K-Innovation ODA project.

1.3 Attendees

■ STEPI Research Team (total of 11)

Names	Position/ organization	Roles/ area of expertise
Dr. Wangdong Kim	Senior Research Fellow/ Office of SDGs Innovation Research, STEPI	Project manager
Dr. Myung-Do Oh	Vice-President/ University of Seoul	Senior advisor
Dr. Mean Sun Noh	Head, Div. of Future Strategy Research/ Korea Small Business Institute (KOSBI)	Tax & finance support Human resources support
Dr. Young-eun Wee	Professor/ Korea Polytechnics	Human resources support
Mr. Sang Gon Kim	Director/ Gyeongbuk TP	Technology development & information support
Dr. Yang-jin Kim	Team manager/ Gyeongbuk TP	Tech transfer & commercialization support
Dr. Jong-Woo Ryu	Manager/ Gyeongbuk TP	Startup & incubation support
Dr. Young-Bum Kim	CEO/ Intellectual Property On	Intellectual property management support
Dr. Byoung Gun Kim	Research Fellow/ Korea Institute of Procurement	Purchase support (Public procurement support)
Dr. Sangho Choi	Director/ Jeonbuk Digital Convergence Center	Accreditation & certification support
Dr. Jong-in Choi	Vice President, Industry-University Cooperation Foundation/ Hanbat National University	I-U-R collaboration support
Ms. Seohee Yoon	Researcher/ STEPI	Coordinator

■ Ghana Project Team and Stakeholders (total of 20)

Names	Position/ organization	Roles/ area of expertise
Dr. Kwaku Afriyie	Minister/ MESTI	
Mr. Oliver Boachie	Special Advisor to Minister/ MESTI	Project manager
Ms. Cynthia Asare Bediako	Chief Director/ MESTI	
Mr. Cephas Adjei Mensah	Deputy Director/ MESTI	Public policy, STI/ biochemistry, public administration
Ms. Judith Awo Semabia	Program Officer/ MESTI	
Mr. Kwaku Ofori	Deputy Director General/ National Information Technology Agency	
Mr. Francis Boateng	Director/ CSIR	
Mr. Della Gordon	Representative/ CSIR-INSTI	
Dr. Wilhemina Quaye	Director/ CSIR-STEPRI	Tech transfer policy & IPR management
Ms. Sheila Frimpong	Acting Director of Communication and Commercialization/ Atomic Energy Commission	
Ms. Rhoda Appiah	Head of Communications & Public Education/ Food & Drugs Authority	
Dr. Yaw Ansu	Snr. Advisor to Minister of Finance/ Ministry of Finance	Tax & finance support
Pf. Ellis Owusu-Dabo	Pro-Vice-Chancellor/ KNUST	Technology commercialization
Dr. Gordon Adomdza	Associate Professor/ Ashesi University	Entrepreneurship & innovation
Pf. Nunoo	Vice Dean for Office of Research, Innovation, & Consultancy/ Univ. of Cape Coast	
Ms. Mammie Nortey	Research Development Officer/ ORID-UG (Univ. of Ghana)	Technology transfer
Mr. Bossman Akanluk Bastimi	Research Analyst/ Heritors People Limited	Technology & innovation
Mr. Derrydean Dadzi	CEO/ Heritors People Limited	Digital transformation, digital policy & services development
Mr. Brian Dzansi	Representative/ GIZ Ghana	
Mr. Jonas Amoako	Digital Transformation Centre, GIZ Ghana	

1.4 Main Activities and Agenda

■ Opening / Closing Ceremony

- Opening Remarks: Prof. Myung-Do Oh (University of Seoul)
- Congratulatory Remarks: H.E. Minister, Dr. Kwaku Afriyie (MESTI)
- Closing Remarks: Dr. Wangdong Kim (STEPI)

■ Kick-off Session

- Introduction to participating organizations, project details, and project procedure

■ Project Experts Interview

- (Procedure) Presentation from Ghana, Interview between Korean and Ghanaian experts for each sub-topic
- (Session composition) Introduction of Ghana TIS/ TISPs, presentation of the GIRC Centre status, Q&A interview

1.4.1 Schedule

Kick-off Meeting			
09:00-09:05(GH) 18:00-18:05(KR)	Kick-off session	Dr. Wangdong Kim	Project Manager Senior Research Fellow, Division of Global Innovation Strategy, STEPI
09:05-09:10(GH) 18:05-18:10(KR)	Opening remarks	Dr. Myung-Do Oh	Senior Advisor Vice President, University of Seoul
09:10-09:15(GH) 18:10-18:15(KR)	Congratulatory remarks	Dr. Kwaku Afriyie	Minister, MESTI
09:15-09:30(GH) 18:15-18:30(KR)	Project overview	Dr. Wangdong Kim	Senior Research Fellow, Division of Global Innovation Strategy, STEPI
Field Interview Session			
09:35-09:45(GH) 18:35-18:45(KR)	Introduction of project teams	Dr. Wangdong Kim Mr. Oliver Boachie	Project Manager, STEPI Special Advisor, MESTI

09:45-10:00(GH) 18:45-19:00(KR)	Introduction of TIS/TISPs in Ghana	Dr. Wilhelmina Quaye	Director, Science & Technology Policy Research Institute of CSIR (CSIR-STEPRI)
10:00-10:15(GH) 19:00-19:15(KR)	Current status of the GIRC Centre	Mr. Oliver Boachie	Special Advisor, MESTI
10:15-10:45(GH) 19:15-19:45(KR)	Q&A	Korean Experts Ghanaian Experts	
Closing Session			
10:45-11:00(GH) 19:45-20:00(KR)	Closing remarks	Dr. Wangdong Kim	Project Manager, STEPI

1.5 Interview Results

1.5.1 Ghana STI ecosystem diagnosis & the GIRC Centre

■ Incubation and Innovation Hub SWOT Analysis

- Opportunities: ICT education for higher education graduates, software development to improve employee professionalism, coding program for children, B2B matching support
- Threats: Development of STI's business aspects across tourism, health, mobile banking, clean transport and all sectors, cybersecurity and internet fraud, high power cost, influx of competing products from overseas, & pressure to lower prices of locally produced products & services, and delays in the certification process by regulatory bodies

■ Problem AnalysisConclusion

Problems with public universities and research institutes, such as limited funds, lack of infrastructure, and weak links with the private sector exist. For incubation and innovation hubs, the challenge is to secure funding for startups, digital infrastructure & industry capabilities, and market penetration issues.

It is recommended that the GIRC-Centre should be carefully established with a clear plan for a sustainable funding model and strong focus on research and innovation with strong research-private sector links and high commercial applicability.

■ Introduction of the GIRC Centre

Despite huge investments in research infrastructure and human resource development since Ghana's independence in 1957, weak linkages and limited private sector support for knowledge creation have led to international disparities in science, research, and innovation. The collaboration among various stakeholders to promote knowledge exchange and intentional investment to promote technology transfers and commercialization are necessary.

The GIRC Centre was established to plan the foundation for socio-economic development of Ghana through STI and establish partnerships between government, public research institutes, academics, & private sector. The major areas of interest include agricultural processing, environment, waste management & circular economy, manufacturing, ITC, renewable energy, healthcare & pharmaceuticals, mining & mineral processing, and oil & gas sectors. Diverse projects have been implemented for the foundation of the GIRC Centre such as the German-Israel Triangular Cooperation Program, the Vice President's Office-Digital Conversion Project, Mastercard Foundation Project, and World Bank Project.

For the funding, it is planned to utilize the Ghana Research and Innovation Fund (GRIF), which is a plan to secure an innovation ecosystem activity support fund by using a fund allocated more than 1 percent of GDP. Based on GRIF, funding sources provided by other development partners will also be used, but the role of private companies in the ecosystem would be more important.

As of now, only the renovation of the current office building has been completed and a small team was formed to perform administrative functions, led by Dr. Gorden from Asheshi University. Based on the K-Innovation Partnership Program, MESTI is planning to look at various activities to be carried out with consultants, but there aren't any technical activities as of yet.

■ Strategic Technology Center (STC)

STC will be established for each strategic and technical field in Ghana as an affiliated organization of the GIRC Centre, providing equipment for training and technology development equipment, consulting department, prototyping, pilot, limited production and market testing. The government will provide infrastructure and basic funds, such as a place, and is searching for a private cooperative partner for STC's own fund-creation activities. The Manufacturing Technology Center in Innovate U.K. is a model similar to the case of a South African public university.

The Foundry and Machine Tool Complex Center is a center for designing and producing basic tools for industries such as agriculture and automobiles, scheduled to be established at the site of the Ghana Atomic Energy Commission.

1.5.2 Q&A between Korean and Ghanaian Experts

■ Technology Demand for Development in Ghana

Agriculture-oriented manufacturing is the priority judging that Ghana has a competitive advantage in the agricultural sector, where 60~70 percent of Ghana's population is engaged, so we intend to focus on agricultural initiatives as a field to start the industrialization agenda. However, since agriculture cannot be the sole solution for the economy, manufacturing must be an essential part.

■ Industry-University-Research Collaboration and Activities

Most investors have a technology transfer center, including the Oragne Enterprise Center at the University of Ghana and Technology Development and Transfer Center to bridge the research-industry gap in the CSIR. In the future, Ghana will focus on technology development and try to bridge the gap between industry and research.

Entrepreneurship and innovation-related departments exist within the University of Ghana, such as the Office of Grants and Research and the Enterprise Development Department within the Kwame Nkrumah University of Science and Technology (KNUST).

The University of Ghana prioritizes external industry-university partnerships based on a strategic plan to close the gap. The Institute of Applied Science and Technology at Ghana University produced more than 200 end-users over the past 5 years, and delivered knowledge, technology, and innovation to various stakeholders. Many studies at Ghanaian universities are collecting information based on the challenges faced by the industry through a demand-driven approach at the proposal stage, and protecting intellectual property rights.

■ Human Resources Training Status

STEM education reinforcement goal: Ghana's education system focuses on non-STEM fields, and about 70 percent of graduates are from the humanities other than science and engineering and mathematics. STC will be utilized so that engineering college graduates can have expertise in the field through onsite training.

Curriculum and certification: In accreditation bodies such as the Ghana Standards Authority, higher education committees are established to review curriculum standards. When the GIRC Centre is operated, we will cooperate with them to define the certification system based on national needs and reflect the opinions of the field.

■ Demand for Public Procurement

Although public procurement is still a new concept in Ghana, we would like to include it in the development of the master plan as follows: Utilizing public procurement system to certify whether products in new innovative fields (i.e. plastic waste combined construction materials) are eco-friendly; seeking ways to generate ideas and commercialize products through public procurement with the government or stakeholders in the ecosystem and innovative entrepreneurs.

1.6 Photos

Group Photo



Interview



2. STI Capacity Development Workshop

2.1 Background

As the first stage of the project, the Korean experts specified the current status of 9 sub-topics of Ghana's TISPs through the Kick-off Workshop and Field interview with the relevant Ghanaian experts. They also answered the questionnaire for each topic which is also reflected in Chapter 2 of this report.

In order to prepare the establishment of the GIRC Centre and enhance the STI capacity of Ghana at an institutional and national level, STEPI analyzed the need for capacity development in the nine fields of TISPs. Based on the analysis of Ghana's current status, the Korean team shared the related experience of Korea and came up with policy recommendations for Ghana including the roadmap for each topic.

2.2 Overview of the Capacity Development Workshop

- **Title: 2021 K-Innovation ODA Program (Ghana) Capacity Development Workshop Program**
- **Date: September 14 (Tue.) – 16 (Thur.), 2021**
- **Venue: Online Conference (Zoom)**
- **Procedure: 3 sessions for 3 days**

(Day 1) In order to identify and analyze the status of TISPs on the Ghanaian side, Ghana experts were invited for presentations and a Q&A session based on 9 topics.

(Day 2 & 3) Policy proposals based on the experience of TISPs in Korea and the current situation in Ghana were presented and a Q&A session was conducted based on the following 9 themes △ Tax and Financial Support △ Human Resources Support △ Public Procurement Support △ Certification & Accrediation Support △ I-U-R Collaboration Support

△Technology Development & Information-Sharing Support △Technology Transfer & Commercialization Support △IPR Management Support

■ STEPI Research Team (total of 11)

Name	Position/ organization	Roles/ area of expertise
Dr. Wangdong Kim	Senior Research Fellow/ Office of SDGs Innovation Research, STEPI	Project manager
Dr. Myung-Do Oh	Vice-President/ University of Seoul	Senior advisor
Dr. Mean Sun Noh	Head, Div. of Future Strategy Research/ Korea Small Business Institute (KOSBI)	Tax & finance support
Dr. Young-eun Wee	Professor/Korea Polytechnics	Human resources support
Mr. Sang Gon Kim	Director/ Gyeongbuk TP	Technology development & information support
Dr. Yang-jin Kim	Team manager/ Gyeongbuk TP	Tech transfer & commercialization support
Dr. Jong-Woo Ryu	Manager/ Gyeongbuk TP	Startup & incubation support
Dr. Young-Bum Kim	CEO/Intellectual Property On	Intellectual property management support
Dr. Byoung Gun Kim	Research Fellow/ Korea Institute of Procurement	Purchase support (Public procurement support)
Dr. Sangho Choi	Director/ Jeonbuk Digital Convergence Center	Accreditation & certification support
Dr. Jong-in Choi	Vice President, Industry-University Cooperation Foundation/ Hanbat National University	I-U-R collaboration support
Ms. Seohee Yoon	Researcher/ STEPI	Coordinator,

■ Ghana Project Team & Stakeholders (total of 16)

Name	Position/ organization	Roles/ area of expertise
Dr. Kwaku Afriyie	Minister/ MESTI	-
Mr. Oliver Boachie	Special Advisor to Minister/ MESTI	Project manager
Mr. Kwaku Ofori	Deputy Director General/ National Information Technology Agency	-
Mr. Osei Oteng-Asante	Office of Coordinating Director/ Ministry of Finance	Tax & finance support
Mr. Franklin Karikari	Director of Business Support / National Entrepreneurship & Innovation Programme (NEIP)	Startup & incubation support
Dr. Boateng Agyenim	Director/ CSIR-IIR	Technology development & information support
Dr. Wilhemina Quaye	Director/ CSIR-STEPRI	IPR management
Ms. Delese Mama Darko	Chief Executive Officer/ Food & Drugs Authority	Accreditation & certification support
Mr. Douglass Boaten	Board Chair/ Ghana Public Procurement Authority	Purchase support
Pf. Ellis Owusu-Dabo	Pro-Vice-Chancellor/ KNUST	I-U-R linkage support
Pf. Angela Owusu-Ansah	Ashesi University	Human resources support
Pf. George Obeng-Adjei	Director of Research/ ORID-UG (Univ. of Ghana)	Technology transfer & commercialization support
Ms. Diana Adobea Owusu Antwi	Research Development Officer/ ORID-UG (Univ. of Ghana)	
Mr. Rudolph Djirackor	-	-
Mr. Derrydean Dadzi	CEO/ Heritors People Limited	Digital transformation, digital policy & services development
Mr. Kafui Leslie Adoboe	Heritors People Limited	-

2.3 Schedule

■ [Day 1] September 14 (Tue)

Opening Ceremony			
09:00-09:05(GH) 18:00-18:05(KR)	Kick-off session	Dr. Wangdong Kim	Project Manager Senior Research Fellow, Division of Global Innovation Strategy, STEPI
09:05-09:10(GH) 18:05-18:10(KR)	Opening remarks	Dr. Myung-Do Oh	Senior Advisor Vice President, University of Seoul
09:10-09:15(GH) 18:10-18:15(KR)	Congratulatory remarks	Dr. Kwaku Afriyie	Minister, MESTI
Ghana Presentation Session			
09:15-09:35(GH) 18:15-18:35(KR)	Presentation of TISPs in Ghana - Tax & Finance Support (10mins) - Q&A (10mins)	Dr. Osei Oteng-Asante	Office of the Coordinating Director Ministry of Finance
09:35-09:55(GH) 18:35-18:55(KR)	Presentation of TISPs in Ghana - Human Resources Support (10mins) - Q&A (10mins)	Dr. Angela Owusu-Ansah	Professor & Provost Ashesi University
09:55-10:15(GH) 18:55-19:15(KR)	Presentation of TISPs in Ghana - Purchase Support (10mins) - Q&A (10mins)	Dr. Douglass Boateng	Board Chair Public Procurement Authority
10:15-10:35(GH) 19:15-19:35(KR)	Presentation of TISPs in Ghana - Accreditation & Certification Support (10mins) - Q&A (10mins)	Mrs. Delese Mama Darko	Chief Executive Officer Food & Drugs Board (FDA)

2021 K-Innovation ODA Program with Ghana

10:35-10:55(GH) 19:35-19:55(KR)	Presentation of TISPs in Ghana - Technology Development & Information Support (10mins) - Q&A (10mins)	Dr. Boateng Agyenim Director Industrial Research Institute Council for Scientific & Industrial Research (CSIR-IIR)
10:55-11:05(GH) 19:55-20:05(KR)	Break Time (10 mins)	
11:05-11:25(GH) 20:05-20:25(KR)	Presentation of TISPs in Ghana - Technology Transfer & Commercialization Support (10mins) - Q&A (10mins)	Prof. George Obeng-Adjei Director of Research Office of Research, Innovation & Development (ORID) University of Ghana
11:25-11:45(GH) 20:25-20:45(KR)	Presentation of TISPs in Ghana - Startup & Incubation Support (10mins) - Q&A (10mins)	Mr. Franklin Karikari Director National Entrepreneurship & Innovation Programme (NEIP)
11:45-12:05 (GH) 20:45-21:05 (KR)	Presentation of TISPs in Ghana - Intellectual Property Management Support (10mins) - Q&A (10mins)	Dr. (Mrs.) Wilhemina Quaye Director Science & Technology Policy Institute (CSIR-STEPRI)
12:25-12:45(GH) 21:25-21:45(KR)	Presentation of TISPs in Ghana - I-U-R Linkage Support (10mins) - Q&A (10mins)	Prof. Ellis Owusu-Dabo Pro Vice Chancellor Kwame Nkrumah University of Science & Technology (KNUST)
12:45-12:50(GH) 21:45-21:50(KR)	Wrap-up	Mr. Oliver Boachie Special Advisor to Minister, MESTI

■ [Day 2] September 15 (Wed)

Korea Presentation Session		
09:00-09:05(GH) 18:00-18:05(KR)	Introduction Dr. Wangdong Kim	Project Manager Senior Research Fellow, Division of Global Innovation Strategy, STEPI
09:05-09:45(GH) 18:05-18:45(KR)	Presentation of TISPs in Korea - Tax & Finance Support Dr. Mean Sun Noh (30mins) - Q&A (10mins)	Head, Div. of Future Strategy Research, Korea Small Business Institute(KOSBI)
09:45-10:25(GH) 18:45-19:25(KR)	Presentation of TISPs in Korea - Human Resources Support Pf. Youngeun Wee (30mins) - Q&A (10mins)	Professor, Korea Polytechnics
10:25-11:05(GH) 19:25-20:05(KR)	Presentation of TISPs in Korea - Purchase Support Dr. Byoung Gun Kim (30mins) - Q&A (10mins)	Research Fellow, Korea Institute of Procurement
11:05-11:15(GH) 20:05-20:15(KR)	Break Time (10 mins)	
11:15-11:55(GH) 20:15-20:55(KR)	Presentation of TISPs in Korea - I-U-R Linkage Support Dr. Jong-in Choi (30mins) - Q&A (10mins)	Vice President, Industry-University Cooperation Foundation, Hanbat National University
11:55-12:35(GH) 20:55-21:35(KR)	Presentation of TISPs in Korea - Accreditation & Certification Support Dr. Sangho Choi (30mins) - Q&A (10mins)	Director, Jeonbuk Digital Convergence Center
12:35-12:40(GH) 21:35-21:40(KR)	Wrap-up Mr. Oliver Boachie	Special Advisor to Minister, MESTI

■ [Day 3] September 16 (Thurs)

Korea Presentation Session		
09:00-09:05(GH) 18:00-18:05(KR)	Introduction Dr. Wangdong Kim	Project Manager Senior Research Fellow, Division of Global Innovation Strategy, STEPI
09:05-09:45(GH) 18:05-18:45(KR)	Presentation of TISPs in Korea - Tax & Finance Support Dr. Mean Sun Noh (30mins) - Q&A (10mins)	Head, Div. of Future Strategy Research, Korea Small Business Institute(KOSBI)
09:45-10:25(GH) 18:45-19:25(KR)	Presentation of TISPs in Korea - Human Resources Support Pf. Youngeun Wee (30mins) - Q&A (10mins)	Professor, Korea Polytechnics
10:25-11:05(GH) 19:25-20:05(KR)	Presentation of TISPs in Korea - Purchase Support Dr. Byoung Gun Kim (30mins) - Q&A (10mins)	Research Fellow, Korea Institute of Procurement
11:05-11:15(GH) 20:05-20:15(KR)	Break Time (10 mins)	
11:15-11:55(GH) 20:15-20:55(KR)	Presentation of TISPs in Korea - I-U-R Linkage Support Dr. Jong-in Choi (30mins) - Q&A (10mins)	Vice President, Industry-University Cooperation Foundation, Hanbat National University
11:55-12:35(GH) 20:55-21:35(KR)	Presentation of TISPs in Korea - Accreditation & Certification Support Dr. Sangho Choi (30mins) - Q&A (10mins)	Director, Jeonbuk Digital Convergence Center
12:35-12:40(GH) 21:35-21:40(KR)	Wrap-up Mr. Oliver Boachie	Special Advisor to Minister, MESTI

2.4 Minutes

2.4.1 Current Status of Ghana TISPs

2.4.2 Sharing Korea's TISPs status and experience

■ Tax and Financial Aid Programs

- Introduction to the transition process of tax and financial support in Korea.
- Introduction of tax support programs such as R&D reserve deduction, R&D facility investment tax credit, technology transfer and technology rental tax benefit, technology acquisition tax benefit.
- Introduction of financial support programs such as Small and Medium Business Bank, SME loan mandatory ratio system, Technology Guarantee Fund, and financial intermediary support loan system, etc.

■ Human Resources Support Program

- Introduction to the transition process of Korea's human resources support policy.
- Introduction of policies for nurturing skilled technicians such as specialized high schools, meister high schools, and polytechnic colleges.
- Introduction of policies for nurturing experts in science and engineering fields such as establishment of specialized universities in science and technology fields, and other non-profit organizations and organizations that foster professional technical human resources.
- Introduction of policies to secure and support domestic scientists and engineers, such as the Ministry of SMEs and Startups' wage subsidy and the research human resources support program for experienced workers.
- Introduction of foreign science and technology experts invitation policies, such as the Brain Pool program and foreign professional recruitment program.

■ **Technology Development & Information Support Program**

- Introduction of Korea's technology development and information support program transition process.
- Introduction of private-led R&D projects such as public-led R&D project description evaluation, regional specialized industry nurturing PLUS project, and world class 300 project.
- Introduction of information-sharing infrastructure such as National Science and Technology Knowledge Information Service (NTIS).

■ **Technology Transfer & Commercialization Support Program**

- Introduction of technology transfer and commercialization policies in Korea, such as the Technology Transfer Act, Korea Institute of Industrial Technology Promotion, Korea University Technology Transfer Association, and National Technology Bank.
- Introduce the policies of the Ministry of Trade, Industry and Energy, such as activation of the technological innovation-type M&A system and expansion of the B&D business model.
- Introduction of policies of the Ministry of Science and ICT such as provision of written technology purchase information between consumers and suppliers through activation of technology evaluation, operation of a technology holding company, and establishment of a research institute and fostering system
- Introduction of the policies of the Ministry of SMEs and Startups, such as the technology startup scout system in universities, etc.

■ **Startup Incubation Support Program**

- Introduction of the transition process of support for startup incubation such as startup net, startup support portal, plan to take another leap forward for venture business, and plan to improve efficiency of startup incubation businesses.
- Introduction of Korean business incubation policies such as the preliminary initial-jump startup package, business incubation manager, business incubation center, etc.

■ Intellectual Property Management Program

- Introduction of Korea's intellectual property management transition process such as Chamber of Commerce and Industry, Invention Promotion Project, Patent Information Service, Regional Intellectual Property Center, etc.
- Introduction of policies to establish a virtuous cycle system of technological innovation through the creation of early intellectual property rights such as the establishment of a patent department in public institutions and invention interview system.
- Introduction of policies to provide information to strengthen patent creation and research prior art through the Patent Information Service.
- Introduction to the establishment of an intellectual property specialized institution for the utilization and advancement of public technology.
- Introduction of technology transaction market and ecosystem creation policy through the Technology Transfer Act and the commercialization of public technology, etc.

■ Purchasing & Public Procurement Support Programs

- Comparison of Ghana's distributed procurement and Korea's centralized joint procurement system.
- Introduction of Korea's public procurement system and policy utilization features such as purchase contract priority, certification and designation, and priority purchase system for technology developed products.
- Introduction of changes in public procurement policies such as efficient use of foreign aid materials, intensive support for SMEs, and mandatory purchase of eco-friendly products, etc.

■ Accreditation and Certification Support Program

- Introduction to the certification transition process in Korea.
- Introduction of the current status of Korean certification system laws, such as the Basic Law on National Standards and Conformity Assessment Law.
- Current status of conformity assessment systems and accreditation organizations, such as the testing institute accreditation system (KOLAS), the product certification body

accreditation system (KAS), and the quality management and environmental management system field accreditation system (KAB).

- Introduction of the national integrated certification mark (KC mark) and KS (Korea Standard) certification system for conformity with industry standards.
- Introduction of new technology and new product certification system in Korea.
- Introduction of certification systems for companies, such as the Innobiz certification system and the venture business verification system, etc.

■ Industry-University-Research Cooperation Support Program

- Introduction of Korea's industry-academia-research cooperation transition course such as Daedeok Science Park (currently Innopolis) and LINC.
- Introduction of the definition and contents of industry-university-research cooperation through the Act on Promotion of Industrial Education and Promotion of Industry-University-Research Cooperation.
- Introduction of the policy of the National Industry-University-Research Cooperation Committee and 'Five-Year (2019-2023) Basic Plan for Industrial Education and Industry-University-Research Cooperation'.
- Introduction of industry-university-research cooperation budgets of the Ministry of Education and Ministry of Science and ICT such as LINC+ projects, etc.

2.5 Photos

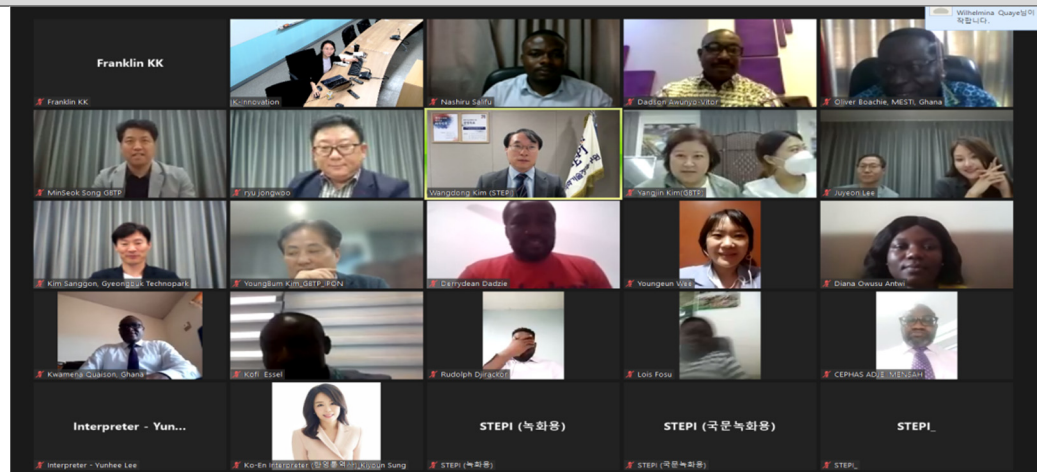
Group Photo (Day 1)



Group Photo (Day 2)



Group Photo (Day 3)



3. The Dissemination Work shop

3.1 Overview of the Dissemination Workshop

- **Title:** 2021 Ghana K-Innovation ODA Program Dissemination Workshop
- **Date:** December 16 (Thur.), 2021
- **Venue:** Seoul Dragon City Hotel (Korea), Fiesta Royale Hotel (Ghana)

3.1.1 Attendees

■ STEPI Research Team

Names	Positions/ organizations	Roles/ area of expertise
Dr. Wangdong Kim	Senior Research Fellow/ Office of SDGs Innovation Research, STEPI	Project manager
Dr. Myung-Do Oh	Vice-President/ University of Seoul	Senior advisor
Dr. Mean Sun Noh	Head, Div. of Future Strategy Research/ Korea Small Business Institute (KOSBI)	Tax & finance support
Dr. Young-eun Wee	Professor/ Korea Polytechnics	Human resources support
Mr. Sang Gon Kim	Director/ Gyeongbuk TP	Technology development & information support
Dr. Yang-jin Kim	Team manager/ Gyeongbuk TP	Tech transfer & commercialization support
Dr. Jong-Woo Ryu	Manager/ Gyeongbuk TP	Startup & incubation support
Dr. Young-Bum Kim	CEO/ Intellectual Property On	Intellectual property management support
Dr. Byoung Gun Kim	Research Fellow/ Korea Institute of Procurement	Purchase support (Public procurement support)
Dr. Sangho Choi	Director/ Jeonbuk Digital Convergence Center	Accreditation & certification support
Dr. Jong-in Choi	Vice President, Industry-University Cooperation Foundation/ Hanbat National University	I-U-R collaboration support
Ms. Seohee Yoon	Researcher/ STEPI	Coordinator

■ Ghana

Names	Positions	Organizations
Cynthia Asare Bediako	Chief Director	MESTI
Mr. Oliver Boachie	Senior Technical Advisor to the Minister	MESTI
Kwamena Quaison		MESTI
Cephas Adjei-Mensah		MESTI
Nashiru Salifu		MESTI
Prof. Appiah-Adu	Head of Delivery Unit, Office of the Vice President	Office of Vice President
Mr. Augustine Blay	Secretary to the Vice President	Office of Vice President
Richard Okyere-Fosu	Director General	NITA
Dr. Osei Oteng-Asante		Ministry of Finance
Mrs. Delese Mimi Darko	CEO	FDA
Dr. Douglass Boateng	Board Chair	Public Procurement Authority
Prof. Angela Owusu-Ansah	Provost	Ashesi University
Prof. George Obeng-Adjei.		University of Ghana
Ms. Mammie Hutchful Nortey		University of Ghana
Mr. Franklin Karikari		National Entrepreneurship and Innovation Programme (NEIP)
Dr. (Mrs.) Wilhemina Quaye.		CSIR-STEPRI
Prof. Ellis Owusu-Dabo		KNUST
Yofi Grant	CEO	GIPC
Prof. Fred McBagonluri	President	Academic City College
Prof. Jerry John Kponyo		KNUST
Prof. Emmanuel Akowuah		KNUST
Prof. Sam Bonsu	Rector	GIMPA

2021 K-Innovation ODA Program with Ghana

Names	Positions	Organizations
Johanna Hartung	Component Head	GIZ
Magdalene Apenteng	Chief Director	Ministry of Communications and Digitalisation
Augustine Owusu/ Richard Sandall		Department of International Trade, UK
Gloria Amoo (or) Ambassador of Israel		Mashav
		Ministry of Education
		Ministry of Agriculture
David Ofori		Accra Digital Centre
Alexander Dodoo	CEO	Ghana Standards Authority
Jemima Oware	Registrar General	Registrar General's Department
Herbert Krapa	Deputy Minister	Ministry of Trade and Industry
		Ministry of Health
Mustapha Ussif	Minister	Ministry of Youth and Sports
Sam Peprah	SC Ventures, Ghana	Standard Chartered Bank
Paulina Adjei		IDIA
Dr. Agyenim Boateng		CSIR-IIR
Humphrey Ayim-Darke		Association of Ghana Industries
Yaw Adu-Gyamfi		Ghana Hubs Networks
Dr. Collins Yeboah-Afari	Director General	AITI-KACE
Dr. Ken Ashigbey	CEO	Ghana Chamber of Telecommunications
Augustina Odame	CEO	Ghana Chamber of Technology

3.2 Schedule



2021
K-Innovation ODA Partnership Program (Ghana)
Dissemination Workshop
 K-Innovation Partnership Program



December 16, Thursday at 18:00-22:00 (KST) / Seoul Dragon City, Baekje Rooms(4F)

The Ministry of Environment, Science, Technology & Innovation (MESTI) in Ghana officially announced its plan to establish the Ghana Innovation & Research Commercialization Center (GIRC Center) in 2018. With the request from the Ghanaian government, the K-Innovation ODA Program from the Science and Technology Policy Institute(STePI) launched a three-year project (2021-2023) called the "Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Center". In 2021, the first year of the project, all the project related activities were performed online due to cross-border travel restrictions amid the COVID-19 pandemic. We would like to share the fruitful results that we have achieved through the meaningful cooperation between Korea and Ghana by inviting all the delegates to the 2021 K-Innovation ODA Program Dissemination Workshop with Ghana.

Date

December 16, Thursday at 18:00-22:00 (KST)

Venue

Seoul Dragon City, Baekje Rooms(4F)

Time	Section	Presenters and Panels
Opening Ceremony		
09:00-09:05(GH) 18:00-18:05(KR)	Opening Remarks	Dr. Myung-Do Oh Senior Advisor Vice President, University of Seoul
09:05-09:10(GH) 18:05-18:10(KR)	Welcoming Remarks	Dr. Kwaku Afriyie Minister, MESTI
09:10-09:15(GH) 18:10-18:15(KR)	Congratulatory Remarks	Ambassador, Jung-taek Lim Ambassador, Embassy of the RoK in Ghana
09:15-09:20(GH) 18:15-18:20(KR)	Group Photo	
Policy Suggestion for Ghana Session		
09:25-09:40(GH) 18:25-18:40(KR)	Policy Suggestion for Ghana Tax&Finance Support	Dr. Mean Sun Noh Head, Div. of Future Strategy Research, Korea Small Business Institute(KOSBI)
09:40-09:55(GH) 18:40-18:55(KR)	Policy Suggestion for Ghana Manpower Support	Pf. Youngeun Wee Professor, Korea Polytechnics
09:55-10:10(GH) 18:55-19:10(KR)	Policy Suggestion for Ghana Purchase Support	Dr. Byoung Gun Kim Research Fellow, Korea Institute of Procurement
10:10-10:25(GH) 19:10-19:25(KR)	Policy Suggestion for Ghana I-U-R Collaboration Support	Dr. Jong-in Choi Vice President, Industry-University Cooperation Foundation, Hanbat National University
10:25-10:40(GH) 19:25-19:40(KR)	Policy Suggestion for Ghana Accreditation & Certification Support	Dr. Sangho Choi Director, Jeonbuk Digital Convergence Center
10:40-10:50(GH) 19:40-19:50(KR)	Break Time	
10:50-11:05(GH) 19:50-20:05(KR)	Policy Suggestion for Ghana Technology Transfer & Commercialization Support	Dr. Min-Seok Song Director, Gyeongbuk TP
11:05-11:20(GH) 20:05-20:20(KR)	Policy Suggestion for Ghana Start-up & Incubation Support	Dr. Jong-Woo Ryu Manager, Gyeongbuk TP
11:20-11:35(GH) 20:20-20:35(KR)	Policy Suggestion for Ghana Intellectual Property Management Support	Dr. Young-Bum Kim CEO, Intellectual Property On
11:35-11:50(GH) 20:35-20:50(KR)	Policy Suggestion for Ghana Technology Development & Information Support	Dr. Sang Gon Kim Director, Gyeongbuk TP
Discussion and Closing Ceremony		
11:50-12:35(GH) 20:50-21:35(KR)	Discussion and Q&A	
12:35-12:45(GH) 21:35-21:45(KR)	Closing Remarks	Mr. Oliver Boachie Special Advisor to Minister, MESTI Dr. Wangdong Kim Project Manager Senior Research Fellow, Division of Global Innovation Strategy, STEPI

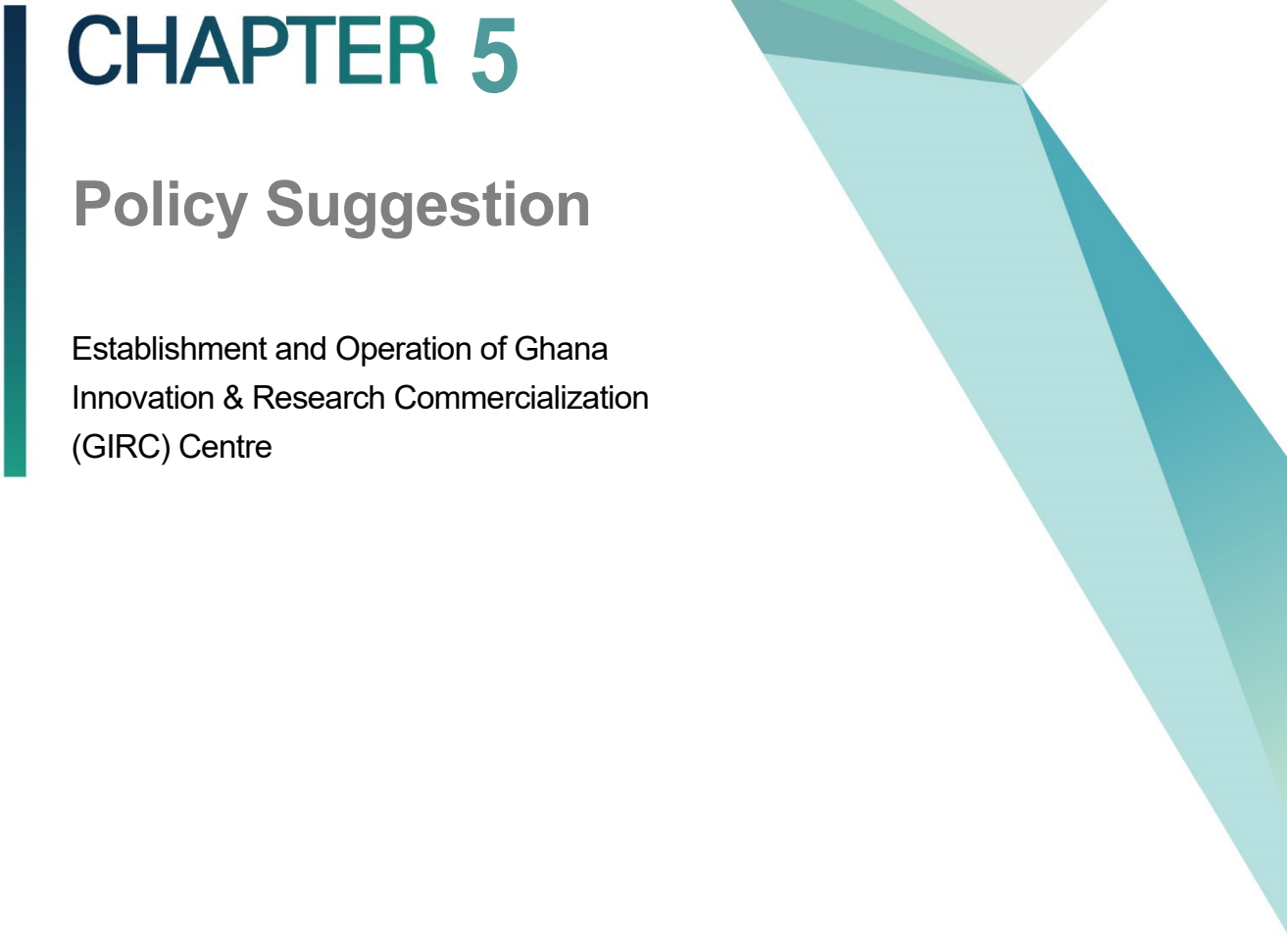
* Simultaneous interpretation (Kor ↔ Eng) will be provided for Policy Suggestion and Discussion session, so please make sure to use the Interpretation function (English Channel).

** The workshop will be presided over by Mr. Seohye Yoon (Opening Ceremony) and Dr. Wangdong Kim.

*** Please take a look at the link below before the Workshop : How to use the Interpretation function in Zoom
<https://premc.org/how-to-use-interpretation-in-zoom/>





A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing left, and a series of overlapping triangles in various shades of teal and light blue pointing right.

CHAPTER 5

Policy Suggestion

Establishment and Operation of Ghana
Innovation & Research Commercialization
(GIRC) Centre



Chapter 5. Policy Suggestions

While Chapter 2 reviewed the current state of Ghana's nine TISPs, Chapter 3 presented Korean experiences on each of the nine themes. Based on these reviews, Chapter 5 proposes potential policies and programs that can be implemented by the government of Ghana.

1. Tax & Finance Support Programs for Ghana

1.1 Tax Support

1.1.1 R&D Reserve Fund Deduction Program

It is recommended to institute the R&D Reserve Fund Deduction Program to boost corporate R&D investment. This program indirectly supports corporate R&D activities by deferring companies' tax payment for a certain period. This program practically benefits companies by exempting them from the interest payments during the carryover period.

In case where a company has secured the reserve fund to cover its R&D expenses, it is allowed to deduct its R&D reserve fund within a range of up to 5 percent of sales (the rate set at the time of the program introduction). For the following three years after the termination of the taxable year of the deduction, the R&D reserve fund is spent and treated as R&D expenditure.

Costs incurred for technology acquisitions from abroad through contracts can also be considered expenditures. The leftover or unused amount of R&D expenditure is reversed as gains for the taxable year of the day after that when three years have passed. In cases such

as business closure or merger, the unused R&D expenditure is reversed as gains of the current year.

Companies who want to take advantage of the R&D Reserve Fund Deduction Program will be required to submit a detailed description of their R&D reserve fund to the tax authority when they file tax reporting every year. When the use of R&D reserve fund is terminated, companies will also be required to submit a detailed description of the use of the fund to the tax authority.

1.1.2 R&D Tax Credit Program

It is also recommended to introduce a tax credit program for R&D related current expenses and facility investment. R&D related current expenses include i) labor costs of employees directly engaged in R&D activities, ii) costs of raw materials, parts, sample products, & test reagents used for R&D, and iii) R&D outsourcing expenses paid to industries and universities. A certain portion of these R&D related current expenses can be deducted from corporate or income tax. In the meantime, an R&D facility investment refers to investments in facilities employed for research and testing (excluding investment in used facilities). A certain portion of the R&D facility investment will also be deducted from corporate or income tax.

Companies wishing to take advantage of the R&D Tax Credit Program R&D are required to submit a tax credit application form to the tax authority at the time of tax reporting. This program is expected to enhance corporate R&D capability and strengthen the national competitiveness of Ghana's related industries.

1.1.3 Tax Support Program for Open Innovation R&D Activities

It is necessary to boost R&D collaboration among local and foreign innovation actors and strengthen corporate R&D capacity by introducing a tax support program for open innovation R&D activities.

This program provides tax support for three categories of open innovation R&D activities such as i) technology acquisition, ii) technology transfer, and iii) technology introduction. In relation to technology acquisition, this program provides corporate or income tax deduction for a certain portion of expenses paid to purchase intellectual property rights including patents and model utility invented and possessed by Ghanaian nationals. The tax support for technology transfers allows corporate or income tax deduction for a certain portion of income generated from the transfer of intellectual property rights including patents and model utility rights to Ghanaian nationals. When introducing technologies required for strengthening Ghana's industrial competitiveness, a certain rate of corporate or income tax deduction can be offered to foreign parties who provide needed technologies for the Ghanaian counterparts through technology contracts over a five-year period.

Applicants for this program are required to submit a tax credit (or exemption) application form to the tax authority at the time of tax reporting. This program is expected to enhance corporate R&D capabilities and strengthen the national competitiveness of Ghana's related industries.

1.2 Financial Support

1.2.1 Long-term Loan Program for SMEs' New Technology Development To Be Newly Launched

This program is also recommended as it helps promote corporate R&D investment by providing SMEs with working capital and facility loans over a maximum period of 20 years at an interest rate far below the market rate. This program will be managed and administered by the Development Bank of Ghana, which will soon be established. The program is expected to enhance corporate R&D capabilities and strengthen the national competitiveness of the related industries.

1.2.2 Lending Quota Program for SMEs To Be Introduced

It is recommended to introduce a lending quota program for SMEs to ensure a stable and seamless supply of funds for them. Through the program, the Central Bank mandates commercial banks to allocate a certain portion of their increased loans to SMEs. For loans extended to SMEs by qualified commercial banks, the Central Bank supports them with an interest rate far below the market rate.

This program will be managed by the Bank of Ghana. It is expected to facilitate an uninterrupted supply of funds to SMEs which have deficiencies in credit and collateral capacity.

1.2.3 Technology Guarantee System To Be Introduced

It is necessary to resolve the problem of asymmetrical imbalance on loan-related information between SMEs and financial institutions, which can be addressed through the introduction of the Technology Guarantee System.

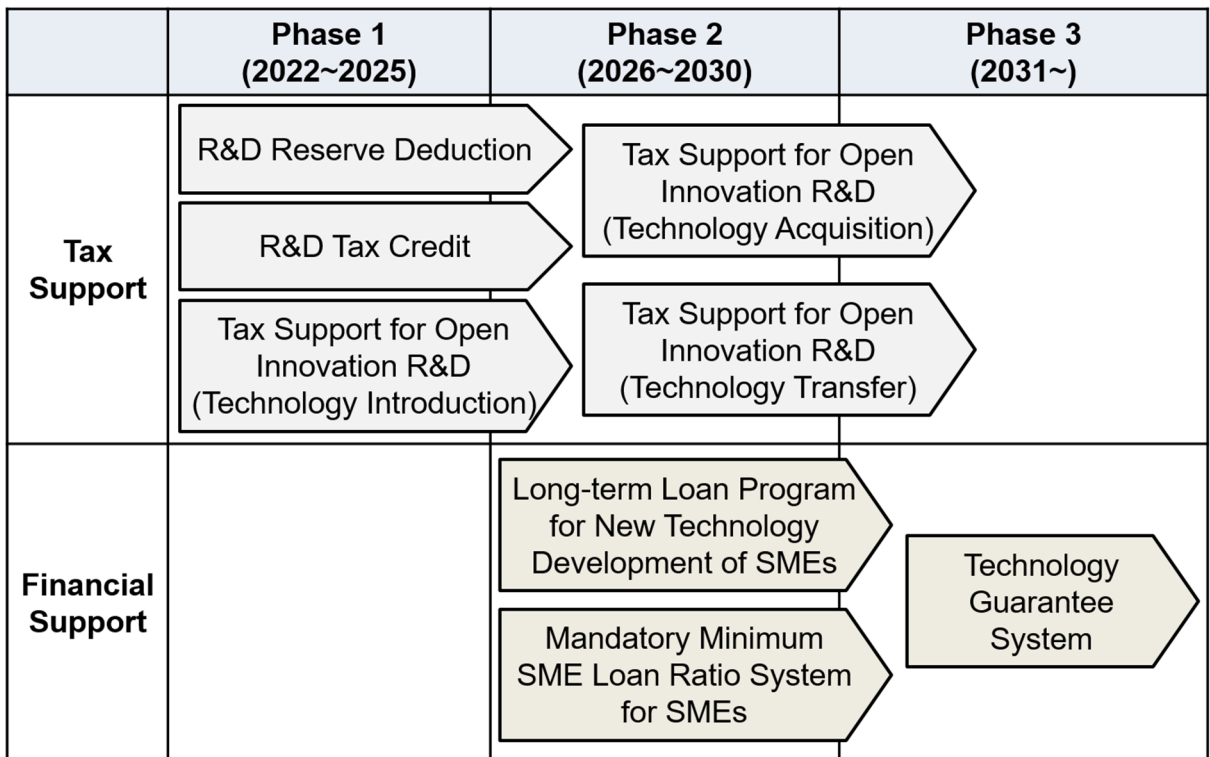
The Technology Guarantee System can be composed of i) a guarantee for loans, ii) guarantee for commercial bills, and iii) guarantee for contract performance. The loan guarantee provides guarantees for loans extended to companies by financial institutions. Through a guarantee for commercial bills, the program guarantees the payment of the amount on the promissory notes issued by companies in connection with their commercial transactions. The guarantee for contract performance extends guarantees for contractors' liability to pay various deposits as a security to ensure the faithful performance of a contract. Included in this category are various deposits, such as bid, contract, payment, and maintenance deposits to be paid in relation to construction contracts and a supply of goods and services.

This program can be managed by the Development Bank of Ghana, which will soon be established, or a new entity can be established specifically for the purpose of running this program. The program is expected to ensure efficient and uninterrupted funding from

financial institutions by issuing credit guarantees for SMEs based on the evaluation of their future values through technology and business viability assessments.

1.3 Roadmap

The following figure shows the roadmap for implementing tax and finance support programs at a national level.



[Figure 5-1] Roadmap for Tax & Finance Support

2. Human Resources Support Programs for Ghana

While Ghana has recently witnessed an increase in doctoral-level R&D human resources, such increases is concentrated disproportionately on the higher education sector. It is vital that the country's human resources development is linked to economic and employment opportunities its manufacturing and agricultural industries offer. In order to ensure such linkages, the government of Ghana needs to implement human resources support programs that can increase the supply of R&D human resources including technicians. Ghana's human resources support programs also need to help translate R&D outcomes into products and services that can be readily commercialized in the private sector through a customized approach to Ghana's industrial needs.

2.1 Expanding Training Centers for Developing Skilled Technicians & Engineers

The government of Ghana has been building up the competencies of Ghanaian technicians through various technical & vocational education and training initiatives as part of the country's technology development projects in the economic sector, R&D support programs for the private sector, industrial technology development projects, and national training assessment programs. In order for Ghana to accelerate manufacturing sector-led industrialization and promote job creation, a stable pool of skilled technicians rather than high-caliber researchers needs to be ensured. This requires Ghana to improve its vocational education at secondary schools and vocational colleges.

For this purpose, high schools need to be divided into i) preparatory high schools which prepare students for college entrance and ii) vocational high schools which offer technical training for students and link vocational training with students' future career path. Vocational high schools develop and train students, so that they can be placed in the field jobs of relevant industries upon graduation. In addition, it is recommended that the government of Ghana restructure two-year colleges including polytechnics and 4-year

universities into vocational colleges dedicated to technical training and provide funding for them. The government also needs to actively respond to the workforce demand required for the country's economic and industrial development. This requires the establishment of training centers that can educate such workforces. Korea's specialized high schools, meister high schools and Korea Polytechnic could serve as prime examples for the government of Ghana's approach to build such training centers.

2.2 Establishing the National Institute of Science & Technology to Develop Scientists & Engineers

Ghana's economic development can be achieved only when Ghanaian S&T talents with high-growth potential are provided with opportunities to further their career development. Considering that the size of Ghana's R&D workforce is less than half of that of Sub-Saharan Africa, the government of Ghana needs to step up its efforts to develop Ghanaian S&T professionals and consider establishing the tentatively called 'National Institute of Science & Technology'.

It is recommended that the institute hire professors with doctoral degrees and overseas educational backgrounds, who can teach theories of state-of-the-art technologies and introduce the recent trends of such technologies to Ghanaian students. The institute also needs to contribute to Ghana's development by conducting various national R&D projects. In addition, Ghana's S&T talents need to be educated and trained in leading countries in science and technology such as the U.S., Japan and EU countries, so that they can contribute to enhancing Ghana's national competitiveness.

The government of Ghana may consider receiving funding from international organizations or foreign governments to establish S&T-specialized universities by referring to Korea's prior experience of establishing institutions for vocational training and S&T human resources development in the 1960s and 1970s based on funding from the UN and some foreign governments (i.e. the U.S. and Germany).

2.3 Inviting Ghanaian Expatriate Scientists and Engineers

The government of Ghana also needs to consider programs to invite Ghanaian expatriate scientists and engineers, both native-born Ghanians and second-generation Ghanaians born overseas, to Ghana. These Ghanaian expatriate scientists and engineers have an advantage over foreign counterparts who have no Ghanaian heritage as the former often have a better understanding of the Ghanaian economy and society. The Ghanaian government needs to provide active support for these Ghanaian expatriate scientists and engineers, so that they can provide useful inputs for developing Ghana's S&T policy. They can also be assigned to important roles and responsibilities at local universities and research institutes. The salary to be paid to these expatriate experts, however, may be substantially lower than that of foreign scientists and technicians. Nonetheless, the government of Ghana may bestow them with an honorable title of 'Nationally Revered Scientist', thereby showing due respect to them for returning to their native land and making contributions to their motherland.

As Korea's experience shows, Korean expatriate scientists and engineers who returned from overseas played a critical role in enhancing the national competence and competitiveness. During the early stages of Korea's economic development, these expatriate scientists and engineers returned home to help the country's S&T innovations, but they had to make a sacrifice by accepting low wages. Nevertheless, the Korean government paid the highest respect to them and even the national president paid his personal attention to them, making them feel proud of their contributions to the nation's development.

2.4 Program to Invite Foreign Scientists and Engineers

It is recommended that Ghana invite foreign scientists and technicians who can help the country activate its manufacturing sector and achieve industrial and economic development. It is necessary to strategically assess and determine industries and technologies to be prioritized for national economic development. In doing so, a practical approach would be to invite experts in the prioritized fields. In addition, it is also necessary to promote R&D cooperation between industries, universities, and research institutes through these experts. Foreign scientists are expected to play a pivotal role in strengthening Ghana's R&D capability and developing high-caliber talents at universities.

The Ghanaian government needs to be actively engaged in the invitation of foreign experts by taking advantage of various international exchange programs hosted by major countries (e.g. the expert exchange program hosted by the Worlds Friends Korea (WFK)). The World Friends Korea Senior Expert Dispatching Program (WFK-KSE) sent a total of 489 experts to governments and public institutions of 39 developing countries between 2010 and 2015.

Foreign applicants eligible for invitation are required to have prior field experience at ICT companies, research institutes, or universities (including graduate programs) in their home countries. The government of Ghana needs to come up with measures that can ensure the invitees' active participation in short- and long-term programs regardless of their educational background, age, and nationality.

2.5 Providing Tax Exemption for Foreign Scientists and Engineers

Countries around the world are offering various kinds of benefits to foreign experts to foster R&D activities within their borders, and thereby enhance their national R&D capability. Ghana needs to offer various incentives to foreign experts, so that the country can attract as many foreign experts as possible in certain priority areas such as R&D. For example, the government of Ghana can introduce a tax incentive program for foreign researchers by providing them with income tax exemptions during their stay in Ghana.

Foreign scientists and engineers to be invited to Ghana may work for local Ghanaian companies or they can also work for government institutions, universities, and research institutes in the R&D field.

2.6 Operating a Program to Support the Job Placement Center for Scientists and Engineers

The government of Ghana needs to immediately implement policies that can readily translate the fruits of R&D into actual products and services and link the private sector and educational institutions to promote the country's economic and industrial development. In doing so, the government of Ghana needs to enact a special act for supporting scientists and

engineers in the mid and long term, thereby providing useful job information for them and instituting a support mechanism that would connect job seekers with science & engineering degree programs at universities.

The Job Placement Center for Scientists & Engineers needs to be established to analyze and provide job information for scientists and engineers and introduce them to job opportunities offered by R&D-focused businesses. The center can also build and operate a career web portal for scientists and engineers, through which job seekers, science & engineering graduates, will be matched with potential employers. The web portal will also provide job information and customized job placement services for its users.

As a part of its efforts to provide customized job placement services, the center can hold job fairs for leading scientists and engineers, host job seminars for job seekers, and sponsor some recruitment-related conferences and lectures. Through these activities, the center will be able to arrange job-related counseling by HR managers and recruitment specialists, collect statistical data on recruitment and employment, and provide statistical data on corporate R&D centers, scientists, and engineers.

2.7 Roadmap

Based on the analyses above, the following figure presents a three-phased roadmap for building Ghana's human resources development system and developing human resources support programs: i) Phase 1 (short term) between 2022 and 2025, ii) Phase 2 (mid term) between 2026 and 2030, and iii) Phase 3 (long term) from 2031 and onwards.

In the short term, the government of Ghana is recommended to focus on policies or programs for developing skilled technicians and scientists and engineers. It can also consider instituting a program to invite Ghanaian expatriate scientists and engineers.

In the mid term, the government of Ghana is recommended to build an educational system or institution such as the National Institute of Science and Technology for bringing up high-caliber scientists and engineers. During this period, it can also implement programs to invite foreign scientists and engineers and provide tax incentives for them.

In the long term, the government of Ghana needs to build a human resources support system that can effectively link scientists, engineers, and researchers to employment opportunities and provide them with online career support services.

	Phase 1 (2022~2025)	Phase 2 (2026~2030)	Phase 3 (2031~)
Manpower System	Establish an institute to train skilled engineers	Establish a national science & technology institute	
Manpower Support	Invite Ghana-born or second generation Ghanaian migrants	Invitation program for foreign scientists Offer income tax exemptions for foreign scientists and engineers	Support program for the science and engineering job placement center

[Figure 5-2] Roadmap for Human Resources System and Support

3. Technology Development & Information Support Programs for Ghana

3.1 Technology Development Support

3.1.1 Public Sector-Led R&D Projects

Referring to the case of Korea, for each project determined to require preliminary feasibility study in promoting public-led R&D projects, it is recommended to conduct preliminary feasibility study target selection (technical evaluation) as well as a preliminary feasibility study for national R&D projects only for projects judged as 'suitable', and then notify the results to the business department and related organizations. In order for the project to be successfully carried out, there are two prerequisites. Firstly, it is necessary to establish clear evaluation criteria for the project subject to a preliminary feasibility study, taking account of the Ghanaian environment. As a primary criterion, given that the government R&D expenditure ratio of Korea to Ghana is 180:1, among the new national R&D projects in Ghana, with a total project cost of more than USD 300,000, national financial support of more than USD 200,000, and the science and technology R&D projects of high national importance are recognized as projects requiring a preliminary feasibility study and could be included to be subject to the technology evaluation. Next, as the second criterion, only projects judged as 'suitable' by developing appropriate evaluation items (e.g. necessity and urgency of scientific and technological development, suitability of national support, differentiation from existing projects and linkages, specificity of project plans, etc.) and evaluation indicators for technical evaluation by project should be included in the preliminary feasibility study. Secondly, it is necessary to design survey items reflecting the atypical characteristics of R&D projects and develop weighted values according to project types to conduct the preliminary feasibility study suitable for the situation in Ghana. In relation to the survey items, detailed evaluation indicators are developed based on scientific and technological feasibility analysis, policy feasibility analysis, and economic feasibility analysis. For example, in the case of scientific and technological feasibility analysis, the appropriateness of deriving problems and issues, the appropriateness of the project goals,

the appropriateness of detailed activities and promotion strategies, etc. can be considered. For policy feasibility analysis, in order to evaluate policy consistency and implementation system, conformity with higher-level plans, project implementation system, and willingness to push forward can be considered as the evaluation indicators. In the case of economic feasibility analysis, economic feasibility can be considered as an evaluation index. Also, how to assign weighted value to each survey item according to the project type shall be set in detail. For example, in the case of a challenging innovation type project, a relatively higher weighted value could be given to a scientific & technological feasibility analysis, while for a growth business, the same applies to a economic feasibility analysis, and for a foundation-forming project, the same applies to an analysis of the policy feasibility.

As for the implementation governance of the project, the Ministry of Environment, Science, Technology and Innovation (MESTI) of Ghana is in charge of overall operation of a preliminary feasibility study for public-led R&D projects, and a public-led R&D project evaluation general committee is formed to provide advice on the operation of the target project, investigation results, and system operation, and to deliberate and adjust important matters related to a preliminary feasibility study. In addition, by selecting a public-led R&D project preliminary feasibility implementation agency, it becomes responsible for the formation and operation of internal researchers and advisory committees to implement the related tasks for each project. The advisory committee selected for each project draws issues and investigation directions, and performs preliminary feasibility analysis and comprehensive evaluation (AHP) such as a scientific and technological feasibility, policy feasibility, and economic feasibility analysis. If necessary, commissioned research and experts are additionally selected to be in charge of patent investigation, facility construction investigation, and equipment construction feasibility review. As it is possible to selectively invest in promising science and technology R&D through this project, it is expected to enhance Ghana's national competitiveness in science and technology and maximize the efficiency of government R&D financial operation.

3.1.2 Company-Led Commercialization Technology Development Project

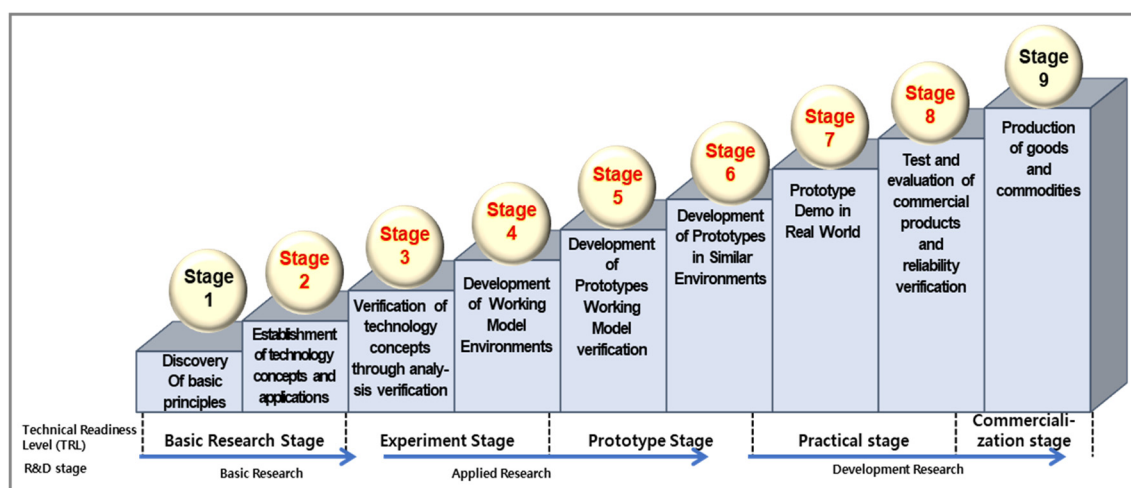
Few private companies exist in Ghana that can lead the national economy. The level of R&D investment is also very low, so it poses a barrier to developing a virtuous cycle of the activated local economy through job creation. Promoting the company-led commercialization technology development projects aims to solidify the groundwork of the national economy and realize stable economic growth by intensively supporting the technology commercialization for project development, and creating jobs for local businesses, nurturing skilled talents that will become a future growth engine.

In promoting private company-led commercialization R&D projects, Ghana's central government needs to designate locally specialized industries by region, referring to relevant Korea's cases, and support commercialization R&D for new product development. A PM (program manager) and PD (program director) in charge should analyze the technical readiness level (TRL) of each project and promote a customized research support project, so that the private company can lead more than 7 levels of research. It will enable the company to pull technology commercialization from the front. Here, it is recommended that Ghana take into account its industry structure and industrial TRL, and managing actor by industry in order to maximize the efficiency of R&D support and differentiate the projects by type in executing them. For the successful implementation of a project, it is necessary to introduce a PM/PD system and TRL. The PM/PD system is an R&D project management by experts. They are familiar with TRL by technology, and supervise and operate a full R&D cycle, which consists of planning a new project, interim and final evaluation, and performance management and distribution. Here, TRL is a measure with which to evaluate the readiness level of core technologies, which aims to estimate the possibility of value-added creation on the continuous line of existing technology and secure technological competitiveness. TRL is evaluated according to the following classification system of up to 9 stages (Stage 1 is the discovery of basic principle and Stage 9 is the production of commercial products). They can also be regrouped into the basic research stage (Stage 1 and 2), experiential stage (Stage 3 and 4), prototype stage (Stages 5 and 6), commercialization stage (Stage 7 and 8), business stage (Stage 9). Based on this, the R&D level of a certain technology is determined to be fundamental research, applied research, or development research. A

PM/PD, who are experts in R&D project management, takes responsibility for the entire process of local industry support R&D projects and advances the effectiveness of supporting an R&D performer (company). Furthermore, the PM/ PD play a critical role in paving the launchpad for R&D that will lead the promising future by selecting R&D performers and project types after considering TRL by industry. As for the implementational governance of these projects, MESTI plans and supervises them, and the local governments (supports the Ghana Innovation Research Commercialization Centre, GIRC) draw and implement a detailed plan according to the guidelines of the central government. However, the performance is evaluated by a designated regional project evaluation agency. Based on the growth of private companies and strengthening of technological competitiveness through these projects, it is expected that Ghana will see national competitiveness rise, the ecology of locally specialized industries become active, and the national economy develop accordingly.

[Table 5-1] Support for Company-Led Commercialization Technology Development in Ghana

Industry	TRL	Main actor of R&D project	Project type
ICT	Fundamental research	Universities	Original technology R&D
Heavy & chemicals industry	Experimental & prototype stages	Government-funded Research institutes	Original technology R&D
Light industry	Commercialization stage	Private firms	Innovative product R&D



[Figure 5-3] TRL Definition by Stage

Material: The Board of Audit and Inspection Audit Research Institute(2018), National R&D project planning and selection evaluation Operational Management Status and Audit Implications

3.2 Information Support

3.2.1 Projects for National R&D Planning and Management Support Service

Ghana lacks an information infrastructure at a national level, which causes a considerable inefficiency to the government R&D investment. R&D researchers can't access knowledge information of science and technology without difficulty and share it among themselves. Therefore, it is critical that Ghana built a comprehensive information management system at a national level that promotes the creation and distribution of quality knowledge of science and technology and realizes national open R&D collaboration. The national R&D

planning and management support service project aims to prepare for a cooperation system across the government departments, then execute NTIS development plans by stage, and improve the efficiency of national R&D investment through adjusting national R&D projects.

The National Science and Technology Advisory Board was established with implementation governance of the project, and the Operational Committee was formed to be responsible for efficient management of knowledge and information related to science and technology and national R&D projects, deliberation of major issues related to distribution, and those related to national R&D project R&D information management. In addition, the NTIS Project Promotion Committee is formed to deliberate and adjust major issues related to the selection, management, and evaluation of research planning tasks for each R&D project. MESTI shall be in charge of establishing and adjusting the business plan and managing the construction projects and maintain a cooperative system with related agencies. The working-level committee, which consists of working-level officials from related ministries, experts recommended by related ministries, and working team leaders of specialized agencies representing the ministries, participates in the process of collecting opinions and consultations and provides technical advice. Finally, the establishment plan of the detailed NTIS system shall be established, and the overall management agency and cooperative research institute of the NTIS project in charge of the establishment and operation of the integrated system shall be designated.

It can be a good idea to benchmark NTIS 1.0., which is the first stage of Korea's NTIS, in building the base information system related to national R&D projects and designing the support system for technology/ industry information and regional technology innovation. However, the success of this project in Ghana should be preconditioned on the improvement of information infrastructure including the access rate and speed of broadband, which are currently low in the country. It is expected that the successful buildup of national R&D planning & management support services will advance the public's access to the knowledge information of science and technology, increase the understanding of the national R&D project status, and popularize science and technology in their daily lives. At the same time, it will advance technological competitiveness throughout the nation as the government, universities, industries, and research institutes share their workforce and

equipment and conduct joint R&D projects. In addition, it will make the national R&D funds more efficient, various information analyses for R&D policy development possible, and R&D projects more transparent.

[Table 5-2] Benchmarking Target Korean NTIS 1.0 Contents

Strategy	Major promotion assignment	Detailed promotion assignment
Foundation related to national R&D business Establishment Of information System	1. Support for national R&D project management systems	① National R&D project management services
	2. Establishment of science and technology human resource information utilization system	② Human resource information services participating in national R&D ③ Evaluation committee recommendation service
	3. Support for management of joint use of equipment and materials	④ Joint use service of equipment and materials from all ministries ⑤ Equipment and materials expert information service
	4. Comprehensive performance management support	⑥ National R&D Performance Information Service ⑦ Excellent and promising technology information service
	5. Establishment of joint foundation for science and technology information	⑧ National R&D Board ⑨ Science and technology statistics service ⑩ Science and technology information distribution service ⑪ National R&D standard information management service ⑫ Information protection system
Establishment of technology/industry information and regional technology innovation support System	6. Establishment of technology/industry information support system	⑬ Technology and industry information service
	7. and regional technology innovation support system	⑭ Regional R&D information services

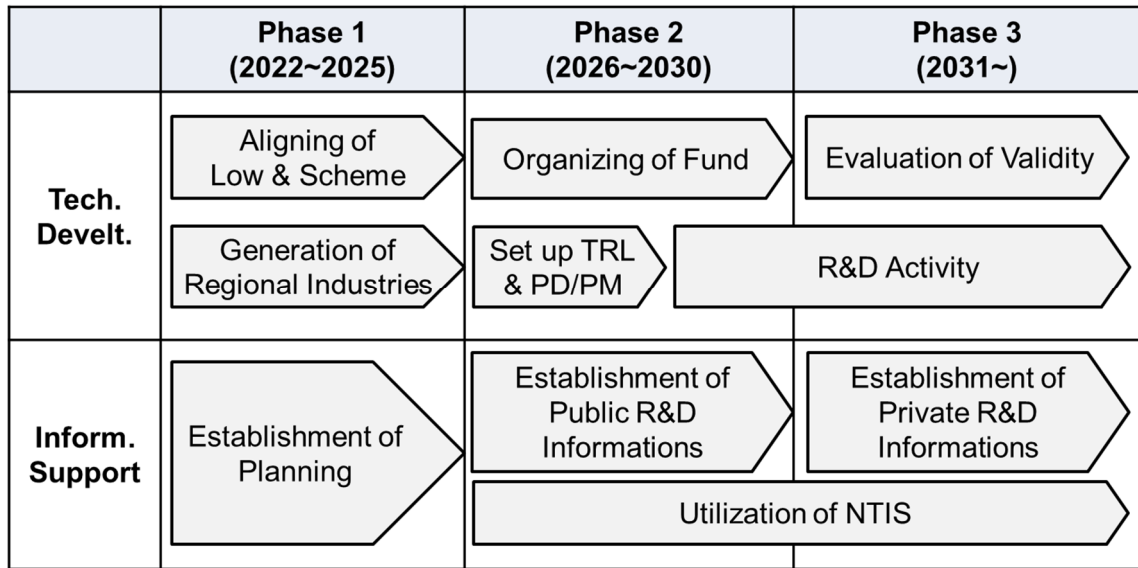
Source: Korea Institute of Science and Technology Information (2015)

3.3 Roadmap

The following is the roadmap for implementing the programs for technology development and information support suitable for the situations in Ghana. Firstly, in Stage 1 (the technology development), it is necessary to arrange and improve the laws and systems for the public sector-led technology development and select local industries for private sector-led technology development. In Stage 2, the funds for technology development need to be raised and TRL and PD/PM systems should be prepared. In Stage 3, it is necessary to evaluate the validity (feasibility) for public sector-led technology development and perform technology development activities by the private sectors.

Secondly, in Stage 1 (information support), it is necessary to make a basic plan for providing information on technology development. In Stage 2, an information system should be built

for the public sector-led technology development. An information system for private sector-led technology development is required to be implemented in Stage 3.



[Figure 5-4] Roadmap for Technology Development and Information Support

4. Technology Transfer & Commercialization Support Programs for Ghana

4.1 Policy Proposal for Ghana's Technology Commercialization

Ghana is -at the early stage of business incubation where its government fosters startup companies based on policies. Accordingly, startup incubation-innovation hubs are expanding to incubate small companies. However, not many startup companies haven't yet entered the stage of stabilization where there can stand on their own beyond the startup stage because of many negative factors such as insufficient budgets and lack of expertise (experts) and capabilities. Considering the current situation in Ghana, we propose some policies to promote technology commercialization and increase companies' capabilities in Ghana as follows:

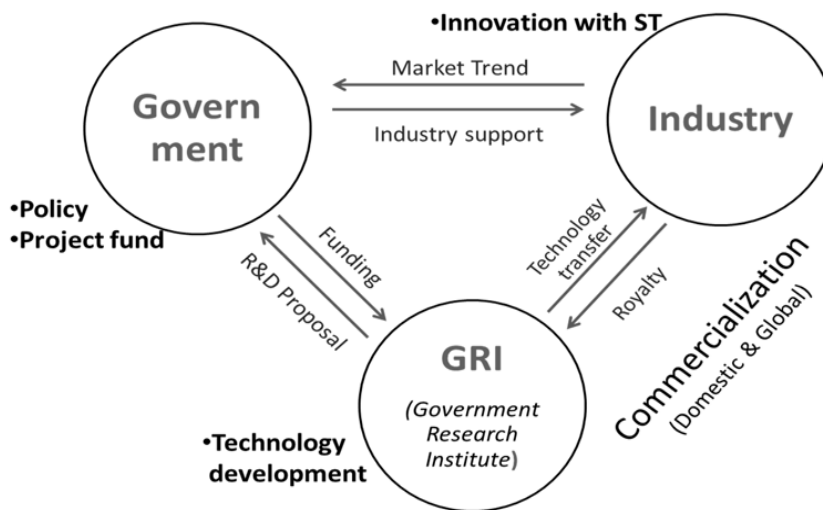
4.1.1 Strengthening the Connection between R&D Organizations and Private Sector

Ghana has a very weak connection between the conventional R&D organizations and private sector. Ghanaian universities receive most of the R&D support funds, but they haven't reached the stage of commercialization or technology transfer while still staying in the early research stage. To commercialize the outputs of research, close interaction and connection should be made among companies, startup incubation-innovation hubs, and other company support agencies. The companies in Ghana have not grown enough yet to have an affiliated research institute inside. Therefore, it is desirable for the conventional research institutes to link to companies and pick out & implement a research project whose output can be commercialized, while encouraging innovation hubs and companies to participate in a joint research project. We recommend that Ghana foster R&D-based research companies in the universities and technology-based venture startup companies.

It is also desirable for the government to lead the link in the initial stage for continuous growth and structural change. The government's intervention such as incentives offering,

system change for promoting innovation, and human capital development is important to enable technology development and for companies to accept transferred technology. The government-funded research institutes (GRIs) such as technology and research development institutes, universities, and technology licensing offices (TLOs) play an important role in the use and accumulation of technology transfers.

In the meantime, the government, industries and research institutes play the role of industry leaders as follows. The government selects the technologies that the industries need, and provides funds to GRIs, which develop new technology and transfer it to the industries. Then, companies develop competitive products, enter a market with them, and make profits, some of which return to GRIs in the form of royalties, and others are paid as an incentive to the developers to motivate them to develop better technology.



[Figure 5-5] R&D Ecosystem and Science, Technology, and Innovation (STI) in Ghana (plan)

4.1.2 Making a Strategy To Incubate Strategic Industry-Intensive Companies

For starters, we recommend that the Ghanaian government select the industrial sectors to nurture intensively and strategically support the companies in those sectors. Let's take a look at the relevant Korean case. The structure of the Korean industry had advanced from the 1st manufacturing to the light industry and then heavy and chemicals-intensive industry.

For Ghana, we suggest that the government first select the most competitive industries strategically and intensively nurture them. The government can consider the policy to cultivate the following industrious sectors first, such as agricultural food processing, textile/garment, minerals, and packaging and distribution service. We also think that the IT industry should be the fundamental domain to develop because it is required in establishing a technology-based industry ecosystem.

4.1.3 Preparation for a Plan to Raise Funds for Scale-Up

Ghana faces the toughest difficulty in creating a technology innovation ecosystem attributed to insufficient funds. The lack of funds for R&D makes it difficult to develop and improve human resources, expand infrastructure, and give experts competitive remuneration to attract them. Therefore, the Ghanaian government should establish institutions in charge of support research commercialization so that the actors working in the innovation ecosystem can be timely supported with proper funds, and thus the companies can stand on their own. To realize this, it should prepare a method for sustainable fundraising, draw a clear plan in advance, review it thoroughly, and set up an institution that provides integrated support.

4.2 Proposal of a Technology Commercialization Support Program for Ghana

4.2.1 Private Investment-Linked R&D Support

A company requires the following to commercialize the outputs of R&D, namely prototype production, certification & acquisition of IPR (Intellectual Property Rights), design & trademark, marketing, consulting, and networking. Therefore, we propose the implementation of private investment-linked R&D support programs (i.e. R&D with an option to purchase). ‘The Ghana Agricultural Technology Transfer(ATT) Project’ realized by USAID’s (United States Agency for International Development) fund (USD 24 million for 5 years (2013-2018)) is the a representative success story. To improve the research and

expansion system regarding the agricultural industry in Ghana, the project contributed to (1) creating a private sector-led agricultural technology transfer mechanism, (2) to connecting producers with research on the development of seed value chains as a market-oriented approach, (3) getting one step closer to important production inputs such as affordable finance, labor-saving technology, and excellent agricultural practice (GAP).

4.2.2 Improvement of the Performance by Startup Companies that Operate Funding & Investment Attraction Programs

Startups and small businesses carry out a variety of programs to attract investment from domestic and overseas investors. The Ghanaian government conducts various investment-inducing programs such as IT pitching, Hackathon, investment attraction contest, and cloud funding. The technology innovation centers attract public and private sector investments, provide startups and small companies with IT pitching and presentation training, and hold investment-pitching events.

4.2.3 Organizing a Pool of Experts in Technology Transfer & Commercialization

The most important for technology transfers is the personnel that participates in technology transfers. In other words, technology transfer is not a product of the system but that of the active engagement of the experts highly familiar with the technology and market concerned. In line with this, an organization that mediates technology transfer in both ways should be positioned between buyers and suppliers. Technology transfers and commercialization are the exact area that requires high expertise and experience, and so it requires more than school learning or elementary training in a company. Technology transfer should be handled by a team of experts who are equipped with specific abilities and experience.

1) Establishment and Management of a Nation-Level Expert DB

It needs to make a DB of the skilled talents specializing in the tasks of intellectual property rights (IPR) such as personnel in charge of a large company's IPR and commercialization, patent lawyers, researchers in technology research institutes, and consultants. In addition,

it is required to actively promote the importance of technology transfer & commercialization tasks, its development potentials, and performance compensation.

2) Support for Technology Discovery and Transaction

Despite the massive amount of information regarding technologies that are traded by each transaction institution, it is not easy for companies to know exactly what technology they need to purchase with the information. Therefore, the technology transfer experts should mediate the whole process of technology transaction, starting with transfer discovery (what the companies want) on their behalf.

3) Commercialization Consultation, Investment, Loaning-Linked Services

The technology transfer experts should facilitate the flow of information between buyers and suppliers from the technology transaction to commercialization stages. They need to support not only fund linkage, which is one of the biggest barriers to commercialization, but also management through consulting service, marketing techniques, and market exploitation.

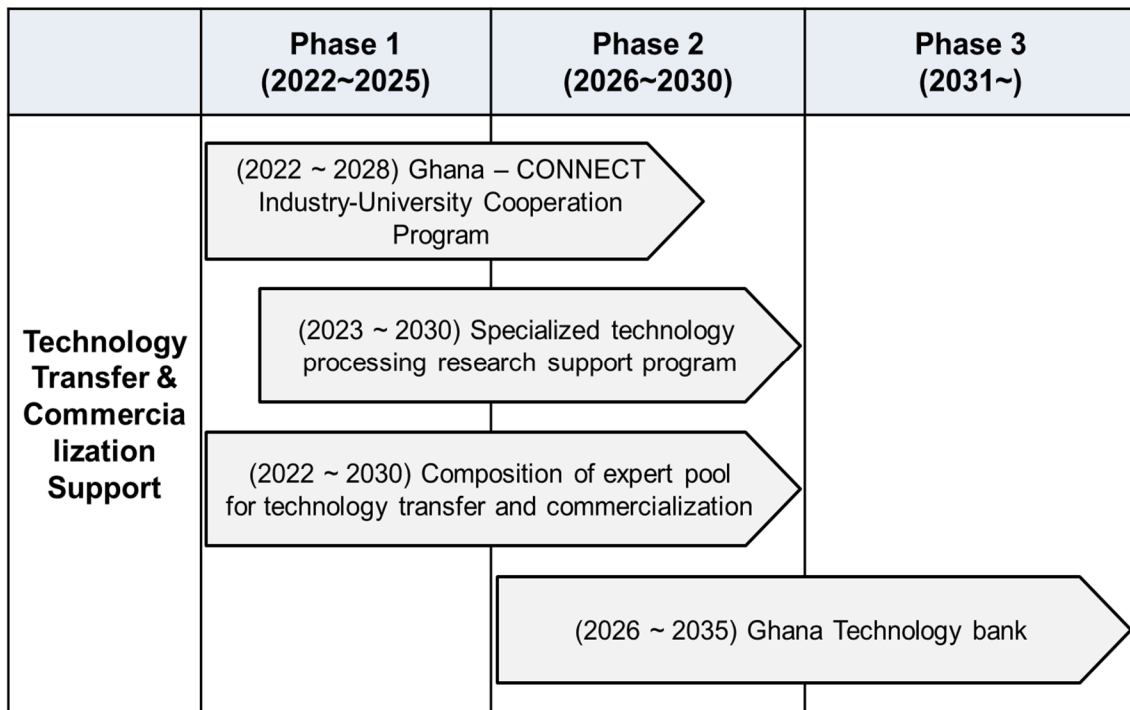
4.3 Roadmap

The most urgent thing to do for technology commercialization in Ghana is to link the conventional research institutes of universities to companies, help carry out researches whose potential for commercialization is high, and encourage the innovation hubs and companies to join R&D activities. Therefore, we recommend that Ghana actively form R&D-based research companies and technology-based venture companies under the universities in the first stage.

In the second stage, when the Ghana Innovation Research Commercialization Centre (GIRC) operates in earnest, as a general supervising entity, it selects the industry sectors and technology that need intensive and prior support, establishes a program to link R&D outputs to commercialization, and is expected to play an active role in realizing technology

commercialization linked to a company. Also, the center also needs to set up a pool of experts in the accumulated technology from each field, and mediate and promote them, so that technology commercialization and transfer can happen actively.

In the third stage, it is necessary to establish a bank of technologies accumulated through STI policy and set up a supervising system for technology transactions to take place among companies, research institutes, developers at a national level. The roadmap to driving technology commercialization can be drawn as shown below.



[Figure 5-6] Roadmap for Technology Transfer & Commercialization Support

5. Startup & Incubation Support Programs for Ghana

5.1 Establishment of Nurturing Strategy for Managers

To promote startups in Ghana, it is necessary to make a startup incubation plan and discover initial startup entrepreneurs. Accordingly, it is urgent to cultivate managers who can support commercialization, startup training, and mentoring. At the same time, there should be nurturing of startup managers who can operate and manage the supporting tasks of startups. ‘Startup support manager’ refers to a person who has a ‘startup support specialist’ certificate issued under the framework act on qualification. Both selecting excellent human resources in Ghana and educating and training them in Korea are also required to nurture the experts in the field. Then these startup managers should play a role in analyzing Ghana's industries and companies, discovering promising companies, fostering youth companies, and operating company support projects.

5.2 Establishment of Startup-Oriented Company Support Strategy

Furthermore, Ghana needs to form managing organizations exclusively for startups, such as KISED, KOSMEs, and Techno Parks, to select, nurture, and process the follow-up management of startup companies. It can be called the ‘Startup Support Center’. The basic fundamentals for the Startup Support Center’s operations are operator, occupant and investor. It is constructed of experts, the steering committee/ board of directors, and advisory committee. In the early startup support in Ghana, we believe that the GIRC, which can integrate all of the aforementioned, is the critical organization that can lead startup incubation support projects and promote the intensive nurturing of the experts very effectively and systematically.

The managing institution in charge of startups should designate and foster specialized industries (by university and region), establish a support strategy by company-growth stage, and provide them with customized growth support.

5.3 Expansion of Technology Innovation Network

In promoting startup policies in Ghana, it is desirable for the Department of Science and Technology to prepare for the support systems and laws for startups. There are many methods and standards for startup support. It seems better to link to youth startups and broadly support them rather than have a specialized few. It is necessary to establish and expand technology innovation networks such as open networks, technology exchange networks, a startup enterprise association, a youth business association, and joint equipment utilization.

To supervise all of these, the roles of the state-run ministries are important. Besides, the system and network of startup-related laws and regulations need establishing to activate and promote startups. Here, the focused will of the Ghanaian government and universities are crucial. For Korea, initial startups are intensively nurtured by the Center for Creative Economy and Innovation, while startup companies whose service year is more than 3 years are by the Techno Park.

5.4 Performance of the Ghanaian Model

The GIRC should create jobs, attract investment, and promote technology investment. To attain this, the IB policy of the government of Ghana should be implemented in a way of providing physical facility funds and operating expenses to IB operators. The government should draw up a Ghana-type startup support model, which reflects the Korean growth model of startup companies and maximizes their output. The figure below summarizes our proposal for Ghana's startup policies.

Start-up Manager Establishment of a development strategy	Start-up centered Establishment of a business support strategy	Technological innovation Network expansion	Ghana Model Performance creation
Fostering of Ghana start-up managers Industry/business analysis Discovery of businesses with potential to grow Fostering of businesses with young entrepreneurs Operation of company supporting businesses	Establishment of start-up facility Designation and fostering of specialized industries (By university and district) Support of developmental stages per company Consulting and Mentoring support Tailored Development support	Open Network Operation of an interchange of technology network Start-up business council Common use of equipment Young entrepreneur council	Ghana! Center of Africa Job creation Activation of investment maintenance Acceleration of technology investment Expansion of a Korean model

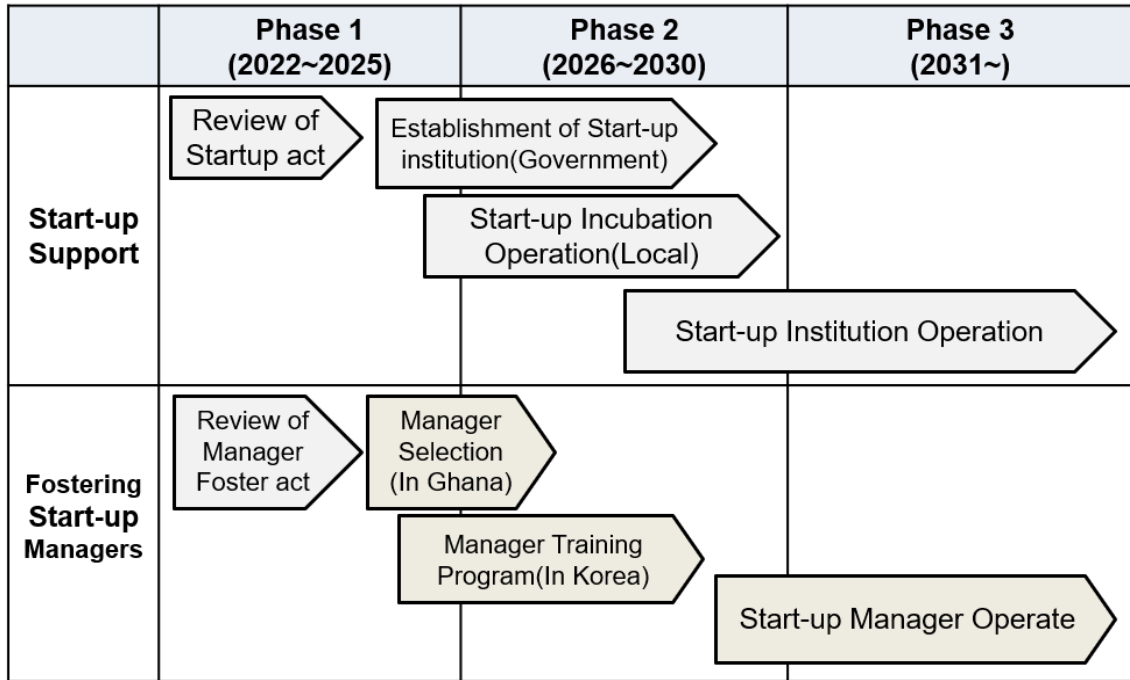
[Figure 5-7] Proposal for Implementing Ghana's IB-Related Projects

5.5 Roadmap

The roadmap to realizing our recommendations for Ghana's startup policies can be drawn as shown below. It divides the policies into the two sections of startup support and manager incubation, but has them be implemented simultaneously.

In the support part section, the current status of startup companies is analyzed in Phase 1 (2022~2044). In Phase 2 (2024~2029), the government-led startup support institutions are founded and local IBs are operated. In Phase 3 (2017~2031 and on), the startup support institutions actively engage in operation.

In the manager fostering section, the current status of the manager is analyzed in Phase 1 (2022~2044). In Phase 2, the managers are selected and they have training in the mid-phase 2. In Phase 3, the managers work in earnest.



[Figure 5-8] Roadmap for Startup & Incubation Support

6. Intellectual Property Management Support Programs for Ghana

6.1 Infrastructure Buildup for Expanding Public/ Private Sector's Patent Creation

As of now, Ghana records 30 million for its population, USD 2,220 for per-capita GDP, and 0.04 percent, for the ratio of GERD to GDP, which is similar to Korea in the 1980s. While the (Ghanaian) Ministry of Justice manages and controls IP rights and prior art search, the public and private sectors hardly do art search and have support programs for patent creation strategy, which are needed for the public and private sectors to secure IP rights. The number of patent applications was 116 in 2016, 21 in 2017, and 15 in 2018. This figure is smaller than that of one university in Korea. Yet, 1,000 trademarks and 500 ~ 800 designs are registered every year, which is similar to those of the Korean Regional Intellectual Property Centers (RIPC). Therefore, the public research institutes and companies need to elevate their awareness of invention and introduce compensation systems for patent creation.

In the beginning, Ghana needs an intensive agency, such as TLO within a university and RIPC in Korea, to help the public and private sectors to create IPs. For this, it is necessary to set up organizations and programs in base regions or 8 Strategic Technology Centers (STCs) in the GIRCs to support IP creation in the fields of technological innovation. In particular, it is required to nurture patent lawyers and patent experts (patent researchers) who can write patent specifications to increase the creation of IPs by the universities and public research institutes. Also, a duty inventions system needs to be introduced to encourage patent researchers to create IPs of their own will.

To help secure IP, it seems essential to establish an intensive agency, which can broadly handle a private firm's trademarks, designs, and brands to patents, in the area of Akura (Ghanian capital) and Kumasi, and organize patent information service agencies for DB infra-buildup, prior art search, and patent map. This idea can be realized through installing an IP managing entity within the GIRCs or expanding the roles of existing TTCs. The successful

organization of these agencies is expected to make the private and public sectors aware more of IP rights and create more IPs. Eventually, it will make it possible to create more IPs at a national level. In addition, the introduction of a duty invention system will improve inventors' income, get the R&D agencies to secure IP, which increases technological competitiveness, enhance the competitiveness of industries and public institutions, and eventually help develop a middle- and long-term research support program for the fields of core capabilities.

6.2 Building IP Information DB and Systems (To utilize IP information)

Ghana is also a member of the WIPO, so it also submits a regional patent application to and receives a local patent license from ARIPO (African Regional Industrial Property Organization). However, it has no online/ offline technology supply network at a national level. On the other hand, Korea has the private sector- and public institution-led IP DB systems. A patent information system such as KIPRIS can let a user check if a researcher or a developer can create IP through the technology that he/she works on. Therefore, Ghana needs to build an internal system for IP information use and a computerized patent administration service as well as an information search service system not only for Ghana but also for the patents, designs, and trademarks worldwide. The establishment of the nation-level patent information DB system is expected to promote the use of IP information and ultimately advance the qualitative growth of IPs. In addition, the computerized patent administration service of patent examination, registration, and trial will improve the efficiency and quality of patent administration and elevate its international competitiveness through IP informationalization.

6.3 IP Advancement Project for Securing Original Technology (To promote IP commercialization)

The GIRC's were established to provide support programs that aim to turn various outputs of R&D by scientists and researchers into products and services through commercialization. Therefore, the outputs attained by the centers should lead to IPs of powerful global competitiveness. To make it happen, universities and public research institutes should

adopt the duty invention system and verification program (invention interview program). These programs will connect technology and patent experts and help set up a strategy to secure excellent IPs. Therefore, it is necessary to incorporate the agencies involved in the management and creation of IPs in the metropolitan areas into the system.

IP management programs include the following: Conducting a patent trend analysis with IP information, drawing a patent map of promising technology fields, planning an R&D strategy, doing a prior-art-research, analyzing technology/ market/ product information, and searching infringement and rejected claims. These programs will be helpful to secure original patents and patents useful for industries. It can also improve the efficiency of research.

[Table 5-3] IP Management Program

IP Management Program	
Patents	Utility Model/ Design/ Brand
• Prior art research & patent trend analysis • Introduction of a patent roadmap for promising technology fields and establishment of R&D direction based on the map • Planning an R&D strategy and an IP application strategy • Patent analysis in case of patent dispute and research into rejected claims • Industry/ products/ market information analysis • Technical valuation	• Prior art (designs/ trademarks) search and securing DB for the purpose • Research into patent infringement and rejected claims • Design/ brand application

6.4 IP Commercialization Programs

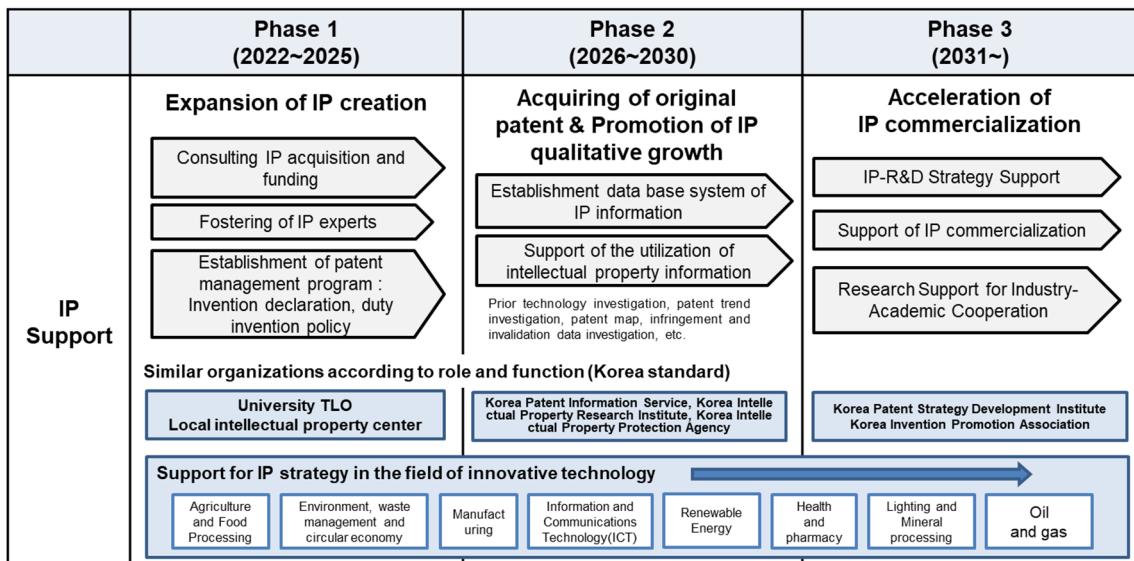
The support projects for patent creation and expansion increase patent productivity. In line with this, various technology commercialization support projects to advance the usability of the created IPs, such as additional R&D, technical feasibility analysis, business model setup, funding, and startup support, help occupy core and original patents in advance. Therefore, creating a powerful patent will enhance a company's competitiveness, and these programs will also stimulate companies to use their R&D performance and connect it to industries,

while attempting to grow based on technology, through which the constitution of the industries throughout the nation will be eventually reformed.

6.5 Roadmap

The level of Ghana's IP creation is similar to that of one local university or a RIPC in Korea. Therefore, Ghana should begin the roadmap from agriculture and food processing where the country has a high industrial capability, and then to heavy industry and high-tech intellectual property industries by fostering the workforce in patent fields. It also needs to introduce a duty invention system, which is expected to increase the needs of the

research institutes and industrial institutions for IP creation. When considering that social and economic concentration happens in the two largest cities in Ghana, the first step would install patent management agencies and nurture patent experts to expand IP creation. In the middle and long term, Ghana needs to operate general IP support projects related to patent information, policy, and protection, while securing original technology, which will help establish the capability for IP industrialization. When it is realized, Ghana's IP will contribute to advancing its national competitiveness and creating various new industries.



[Figure 5-11] Policy & Program Roadmap

7. Purchase Support Programs for Ghana

The distribution of Ghana's GDP across industrial sectors shows the service sector accounts for 59 percent, whereas 25 percent of the country's GDP comes from the manufacturing sector. As for manufactured goods such as cars and home electronics, the country is heavily dependent on imports. Despite being an oil-producing country, Ghana even imports petrochemicals as it is lacking processing technologies.

To address these problems, the Ghanaian government instated the 'One District-One Factory (1D1F)' initiative in 2017 to construct factories in 216 districts nationwide and thereby improve productivity. Recently, the country has developed the 'For Better Business Together' Initiative to make a breakthrough amid the COVID-19 pandemic.

For the purpose of achieving the SDGs (Sustainable Development Goals), this initiative aims to support linking young Ghanaian entrepreneurs with global entrepreneurs and investors around the world, revitalize Ghanaian local businesses, and search for solutions to tackle ongoing problems facing Ghanaian businesses.

7.1 Priority Purchase of Quality-Certified Products for Acquiring Manufacturing Technology

In regard to Ghana's determination to develop its manufacturing industry and promote science and technology, achieving public procurement-driven innovation does not seem to be a particularly challenging task. However, compared to Korean experience, Ghana's weak manufacturing sector that remains below the minimum required level for innovation may serve as a complicating factor.

The Korean model of innovative procurement has been developed through strong linkages with its technology certification scheme centered around manufactured goods because the

public procurement as a means of innovation has been concentrated on priority purchase of certain products.

As aforementioned, Korea's innovative procurement policy started with the preferential treatment of quality-certified products, which was originally introduced for the purposes of quality assurance and productivity improvement. With that said, the quality certifications in the early stage can be understood to be 'standards'.

Nowadays, 'quality-certified' goods refer to goods that have attained a higher level of technological superiority. However, in the early stage of Korea's economic development, quality assurance and productivity improvement were considered key objectives, so the public procurement market acted as a catalyst to induce active purchasing of those quality-certified goods.

In order for Ghana to achieve innovation in the manufacturing sector using public procurement, it is recommended that Ghana first establish a quality certification system for its products, enact laws mandating the purchase of these quality-certified products, and establish related support programs.

For this purpose, the GIRC Centre needs to set quality standards of major goods traded in Ghana and eventually develop a program that can prioritize such products in the public procurement market through cooperation with the Ghana Standards Authority, an institution responsible for conducting conformity assessments.

7.2 Ghana's Purchase-Conditioned Technology Development Program

Korea has implemented the purchase-conditioned technology development program by linking R&D outcomes with public procurement, and so this program is being used as an efficient mechanism for technology commercialization.

As a part of this program, SMEs propose products based on new technologies to public institutions, which then confirm their purchase of such products in advance. Then, these products are designated as the subject of R&D and the buyer institutions become eligible to receive the government support when they purchase the proposed products from SMEs.

Given the relatively small size of Ghana's R&D budget and activities, it may be difficult to implement such programs in a large scale. Instead, it would be possible to implement a pilot in some selective areas in the short run.

This program helps a supplier identify product groups that are well-aligned with its future technology development direction with high demand from the public institutions. Then, the supplier can proceed with R&D of such products based on the condition of pre-arranged purchase. In such a way, the program directly helps the supplier expand its markets for new technology products.

Collective efforts of various stakeholders are key to the success of this kind of program that links R&D projects with public procurement. For starters, a separate track for such a program needs to be established within institutions such as the GIRC Centre. It is also recommended to consider an additional support program for matching public institutions' demand with companies' supply.

In Korea, the government only provides funding support for R&D expenses of the selected projects under such a program. It is the suppliers' job to secure the letter of purchase confirmation from the buyer institutions. When the program is introduced to Ghana, it is important to make necessary changes to the program to meet the country's needs.

7.3 Innovative Procurement Based on Ghana's Current Industrial Foundations

Innovative procurement is not always confined to the sophisticated stage of technology development. The innate value of innovative procurement lies in the use of public purchasing power propelled by government funding in enhancing the quality of public

services and promoting technology and industrial innovation, thereby linking such power with support schemes for businesses.

The first two policy recommendations are considered direct support schemes for the manufacturing industry, though they can be classified as diversified forms of innovative procurement. In the meantime, the third policy recommendation involves developing an institutional and structural framework to use the public procurement mechanism as a means of innovative procurement. At the present, the Korean innovative procurement policy places its emphasis on i) planning or discovering demands by identifying public, national or social problems, ii) purchasing solutions that can address these problems, and iii) linking such demands to R&D.

In Ghana where infrastructure for the manufacturing industry is still underdeveloped, its service sector might offer higher development potential. So public procurement may play an important role in developing knowledge-based service industries such as the IT industry. In the case of Ghana which has a high dependency on imported manufactured goods, another possible approach would be to use a technology alliance or transfer as important procurement means to offset its import dependency.

For this purpose, the government of Ghana needs to identify and designate public demands within the GIRC Centre that have the potential to contribute to the national development of Ghana. In relation to this, various detailed approaches can be explored, such as linking the identified demands with R&D programs, establishing alliances with foreign companies in the areas of target purchases, and transitioning the purchase program into digital technologies including IT.

7.4 Roadmap

The current public procurement system in Ghana is quite different from that of Korea. Since public procurement is a state-controlled market, it is natural that the operational mode of public procurement differs between the two countries, reflecting their differences in policy environments such as their industrial foundations and economic structure.

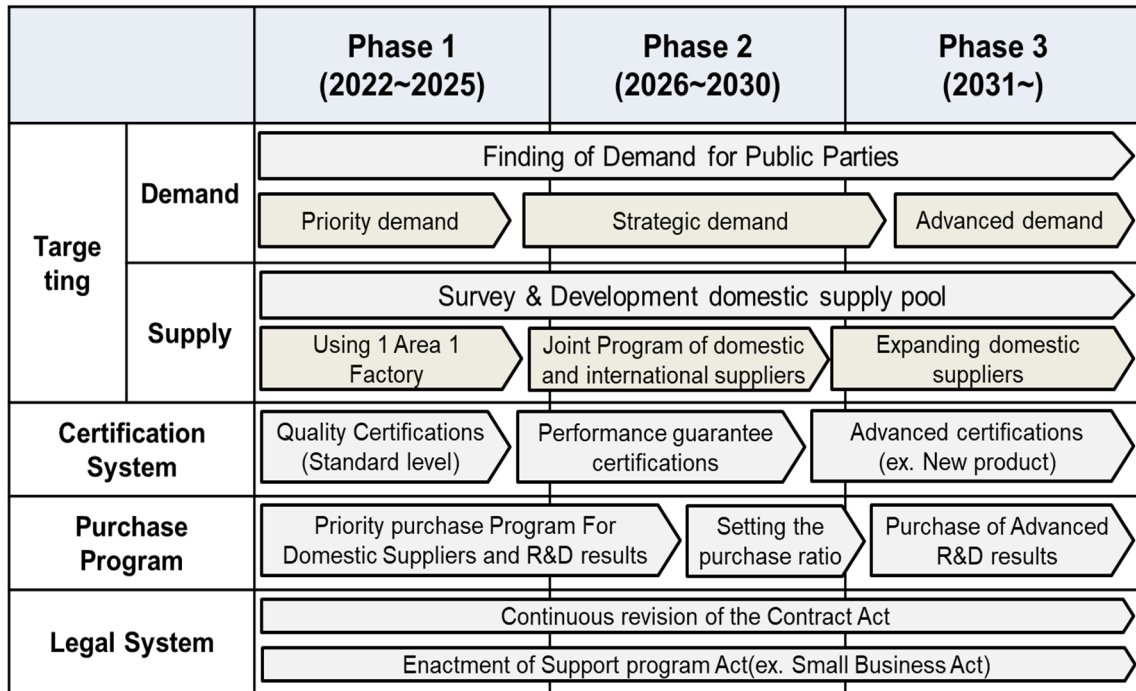
Despite these differences, Korea's experience still provides some useful insights for Ghana as Korea has benchmarked the public procurement system of the developed countries and has modified their system to make it fit for Korea.

The proposed roadmap in this study requires cooperation between different areas involving various stakeholders of public procurement. Thus, it would take a great deal of time to develop such a cooperative system. Therefore, a phased approach is recommended, so that Ghana can focus on its priority areas (i.e. areas with high dependency on foreign companies or areas with strong needs for building local technology & manufacturing capacity) for gradually transforming its public procurement system instead of changing everything at once.

1) Components of the Roadmap

The use of public procurement as a driving force for innovation requires the first step of identifying areas in need of public demands or the government purchasing power. Then, the next step will be matching such demands with local supply networks of locally produced goods.

From an institutional perspective, in order to take advantage of such programs as prioritized purchase or concentrated purchases, it is recommended to first establish a system for quality- or technology-certifications. Then, the government procurement program can be designed based on such a certification system. Lastly, it is recommended to build a legal and regulatory framework for operating the government procurement program.



[Figure 5-12] Proposed Roadmap for Using Ghana's Public Procurement

2) Phase1 (Quality Certification System and Priority Purchase)

Korea also implemented the quality certification system to enhance the competitiveness of the locally manufactured goods before its industries and technologies became advanced. Based on this system, the Korean government actively purchased locally manufactured goods in public procurement.

The key to this process is the establishment of certification and standardization systems. Whether to use international or national standards here is a decision left to the policymakers.

3) Phase2 (Linking R&D with Public Procurement)

Once the manufacturing foundations and quality certification system are established, an approach of linking public procurement with R&D program can be considered, so that R&D outcomes can be efficiently utilized in the public arena. At this stage, it might be necessary

to build an additional certification system for the purpose of assessing product excellence in performance going beyond the basic quality certification.

In this case, quality standards equivalent to global standards can be used or a separate standard scheme can be developed to identify target purchase items.

In addition, a joint venture between domestic and foreign firms can be considered as a policy means to improve the added-value of locally manufactured goods in the Ghanaian public procurement market.

4) Phase3 (Achieving Technological Progress and Solving Public Problems)

Discussions can be made on the ways to achieve the ultimate goal of public procurement and the qualitative improvement of public services. One possible approach would be reflecting technological progress into public services by linking it to innovation policies.

It can only be possible when the domestic industrial foundations are fully established and local companies attain a higher level of technological capabilities, which will also facilitate the linkage between public procurement to national R&D policies.

8. Accreditation & Certification Support Programs for Ghana

Based on the analysis of the current state of Ghana's certification programs in Chapter 2 and Korea's certification-related experience in Chapter 3, the following policy suggestions have been developed for Ghana. For the purpose of spurring S&T innovations in Ghana, this section presents certification-related policies enabling highly efficient certification programs to be implemented by the GIRC Centre with minimum costs.

8.1 Conducting Certification Needs Analyses of Ghana's Technologies and Industries

Before planning its certification support programs, the GIRC Centre needs to accurately identify the certification needs for technologies, products, and services in Ghana's strategic industries prioritized for achieving the country's technology innovation.

Certification is a critical service industry that ensures quality of products, technologies, and services, verifies standards, and guarantees their credibility using standardized specifications. Like the conformity assessments of products mandated by the law, certification requires not only expensive facilities, a large pool of human resources, and quality systems for testing in corresponding fields but also a structure that can generate steady income for the stable operation of certification services. Therefore, it is necessary for GIRC Centre to clearly identify the certification demand for technologies, products, and services in Ghana's strategic industries of the technology and innovation.

It is recommended that a needs analysis be conducted every year in corresponding technologies and industries in tandem with universities, research institutes, and companies which need certifications for their engagement in technology marketing, public procurement, technology evaluation, and technology funding. Along with the certification needs analyses, Ghana also needs to accurately understand the current state and development potential of its technologies as well as its industries.

8.2 Building a Cooperative Network with the Ghanaian Accreditation Body

Currently the government of Ghana is operating a formal accreditation body as a separate entity and is now preparing to sign an MRA or MLA with ILAC/IAF. It is recommended that the Ghanaian accreditation organization take the full responsibility for ensuring the compliance of the country's certification system with international standards while the GIRC Centre becomes in charge of the introduction and operation of the certification programs. During all this, it is necessary to build a quality management system in compliance with ISO/IEC standards to secure the credibility of Ghana's certification programs. For this purpose, it is desirable to build a cooperation mechanism between the GIRC Centre and Ghanaian accreditation bodies. Depending on the certification programs, the GIRC Centre may use the existing accreditation scheme of the accreditation body or build its own accreditation (or designation) program to accredit testing and certification organizations. When operating its own accreditation program, the GIRC Centre needs to build and operate its own quality management system in compliance with ISO/IEC standards. The cooperation between the GIRC Centre and Ghanaian accreditation bodies can be pursued in two ways. In the first approach, the GIRC Centre can commission the accreditation or designation mechanism of its certification programs to the accreditation bodies. The second approach would involve outsourcing of the accreditation or designation system by the GIRC Center through counseling or consulting with the accreditation bodies.

Standards	Fields
ISO/IEC 17011	Quality management system of certification bodies
ISO/IEC 17065	Quality management system of certification bodies
ISO/IEC 17025	Quality management system of certification bodies

8.3 Certification Program for New Technology Innovations in Ghana

The GIRC Centre is expected to emerge as a key enabler of Ghana's S&T innovations, which highlights the need to develop and implement a series of policies for achieving such

innovations. In fact, the development of certification programs is one of the important policy measures that can foster technology innovation in Ghana. However, the operation of certification programs warrants expensive test facilities, a large pool of high-caliber human resources, and quality management system. Considering the limited resources of the GIRC Centre currently under its possession, it would be a bit of a stretch for the GIRC Centre to build and operate certification programs on its own initiative. Therefore, it is necessary to propose certification programs suitable for the GIRC Centre's current conditions.

Certification is generally divided into legally mandatory or optional certifications. While the legally mandatory certification is conducted as a kind of conformity assessment to validate product safety by ministry and industry, the legally optional certification is often provided for those who need certification for their engagement in technology marketing, public procurement, technology evaluation, and technology funding. In addition, there is the voluntary certification run by certain groups such as forums in the private sector for the purpose of product marketing, though it is not legally based.

In the case of the legally mandatory certification conducted for the purpose of guaranteeing the safety of products developed, manufactured, imported or circulated in the market, it is recommended that the certification process be fully handled by the ministry concerned. In the meantime, as more products will be developed in the private sector as a result of accelerating industrial development, the demand for certification is also expected to increase for the purpose of marketing such products. Therefore, it is recommended that the certification of these products be handled voluntarily by stakeholders of the private sector. However, it is desirable for the GIRC Centre to develop and operate certification programs in industries such as agricultural and food processing, environmental waste recycling, manufacturing, ICT, and new recyclable energy, which will facilitate technology innovation in these critical industries. There may already exist numerous legal certification programs in Ghana that can possibly support such certification needs of the GIRC Centre, but these certification programs may not be effective in supporting the GIRC Centre due to their limited budget. Instead, the most desirable approach that also serves the objectives of the GIRC Centre would be the development and operation of the 'New Technology Certification Scheme' along with the 'New Product Verification Scheme', which confirms the application

of the certified new technologies to the products. By benchmarking the Korean KS Certification Program, the government of Ghana may consider the operation of a Ghanaian standard certification program to circulate products in compliance with Ghana's national or group standards. In such a case, it is also recommended that the operation of the program be fully handled by industry-related ministries. In addition, legally optional certification programs can be separately operated in relevant industries. Again, it is recommended that the operation of these programs be handled by industry-related ministries.

The New Technology Certification Scheme can be applied to technologies whose development has been successfully completed after securing quantitative evaluation data through manufacturing, testing, and operation of prototypes based on the established technological theories. Technologies from any industries are subject to this scheme, and so for the certification of these technologies, a set of evaluation criteria will be developed and a review committee will be set up with experts in corresponding technologies. Considering such a wide scope of technologies and high level of technological expertise in evaluation, this scheme has an advantage of spurring major technology innovations with limited budgets. The New Product Verification Scheme is a kind of follow-up program of the New Technology Certification Scheme as it certifies products that apply new technologies certified by the New Technology Certification Scheme. Since this section only presents policy suggestions for the New Technology Certification Scheme, further discussions need to be made on these two schemes in the future through consultation with the GRIC Centre. In addition to these two schemes, certification programs for venture businesses and innovative companies can also be considered.

8.4 Establishment of a Cooperative Mechanism with Universities and Research Institutes

Certification warrants a large pool of technologies and human resources, while the level of secured technologies and human resources may vary across different certification programs, but they all need to meet the requirements of the ISO/IEC standards. For the successful

operation of the aforementioned New Technology Certification Scheme and New Product Verification Scheme, it is critical to secure not only a pool of high-caliber human resources who will run these schemes but also a group of experts who will serve as members of a review committee.

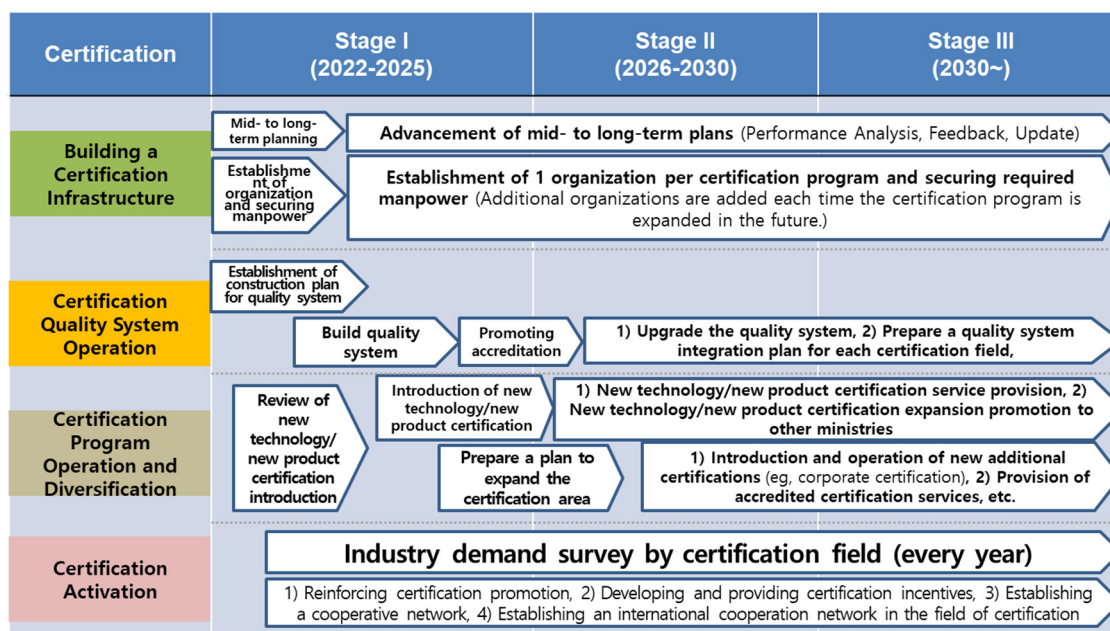
For this purpose, it is recommended to secure human resources from universities and research institutes available for cooperation with the GIRC Centre. Considering the rising need for more experts who can support the development of additional certification programs, the expansion of certification services, and development & education of certification-related human resources, it is critical to form this kind of cooperation networks.

8.5 Activating Certification Services

The development and operation of certification programs that can drive Ghana's economic development is extremely important since they help Ghana maintain its competitiveness by promoting technology development and commercialization. As important as the development and operation of these certification programs is the development of a strategy to disseminate those certification programs. A successful dissemination strategy needs to offer a wide range of benefits applicable to the certified technologies, products, and services. For this purpose, the GIRC Centre needs to maintain close cooperation with state-run ministries and other innovation stakeholders. Particularly for the dissemination of products applying new technologies, certification programs need to offer diverse incentives such as the committed purchase by public institutions, preferred treatment when applying for government programs, designation of the certified products as excellent products by the government in public procurement, and preferential treatment when applying for technology funding programs.

8.6 Roadmap

The following is the proposed roadmap for introducing certification programs at the GIRC Centre in Ghana.



Firstly, for the establishment of the certification infrastructure, it is recommended to take the following approach such as; i) Developing a mid- and long-term plan for certification programs under the lead of the GIRC Centre, analyzing the performance of the plan, providing feedback, and continuously upgrading the plan, ii) securing necessary certification bodies and human resources according to the mid- and long-term plan, iii) and preparing for the operation of certification programs. For example, it is necessary to secure one operational unit per certification program. When new certification programs are added later on, additional relevant units will be established and operated.

Secondly, for the operation of the certification quality management system, it is recommended to develop a plan to build the quality management system by mid-2023, start building the system in the later half of 2023, and upgrade the system through cooperation with accreditation bodies, when necessary. Afterwards, the certification quality

management system can be accredited by Ghanaian accreditation bodies and be continuously upgraded to achieve a higher level of excellence.

Thirdly, for the introduction of the certification programs, it is recommended to start the introduction and operation of the New Technology/ New Product Certification Schemes at the GIRC Centre in 2024 after a review process and then continuously explore ways to promote these new certification services. At the same time, foundations need to be built for introducing additional certification programs such as the Corporate Certification.

Lastly, as an attempt to promote certification services, the government of Ghana is recommended to conduct an industrial certification needs analysis every year in relative certification fields, and develop and offer various incentives for each field.

9.

I-U-R Collaboration Support Programs for Ghana

Often, the promotion of I-U-R collaboration requires the government's careful planning and well-designed incentive programs to foster collaboration between these key stakeholders of industries, universities, and research institutes. The government also needs to secure resources capable of conducting effective collaboration. In addition, information necessary for conducting collaboration should be provided. The sustainability of collaboration should also be ensured, so that collaboration is not performed as a one-time event. Moreover, obstacles to I-U-R collaboration need to be removed to increase the synergies between the innovation stakeholders. Since Ghana is not an exception, these key factors affecting the I-U-R collaborations are also applied to Ghana. Therefore, policy suggestions have been developed on each of these five factors critical for promoting I-U-R collaboration.



[Figure 5-13] Five Key Factors Affecting I-U-R Collaboration

9.1 Mid- & Long-term Planning for I-U-R Collaboration by the Ghanaian Government

To be sure, the government of Ghana is fully aware of the importance of I-U-R collaboration and its impact. Thus, it is now necessary for the government of Ghana to carefully design the overall direction, detailed objectives, and budget requirements of the I-U-R collaboration through mid- and long-term planning. This mid- and long-term plan for I-U-R collaboration needs to include components such as the vision, objectives, and detailed strategies for developing creative talents, creating high added value through the transfer and commercialization of potential technologies, and building a startup support system that can help anybody kick-start a business. In addition, the government of Ghana needs to build infrastructure for promoting I-U-R collaboration, and thus enhance the trust-based relationship between its stakeholders as well as the sustainability of their collaboration. It is recommended that the government of Ghana refer to Korea's Five-Year Basic Plan for Industrial Education and I-U-R Collaboration reviewed early on. Especially for supporting the startup activities and promoting a systematic approach to entrepreneurship education, the government of Ghana needs to design the 'Five-Year Plan for Promoting Entrepreneurship Education at Universities'. Within the framework of these plans, it would be important to develop effective policies and action plans that would produce practical and tangible outcomes through extensive participation and cooperation.

In this context, the government of Ghana needs to develop the I-U-R collaboration programs by each ministry, support the implementation of these programs by allocating a sizeable budget, and coordinate and share the outcomes of the programs across ministries. For your reference, the following Table presents a brief overview of the Korean government's I-U-R collaboration programs and its budget allocation by each ministry as of 2021, which has been on the rise recently.

[Table 5-4] Government Budget for I-U-R Collaboration & the No. of I-U-R Collaboration Programs by Ministry in Korea

USD1.5 billion (2021)
42 IUR Collaboration Programs (2021)
Ministry of Education (MOE): 15 programs
Ministry of Science & ICT (MSIT): 10 programs
Ministry of SMEs & Startups (MSS): 6 programs
Ministry of Trade, Industry & Energy (MOTIE): 6 programs
Ministry of National Defense (MND): 2 programs
Ministry of Employment & Labor (MOEL): 1 program

Source: Choi, Jong-in (2021)

9.2 Developing Creative Talents by Diversifying & Improving Industrial Education

Developing creative talents based on the needs of the Ghanaian industries is one of the most important policy challenges facing the government of Ghana. Therefore, it is recommended that the government of Ghana take a series of human resources development initiatives in a virtuous cycle including the development and operation of demand-based curriculums, and the operation of internship programs, Capstone Design Program, and other entrepreneurship courses. At the same time, the government of Ghana also needs to support the training and human resources development needs of the newly emerging industries, so that the country becomes capable of preemptively responding to the current and future restructuring of Ghanaian industries.

Above all, Ghana needs to enhance the capacity of the Ghanaian talents based on the 4Cs (Critical Thinking, Communication, Creativity, & Collaboration), implement self-driven capacity-building projects, and improve educational courses and environment, so that they can contribute to the capacity enhancement of Ghanaian industries. In particular, the internship programs need to be expanded and diversified, so that they can provide students with more opportunities to develop their vocational skills and competencies, while supporting industries' acquisition of skilled talents who can be immediately deployed in the field. For this purpose, various teaching methods can be employed including the project or problem-based learning (PBL) and flipped learning. These approaches would also help students develop their capacity to design their future on their own initiative in the long run.

9.3 Creating High Added Value through Technology Development, Transfer and Commercialization

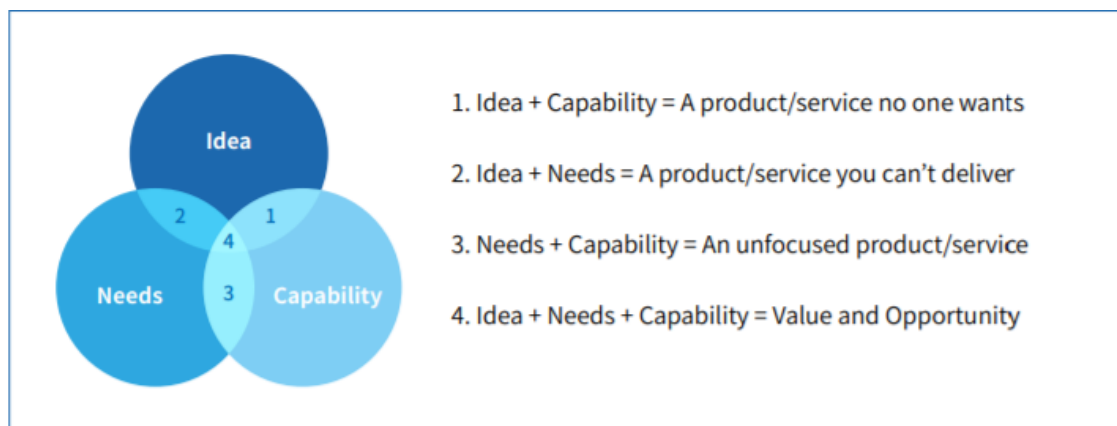
It is necessary to identify issues the Ghanaian industries and companies are grappling with and address the identified issues through open innovation R&D. For example, problems arising in the process of product development or mass production can be addressed by discovering domestic or foreign technology solutions through I-U-R collaborations. The promotion of open innovation R&D or 'buy R&D' will introduce quick remedies and enhance trust between key stakeholders of industries, universities, and research institutes. For effective support of technology transfers and commercialization from universities or government research institutes to industries, it is recommended that the government of Ghana undertake a pilot program for a virtuous cycle by actively pursuing inter-university technology packaging (i.e. interdisciplinary approach to converging multiple technologies), converting ideas proposed by faculties and graduates students into reality, and reinvesting technology fees from technology commercialization into follow-up projects.

Benchmarking international best practices in technology commercialization and applying foreign knowledge and know-how to technology commercialization educational programs hosted by Ghanaian universities and companies will surely make an immediate and visible impact in the government of Ghana's efforts to ramp up technology commercialization.

9.4 Establishing a Startup Support System

It is recommended to establish a lifecycle-based startup support system, enabling future entrepreneurs to learn entrepreneurship, experience entrepreneurial activities, and develop a business mindset in the early years of their life. Building such a system requires the provision of startup-related educational courses and funding support for university students. When students are provided with opportunities of personally experiencing a whole process of discovering their needs or deficiencies, generating their own unique ideas, and developing and testing business models by using related knowledge and capabilities, they are likely to become high-caliber talents with entrepreneurship. The INC model that

stands for the three elements of Ideas, Needs, and Capability can be a good reference method for creating values and realizing opportunities.



Source: Choi, Jong-in (2018)

[Figure 5-14] INC Model

Administrative and funding support that leads to actual startup is also needed. In its efforts to provide a wide range of startup-related hands-on experience for future entrepreneurs, the government of Ghana needs to support the designing and operation of team project-based courses involving experts from industries, actual launching and operation of startups, and expansion of startup-related field experiential programs at home and abroad. The government of Ghana can also provide potential entrepreneurs with unique learning experiences by offering a virtual reality-based startup environment and corporate management courses while supporting the production of prototypes using makerspace.

9.5 Expanding Infrastructure and Budget to Promote I-U-R Collaboration

In this regard, the government of Ghana needs to first establish and operate a dedicated organization for I-U-R collaboration. For example, an institutional framework needs to be established to launch and operate the I-U cooperation centers on university campuses and strengthen their functions. At the same time, it is necessary to build specialty areas of each university through the linkages between companies and university labs by strengthening

the function of I-U cooperation centers as an entity for promoting cooperation with companies. The Ghanaian GRIs are also required to set up and operate internal organizations dedicated to I-U-R collaboration and enhance mutual trust by expanding human resources exchanges with universities and industries.

It is also recommended to diversify the employment status of university faculties. The government of Ghana can consider various options including the introduction of the I-U dedicated professor system and provision of R&D incentives for researchers from GRIs engaged in I-U-R collaboration to take root and disseminate an HR system conducive to I-U-R collaboration. Especially, efforts need to be made to identify and address regulatory obstacles to I-U-R collaboration. For this purpose, various inputs from the frontline stakeholders need to be gathered to identify and lift unnecessary regulations.

At the same time, the government of Ghana needs to set a vision of mutually beneficial development for universities and regional communities, and have universities design and implement a series of I-U collaboration programs aiming to solve various issues of the regional communities, considering the regional context. As for the evaluation of these programs, it is necessary to develop a set of performance indicators, conduct regular assessments using these performance indicators, and provide incentives and rewards for high-performers of I-U collaboration. In fact, building and strengthening trust in I-U collaboration requires budget support, which can be secured from the government of Ghana and companies as well as foreign and international organizations.

9.6 Roadmap

Based on the aforementioned policy suggestions, the following actionable roadmap can be proposed from the short-, mid-, and long-term perspectives. This roadmap includes the key tasks of i) developing creative talents, ii) promoting technology development, transfer, and commercialization, iii) establishing a startup support system, and iv) securing the infrastructure and budget for I-U-R collaboration.

The first task of developing creative talents entails key activities such establishing the entrepreneurial leadership at industries, universities, and research institutes, developing demand-based curriculum, and introducing and promoting internship programs and the Capstone Design Program. These activities will surely support the government of Ghana's efforts to develop creative local talents. In addition, the government of Ghana will also implement such activities as the introduction of the I-U Collaboration Dedicated Professor Program and the expansion of human and physical resources exchanges between industries and universities, which will help establish the I-U collaboration culture at the Ghanaian universities.

Secondly, the government of Ghana needs to actively pursue technology development, transfer, and commercialization. The most wanted technologies by the Ghanaian companies will be developed at universities and research institutes, who will then transfer these technologies to companies in a bid to meet the corporate needs. Along with these efforts, the government of Ghana will also strive to substitute foreign imported technologies with locally developed technologies and pursue the joint development of new technologies. This second task can be accomplished through the implementation of some additional key activities such as conducting open innovation R&D, supporting companies with inter-university technology packaging, promoting technology development through joint international research, promoting investment in advance technology development, and strengthening capacity for global technology commercialization. The expanded scope of technology commercialization will help build a virtuous cycle linking technology, product, and market, and finally generate tangible outcomes.

The third task of establishing a startup support system also involves a series of key activities including the introduction of advanced entrepreneurship educational programs, internalization of lessons learned from these educational programs, and foundation-building for startups led by university faculties and researchers. The early startup funding can be first secured from international organizations, and then funding sources can be expanded to include the government of Ghana and private sector. The government of Ghana can also develop internal entrepreneurship educational programs, develop successful startup cases led by university faculties and researchers, and bring up local angel

investors and venture capitalists. In the long term, the government of Ghana will be able to instate a new startup culture at universities, research institutes, and companies.

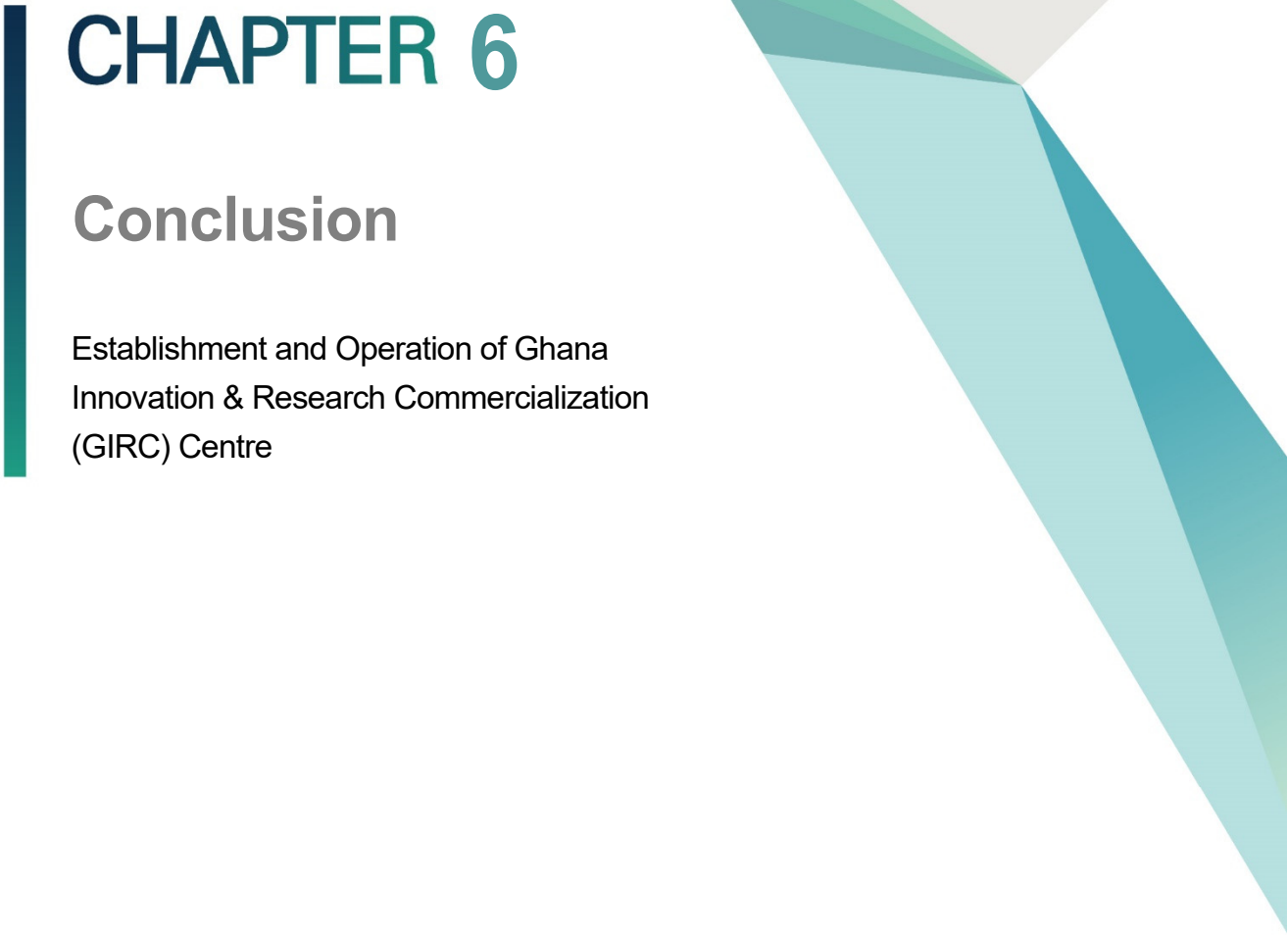
Fourthly, the government of Ghana needs to secure the infrastructure and budget for I-U collaboration. In this regard, the government of Ghana first needs to map out the mid- and long-term development plan for I-U-R collaboration, implement the plan step-by-step, and gradually expand the budget for I-U collaboration by securing funding from the government, international organizations, and local & foreign companies. As a part of this task, the government of Ghana will also develop experts affiliated with I-U cooperation centers on campuses and strengthen trust in I-U collaboration for ultimately achieving the qualitative improvement of these centers. Considering the importance of expanding a pool of I-U collaboration experts, the government of Ghana needs to establish an effective reward system and expand the size of reward for these experts, establish an I-U collaboration-friendly culture, and promote human resources exchange between local and overseas I-U cooperation centers.

[Table 5-5] Roadmap for Promoting I-U-R Collaboration

Task/ Stage	Short term	Mid term	Long term
Developing creative talents	• Immediately establish leadership at industries, universities, & research institutes. • Introduce the I-U-R collaboration-dedicated professor program. • Develop need-based curriculum • Expand short-term internship programs and the Capstone Design Program • Introduce the PBL and flipped learning methods	• Establish entrepreneurial leadership. • Increase the number of the I-U-R collaboration-dedicated professors and restructure their evaluation and compensation system • Develop and operate curriculum based on future needs • Implement long-term internship programs and the Capstone Design Program • Promote human resources exchange by dispatching professors on	• Promote entrepreneurial leadership. • Change the status of full-time faculties into I-U-R collaboration-dedicated professors. • Redesign and complete curriculum based on future needs. • Build a virtuous cycle linking long-term internship programs, Capstone Design Program, & startups. • Promote human resources exchange between universities, research institutes, and industries.

Task/ Stage	Short term	Mid term	Long term
		the sabbatical leave to industries. Introduce the PBL and flipped learning methods.	Adopt the PBL and flipped learning methods and entrepreneurship education.
Promoting technology development, transfer, and commercialization	- Systematically discover corporate needs. - Substitute foreign imported technologies with locally developed technologies. - Pursue technology development and transfer based on market demands. - Build technology commercialization capacity.	- Pursue open innovation R&D. - Promote inter-university technology packaging. - Promote technology development through joint international research. - Promote investment in advanced technology development. - Strengthen capacity for global technology commercialization.	- Secure re-investment opportunities by expanding income from technology transfer fees. - Build a virtuous cycle linking technology, product, and market by expanding the scope of technology commercialization. - Generate tangible outcomes from global technology commercialization.
Establishing a startup support system	- Introduce advanced entrepreneurship educational programs and internalize lessons from these programs. - Lay the foundations for startups led by faculties and researchers. - Source early startup funding from the government and international organizations.	- Develop internal entrepreneurship educational programs. - Develop successful startup cases led by faculties and researchers - Develop local angel investors and venture capitalists - Provide startup field training programs at home and abroad.	- Develop star entrepreneurs. - Establish a virtuous cycle of startup, market success, and exit - Establish a culture of startups led by faculties and researchers.
Expanding I-U collaboration infrastructure & budget	- Develop a mid- and long-term plan for I-U-R collaboration - Gradually increase the government budget - Secure collaboration budget from local	- Implement the mid-and long-term plan for I-U-R collaboration and evaluate its outcomes - Increase the I-U-R budget of various state-run ministries.	- Develop and implement the 2nd phase of the mid-and long-term plan for I-U-R collaboration - Attract corporate R&D centers to university

Task/ Stage	Short term	Mid term	Long term
	companies and overseas institutions through budget linkage • Secure and develop experts for university-affiliated I-U cooperation centers, • Build and strengthen trust in I-U collaboration, • Establish the I-U cooperation centers at universities.	• Secure stable funding from local and foreign industries • Establish a reward system for I-U collaboration experts and expand the size of the reward. • Establish the I-U collaboration culture. • Promote human resources exchange between local and foreign I-U collaboration centers.	campuses and operate open campuses • Open university labs to SMEs to promote cooperation with SMEs • Expand global partnership. • Establish leading I-U cooperation centers and secure local and foreign sponsors of these centers. • Pursue the qualitative improvement of I-U cooperation centers.

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CHAPTER 6

Conclusion

Establishment and Operation of Ghana
Innovation & Research Commercialization
(GIRC) Centre

Chapter 6. Conclusion

1. Summary

This project was carried out as the first year (2021) activities of the three-year program titled the ‘Establishment and Operation of Ghana Innovation & Research Commercialization (GIRC) Centre’. The goal of the first year activities is to analyze the Technological Innovation Support Programs (TISPs) as the boundary conditions affecting the establishment and operation of the GIRC Centre and present important Korean experience-based implications for the government of Ghana. To achieve this goal, the project team divided the Ghanaian TISPs into nine themes and analyzed their current state. Based on these analyses, Korean experts have presented insightful Korean experience-based implications for the government of Ghana.

As presented in Chapter 5, the project team has come up with numerous policy suggestions by theme area. For the effective operation of TISPs based on these policy suggestions, what should be the priorities of the government of Ghana? Firstly, the government of Ghana needs to practice a strong willingness for the realization of ‘STI-led Economic Growth’. Korea’s per capita GDP in the 1960s stood merely at USD 79, which was even lower than that of Ghana or Sudan in the same period. After 36 years under Japanese colonial occupation from 1910 to 1945 and the outbreak of the Korean war in 1950, the country was in complete ruins and its economy was mired among the ranks of the world’s poorest countries. However, in the 60 years since extreme poverty, Korea has ultimately emerged as the world’s tenth largest economy with its per capita GDP reaching USD 31,000 in 2020. Behind this remarkable growth was the strong policy vision and philosophy of the deceased former President Park Chung-hee who laid the foundation for the country’s economic

growth during the 1960s and 1970s. As the only available policy option to overcome poverty, he presented the vision of ‘founding the nation based on science and achieving technological independence’. In the 30 years after this vision was first announced, this vision was finally achieved in the 1990s. Korea has become the first country which has successfully transformed itself from a recipient of international aid to a donor nation. Since 2019, President Nana Akufo-Addo of Ghana has been actively advancing the ‘Ghana Beyond Aid’ Initiative, which the project team praises as a well-received policy direction for the country.

What remains now is the real question of how to achieve this goal of ‘going beyond the aid’. It is our strong belief that the answer to this question can be found in Korea’s experiences, which highlight the importance of the ‘strong political will for STI-led economic growth’ by a country’s leaders along with high-ranking government officials.

Secondly, Ghana needs to intensify its R&D investment. Without proper funding or investment, it would be impossible for a nation to execute its visions, philosophies or policies regardless of their level of excellence. The Korean government set a vision of achieving ‘STI-led economic growth’ in the 1960s and ever since has strived to realize this vision through intensive R&D investment. For example, in the 1960s, the Korean government received aid from USAID in exchange for dispatching its troops to fight in the Vietnamese War and invested the aid money into establishing the Korea Institute of Science & Technology (KIST), the first Korean government-funded research institute. The reparation fund Korea received from Japan for its colonial occupation in 1965 was also used for the construction of POSCO, a steel-making company which served as the stepping stone for Korea’s industrial development. As these examples show, all the available funds the Korean government could mobilize were invested in building STI infrastructure. Since 2009, Korea has emerged as the world’s leader in terms of GERD-to-GDP ratio (namely gross domestic expenditure on R&D as a percentage of GDP). As of 2019, Korea has ranked top with GERD/GDP ratio of 4.64 percent, followed by Israel. In contrast, Ghana’s GERD/GDP ratio remains at a mere 0.4 percent. The government of Ghana has reportedly developed a plan for raising an STI fund of approximately USD 50 million as of 2017, which is believed to be insufficient

to achieve the goal of 'going beyond the aid'. If the government of Ghana plans to pursue 'STI-led economic growth' as a strategy to realize the 'Ghana Beyond Aid' Initiative, it needs to place its priority on strengthening R&D investment. The immediate target for the government of Ghana in realizing this initiative will be its achievement of a GERD/GDP ratio of 1 percent.

Thirdly, the government of Ghana needs to pursue the strategy of 'no more plan, just implementation'. As for the numerous policy suggestions that have been identified based on Korean experiences, one of the most important factors to be taken into account is the 'culture' or the 'job attitude' of the government officials. The most frequently spoken expression that people often hear in Korea is 'hurry, hurry', which characterizes its so-called 'face-paced culture'. In other words, the Korean officials all share the value of 'immediately implementing plans as soon as they are developed.' In contrast, most of the developing countries have shown limitations in implementing plans, though they map out so many plans, which infers that developing countries can get mired in the so-called 'only talk, no action culture'. Then, what is the current situation in Ghana? If the government of Ghana wants to achieve the vision of the 'Ghana Beyond Aid' Initiative, its policy priority should be placed on sharing and disseminating the value of 'immediately implementing plans once they are developed' among its government officials.

2. 2nd Phase Project (2022)

The first phase of the project was focused on analyzing Ghana's TISPs as the boundary conditions affecting the establishment and operation of the GIRC Centre and presenting important implications of Korean experiences for the government of Ghana. In short, the current stage of the project has been focused on identifying policy suggestions that can be referred to by MESTI, the lead ministry in charge of the GIRC Centre, and state-run ministries concerned.

The second phase of the project in 2022 would be focused on developing a master plan for the successful establishment and operation of the GIRC Centre based on the outcomes of the first phase. The second phase would entail 'as-is' analyses of the vision, mission, strategy, structure, budget, human resources, and networking of the GIRC Centre to build its internal system and present its development direction. The second phase activities would also include the development of a roadmap for the Centre's operational program by stage.

Above all, a successful execution of the project requires a clear and accurate understanding of the current state of the GIRC Centre concerning the Centre's establishment and operation. As a result, it is vital to understand the government of Ghana's stance and other relevant information that may have important implications for establishing the Centre including the background & rationale, progress, and current state of the Centre's establishment. During the first year, the project team had to face difficulties due to the travel restrictions imposed by the pandemic. It is our ardent hope that more frequent and close communications be conducted in 2022, the second year of the project, through onsite travels and visits with a fast-paced recovery from the pandemic.

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